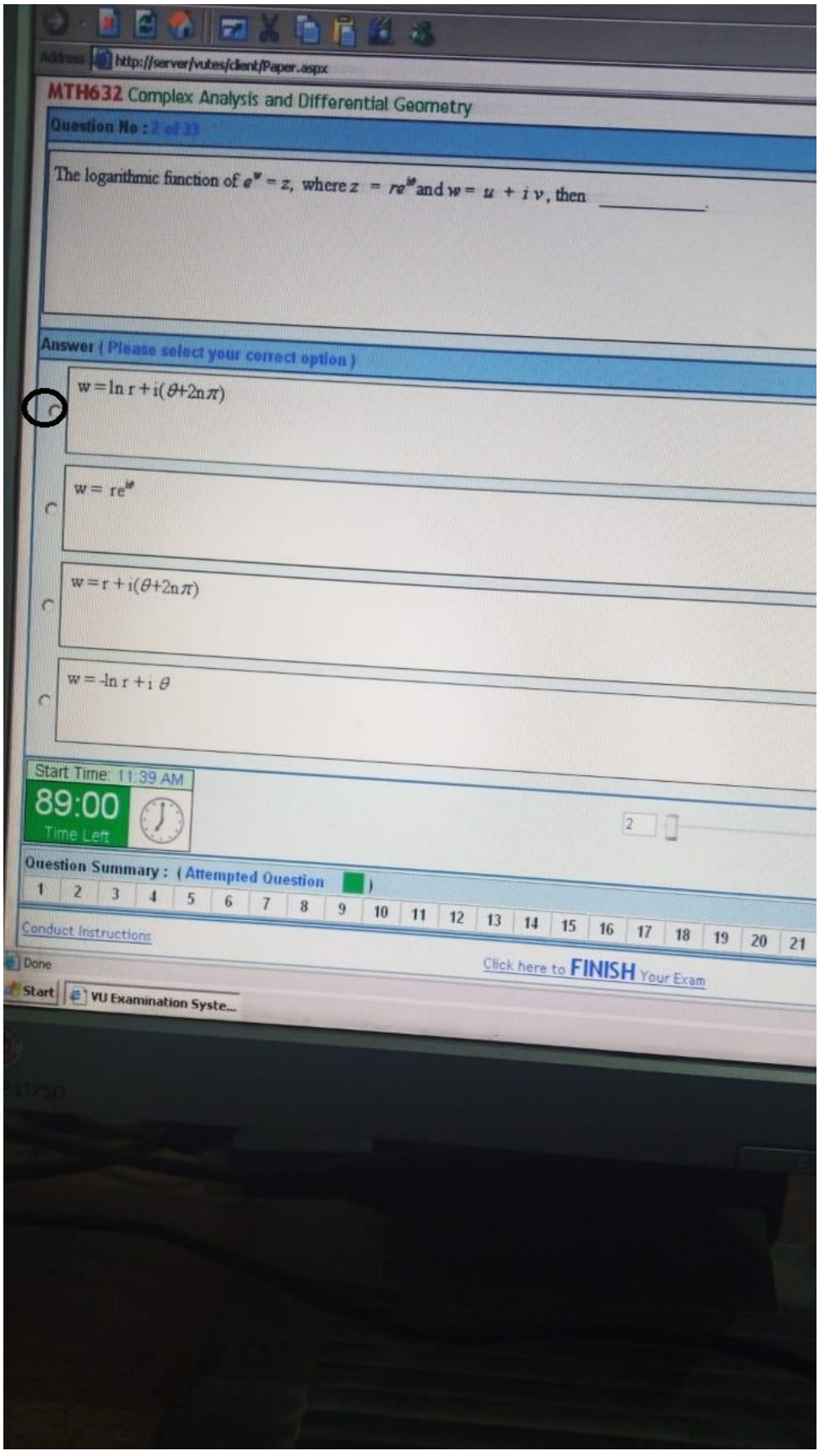
MTH632

6-3-21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MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then _____.
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{i\theta}$
$w = r + i(\theta + 2n\pi)$
$w = -\ln r + i\,\theta$
Start Time: 11:39 AM
89:00
Time Left
Question Summary : (Attempted Question)
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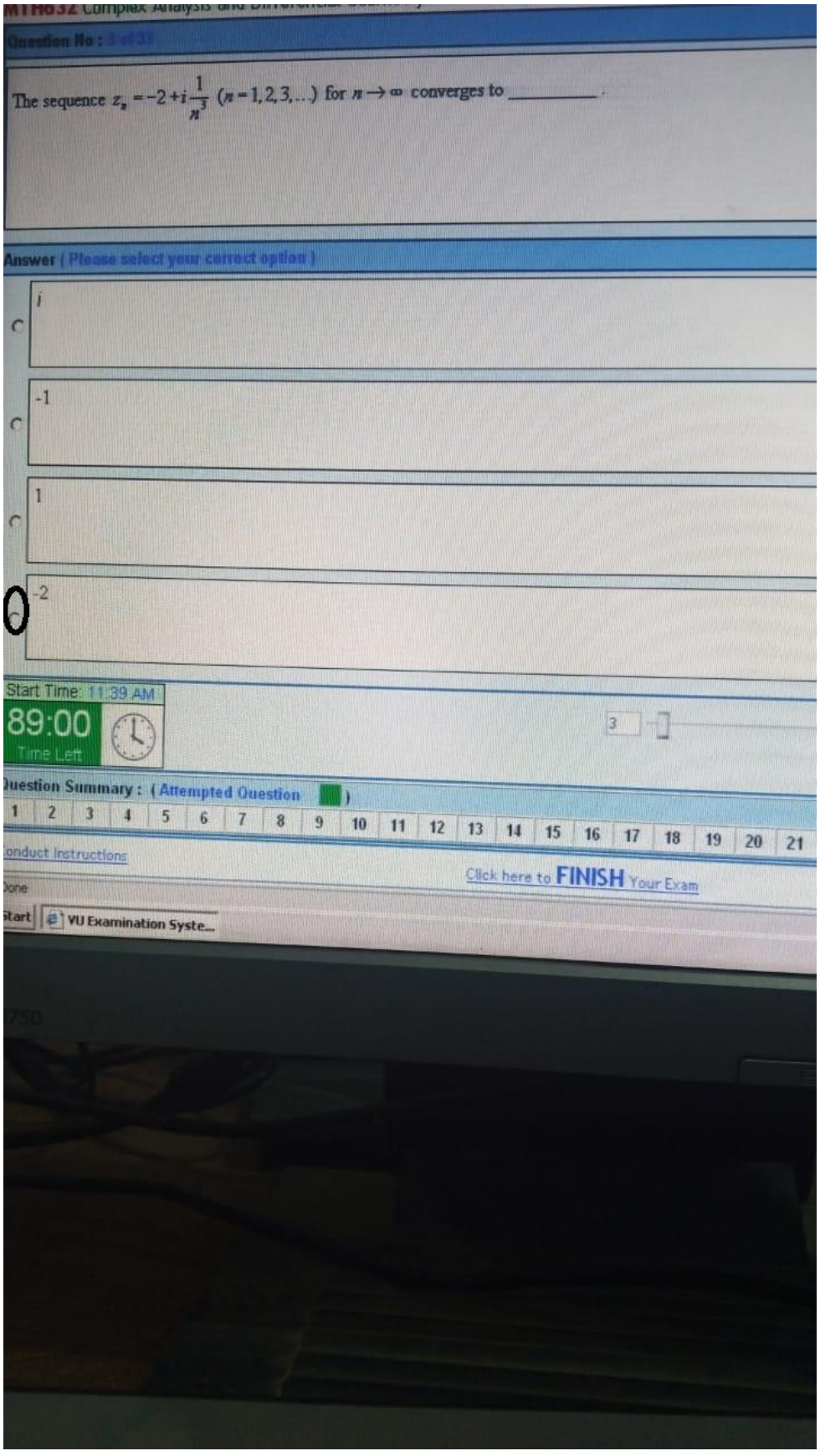
MTH632 Complex Analysis and
Question No :
The sequence $z_n = -2 + i\frac{1}{n^3}$ $(n = 1,2,3,\ldots)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
i
-1
1
-2
Start Time: 11:39 AM
89:00
Time Left
Question Summary : (Attempted Question)
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MTH632 Complex Analysis and Differential Geometry

Question No : 1 of 33

In the complex valued function $U(x, y)$ is harmonic in some domain , if _____.

Answer (Please select your correct option)

- $u_{xx} - u_{yy} = 0$
- $u_{xx} + u_{yy} = 0$
- $u_{xx} = u_{yy} = 0$
- $u_{xx} = u_{yy}$

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Question Summary : (Attempted Question)

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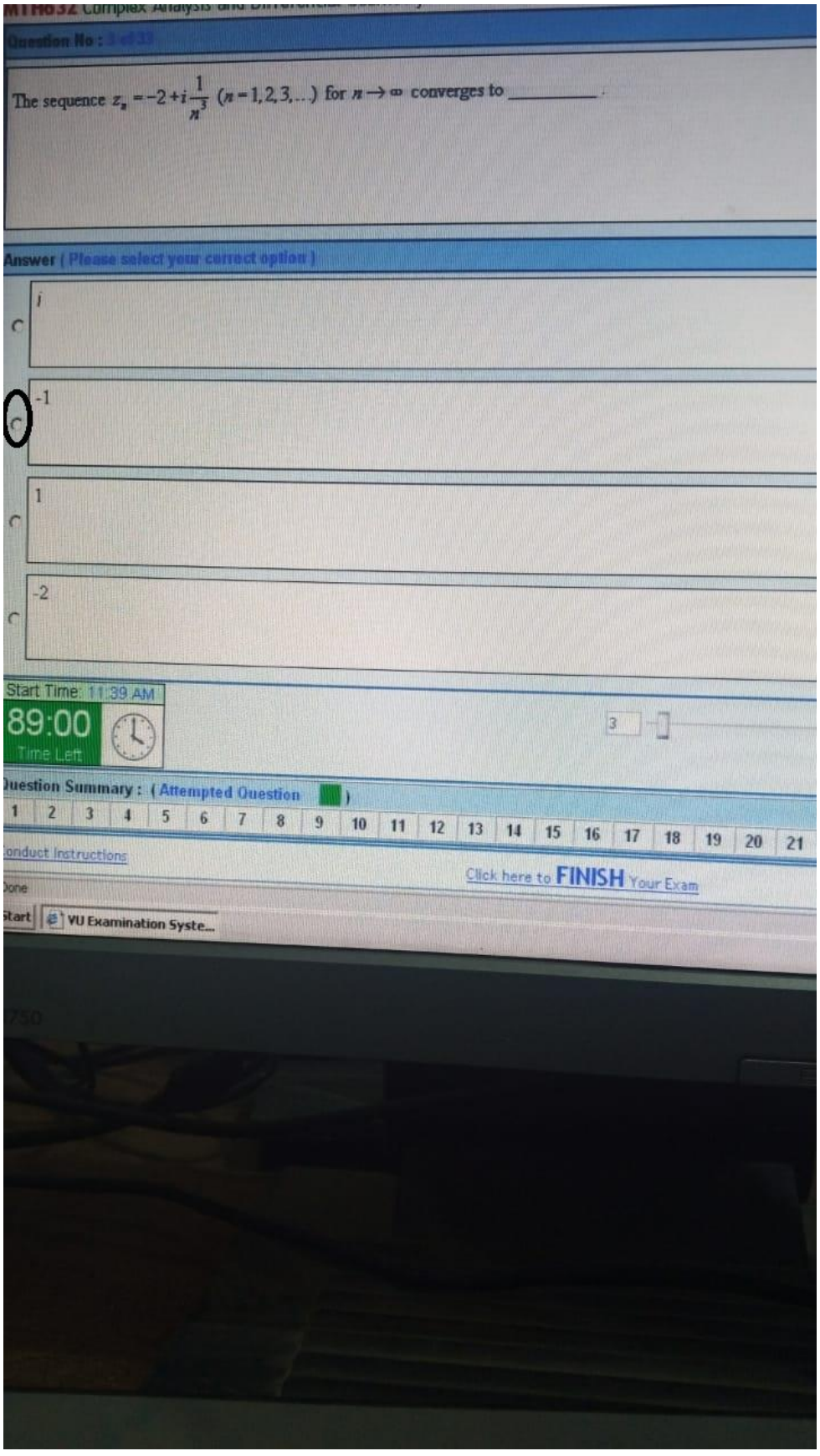
Question No :
The sequence $z_n = -2 + i\frac{1}{n^3}$ $(n = 1,2,3,\ldots)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
i
-1
1
-2
Start Time: 11:39 AM
89:00
Time Left
Question Summary : (Attempted Question)
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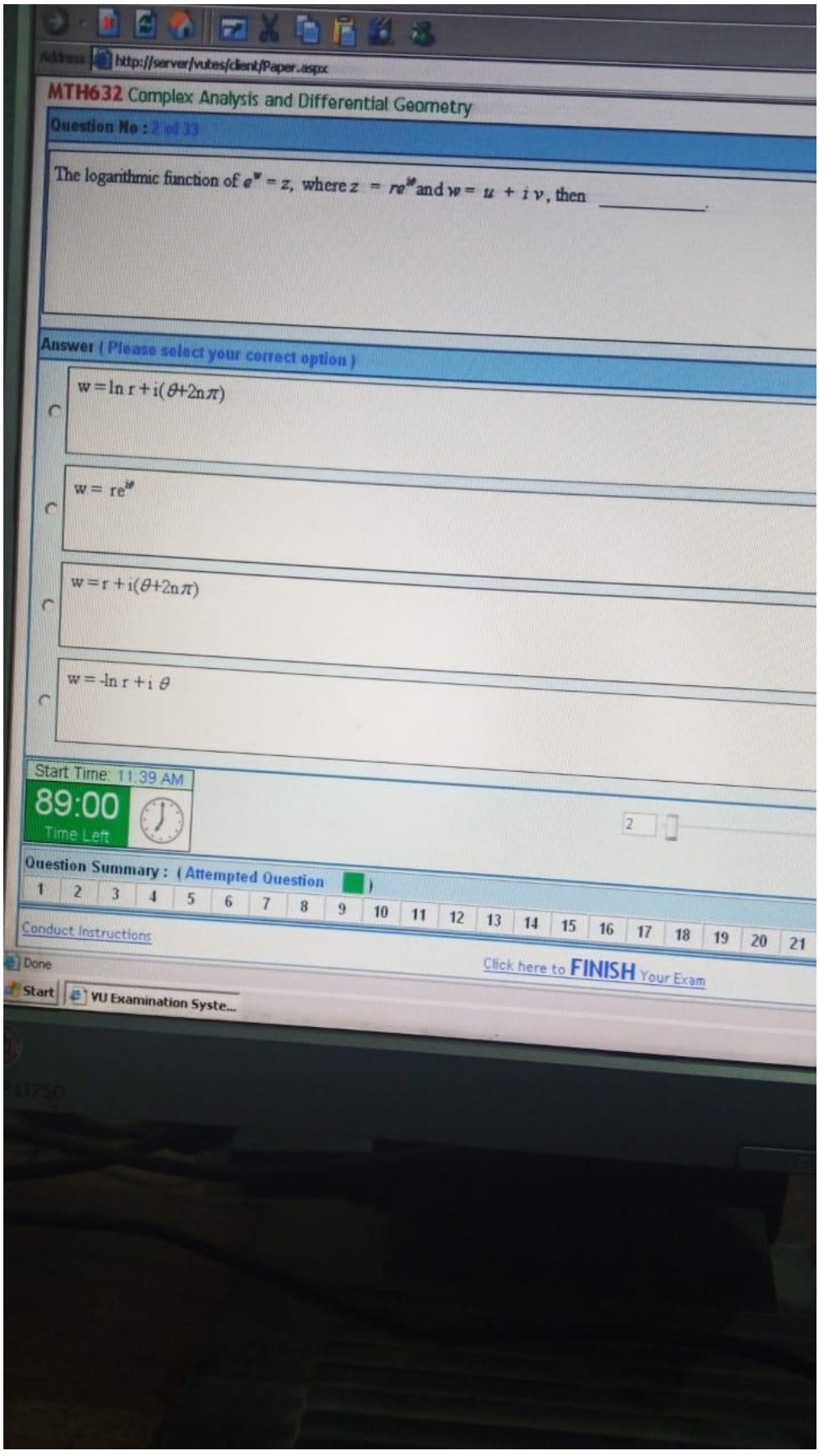
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MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then _____.
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{i\theta}$
$w = r + i(\theta + 2n\pi)$
$w = -\ln r + i\,\theta$
Start Time: 11:39 AM
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Time Left
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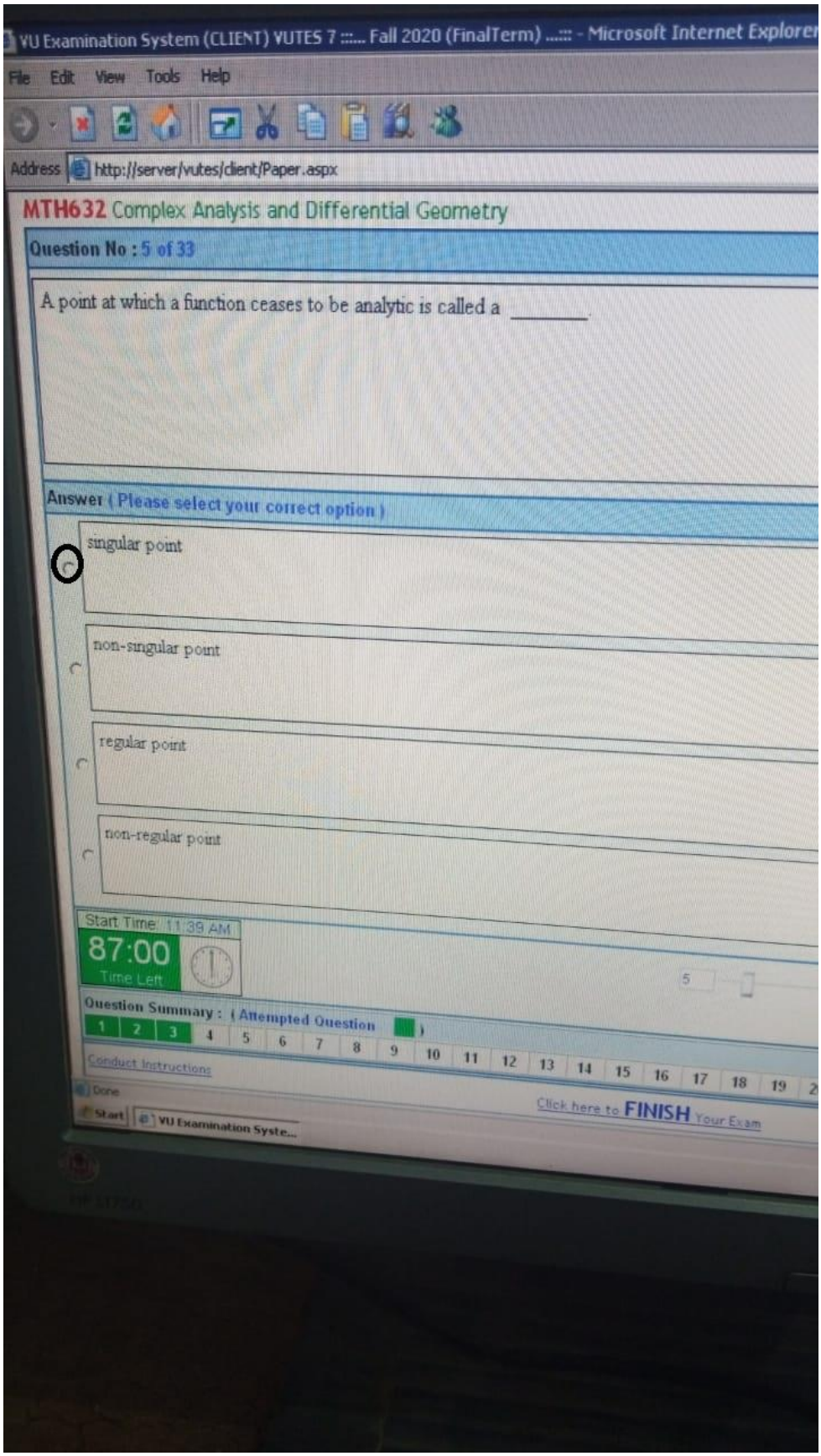
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MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
A point at which a function ceases to be analytic is called a _____.
Answer (Please select your correct option)
singular point
non-singular point
regular point
non-regular point
Start Time 11:39 AM
87:00
Time Left
Question Summary : (Attempted Question)
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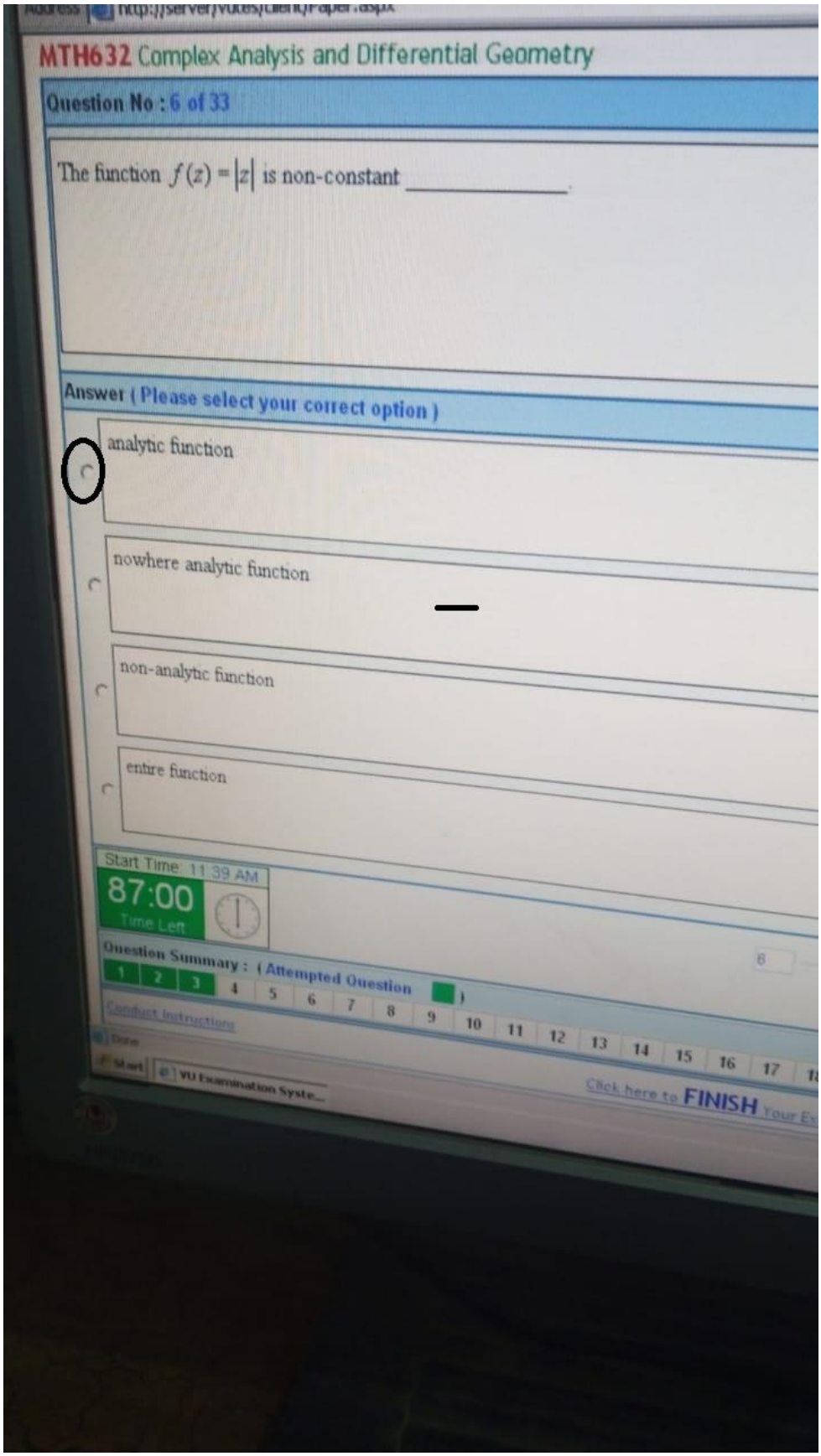
MTH632 Complex Analysis and Differential Geometry
Question No : 6 of 33
The function $f(z) = |z|$ is non-constant _____.
Answer (Please select your correct option)
analytic function
nowhere analytic function
non-analytic function
entire function
Start Time: 11:39 AM
87:00
Time Left
Question Summary : (Attempted Question)
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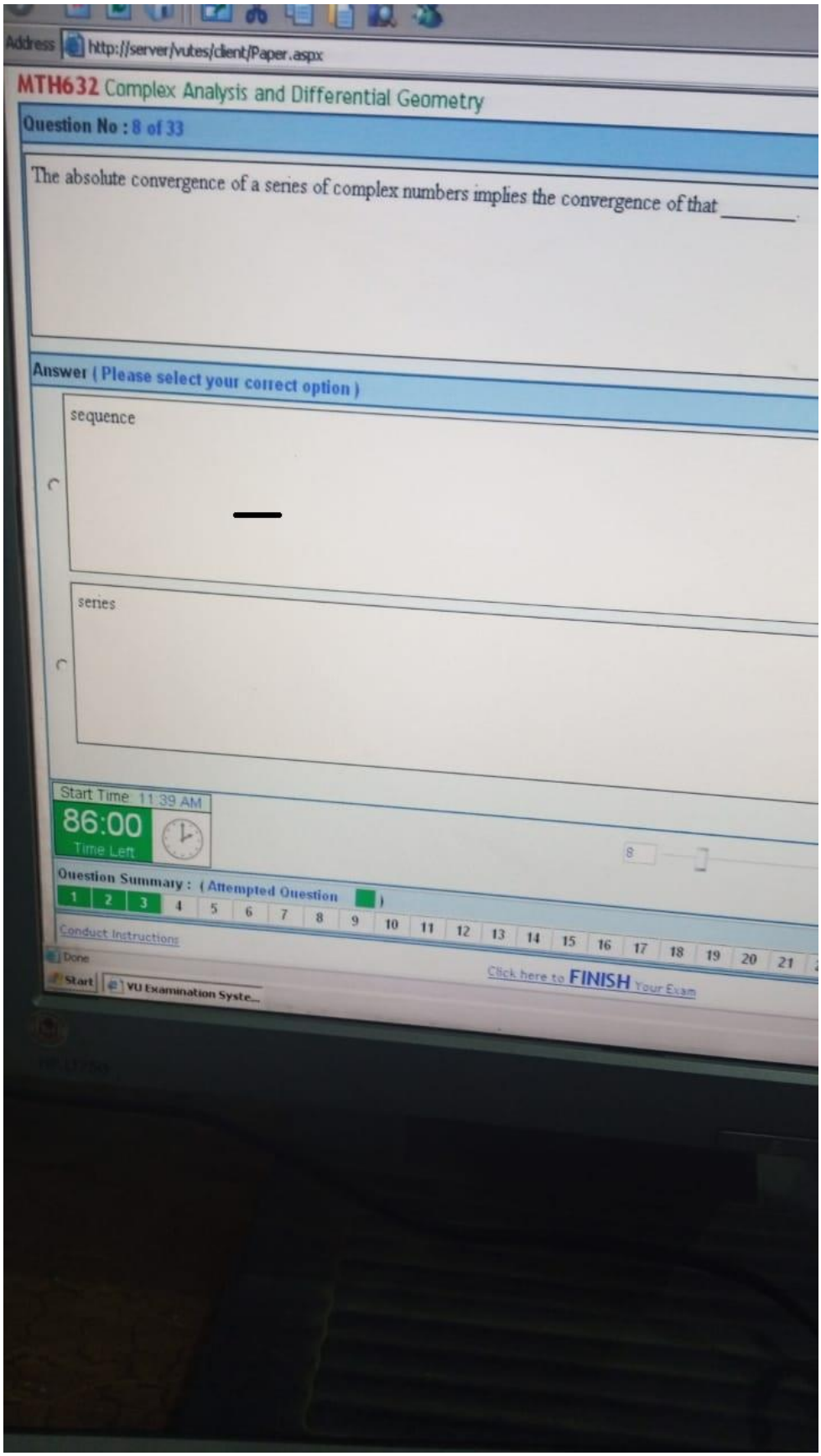
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MTH632 Complex Analysis and Differential Geometry
Question No : 8 of 33
The absolute convergence of a series of complex numbers implies the convergence of that _____.
Answer (Please select your correct option)
sequence
series
Start Time 11:39 AM
86:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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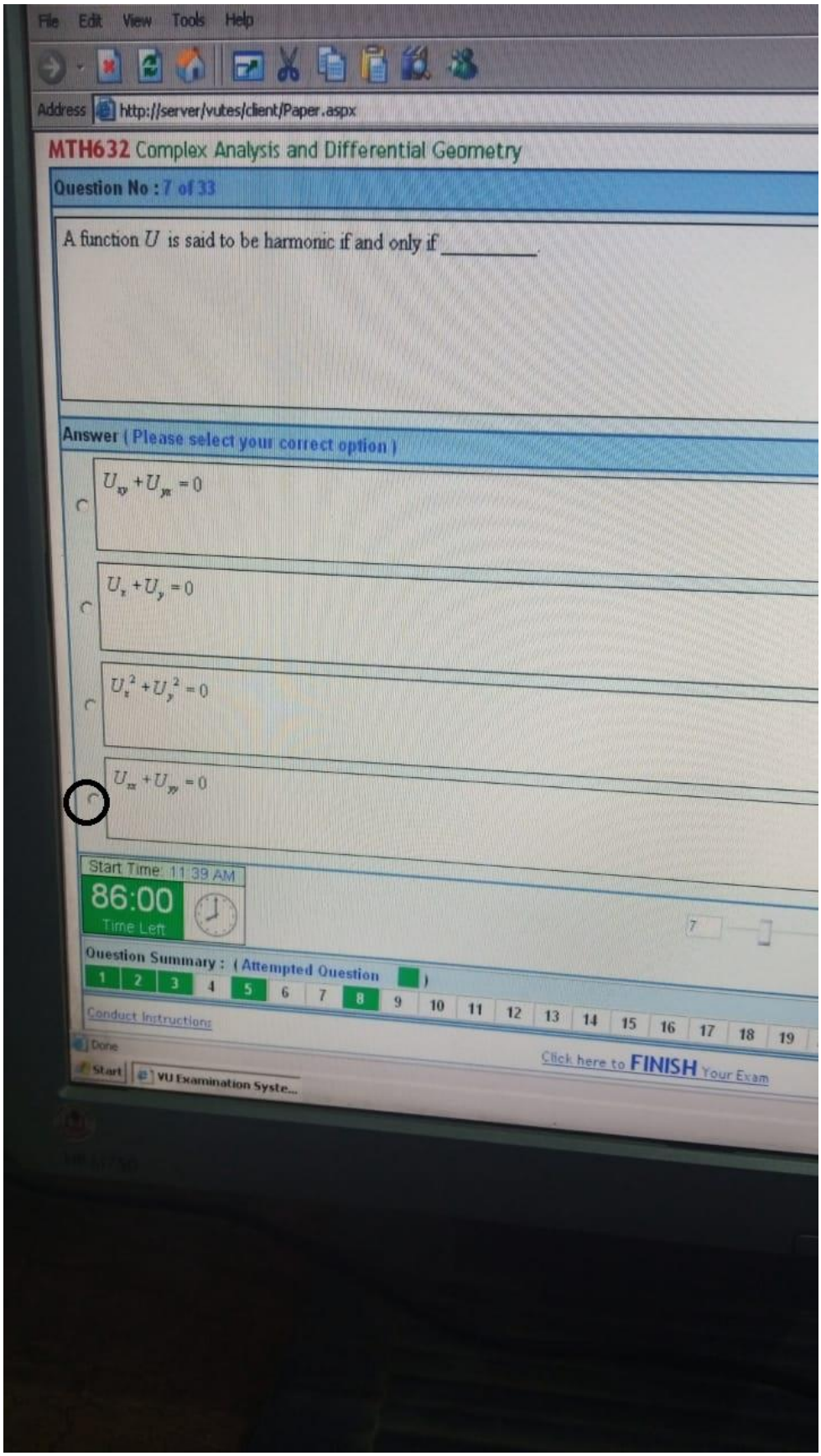
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MTH632 Complex Analysis and Differential Geometry
Question No : 7 of 33
A function U is said to be harmonic if and only if _____.
Answer (Please select your correct option)
$U_{xy} + U_{yx} = 0$
$U_x + U_y = 0$
$U_x^2 + U_y^2 = 0$
$U_{xx} + U_{yy} = 0$
Start Time 11:39 AM
86:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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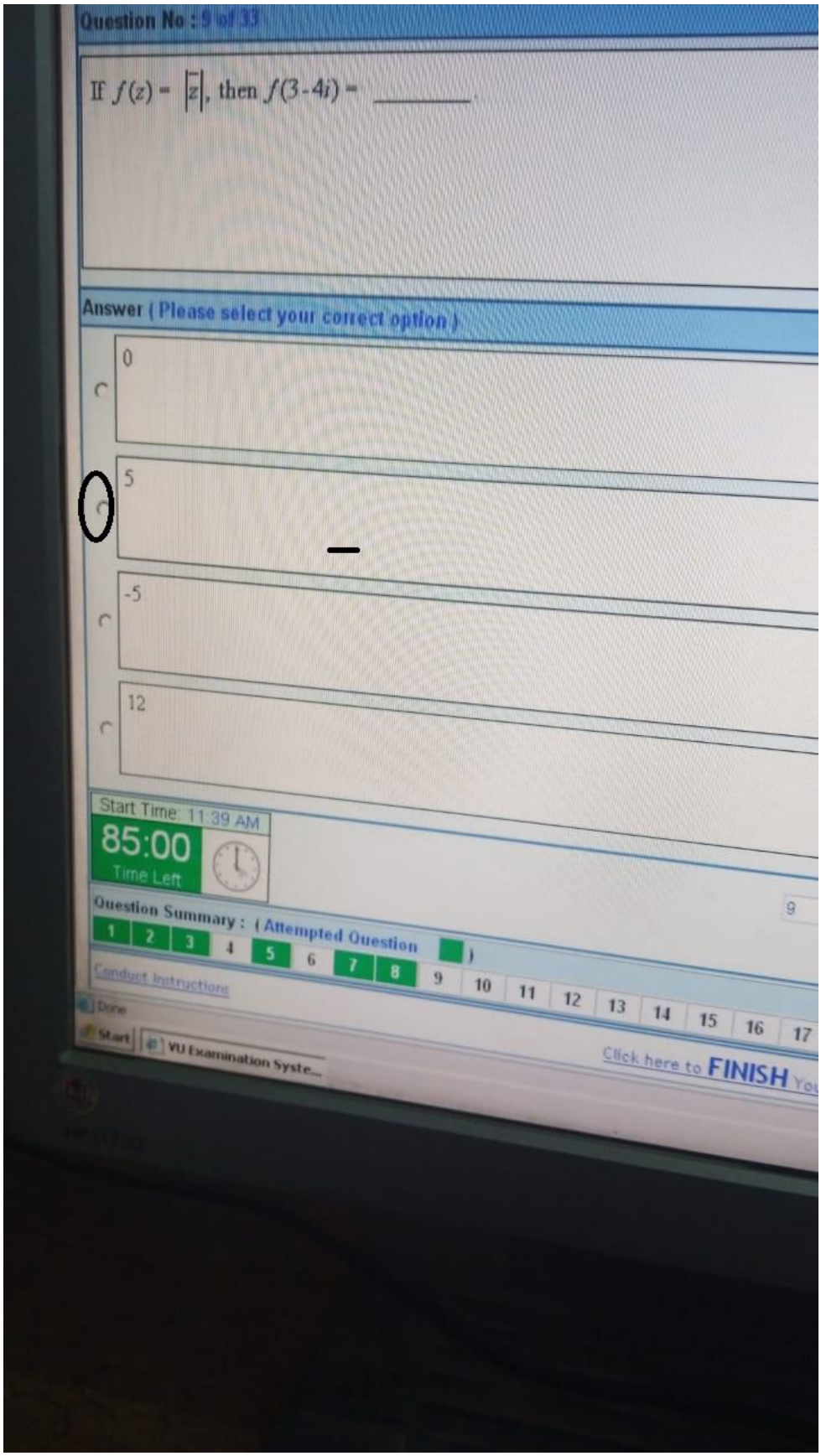
Question No : 9 of 33
If $f(z) = |\bar{z}|$, then $f(3-4i) =$ _____.
Answer (Please select your correct option)
0
5
-5
12
Start Time 11:39 AM
85:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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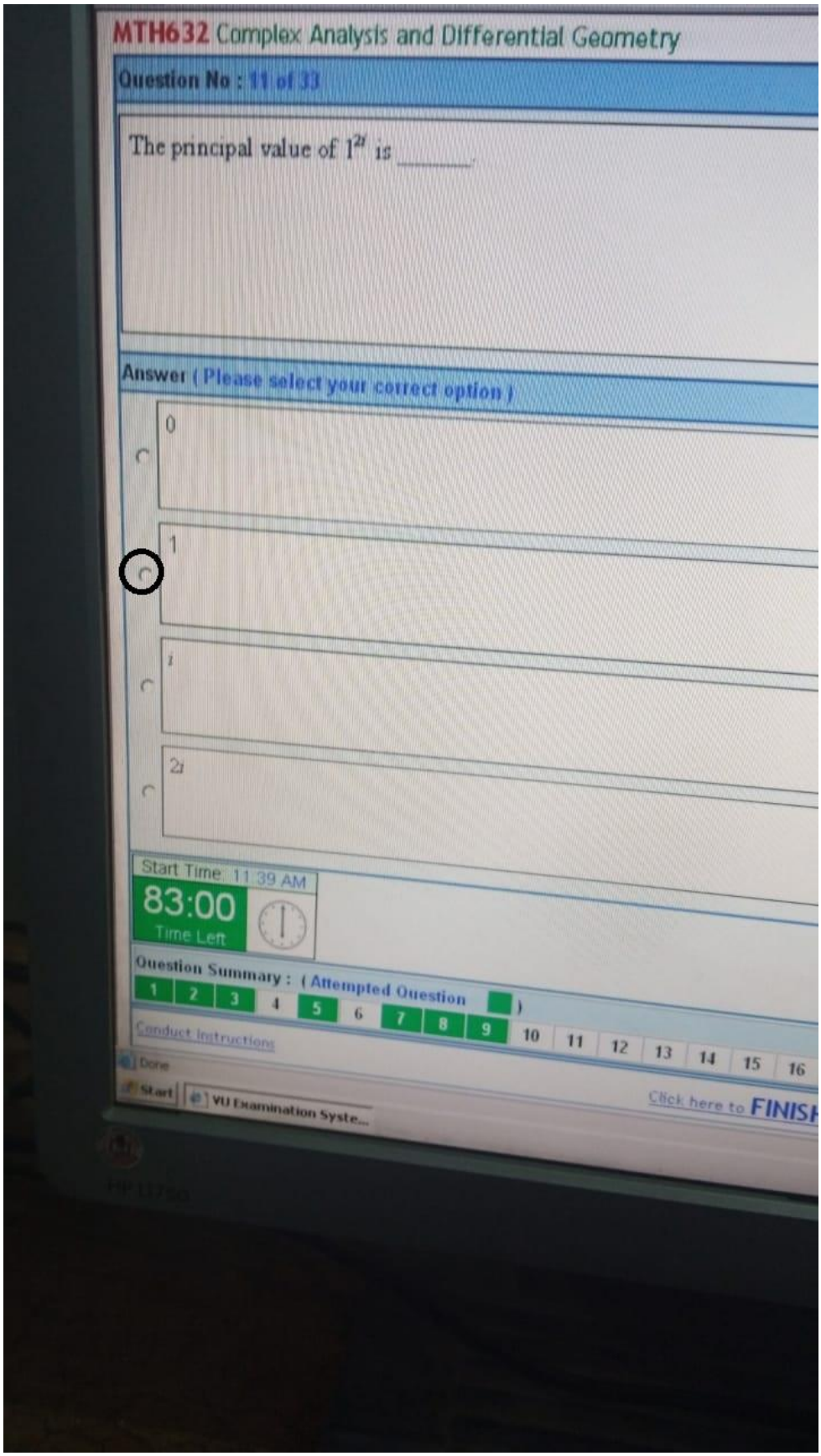
MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 33
The principal value of 1^{2i} is _____.
Answer (Please select your correct option)
0
1
i
2i
Start Time: 11:39 AM
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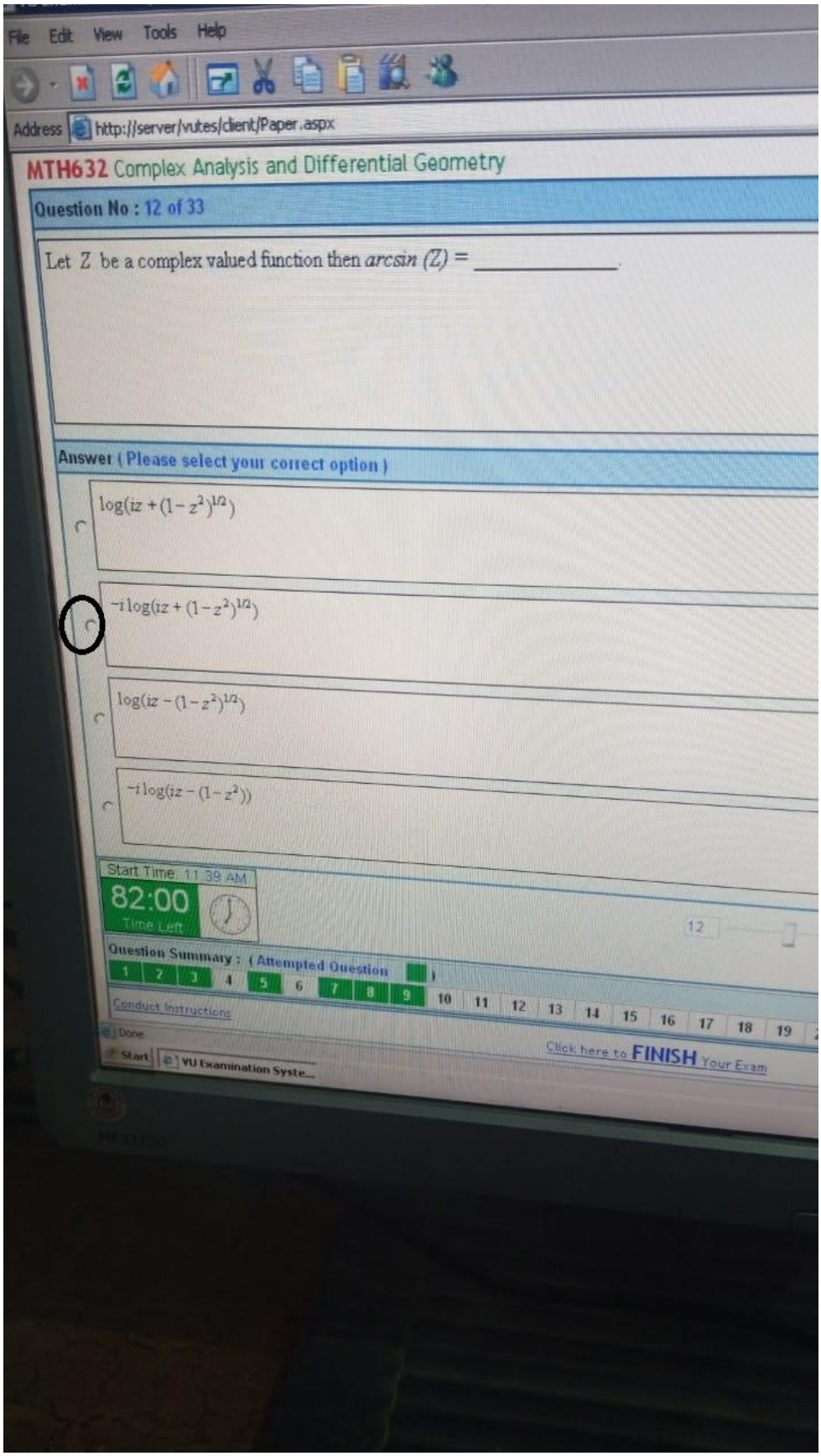
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MTH632 Complex Analysis and Differential Geometry
Question No : 12 of 33
Let Z be a complex valued function then arcsin (Z) = _____.
Answer (Please select your correct option)
log(iz + (1 − z²)^1/2)
−i log(iz + (1 − z²)^1/2)
log(iz − (1 − z²)^1/2)
−i log(iz − (1 − z²))
Start Time: 11:39 AM
82:00
Time Left
12
Question Summary : (Attempted Question)
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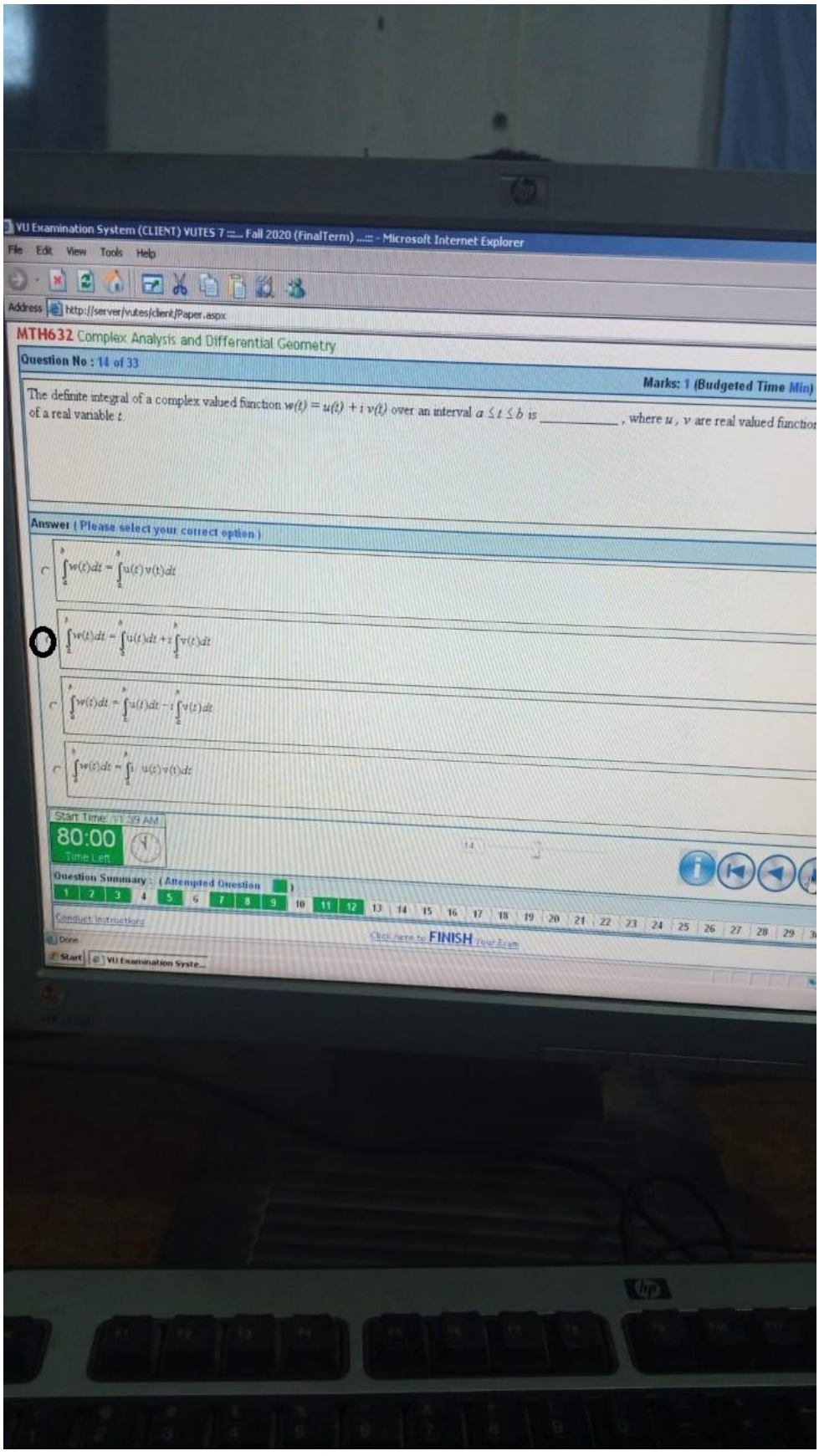
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MTH632 Complex Analysis and Differential Geometry
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functio
of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b i\,u(t)v(t)dt$
Start Time: 11:39 AM
80:00
Time Left
Question Summary : (Attempted Question)
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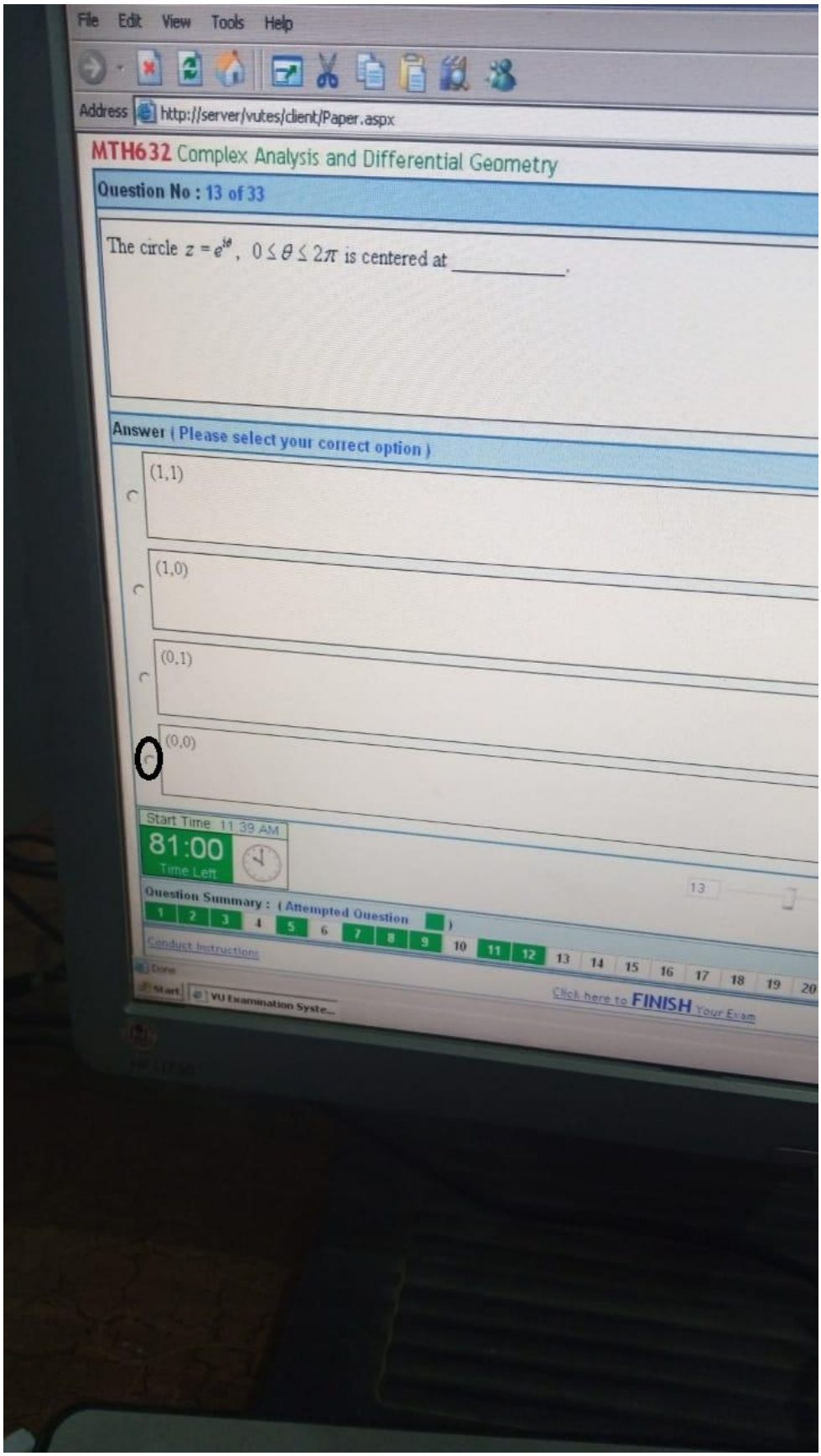
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MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time 11:39 AM
81:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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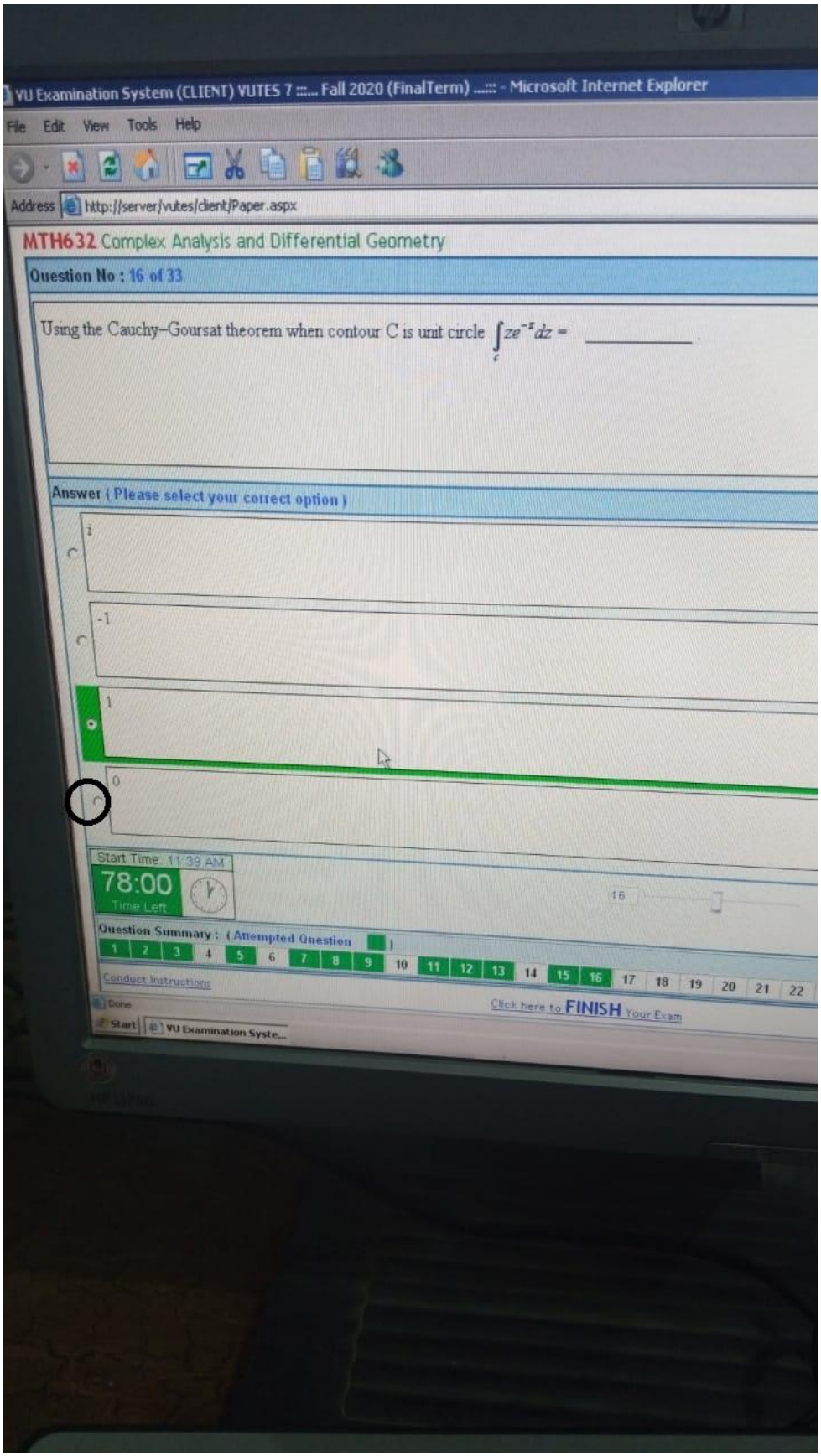
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MTH632 Complex Analysis and Differential Geometry
Question No : 16 of 33
Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$ _____ .
Answer (Please select your correct option)
i
-1
1
0
Start Time 11:39 AM
78:00
Time Left
Question Summary : (Attempted Question)
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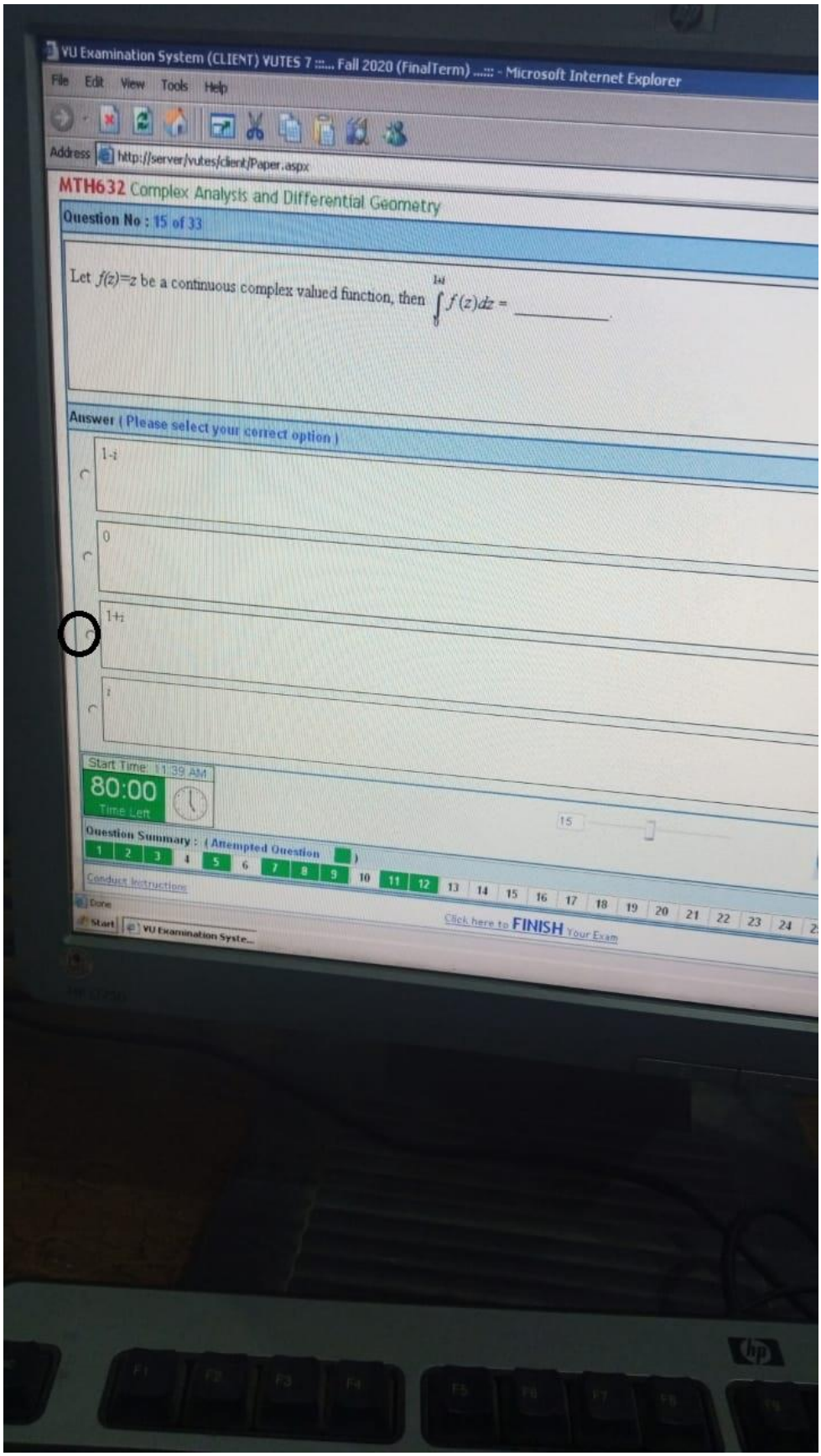
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MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let f(z)=z be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____.
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:39 AM
80:00
Time Left
Question Summary : (Attempted Question)
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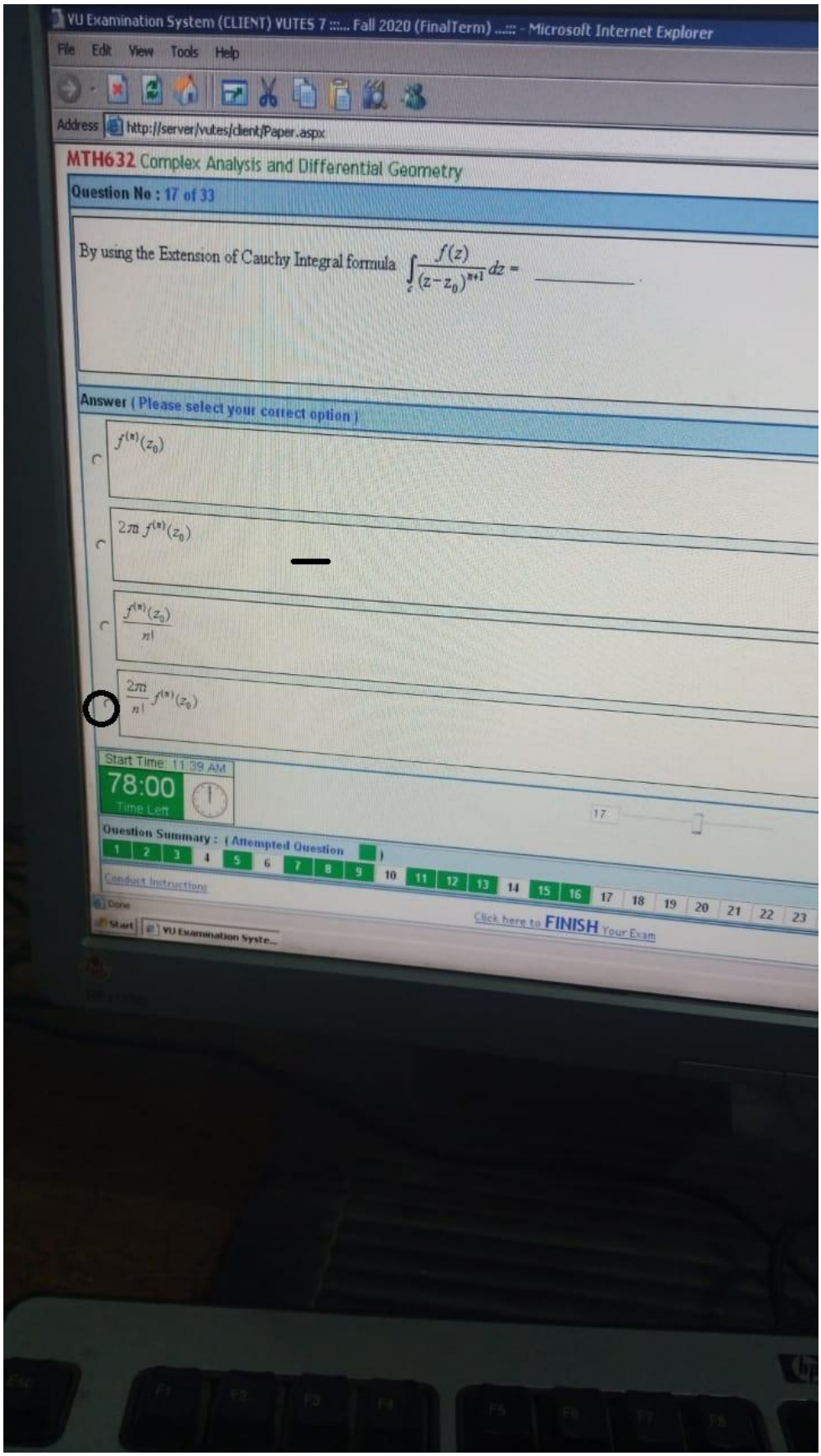
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MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____.
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:39 AM
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Question Summary : (Attempted Question)
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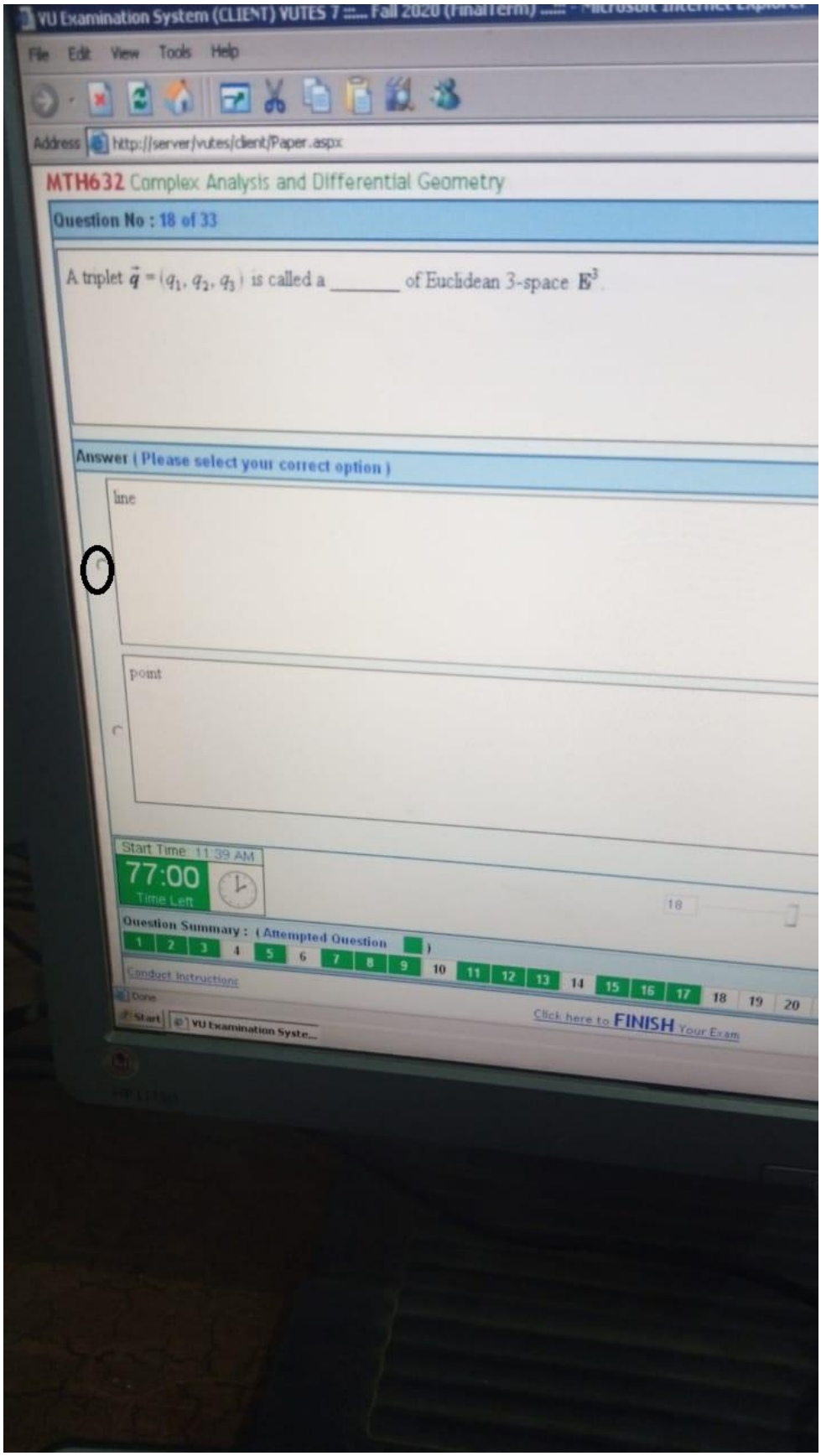
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MTH632 Complex Analysis and Differential Geometry
Question No : 18 of 33
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time 11:39 AM
77:00
Time Left
18
Question Summary : (Attempted Question)
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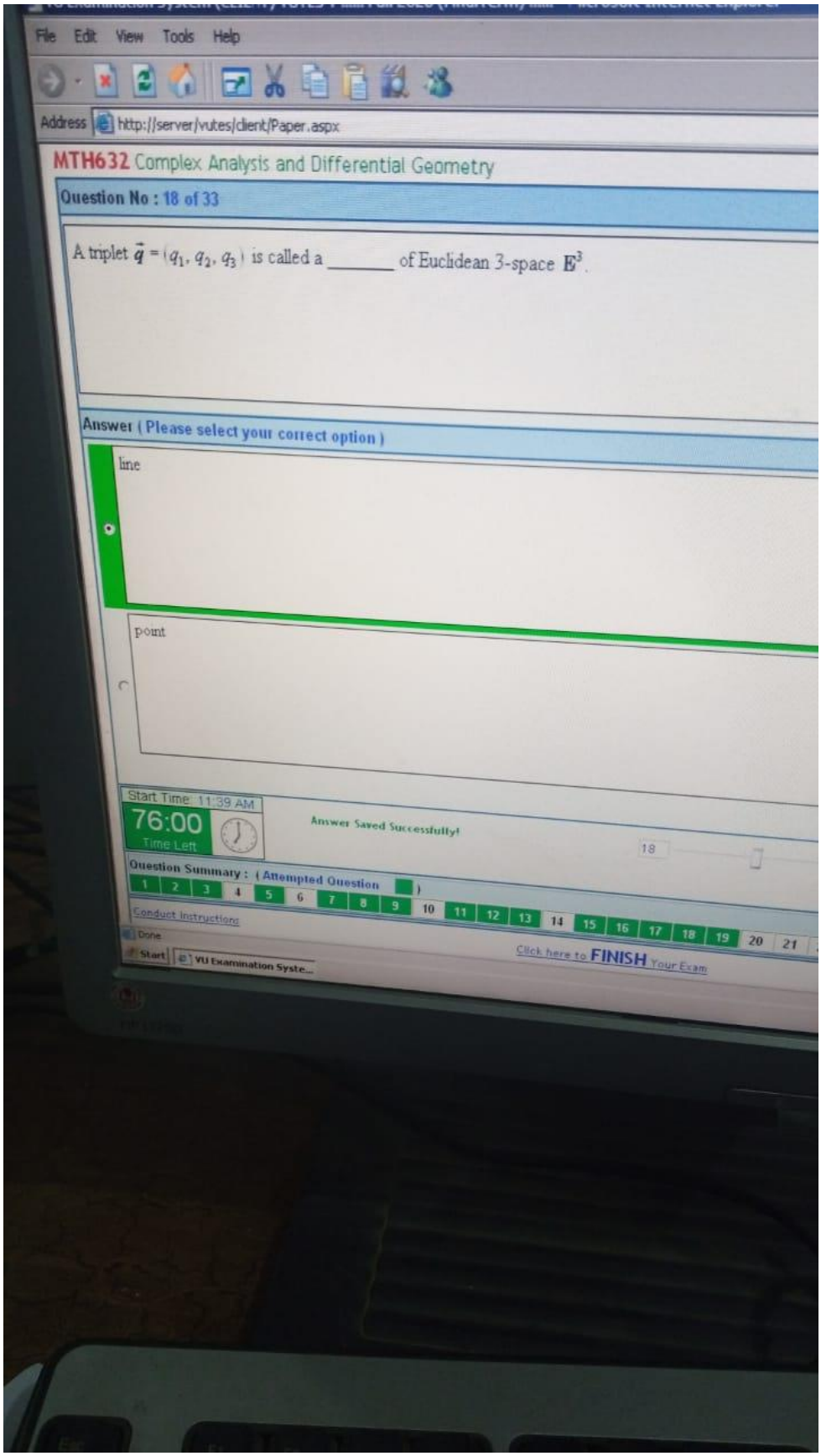
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MTH632 Complex Analysis and Differential Geometry
Question No : 18 of 33
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time 11:39 AM
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Time Left
Answer Saved Successfully!
Question Summary : (Attempted Question)
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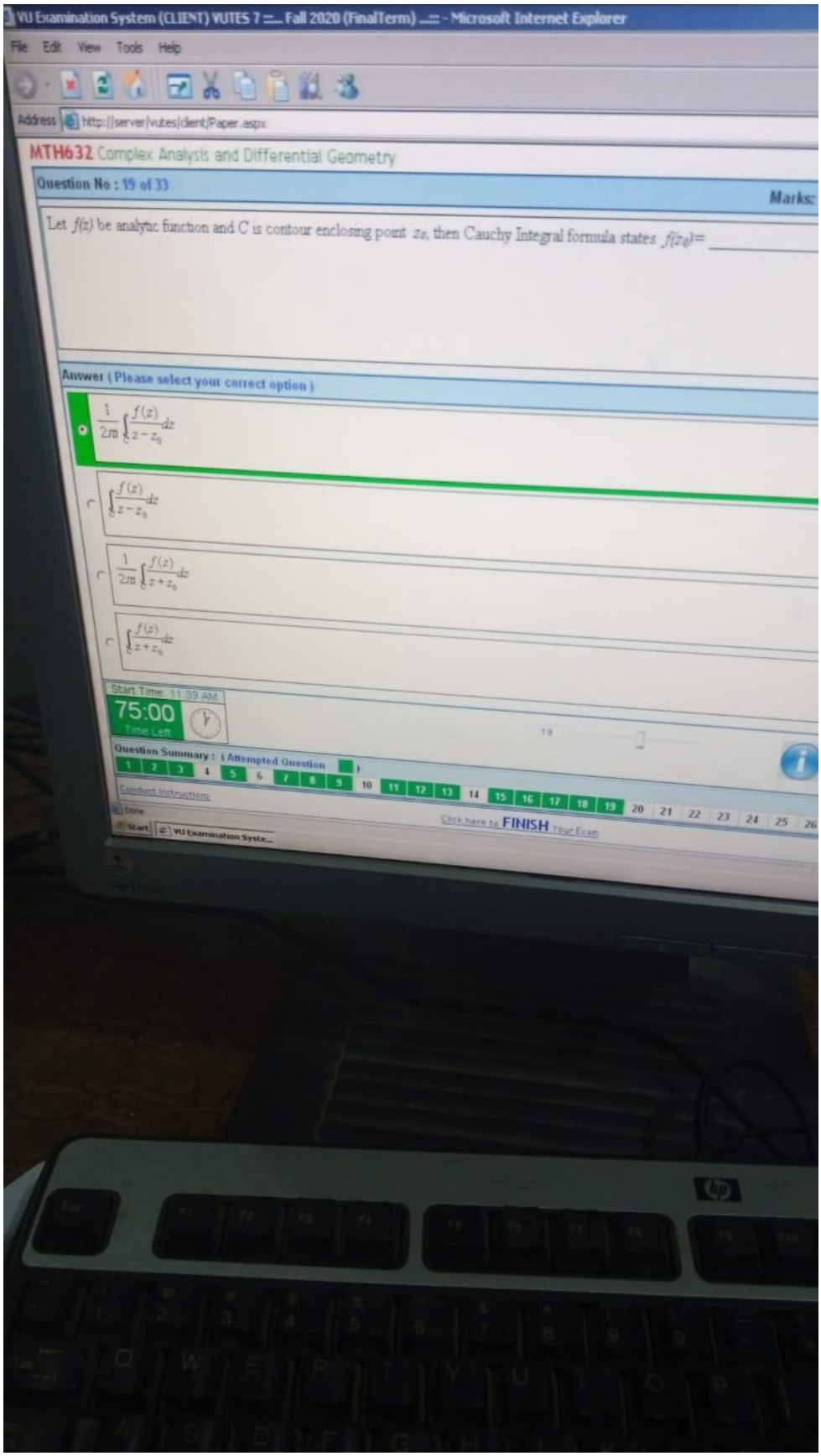
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MTH632 Complex Analysis and Differential Geometry
Question No : 19 of 33
Marks:
Let $f(z)$ be analytic function and C is contour enclosing point z_0, then Cauchy Integral formula states $f(z_0)$= _____
Answer (Please select your correct option)
$\frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}dz$
$\int_C \frac{f(z)}{z-z_0}dz$
$\frac{1}{2\pi i}\int_C \frac{f(z)}{z+z_0}dz$
$\int_C \frac{f(z)}{z+z_0}dz$
Start Time 11:39 AM
75:00
Time Left
Question Summary : (Attempted Question)
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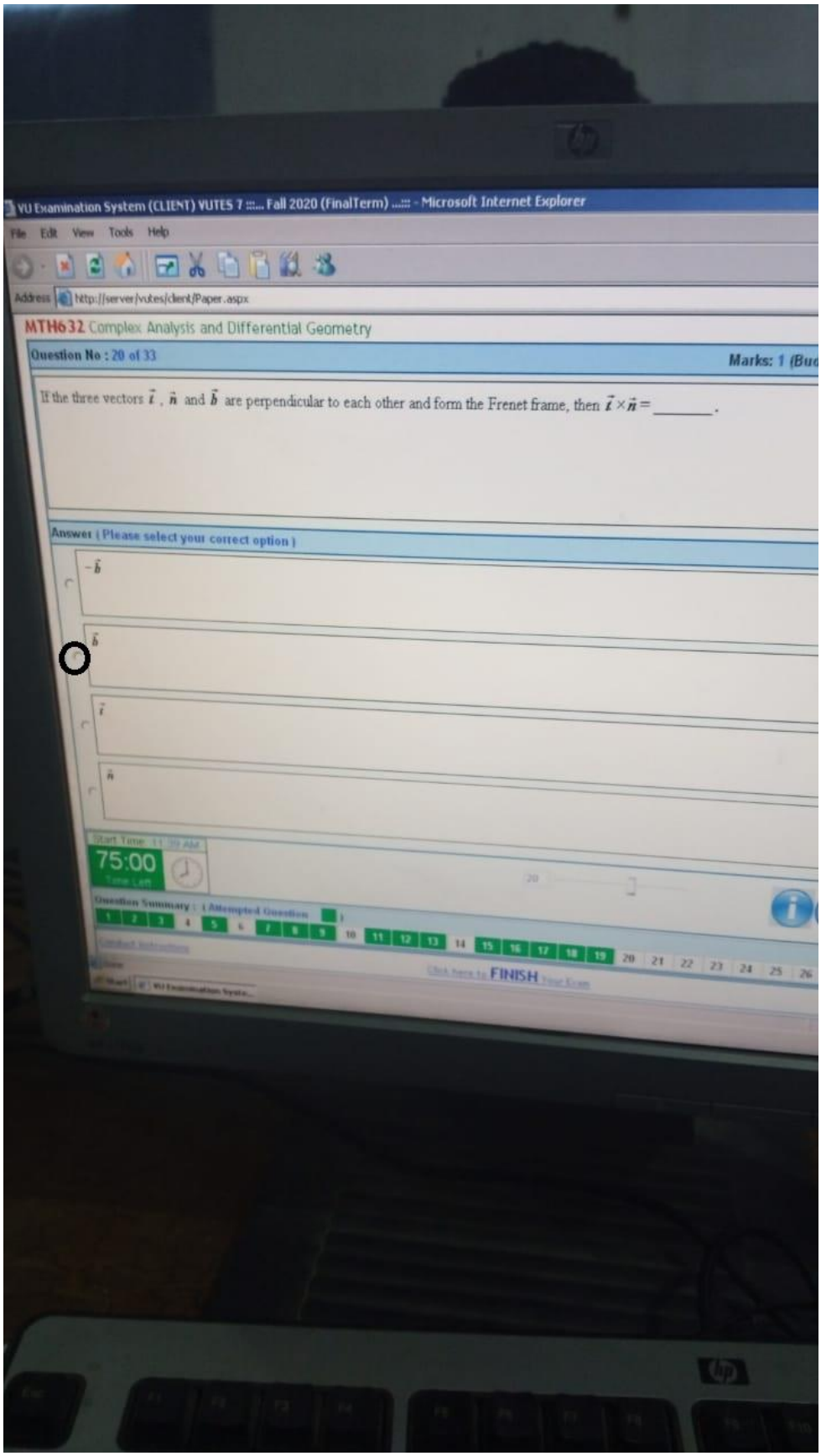
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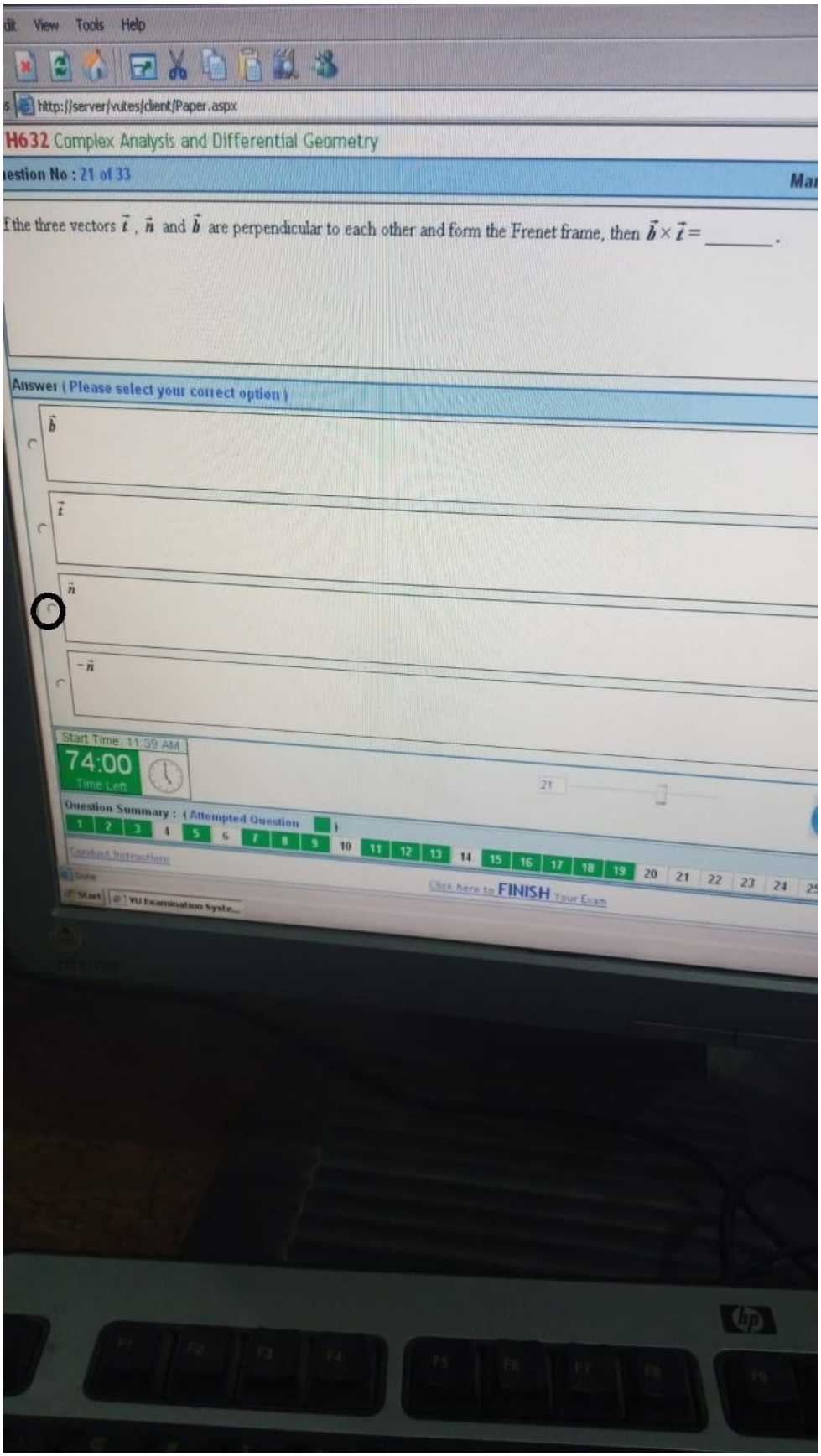
MTH632 Complex Analysis and Differential Geometry
Question No : 20 of 33
Marks: 1 (Bu
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{t} \times \vec{n} =$ _____.
Answer (Please select your correct option)
$-\vec{b}$
$\vec{b}$
$\vec{t}$
$\vec{n}$
75:00
Question Summary : 1 Attempted Question
FINISH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H632 Complex Analysis and Differential Geometry
estion No : 21 of 33
Mar
f the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ _____.
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time 11:39 AM
74:00
Time Left
21
Question Summary : (Attempted Question)
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Conduct Instructions
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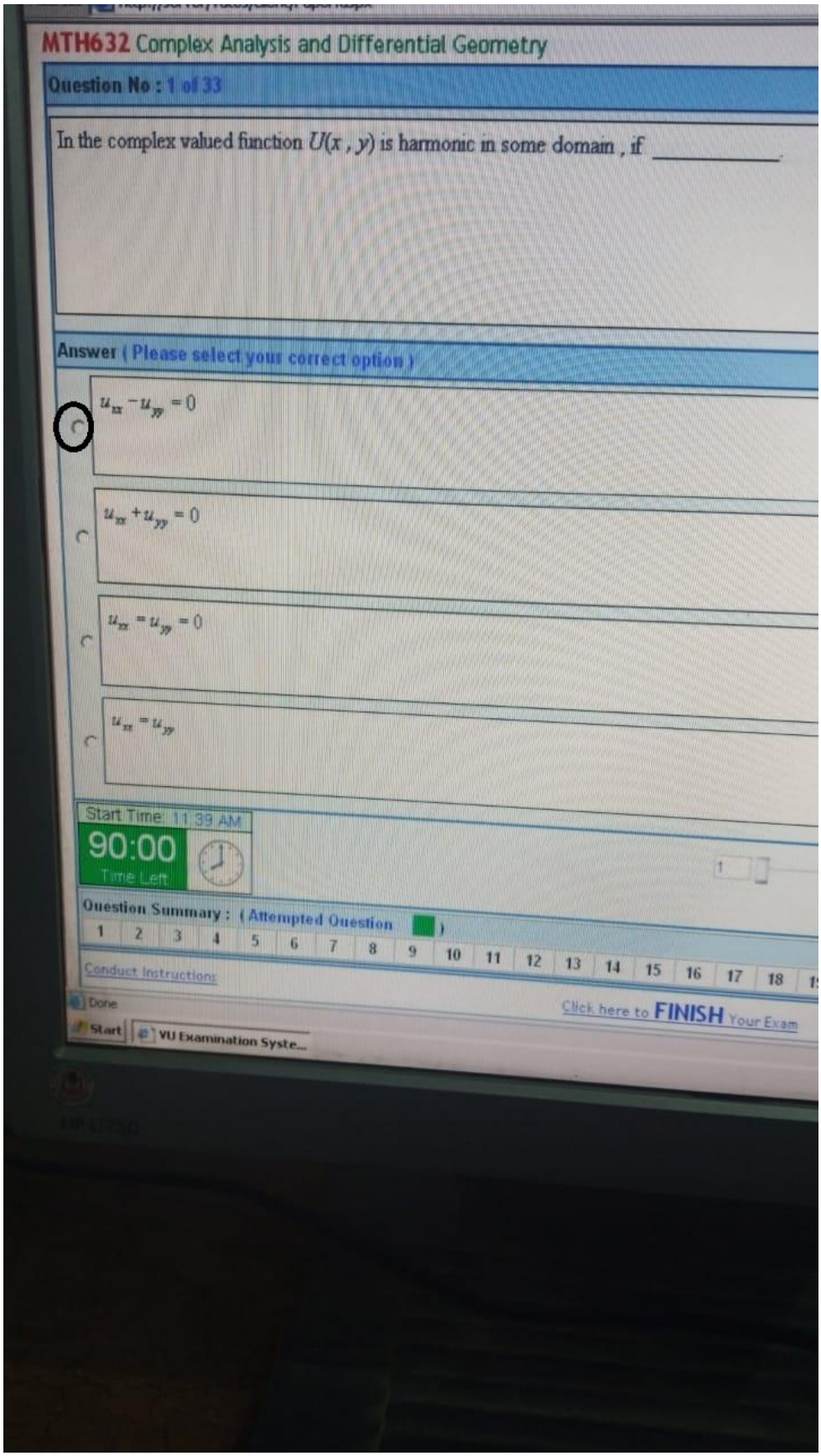
MTH632 Complex Analysis and Differential Geometry
Question No : 1 of 33
In the complex valued function $U(x, y)$ is harmonic in some domain , if _____.
Answer (Please select your correct option)
$u_{xx} - u_{yy} = 0$
$u_{xx} + u_{yy} = 0$
$u_{xx} = u_{yy} = 0$
$u_{xx} = u_{yy}$
Start Time: 11:39 AM
90:00
Time Left
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Conduct Instruction
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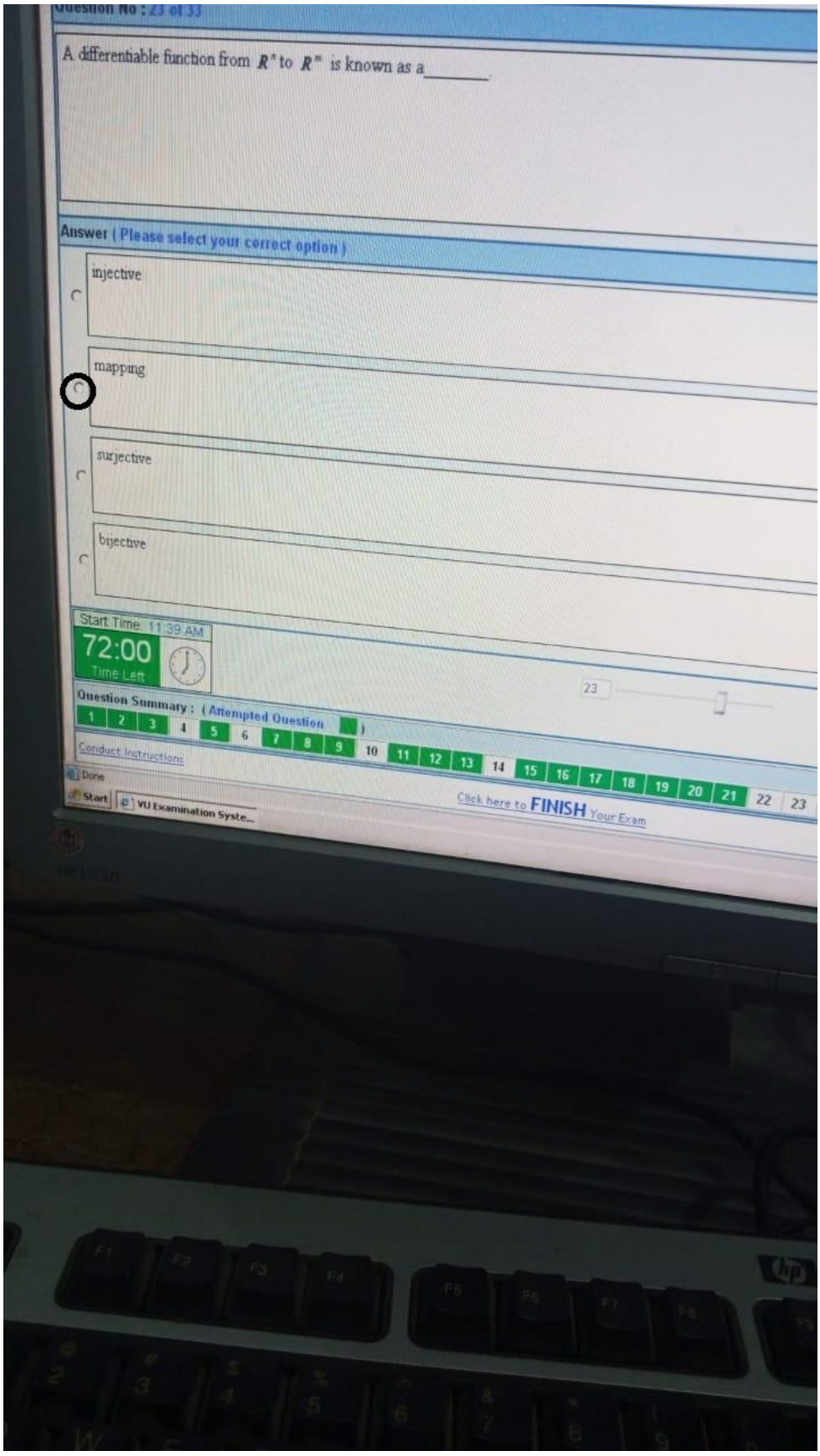
Question No : 23 of 33
A differentiable function from R^n to R^m is known as a_____.
Answer (Please select your correct option)
injective
mapping
surjective
bijective
Start Time 11:39 AM
72:00
Time Left
23
Question Summary : (Attempted Question)
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Conduct Instructions
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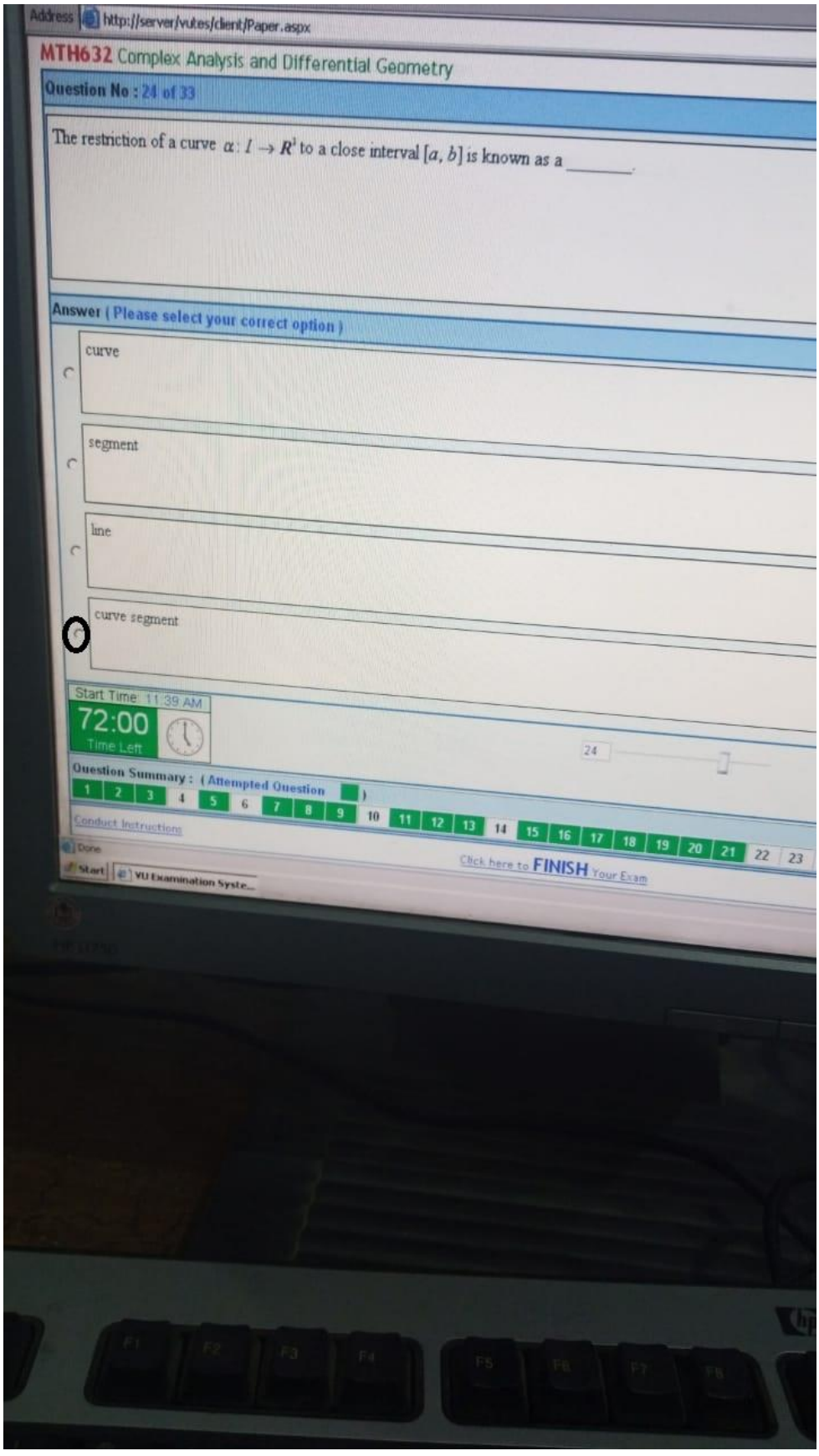
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MTH632 Complex Analysis and Differential Geometry
Question No : 24 of 33
The restriction of a curve $\alpha : I \to R^1$ to a close interval $[a, b]$ is known as a _____.
Answer (Please select your correct option)
curve
segment
line
curve segment
Start Time: 11:39 AM
72:00
Time Left
24
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Conduct Instructions
Done
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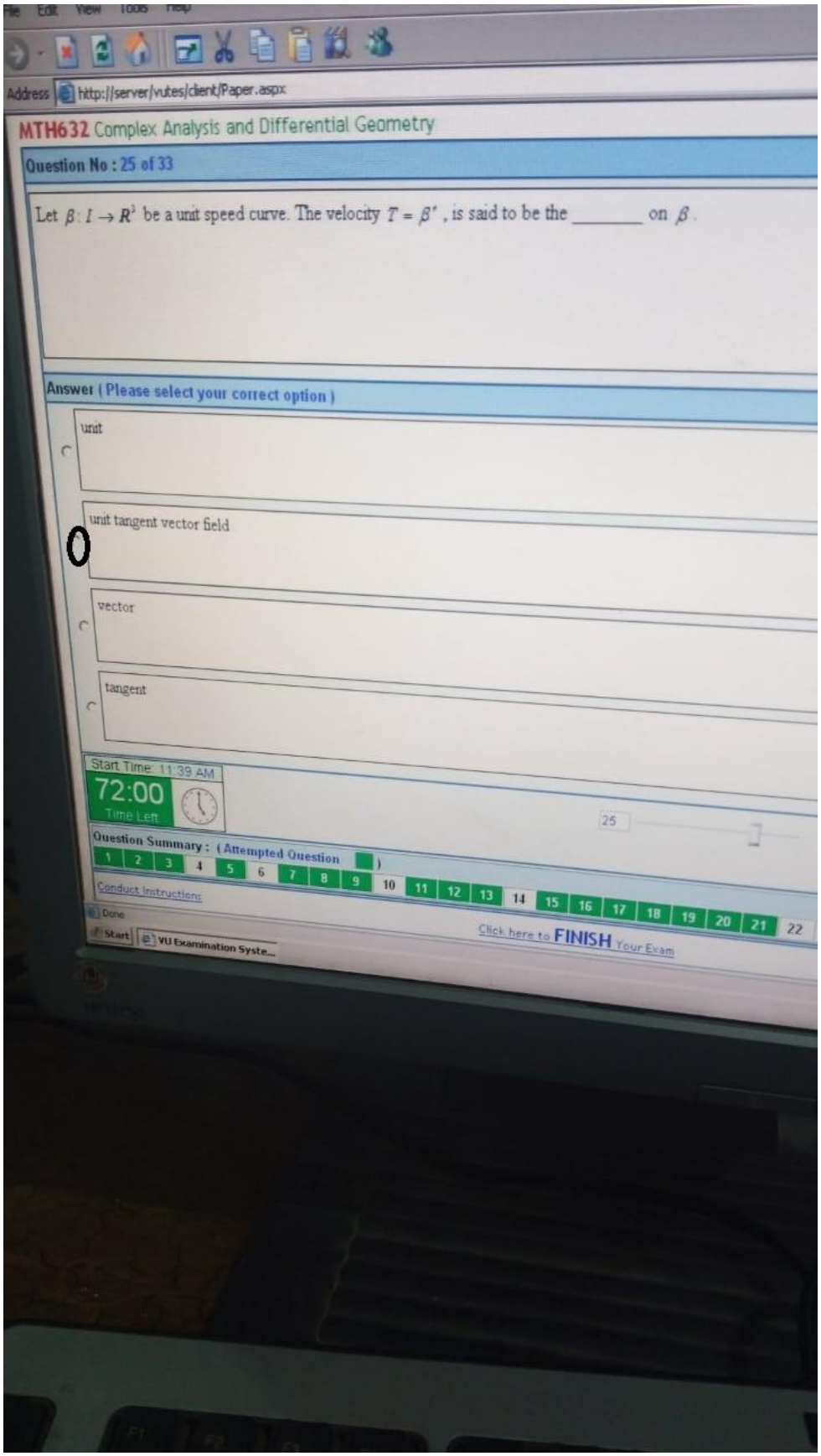
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 25 of 33
Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:39 AM
72:00
Time Left
25
Question Summary : (Attempted Question)
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Conduct Instructions
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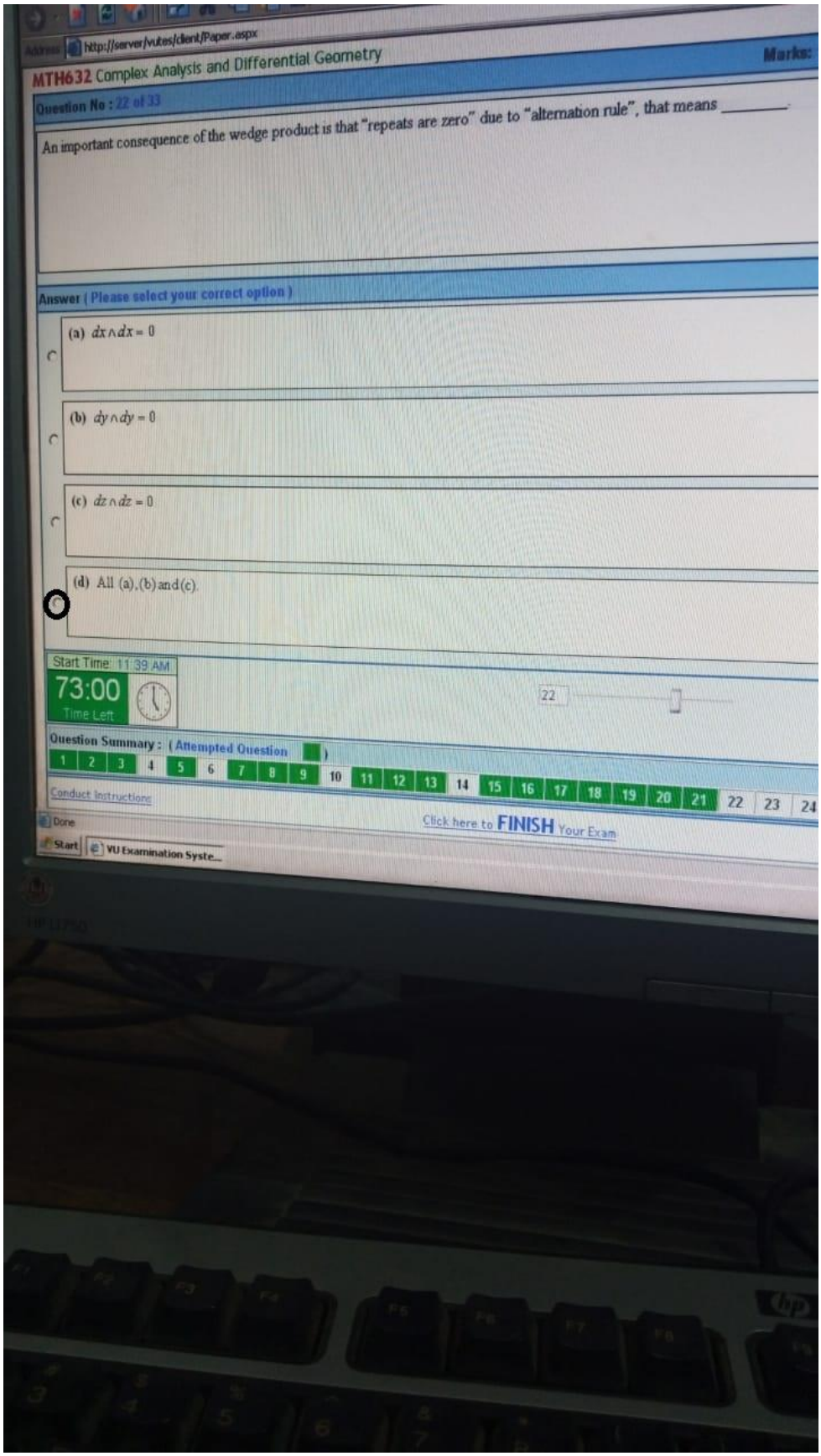
VU Examination Syste...
Click here to FINISH Your Exam


http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Marks:
Question No : 22 of 33
An important consequence of the wedge product is that "repeats are zero" due to "alternation rule", that means _____.
Answer (Please select your correct option)
(a) $dx \wedge dx = 0$
(b) $dy \wedge dy = 0$
(c) $dz \wedge dz = 0$
(d) All (a), (b) and (c).
Start Time: 11:39 AM
73:00
Time Left
22
Question Summary : (Attempted Question)
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Conduct Instructions
Click here to FINISH Your Exam
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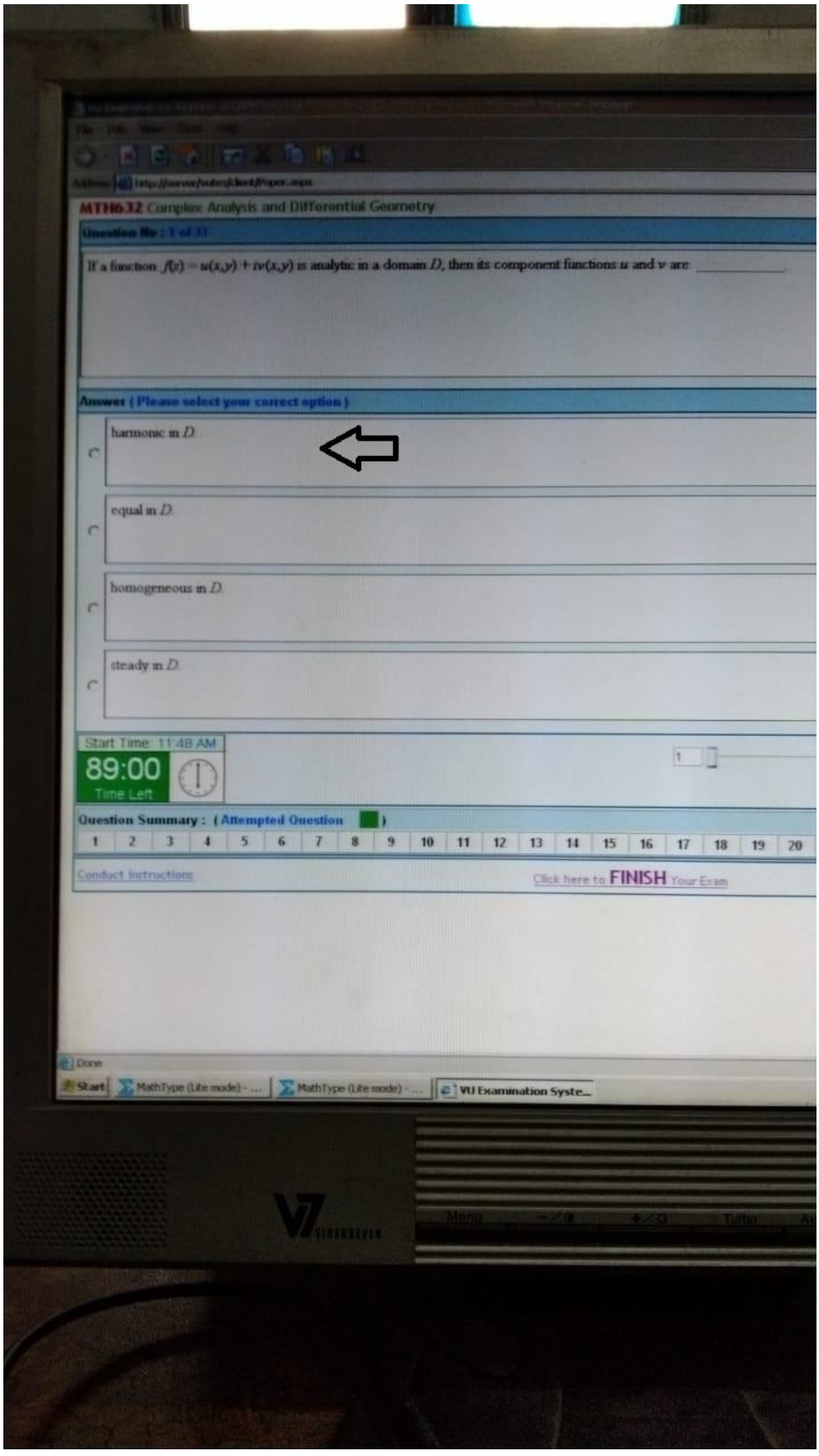
MTH632 Complex Analysis and Differential Geometry
If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____.
Answer (Please select your correct option)
harmonic in D.
equal in D.
homogeneous in D.
steady in D.
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
Click here to FINISH Your Exam
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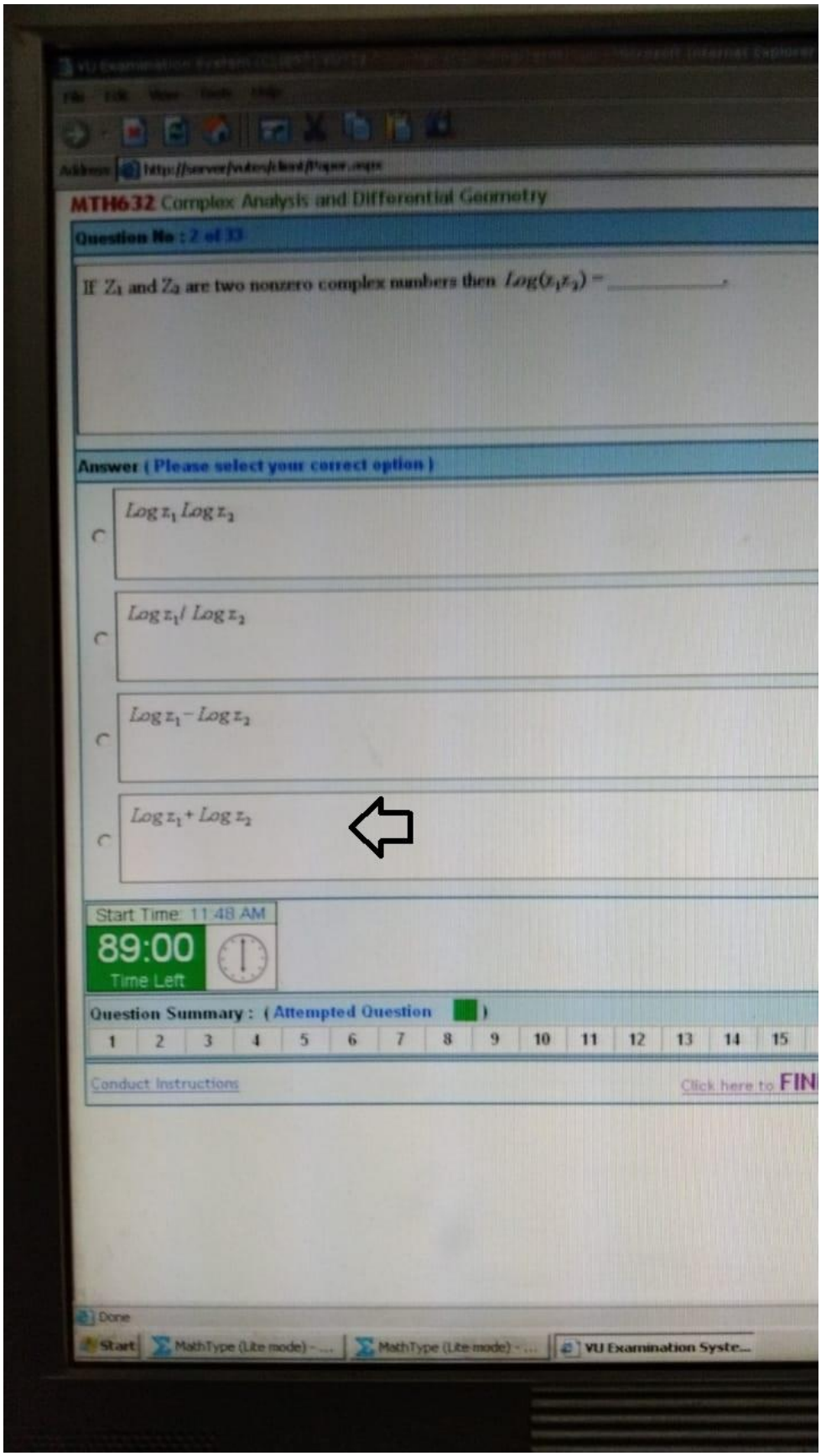
MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
If Z_1 and Z_2 are two nonzero complex numbers then $Log(z_1 z_2)$ = _____.
Answer (Please select your correct option)
$Log\, z_1\, Log\, z_2$
$Log\, z_1 / Log\, z_2$
$Log\, z_1 - Log\, z_2$
$Log\, z_1 + Log\, z_2$
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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Start
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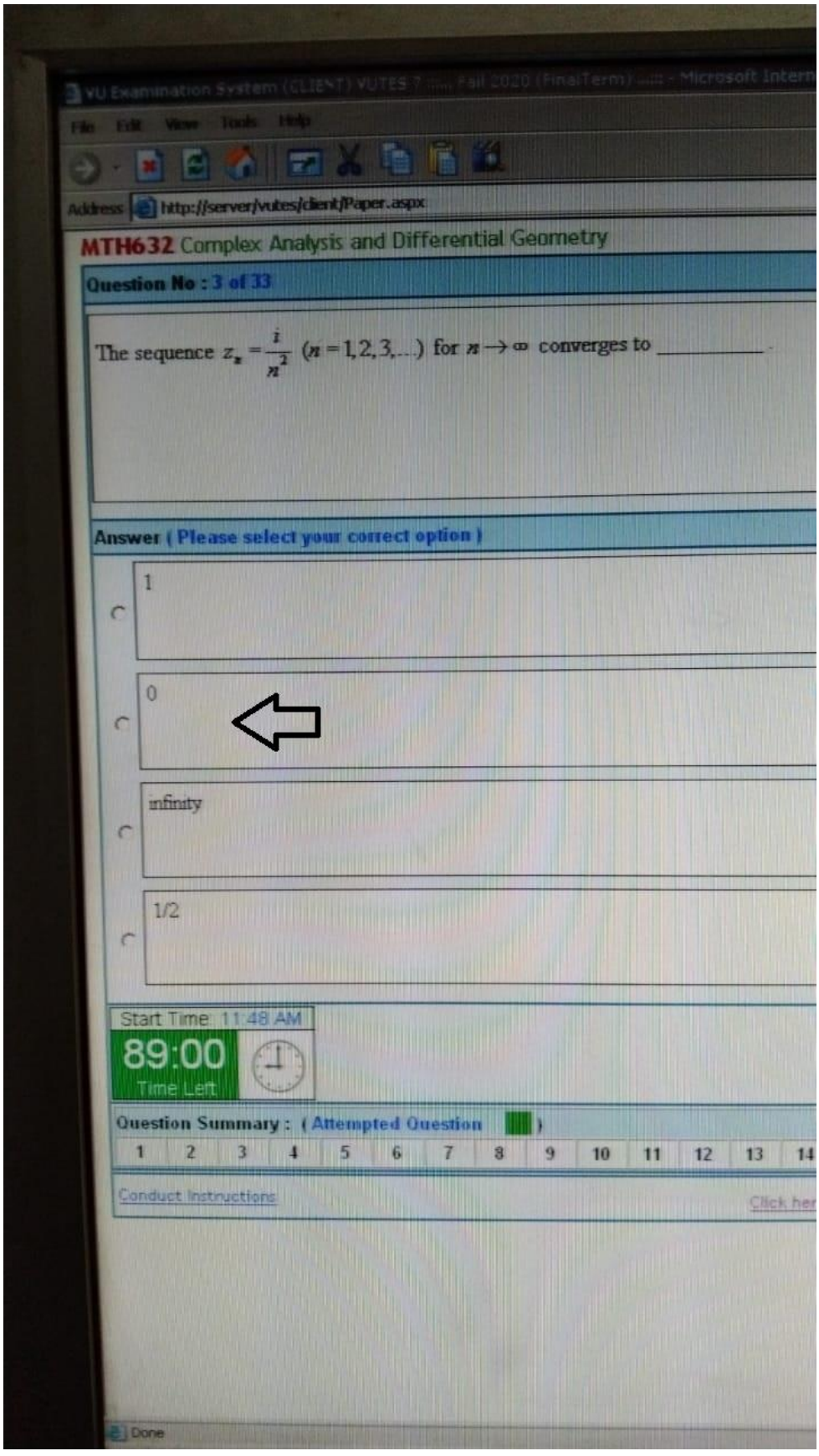
VU Examination System (CLIENT) VUTES 7 ..... Fall 2020 (FinalTerm) ..... - Microsoft Intern
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 3 of 33
The sequence $z_n = \frac{i}{n^2}$ $(n = 1,2,3,...)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
1
0
infinity
1/2
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
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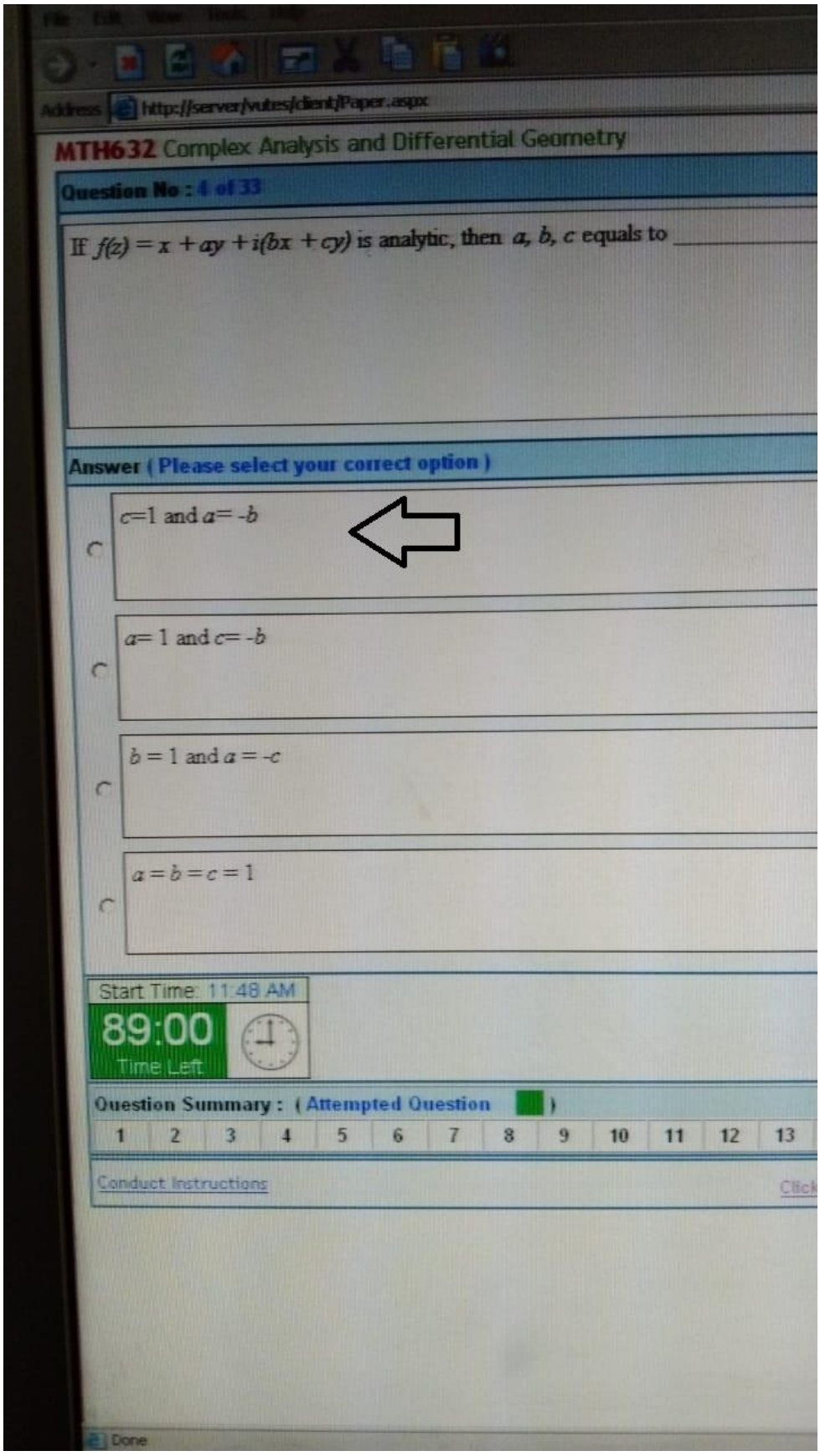
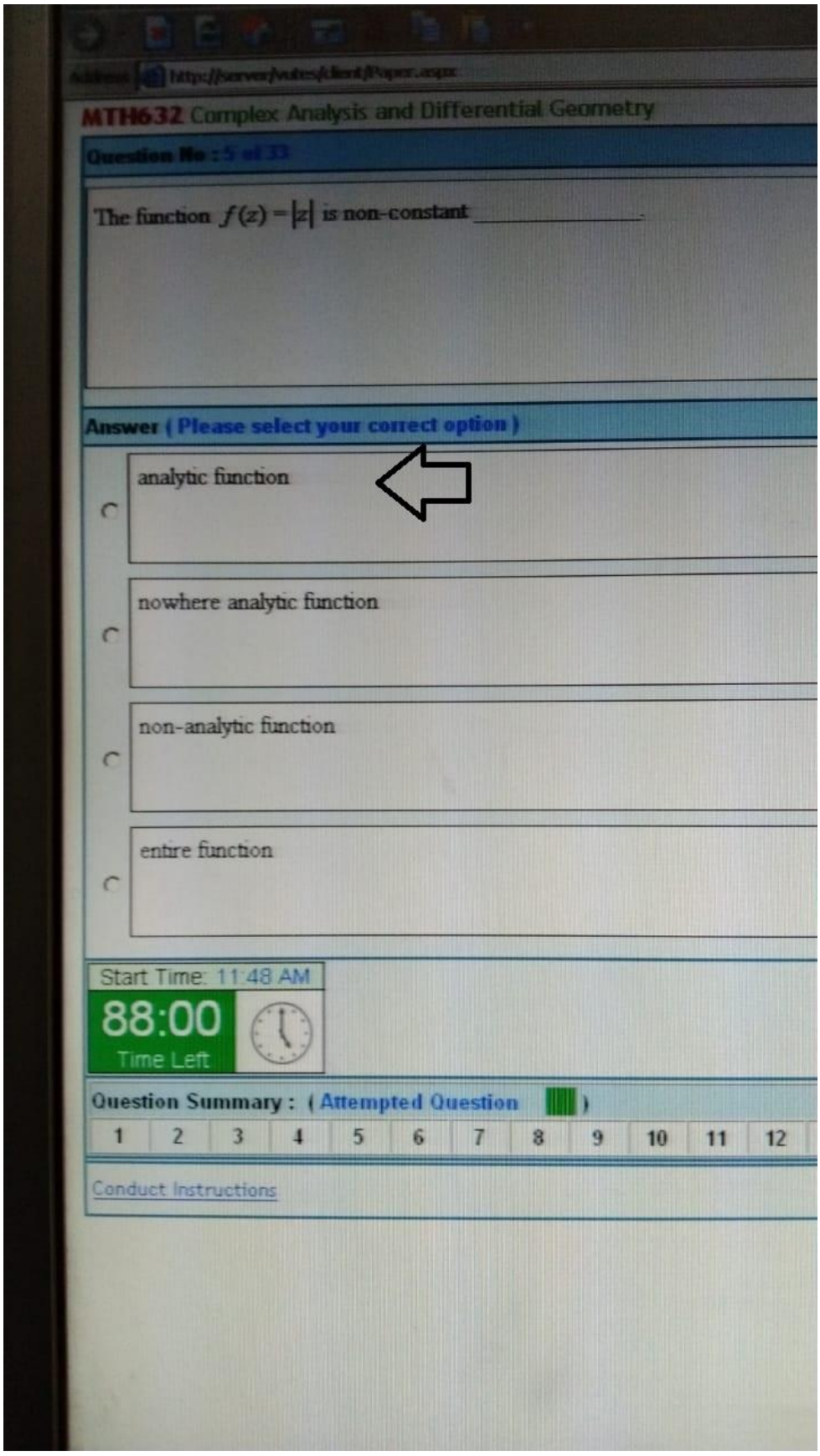
Conduct Instructions
Click her
Done


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 4 of 33
If f(z) = x + ay + i(bx + cy) is analytic, then a, b, c equals to _____
Answer (Please select your correct option)
c=1 and a= -b
a= 1 and c= -b
b = 1 and a = -c
a = b = c = 1
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions
Done


MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
The function $f(z) = |z|$ is non-constant _____.
Answer (Please select your correct option)
analytic function
nowhere analytic function
non-analytic function
entire function
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
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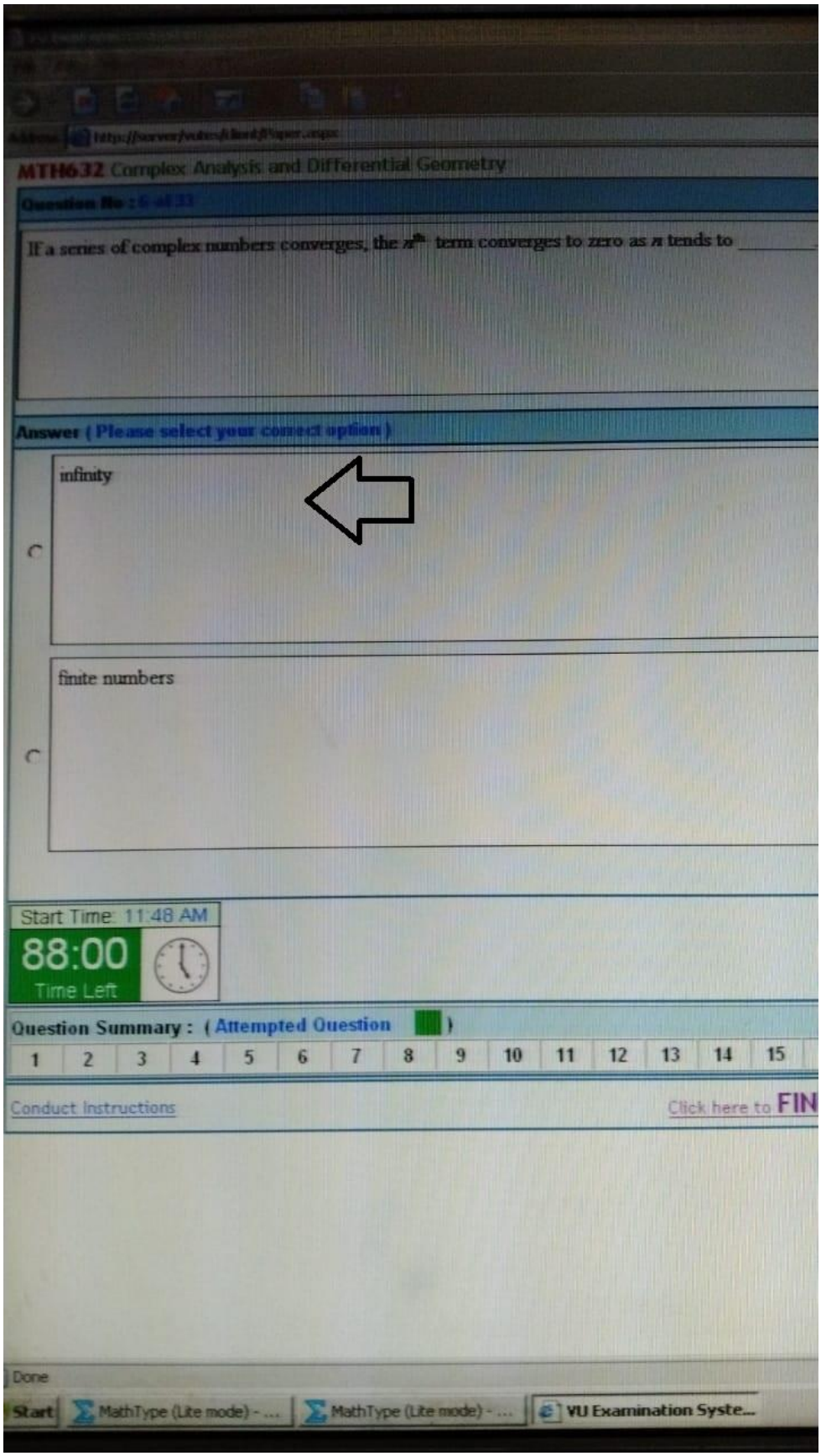
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the n^{th} term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
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Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

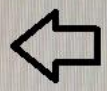
MTH632 Complex Analysis and Differential Geometry

Question No : 7 of 33

If $e^{ax}\cos y$ is harmonic then 'a' is _____.

Answer (Please select your correct option)

- 0
- -1
- 2
- i



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Question Summary : (Attempted Question)

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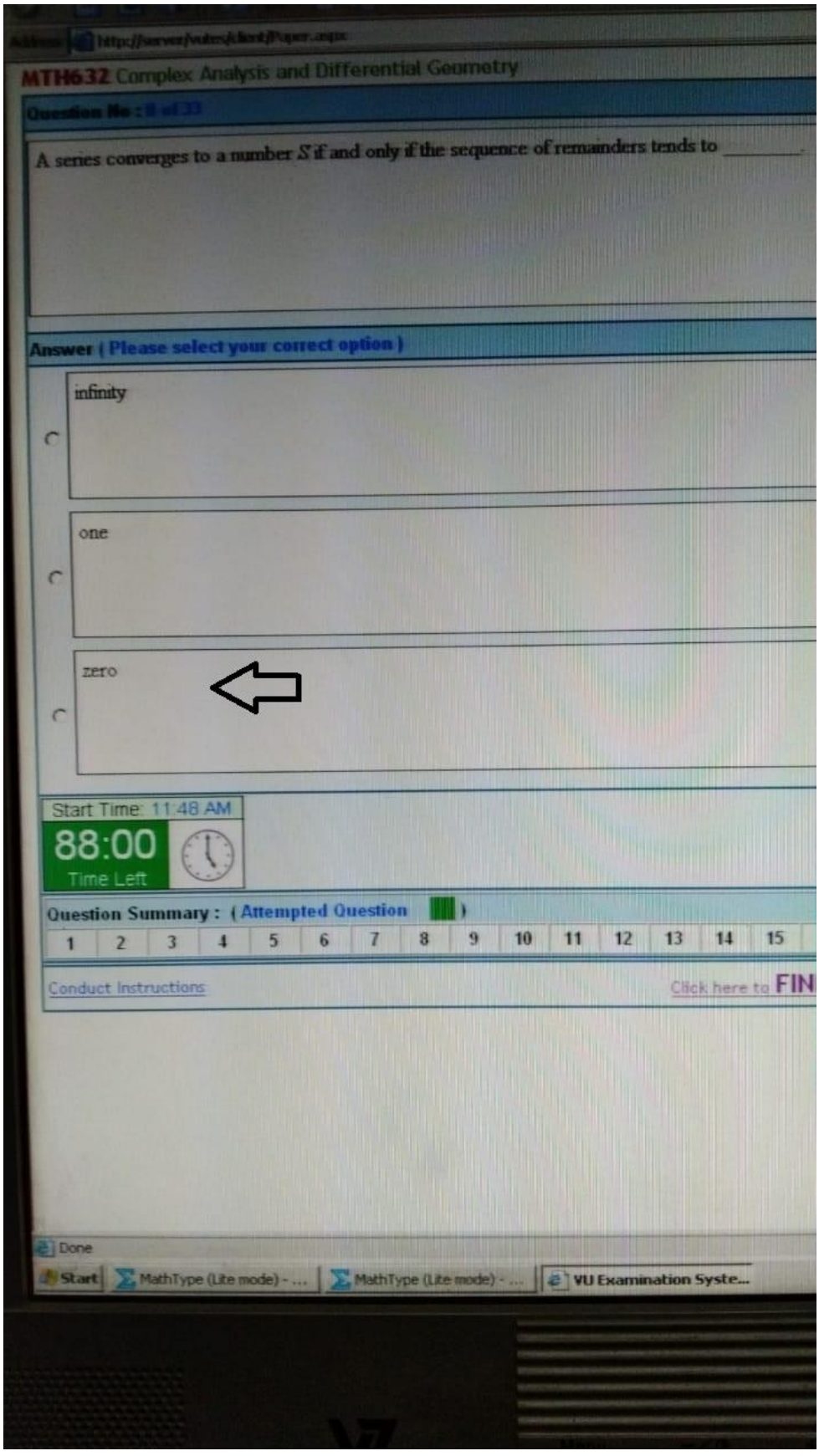
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
A series converges to a number S if and only if the sequence of remainders tends to _____.
Answer (Please select your correct option)
infinity
one
zero
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
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Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
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MTH632 Complex Analysis and Differential Geometry

Question No : 9 of 20

The points at which $f(z) = \frac{1}{z^2+1}$ is not analytic are _____.

Answer (Please select your correct option)

- 1 and -1
- i and $-i$
- 1 and i
- -1 and $-i$

Start Time: 11:48 AM

88:00
Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	1

Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

In the trigonometric functions of complex numbers $\sin z$ x = _____

Answer (Please select your correct option)

- $\frac{e^{ix} - e^{-ix}}{2i}$ ⇦
- $\frac{e^{ix} - e^{-ix}}{2}$
- $\frac{e^{-ix} - e^{ix}}{2i}$
- $\frac{e^{-ix} - e^{ix}}{2}$

Start Time: 11:48 AM

88:00

Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13

Conduct Instructions

Done

MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 20

The principal value of 1^{2i} is _____.

Answer (Please select your correct option)

- 0
- 1
- i
- 2i

Start Time: 11:48 AM

88:00

Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10

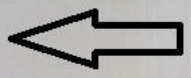
Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

Question No : 2 of 33

If Z_1 and Z_2 are two nonzero complex numbers then $Log(z_1 z_2) =$ _____.

Answer (Please select your correct option)

- $Log\, z_1\, Log\, z_2$
- $Log\, z_1 / Log\, z_2$
- $Log\, z_1 - Log\, z_2$
- $Log\, z_1 + Log\, z_2$ 

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Question Summary : (Attempted Question)

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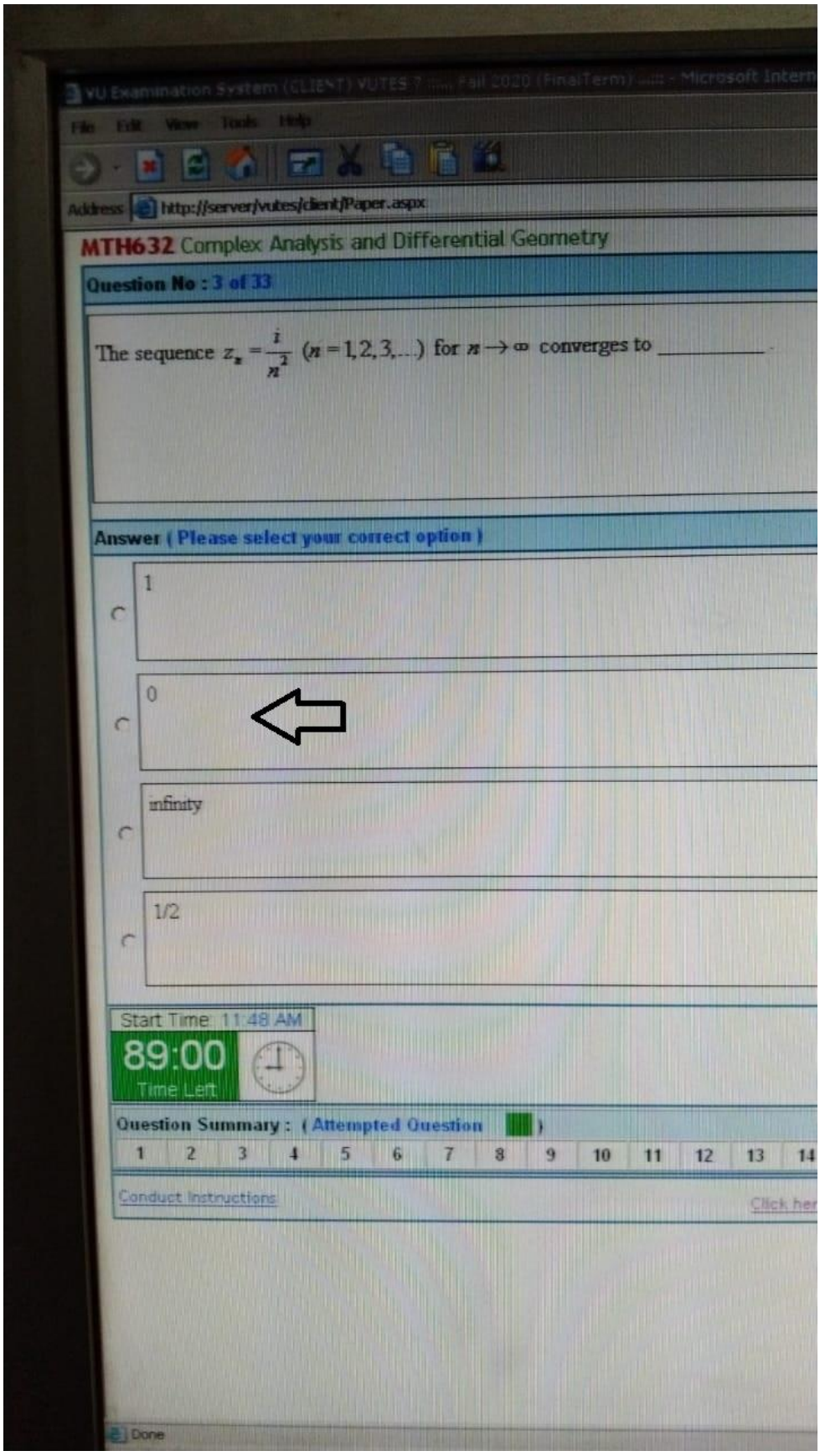
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Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 3 of 33
The sequence $z_n = \frac{i}{n^2}$ $(n = 1,2,3,...)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
1
0
infinity
1/2
Start Time: 11:48 AM
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Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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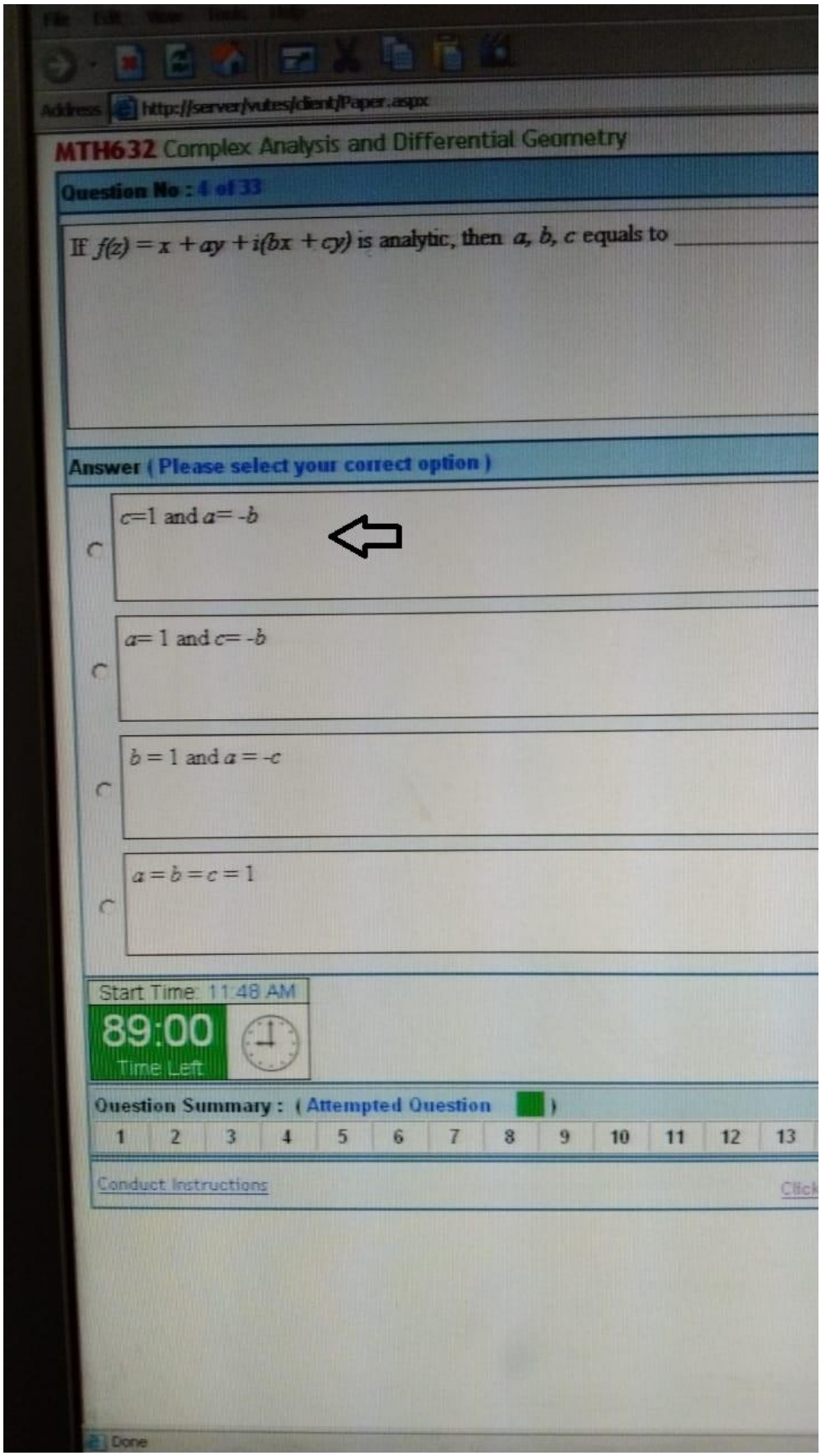
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 4 of 33
If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____
Answer (Please select your correct option)
$c=1$ and $a=-b$
$a=1$ and $c=-b$
$b=1$ and $a=-c$
$a=b=c=1$
Start Time: 11:48 AM
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Question Summary : (Attempted Question)
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Conduct Instructions
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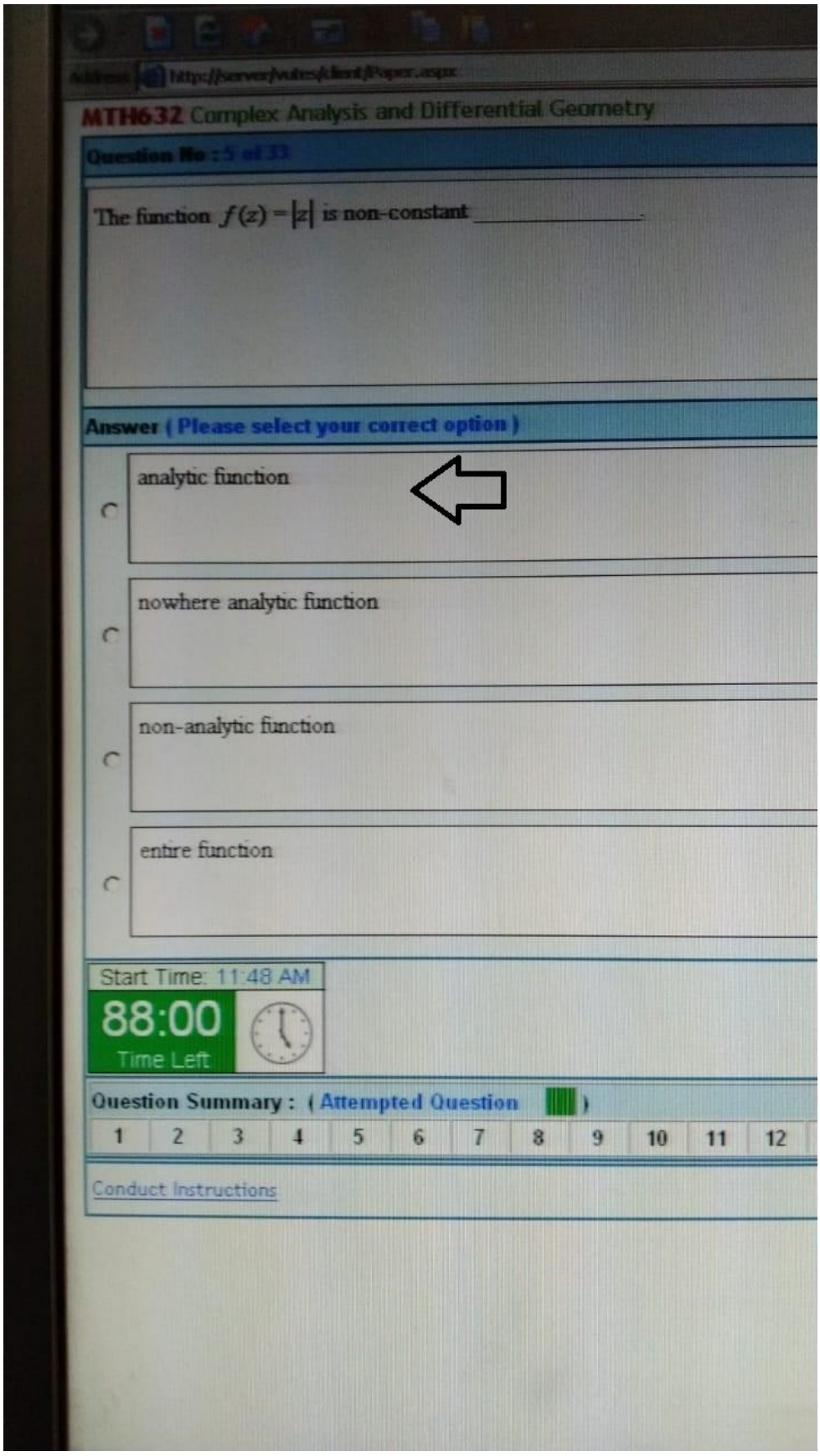
MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
The function $f(z) = |z|$ is non-constant _____.
Answer (Please select your correct option)
analytic function
nowhere analytic function
non-analytic function
entire function
Start Time: 11:48 AM
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Question Summary : (Attempted Question)
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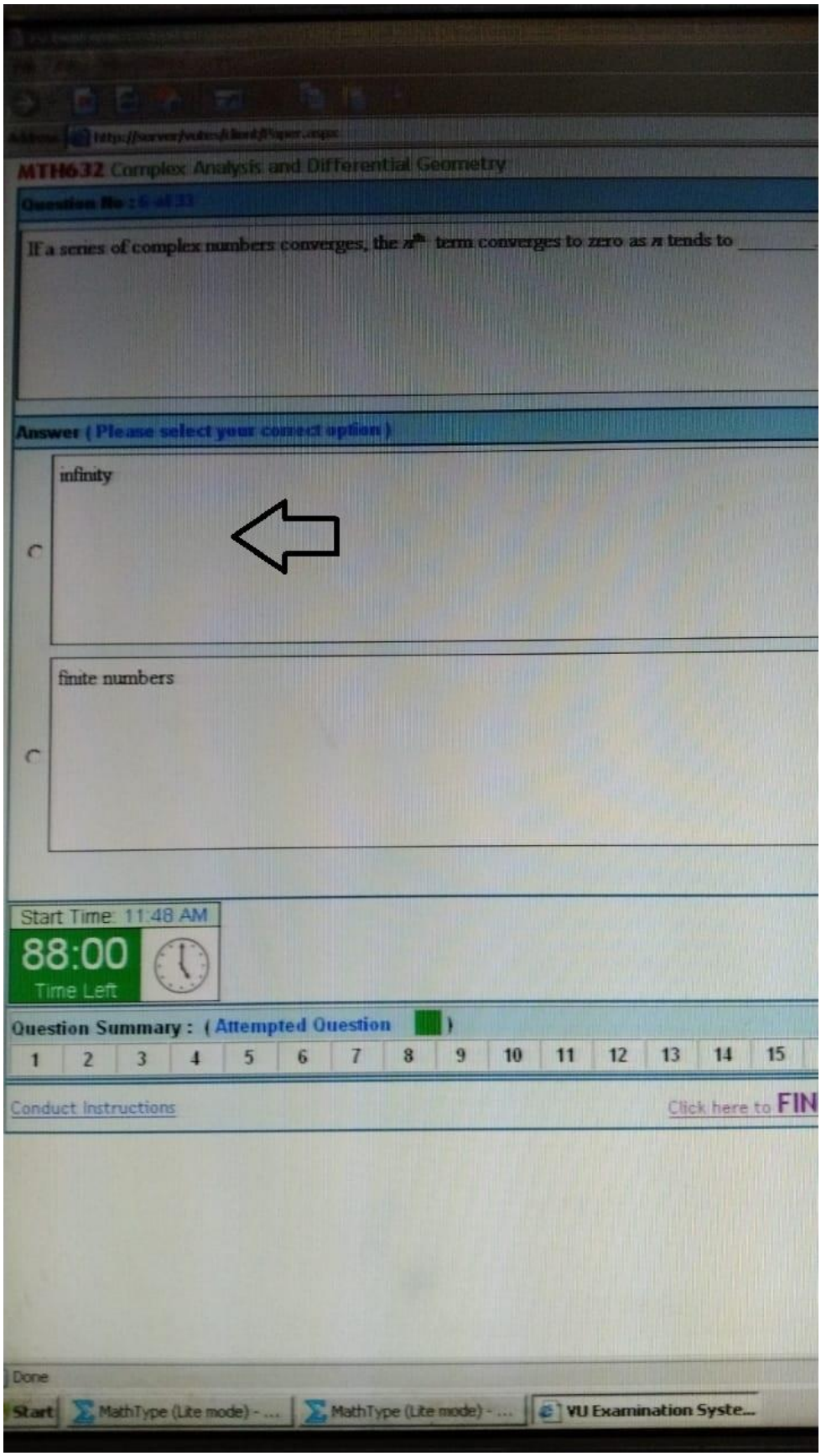
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the n^{th} term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
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Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
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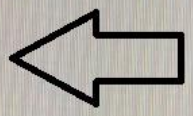
MTH632 Complex Analysis and Differential Geometry

Question No : 7 of 33

If $e^{ax}\cos y$ is harmonic then 'a' is _____.

Answer (Please select your correct option)

- 0
- -1
- 2
- i



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Question Summary : (Attempted Question)

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Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

Question No : 8 of 33

A series converges to a number S if and only if the sequence of remainders tends to _____.

Answer (Please select your correct option)

- infinity
- one
- zero

Start Time: 11:48 AM

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Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Conduct Instructions

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MTH632 Complex Analysis and Differential Geometry

Question No : 9 of 20

The points at which $f(z) = \frac{1}{z^2+1}$ is not analytic are _____.

Answer (Please select your correct option)

- 1 and -1
- i and $-i$
- 1 and i
- -1 and $-i$

Start Time: 11:48 AM

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Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	1

Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

In the trigonometric functions of complex numbers $\sin z$ $x =$ _____

Answer (Please select your correct option)

- $\frac{e^{ix} - e^{-ix}}{2i}$ ⇦
- $\frac{e^{ix} - e^{-ix}}{2}$
- $\frac{e^{-ix} - e^{ix}}{2i}$
- $\frac{e^{-ix} - e^{ix}}{2}$

Start Time: 11:48 AM

88:00

Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13

Conduct Instructions

Done

MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 20

The principal value of 1^{2i} is _____.

Answer (Please select your correct option)

- 0
- 1
- i
- 2i

Start Time: 11:48 AM

88:00

Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10

Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

In the trigonometric functions of complex numbers $\sin z$ $x =$ _____

Answer (Please select your correct option)

- $\frac{e^{ix} - e^{-ix}}{2i}$ ⇦
- $\frac{e^{ix} - e^{-ix}}{2}$
- $\frac{e^{-ix} - e^{ix}}{2i}$
- $\frac{e^{-ix} - e^{ix}}{2}$

Start Time: 11:48 AM

88:00

Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13

Conduct Instructions

Done

MTH632 Complex Analysis and Differential Geometry

In the trigonometric functions of complex numbers $\sin z$ $x =$ _____

Answer (Please select your correct option)

- $\frac{e^{ix} - e^{-ix}}{2i}$
- $\frac{e^{ix} - e^{-ix}}{2}$
- $\frac{e^{-ix} - e^{ix}}{2i}$
- $\frac{e^{-ix} - e^{ix}}{2}$

Start Time: 11:48 AM

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Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13

Conduct Instructions

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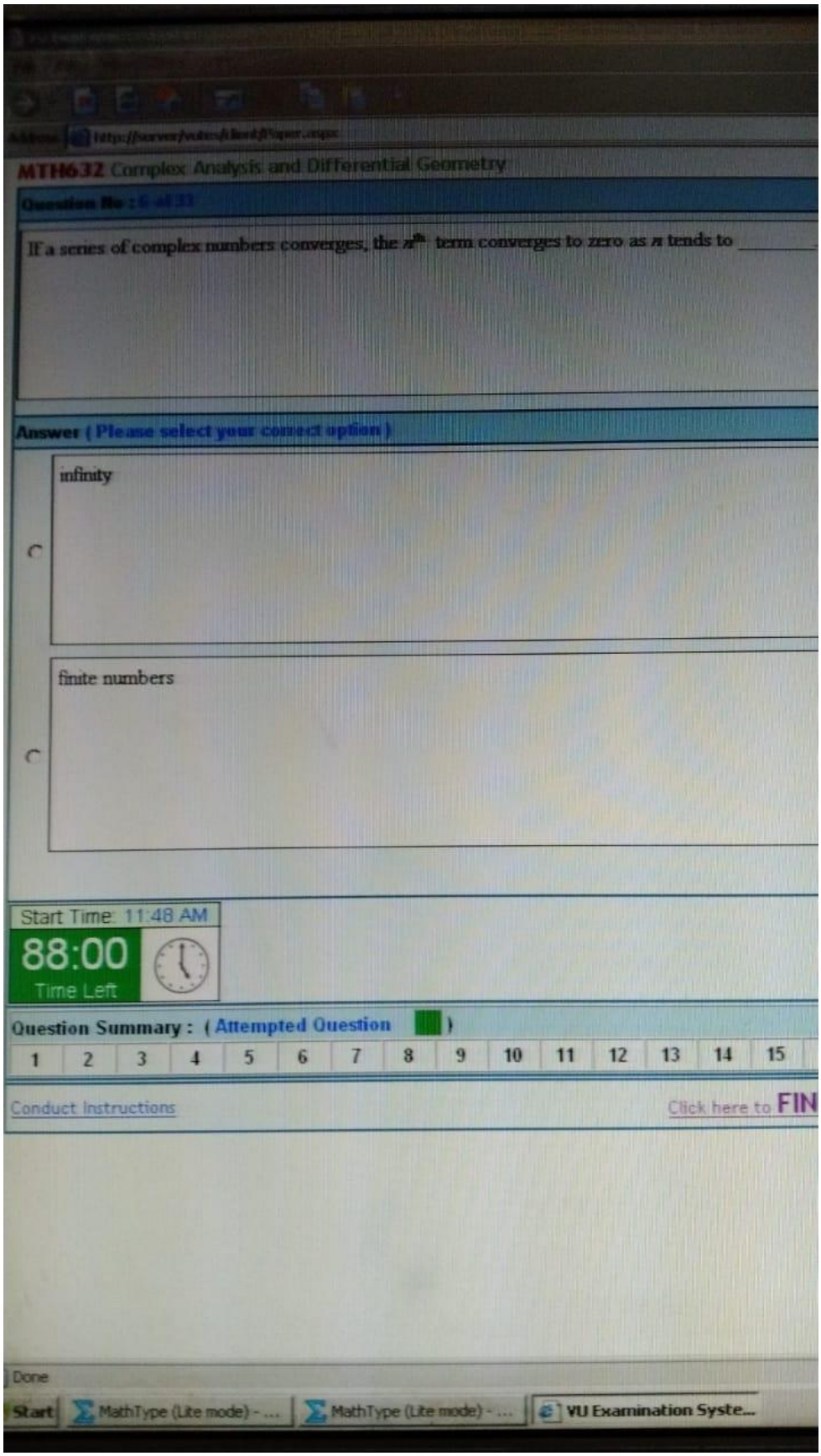
MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the n^{th} term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
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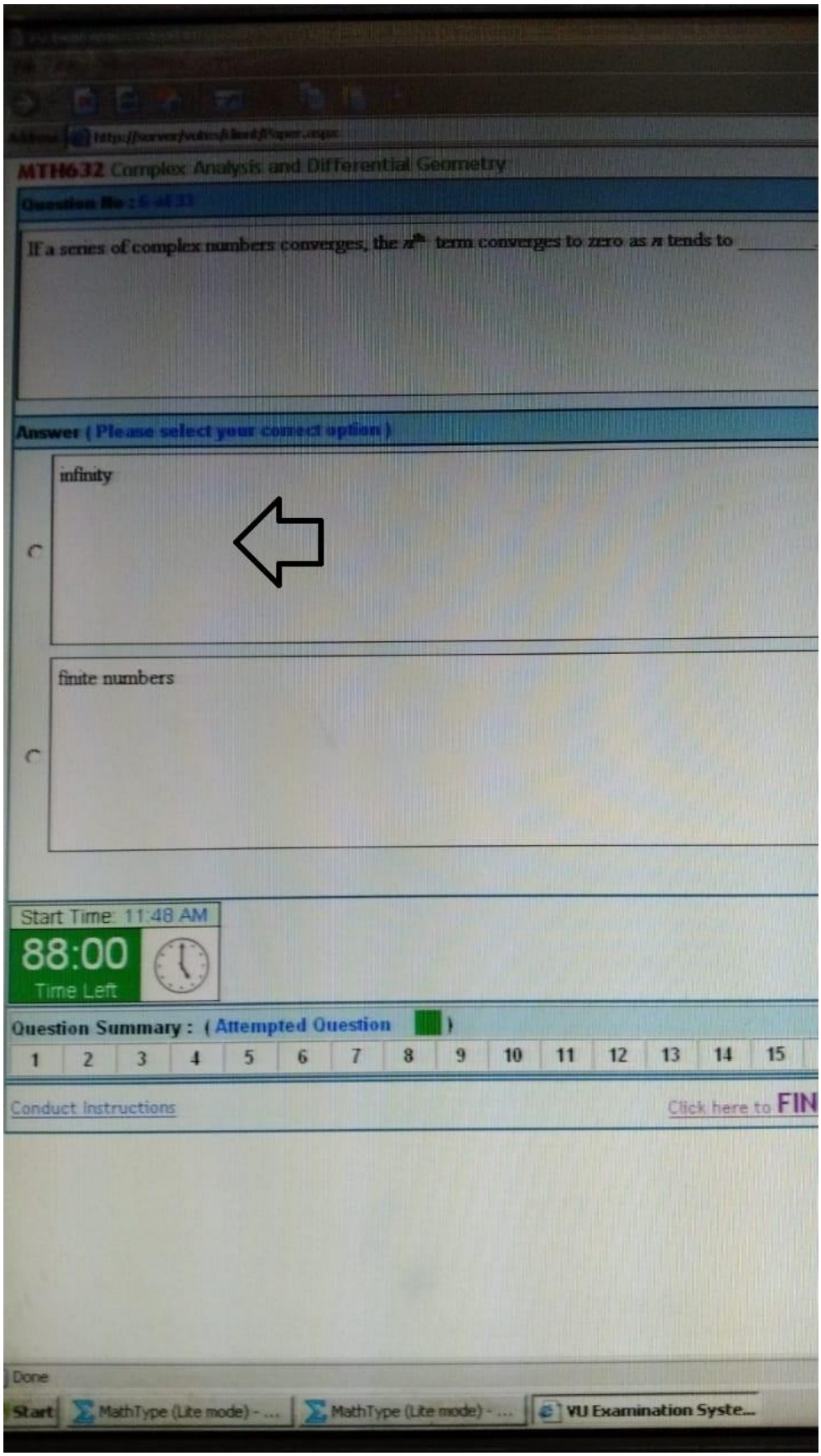
MTH632 Complex Analysis and Differential Geometry
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finite numbers
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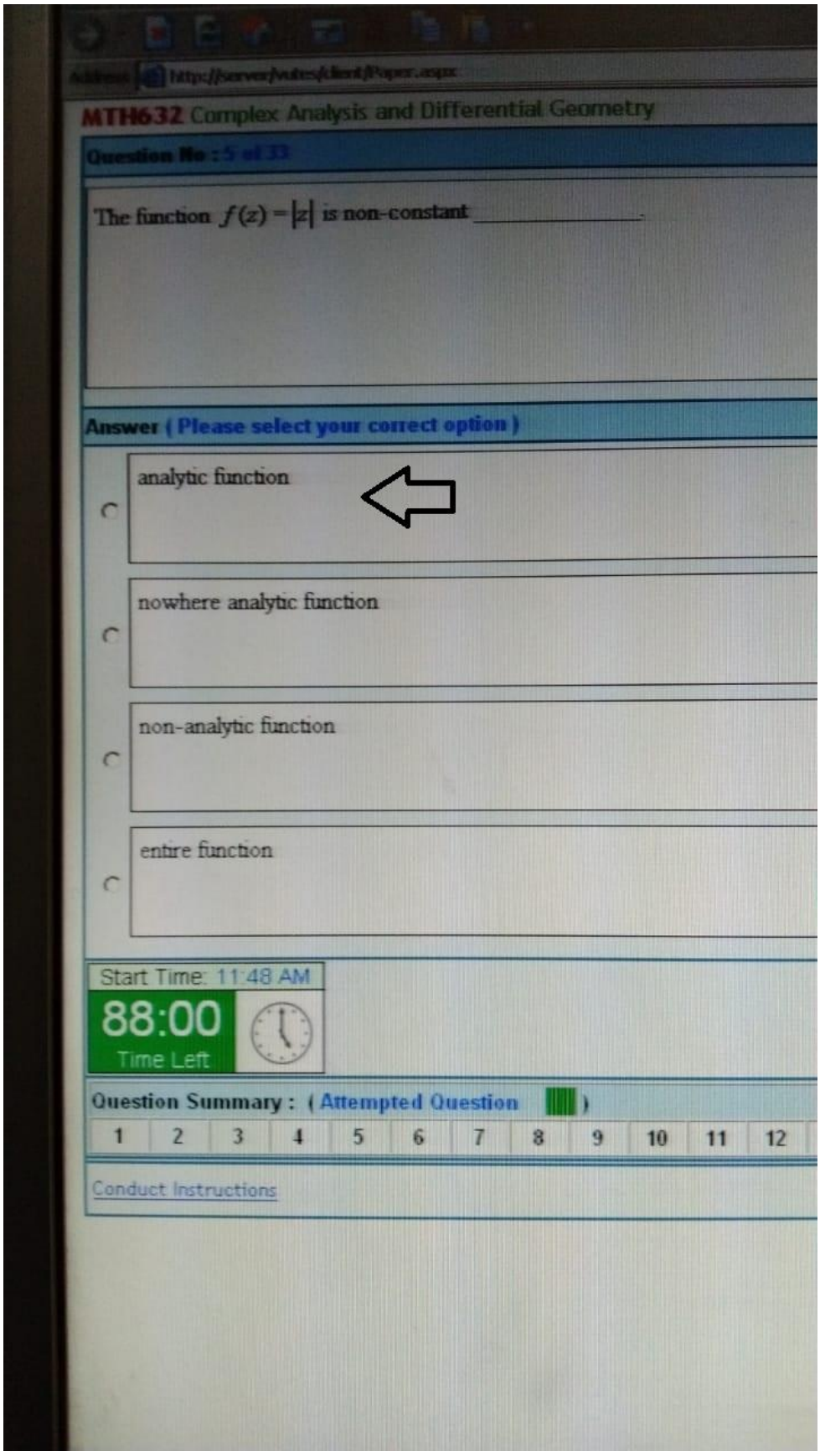
MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
The function $f(z) = |z|$ is non-constant _____.
Answer (Please select your correct option)
analytic function
nowhere analytic function
non-analytic function
entire function
Start Time: 11:48 AM
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Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

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MTH632 Complex Analysis and Differential Geometry

Question No : 4 of 33

If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____

Answer (Please select your correct option)

- $c=1$ and $a=-b$
- $a=1$ and $c=-b$
- $b=1$ and $a=-c$
- $a=b=c=1$

Start Time: 11:48 AM

89:00 Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13

Conduct Instructions

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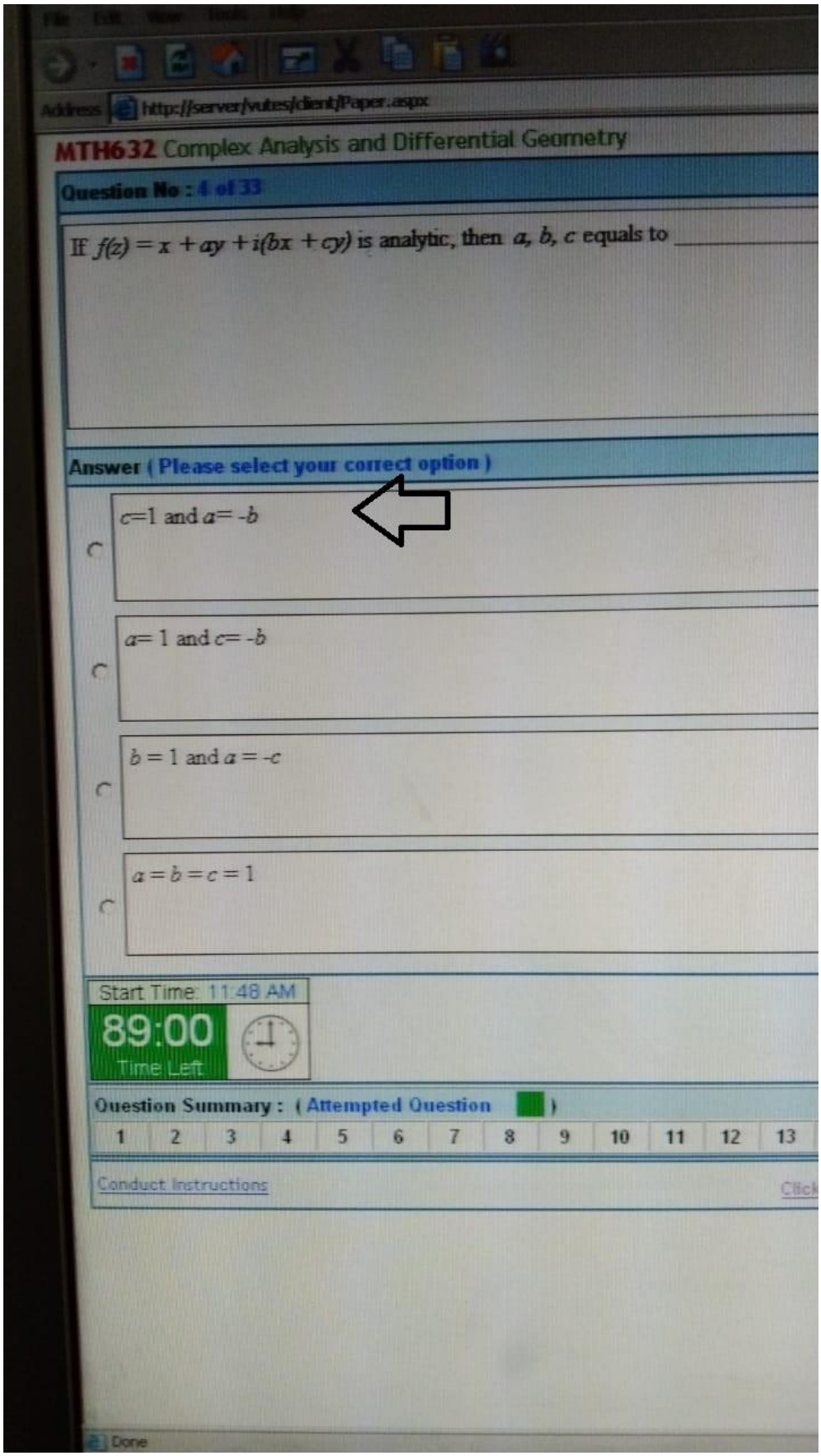
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MTH632 Complex Analysis and Differential Geometry
Question No : 4 of 33
If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____
Answer (Please select your correct option)
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$a=1$ and $c=-b$
$b=1$ and $a=-c$
$a=b=c=1$
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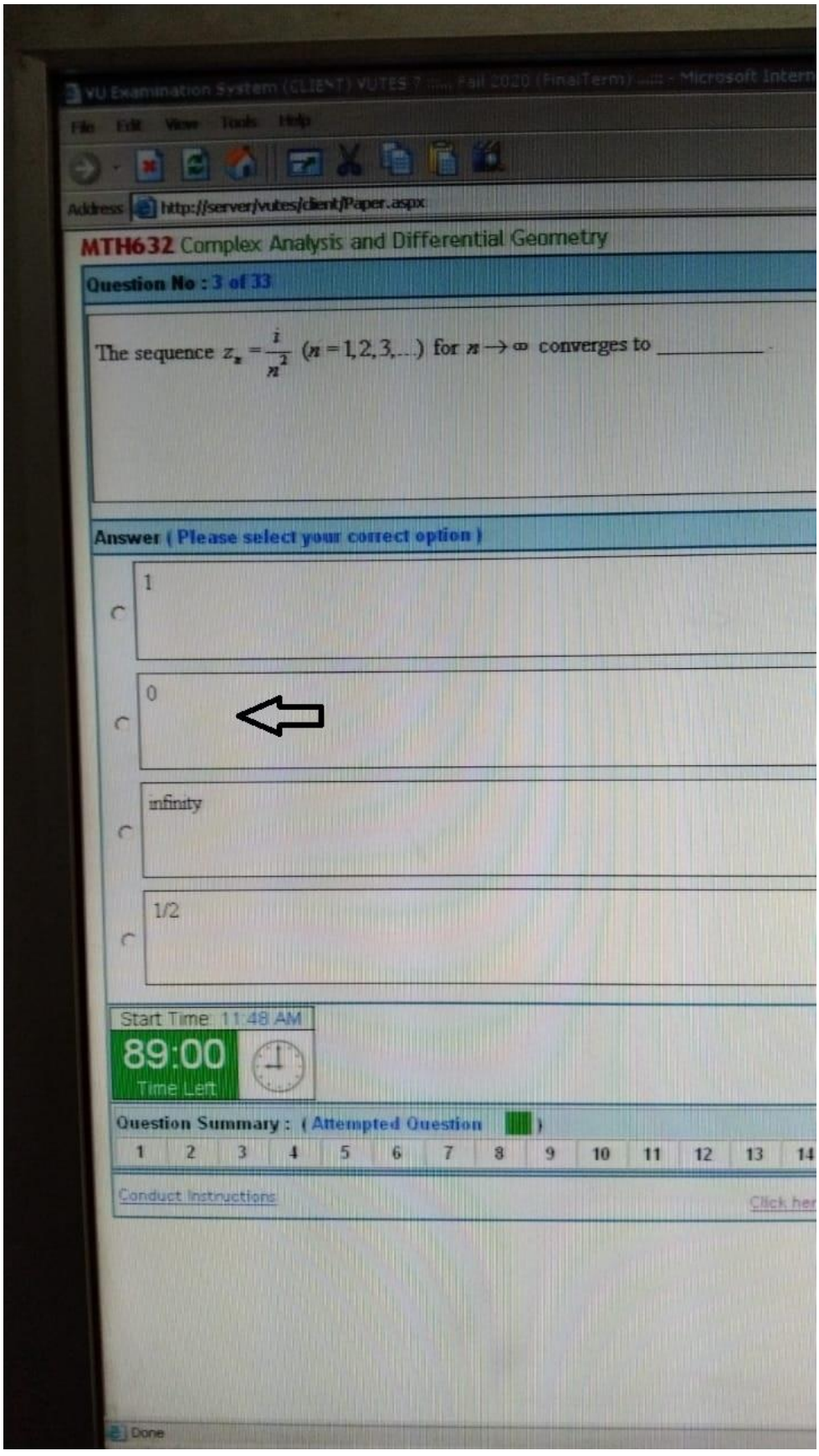
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Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 3 of 33
The sequence $z_n = \frac{i}{n^2}$ $(n = 1,2,3,...)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
1
0
infinity
1/2
Start Time: 11:48 AM
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Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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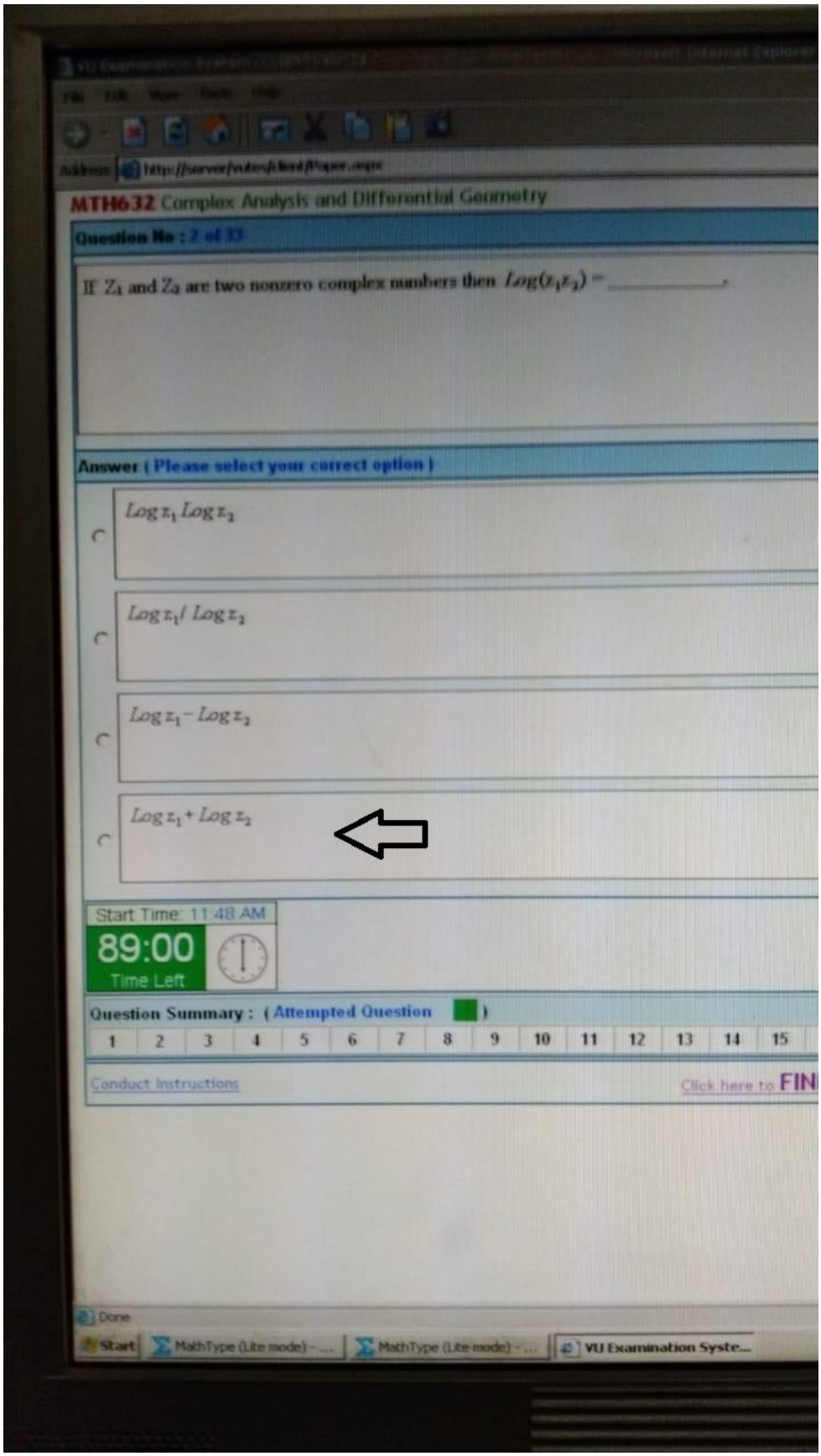
MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
If Z_1 and Z_2 are two nonzero complex numbers then $Log(z_1 z_2) =$ _____.
Answer (Please select your correct option)
$Log\, z_1\, Log\, z_2$
$Log\, z_1 / Log\, z_2$
$Log\, z_1 - Log\, z_2$
$Log\, z_1 + Log\, z_2$
Start Time: 11:48 AM
89:00
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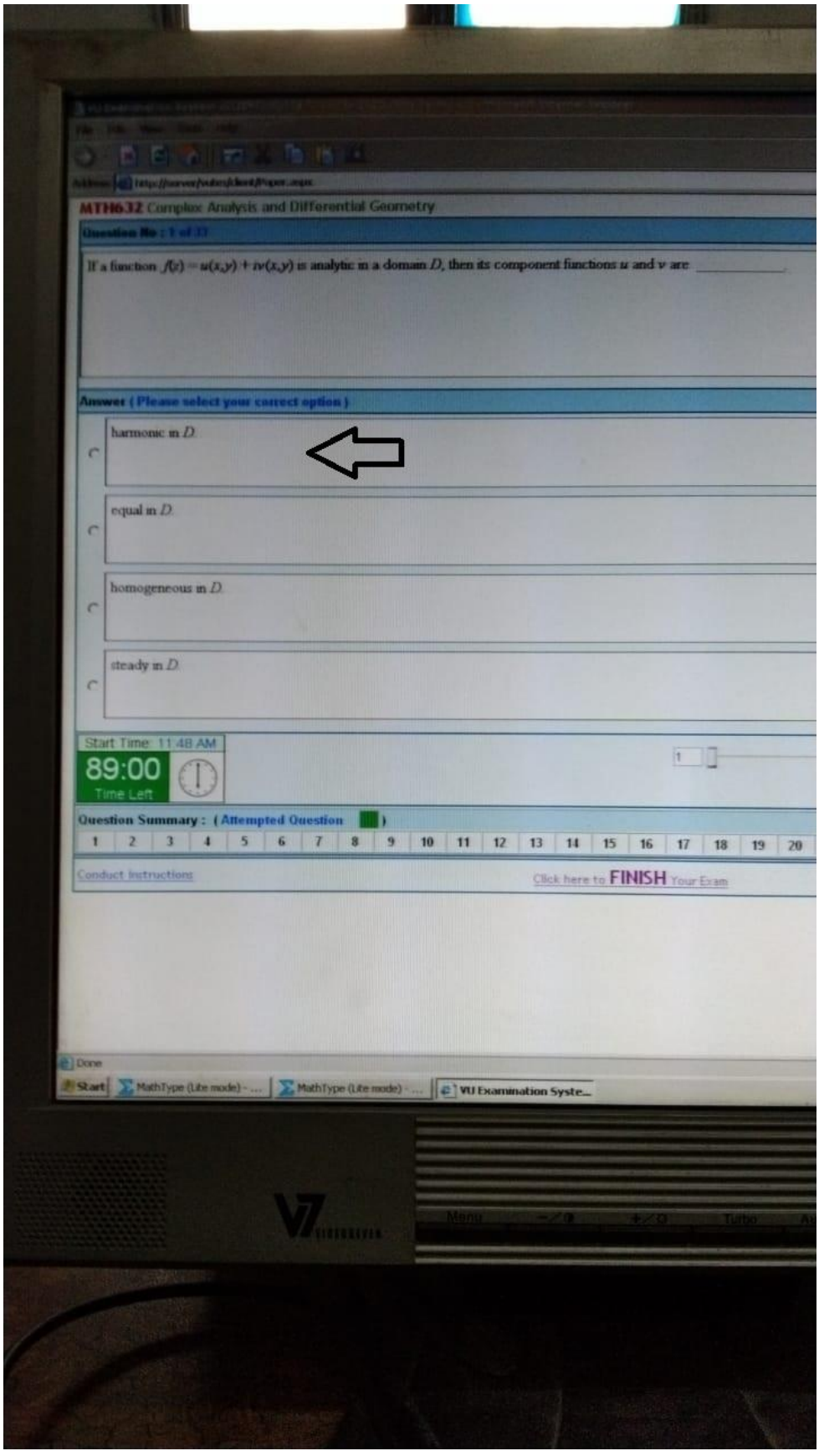
MathType (Lite mode) - ...
VU Examination Syste...


MTH632 Complex Analysis and Differential Geometry
If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____.
Answer (Please select your correct option)
harmonic in D.
equal in D.
homogeneous in D.
steady in D.
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

Address http://server/vutes/client/Paper.aspx

MTH632 Complex Analysis and Differential Geometry

Question No : 12 of 33

Let Z be a complex valued function then $\sinh Z =$ _____

Answer (Please select your correct option)

- $\frac{e^{z} - e^{-z}}{2}$ ⇦
- $\frac{e^{z} + e^{-z}}{2}$
- $\frac{e^{z} - e^{-z}}{2i}$
- $\frac{e^{z} + e^{-z}}{2i}$

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83:00
Time Left

Question Summary : (Attempted Question)

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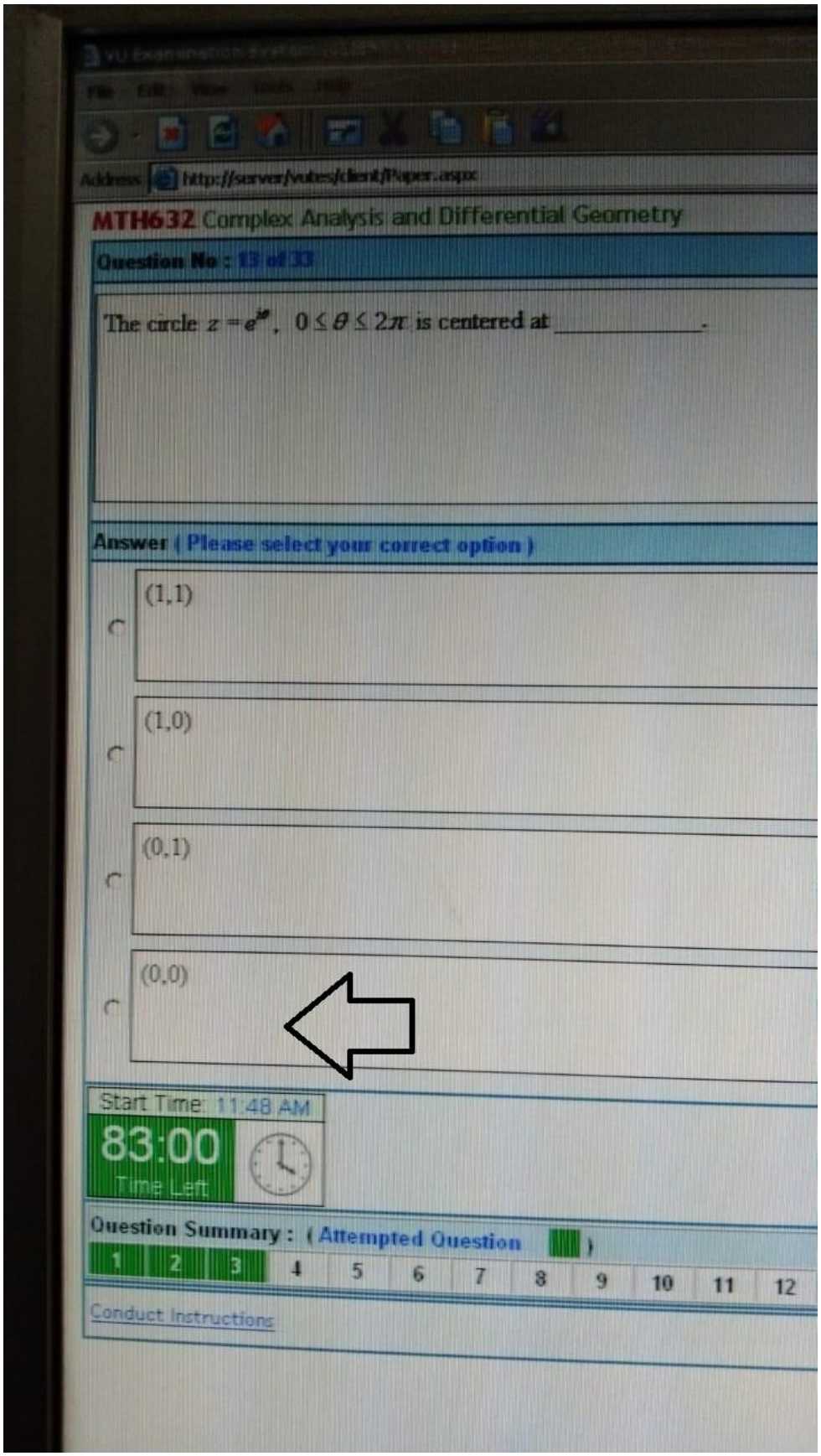
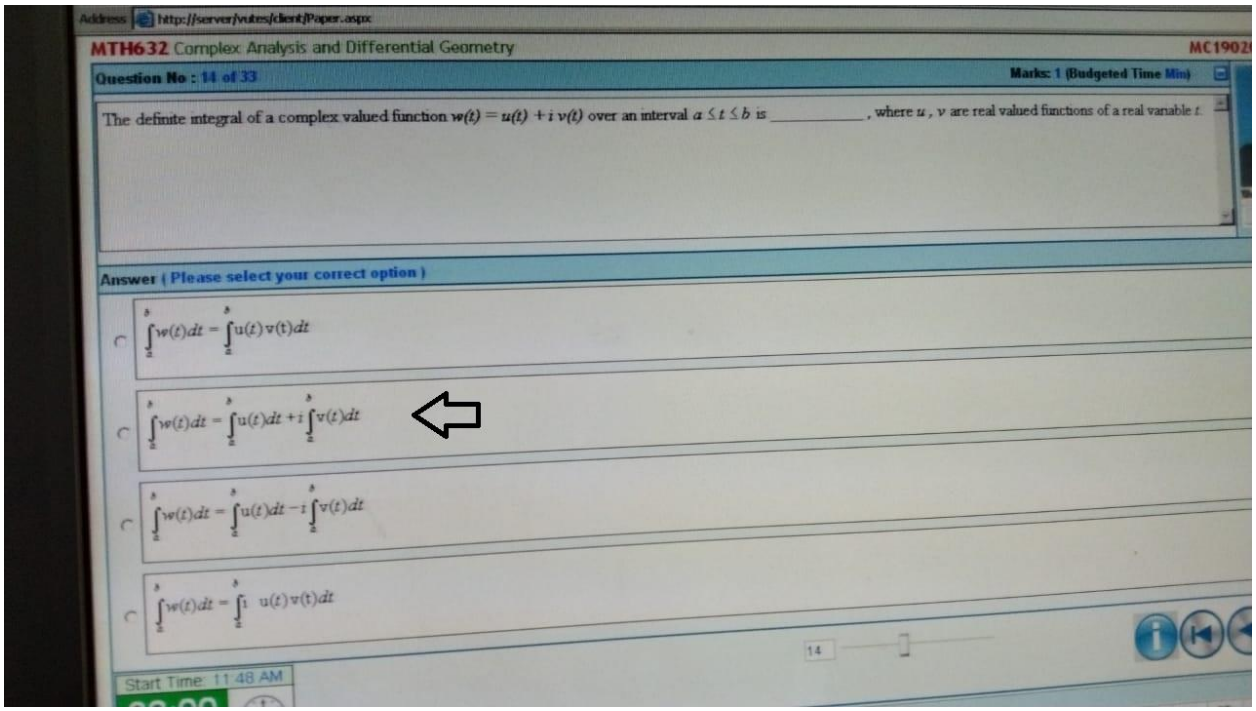
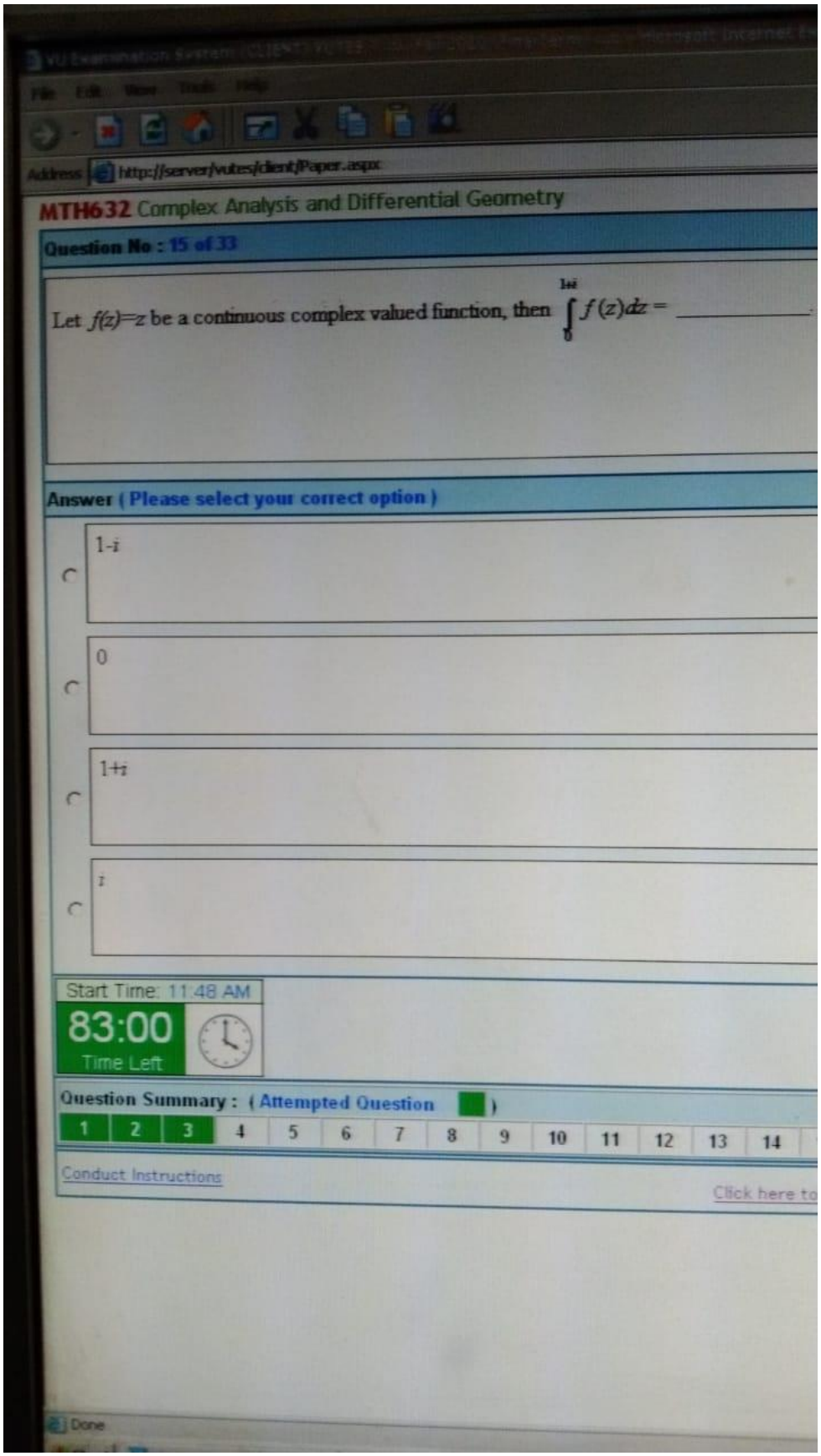
Conduct Instructions


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
MC1902
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functions of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b i\ u(t)v(t)dt$
14
Start Time: 11:48 AM


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let $f(z)=z$ be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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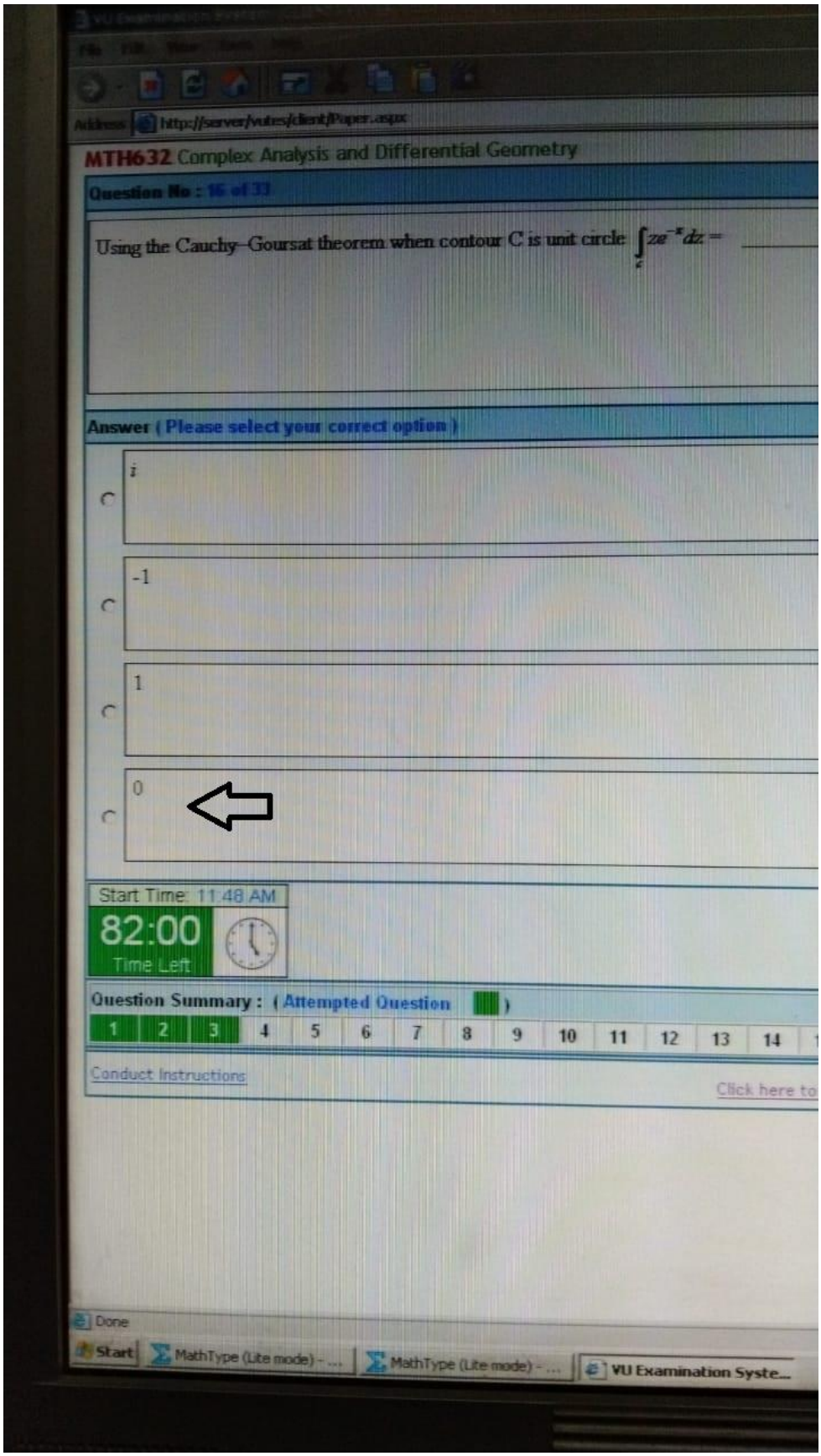
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 16 of 33
Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$ _____
Answer (Please select your correct option)
i
-1
1
0
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
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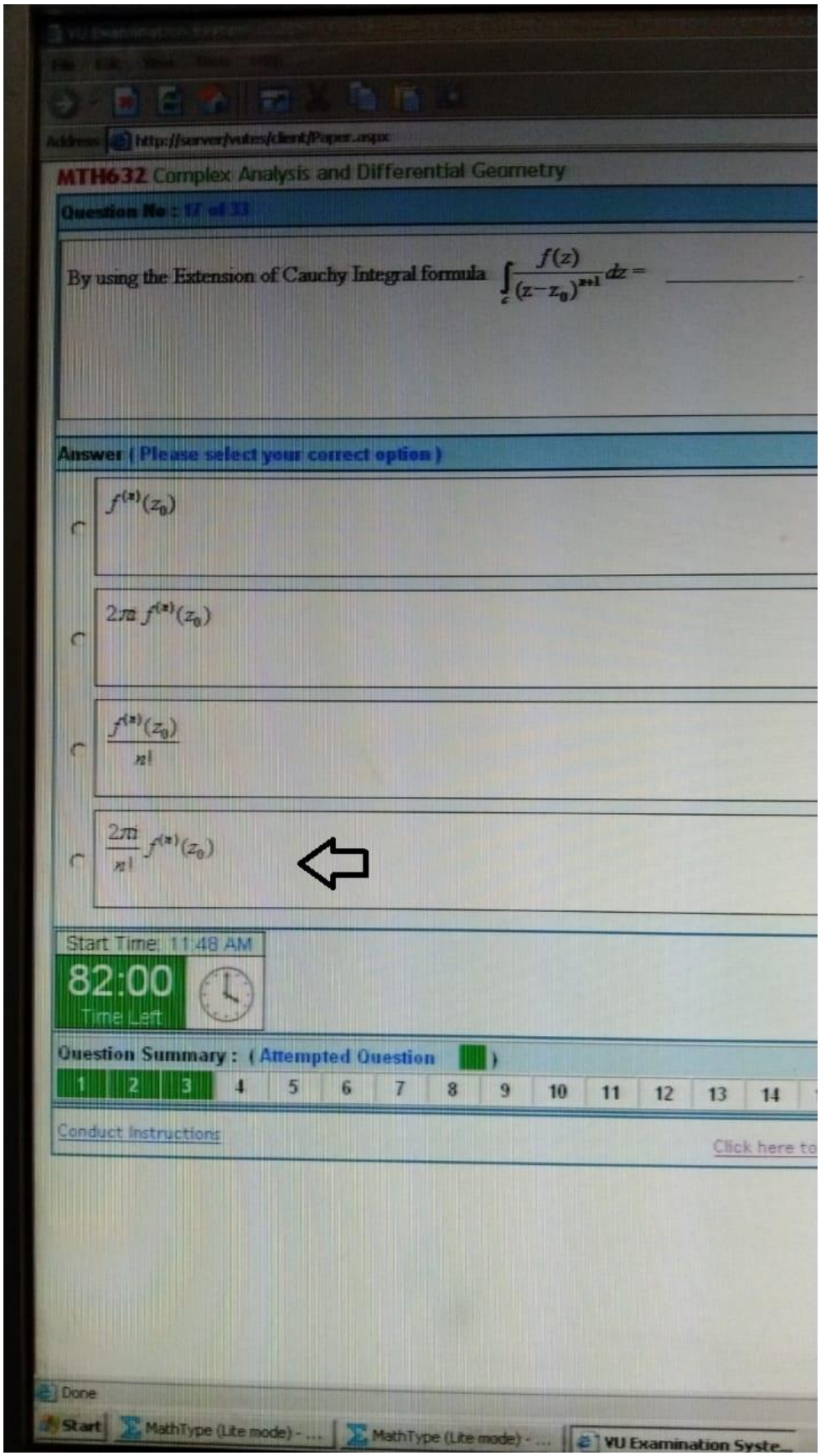
MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
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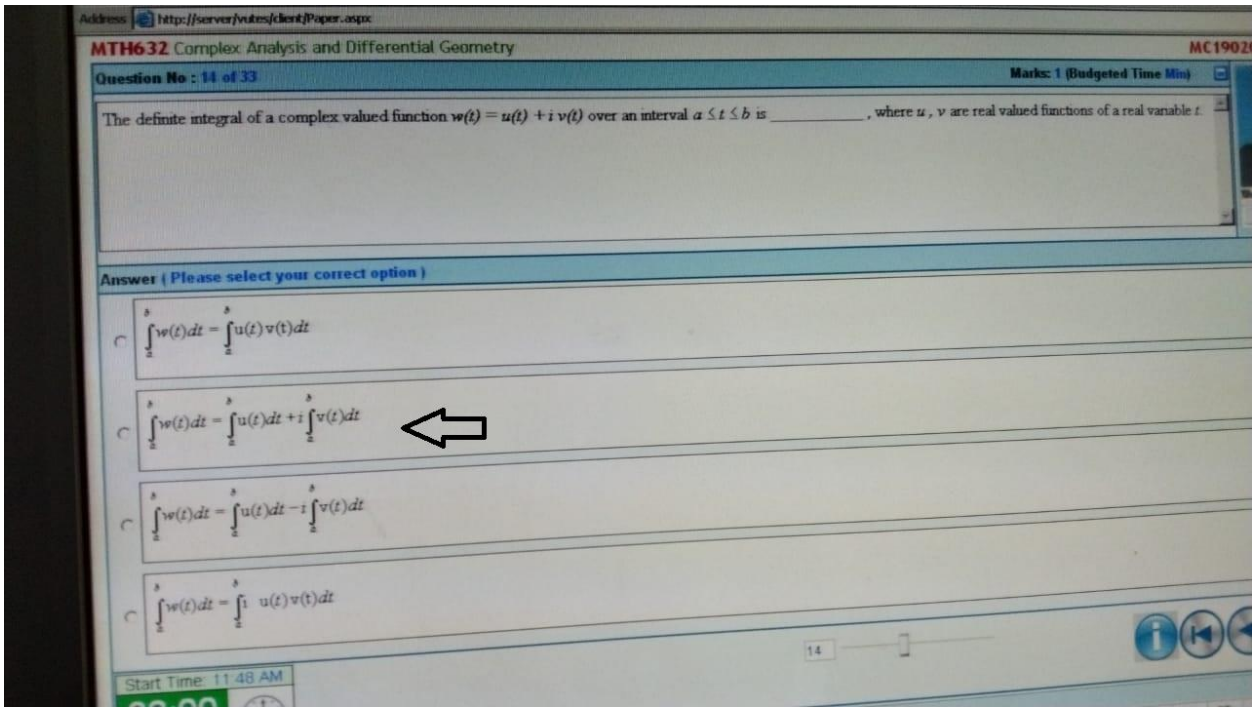
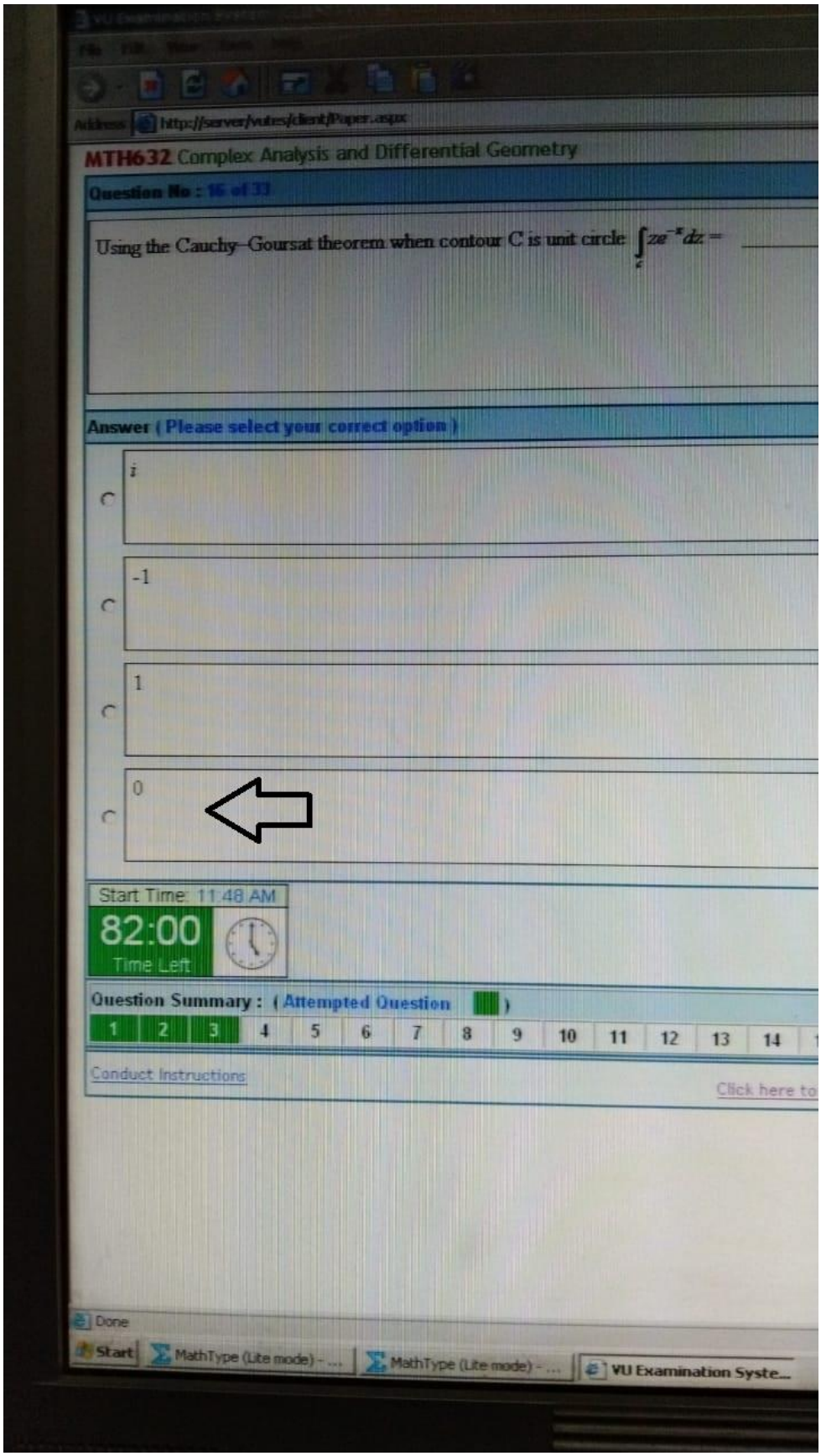
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
MC1902
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functions of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b i\ u(t)v(t)dt$
14
Start Time: 11:48 AM


http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 16 of 33
Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$ _____
Answer (Please select your correct option)
i
-1
1
0
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
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Start
MathType (Lite mode) - ...
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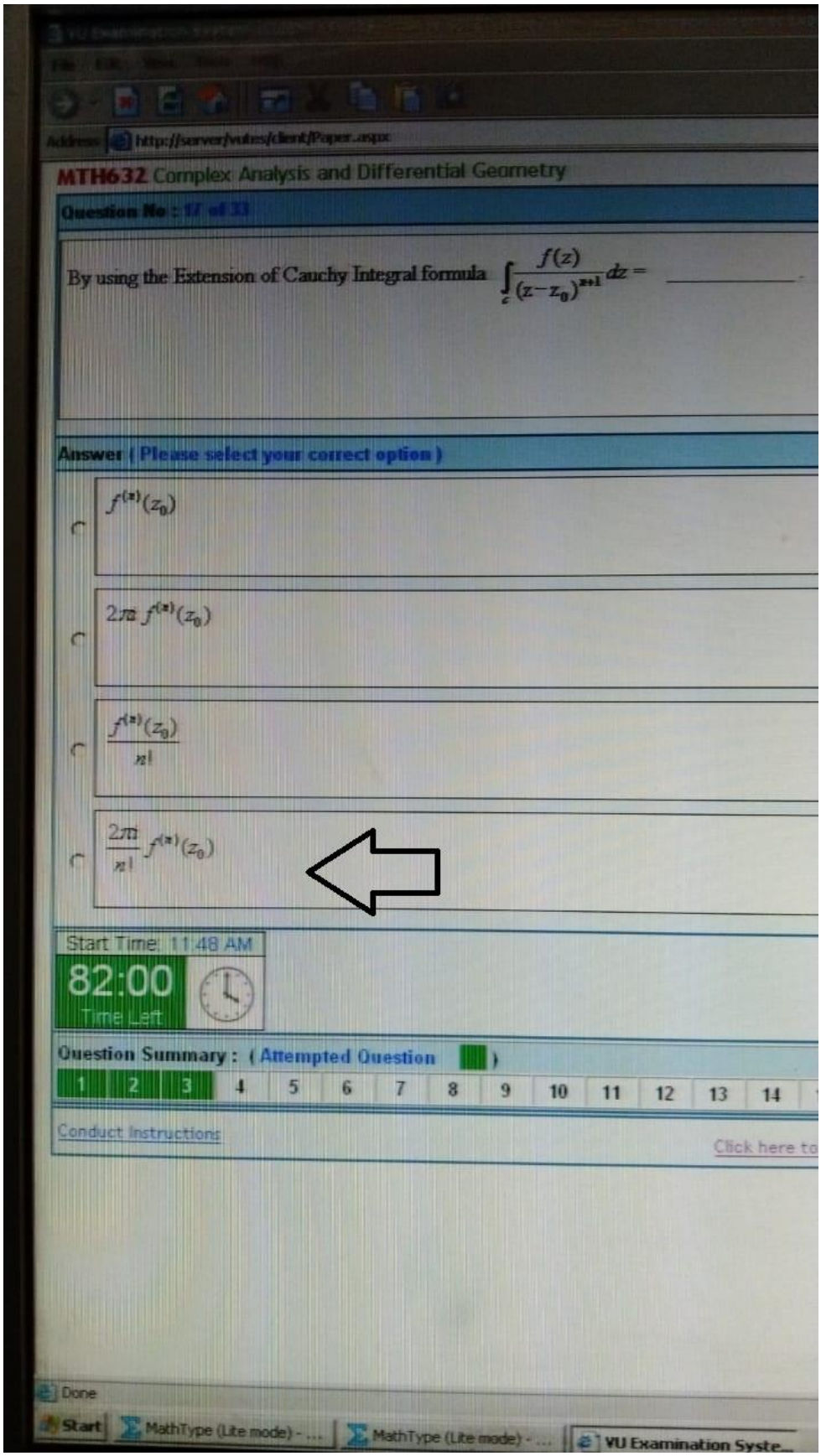
VU Examination Syste...


MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
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Start
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MathType (Lite mode) - ...
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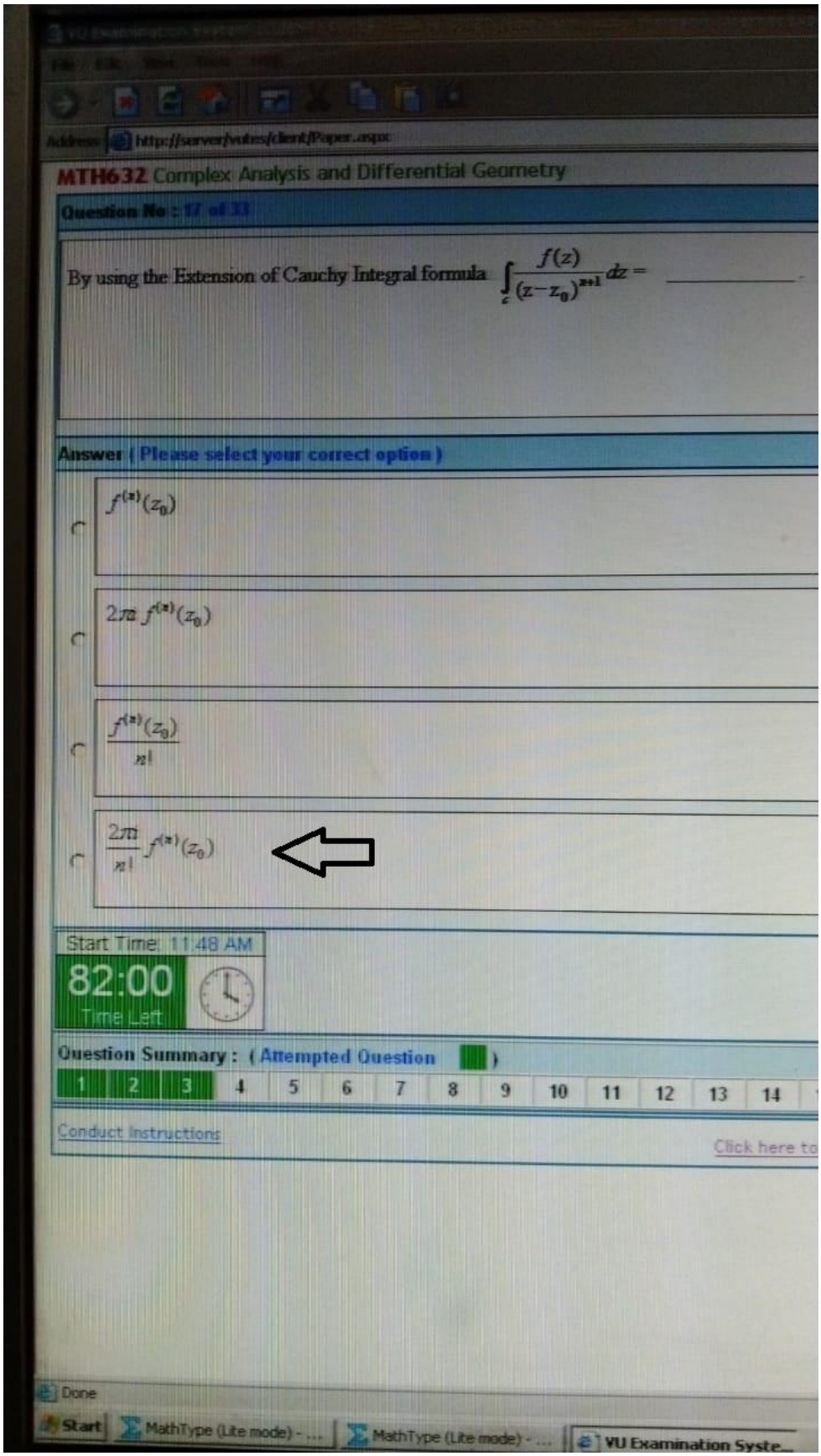
MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
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Start
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MathType (Lite mode) - ...
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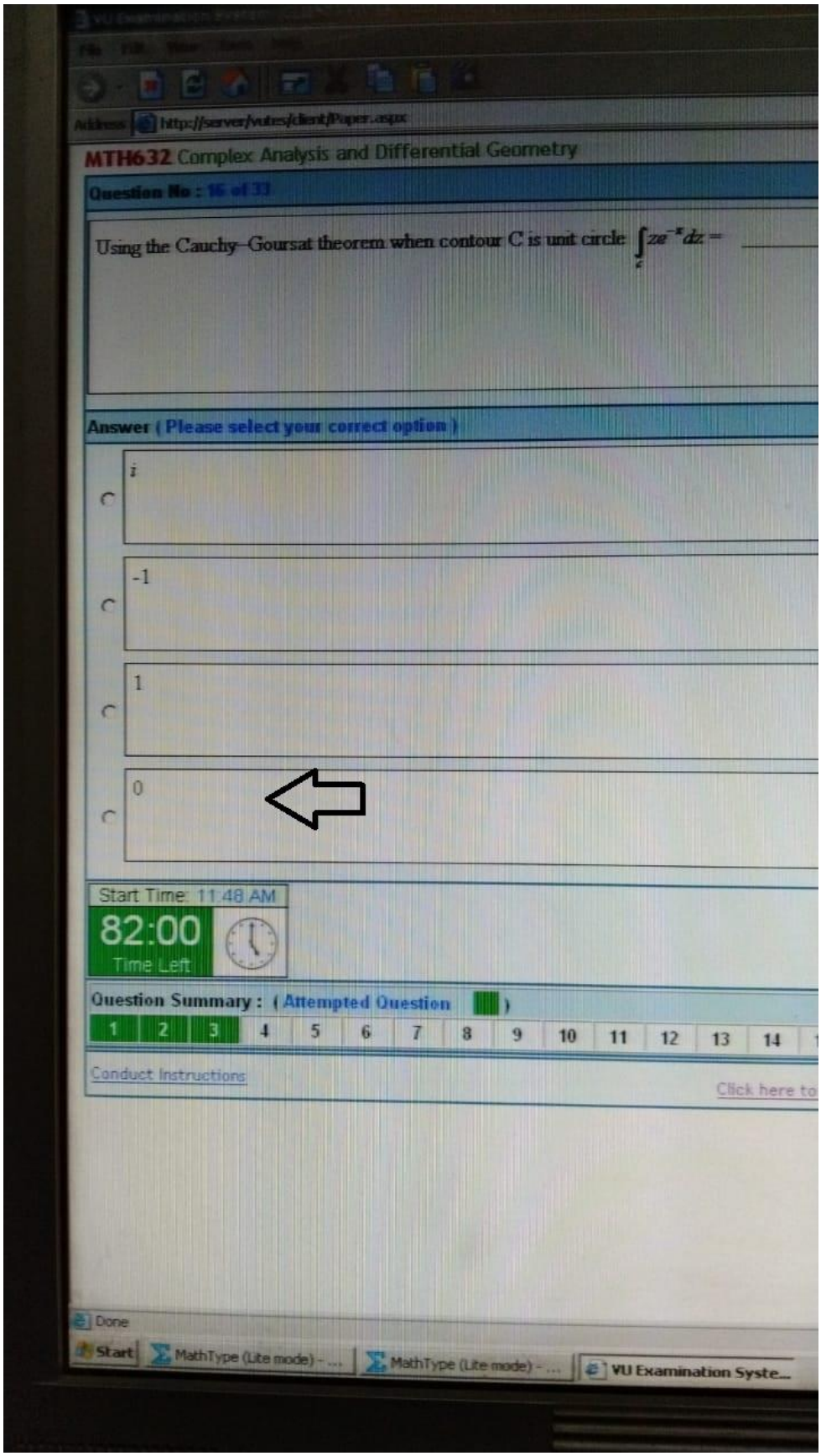
MTH632 Complex Analysis and Differential Geometry
Question No :
Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$
Answer (Please select your correct option)
i
-1
1
0
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Done
Start
MathType (Lite mode) - ...
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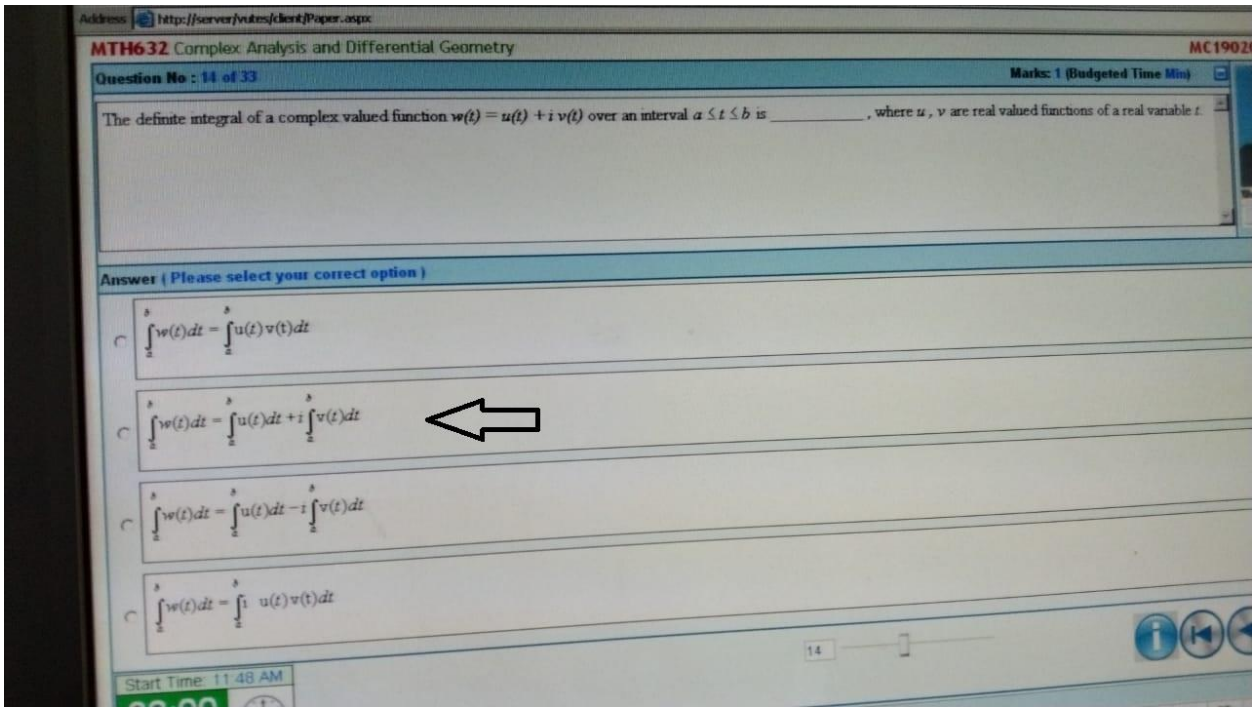
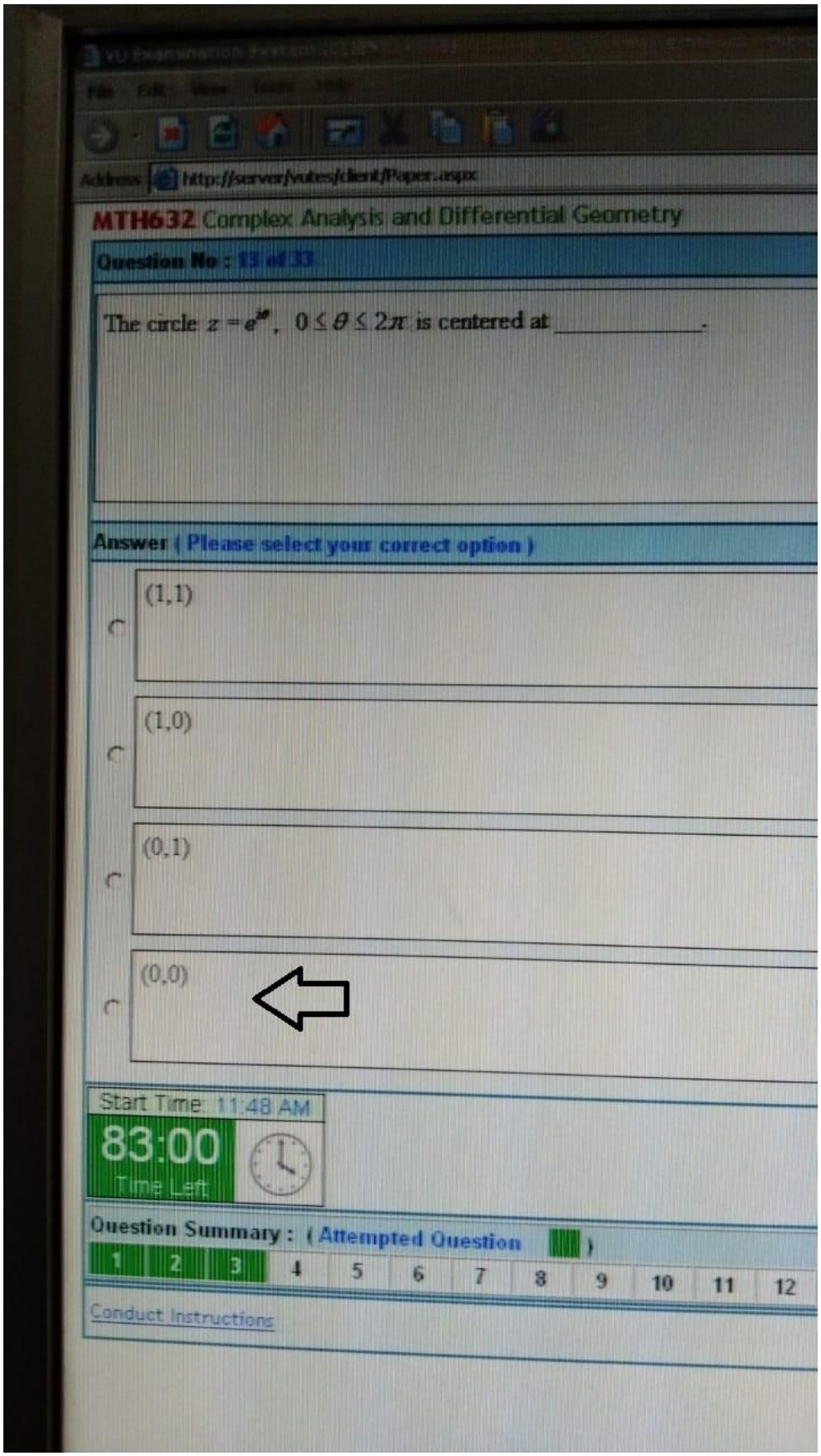

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
MC1902
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functions of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
14
Start Time: 11:48 AM


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

Address http://server/vutes/client/Paper.aspx

MTH632 Complex Analysis and Differential Geometry

Question No : 12 of 33

Let Z be a complex valued function then $\sinh Z =$ _____

Answer (Please select your correct option)

- $\frac{e^{z} - e^{-z}}{2}$ ⇦
- $\frac{e^{z} + e^{-z}}{2}$
- $\frac{e^{z} - e^{-z}}{2i}$
- $\frac{e^{z} + e^{-z}}{2i}$

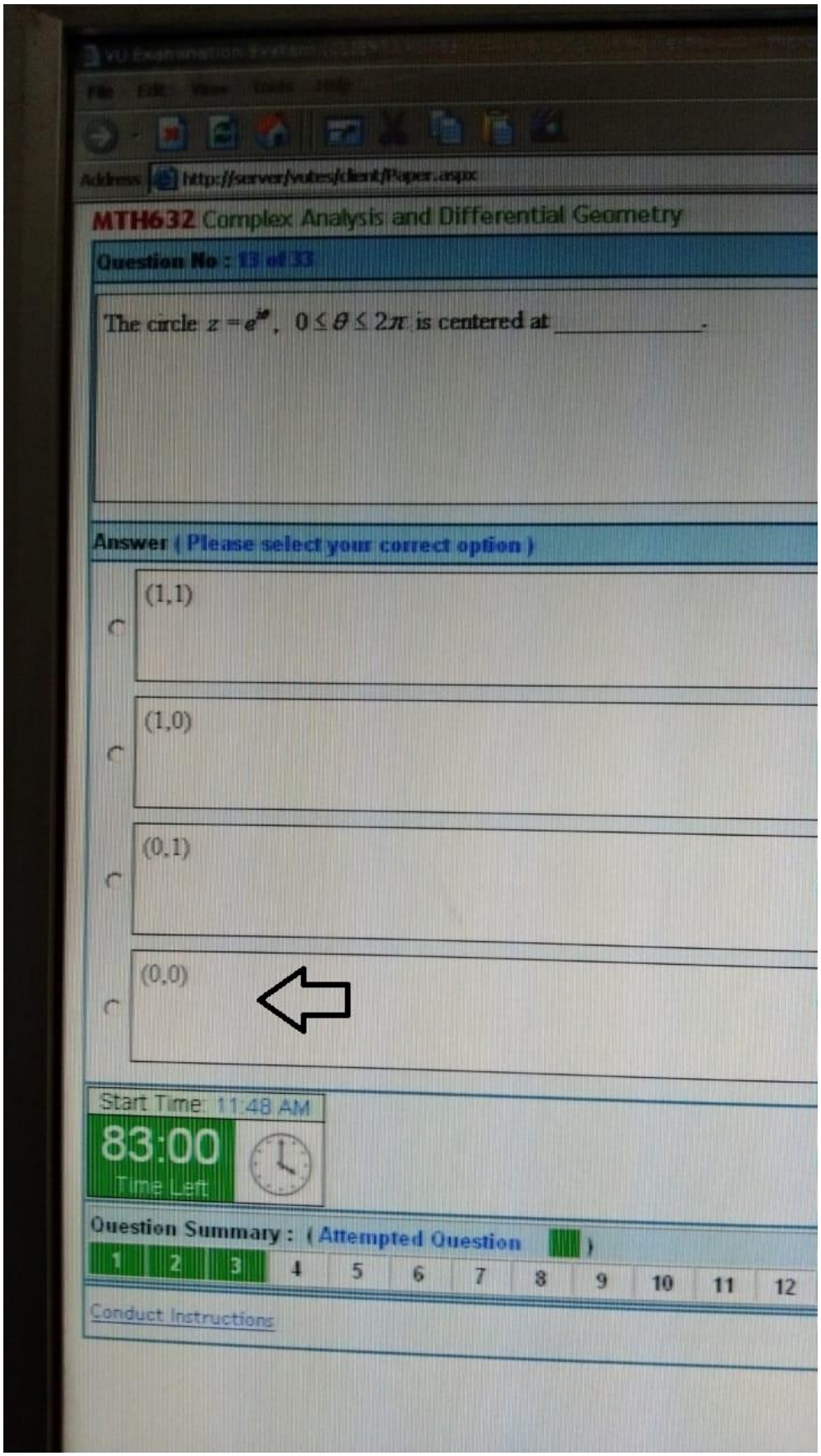
Start Time: 11:48 AM

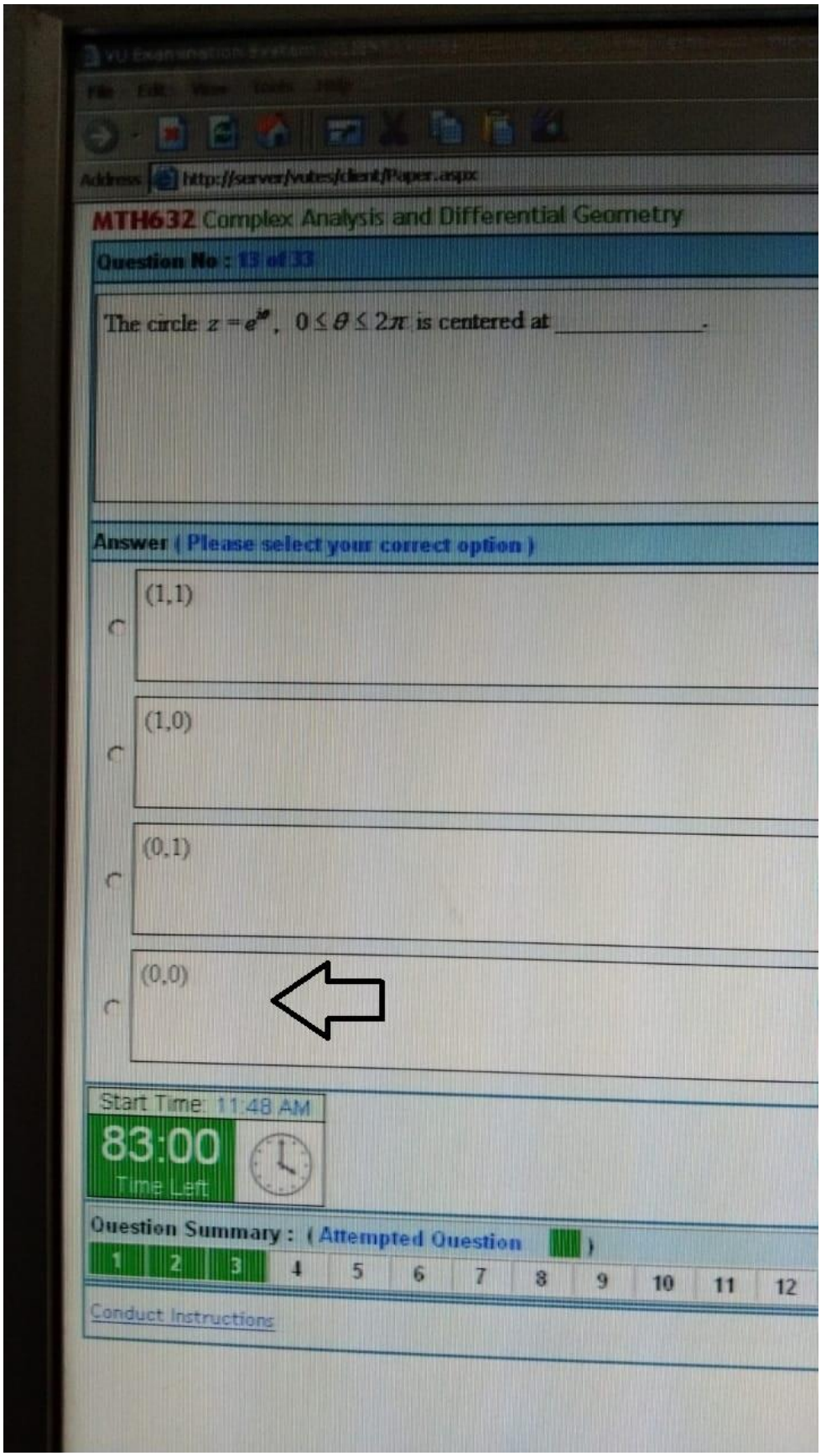
83:00
Time Left

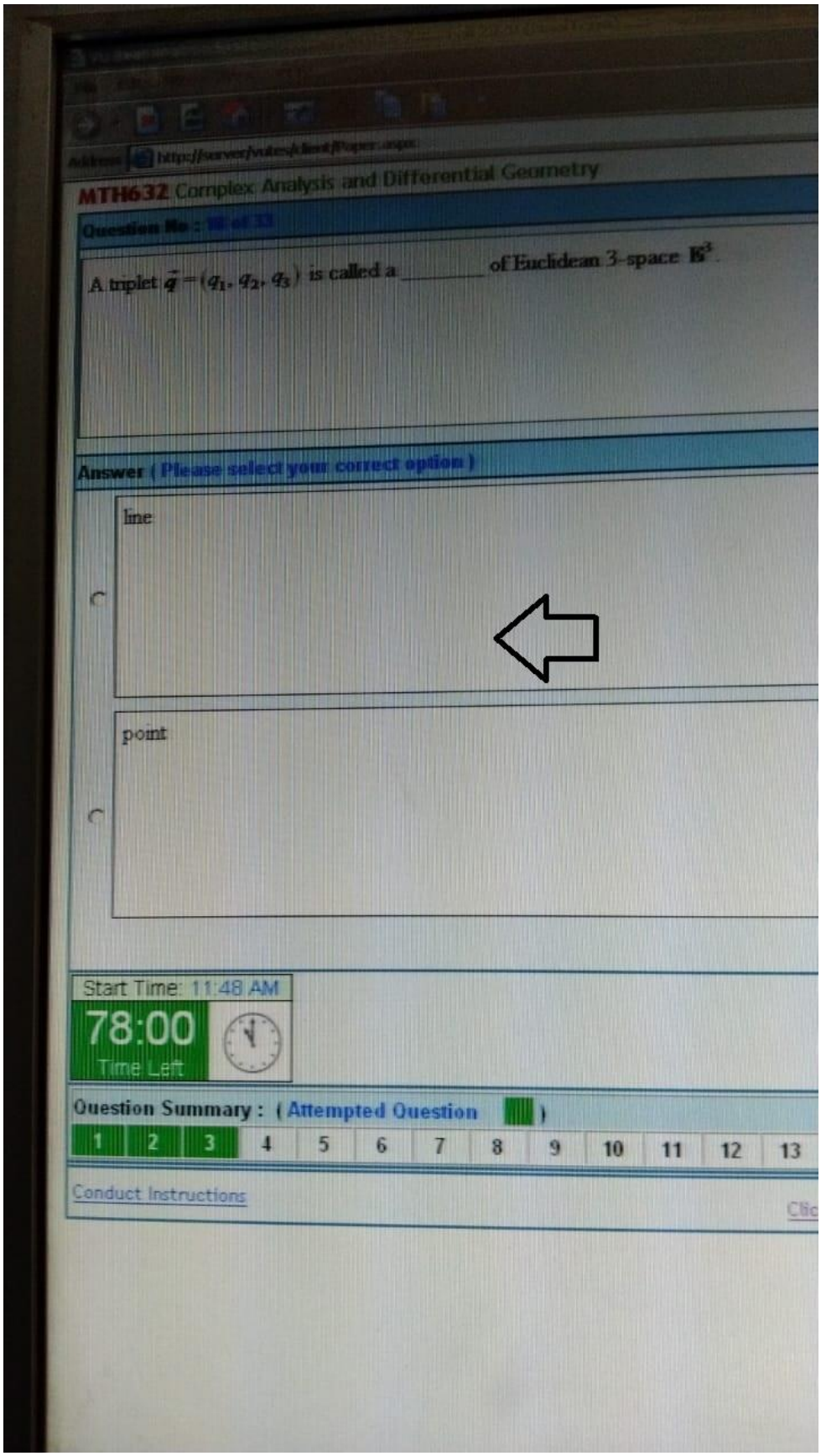
Question Summary : (Attempted Question)

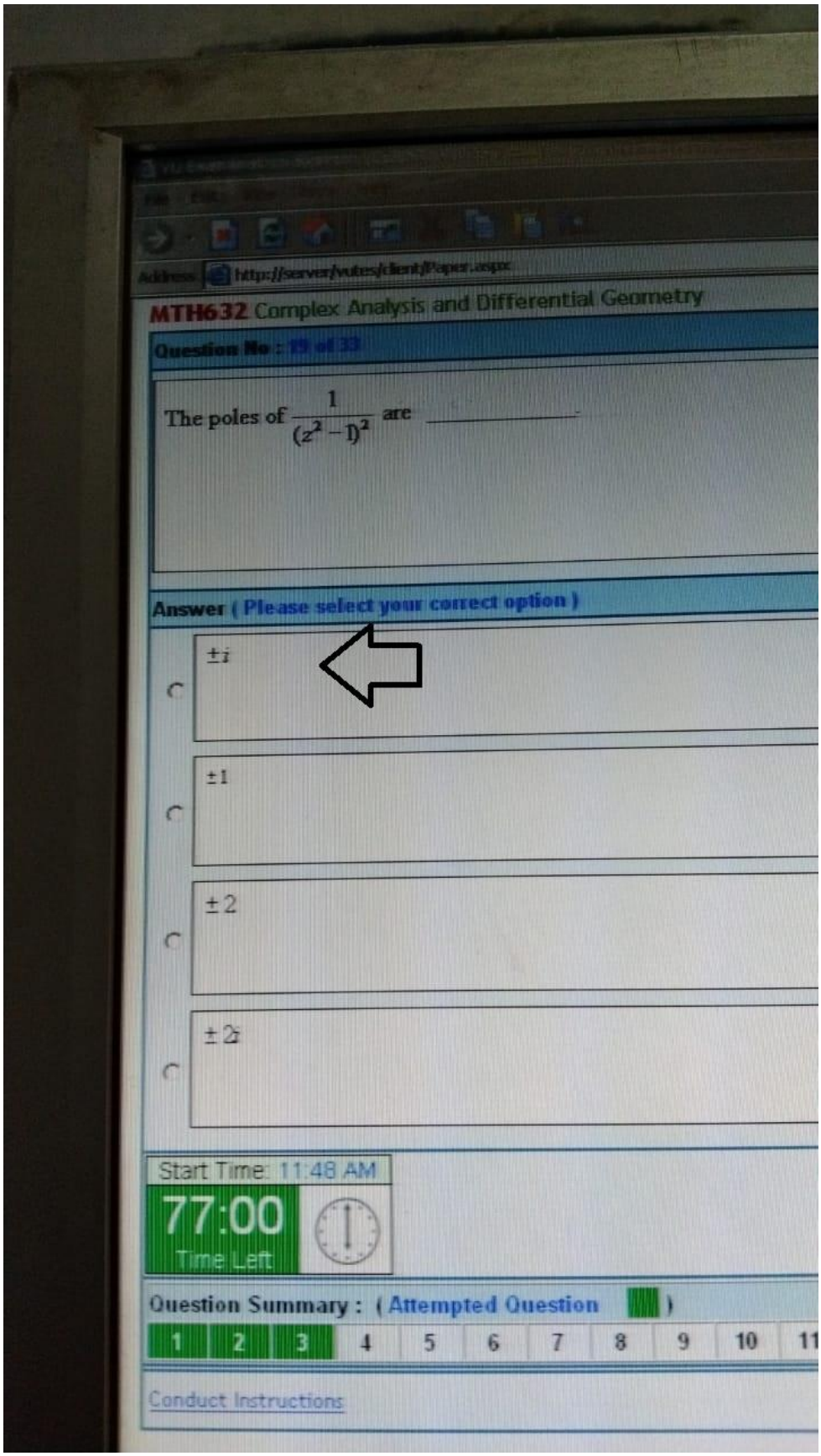
1	2	3	4	5	6	7	8	9	10	11

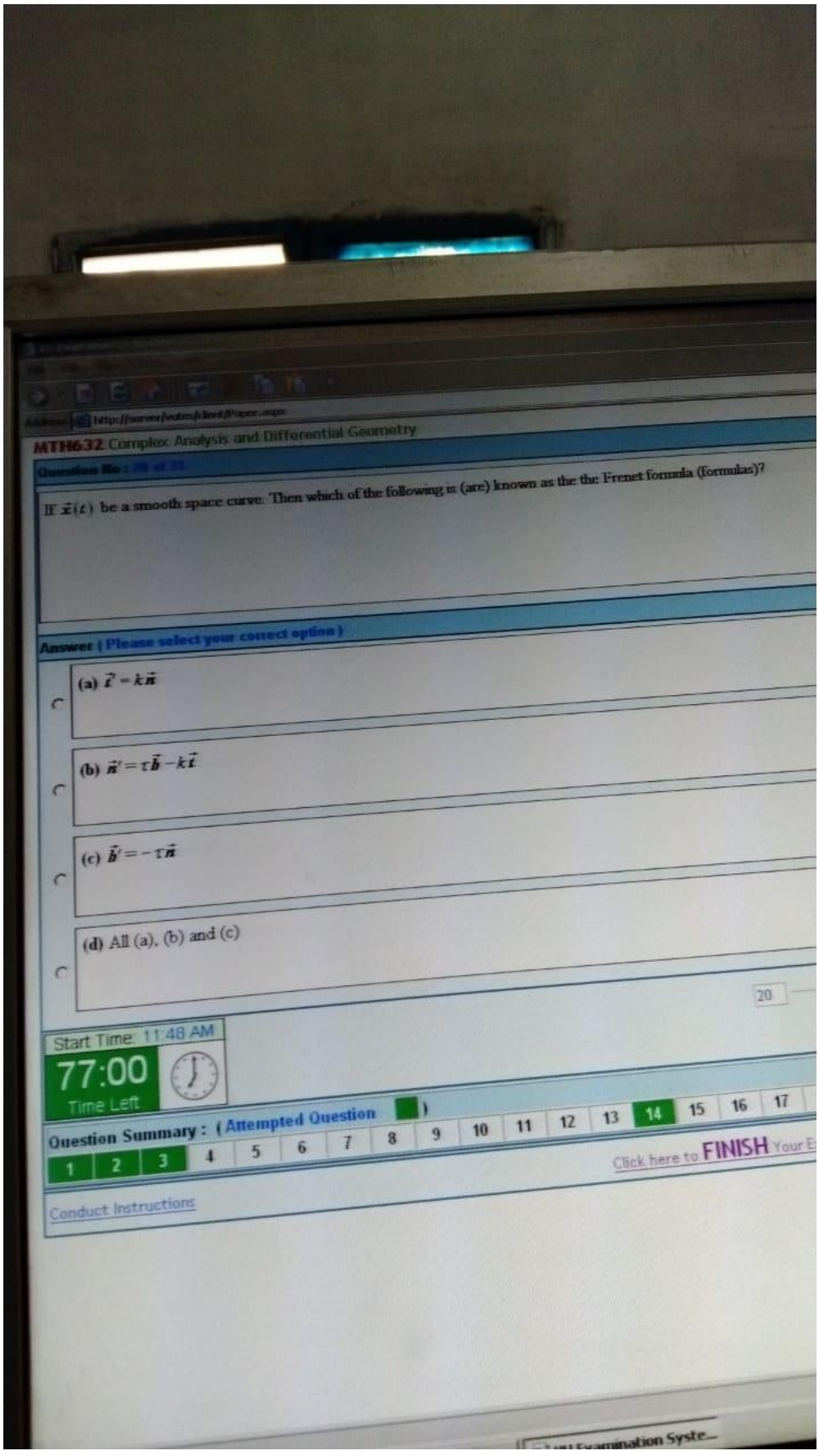
Conduct Instructions

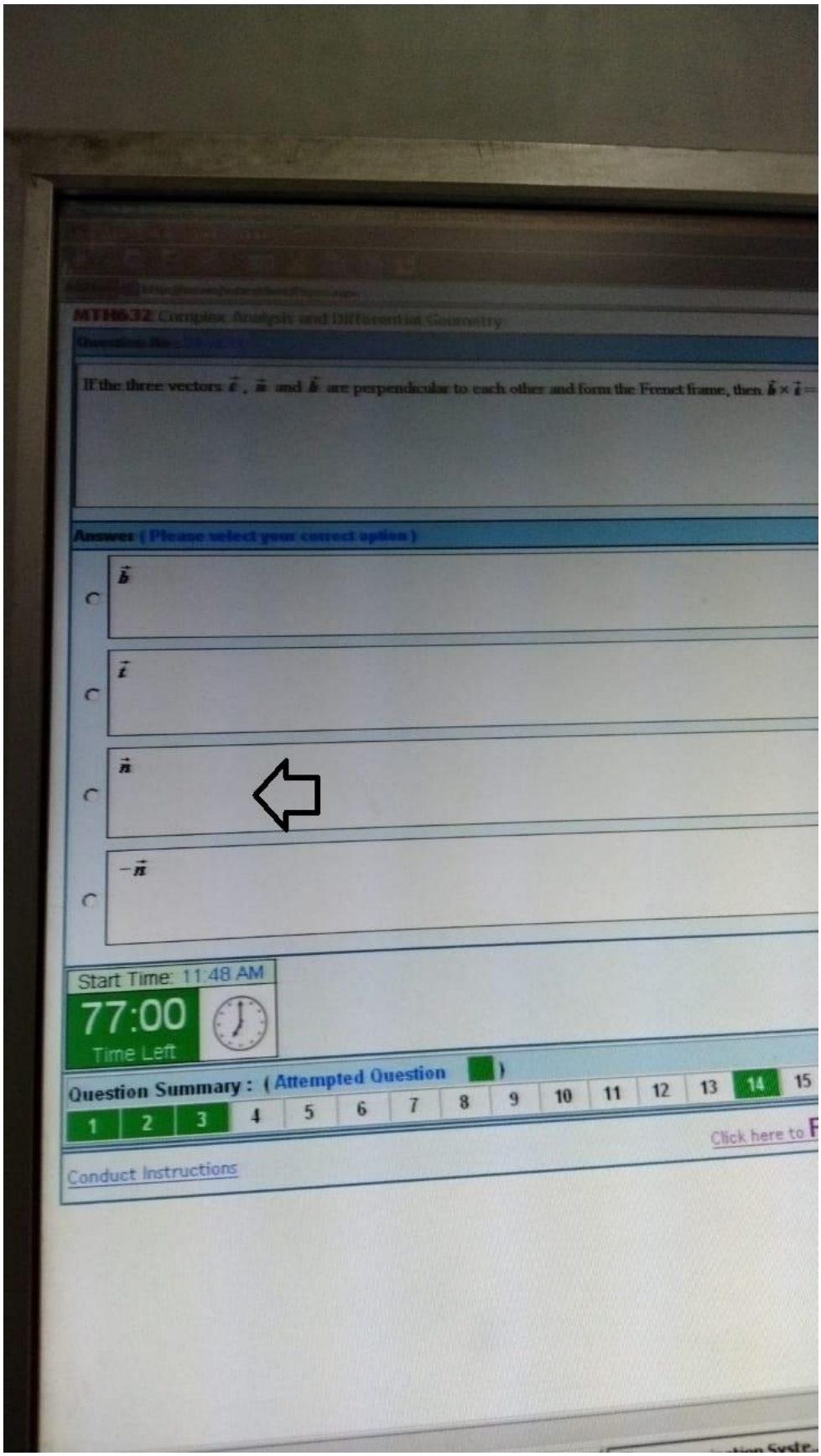

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

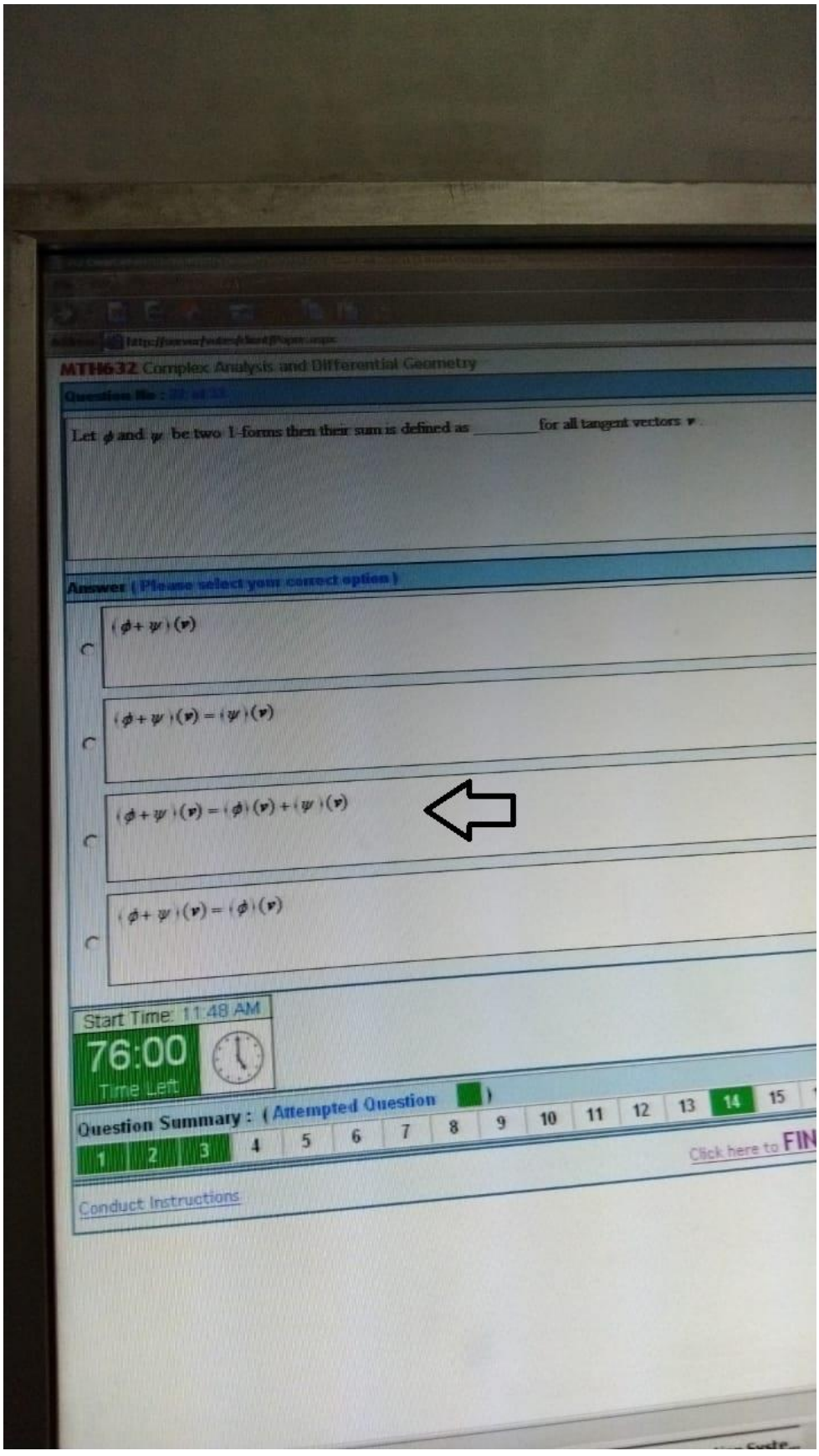

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

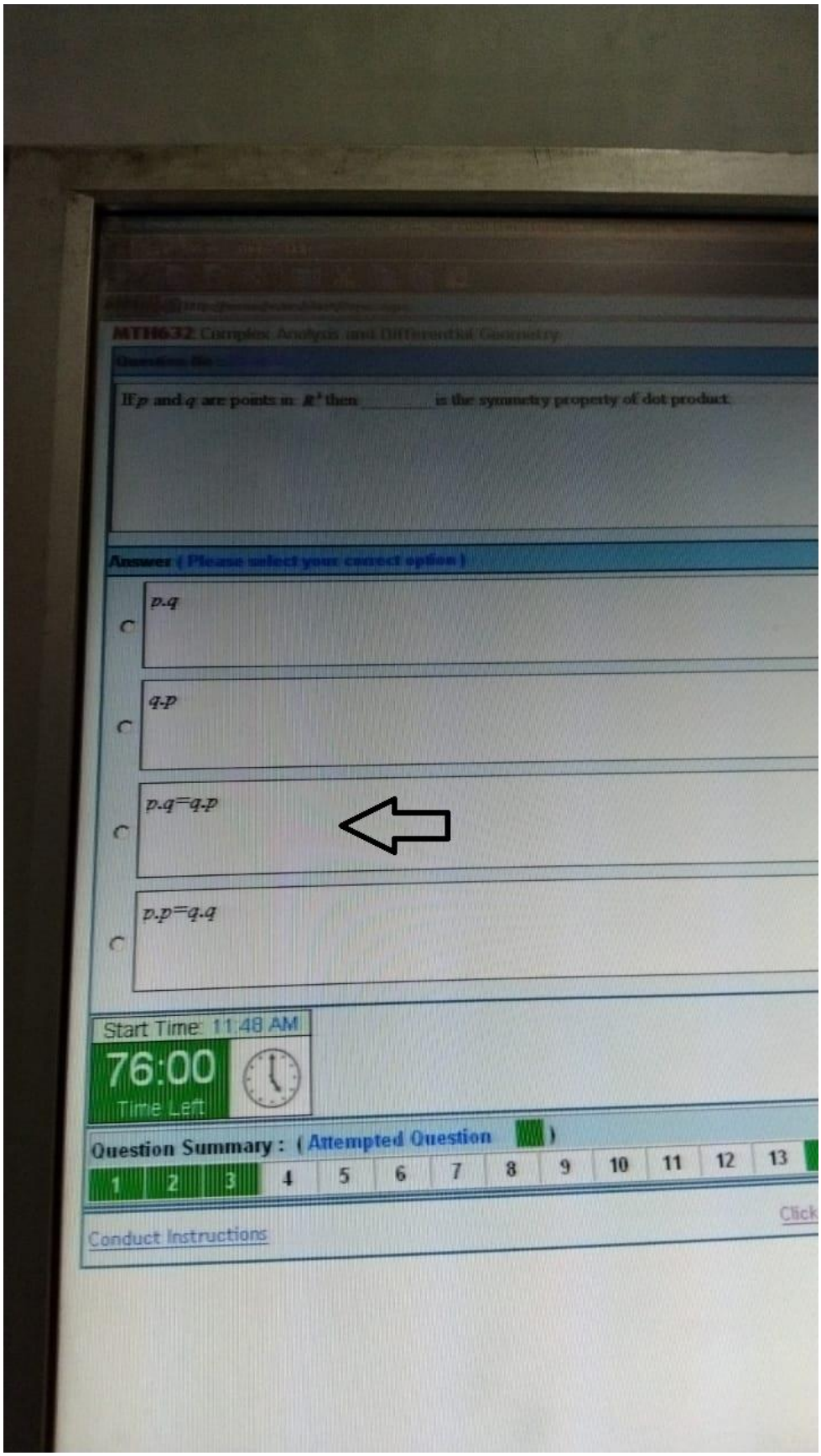

MTH632 Complex Analysis and Differential Geometry
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time: 11:48 AM
78:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

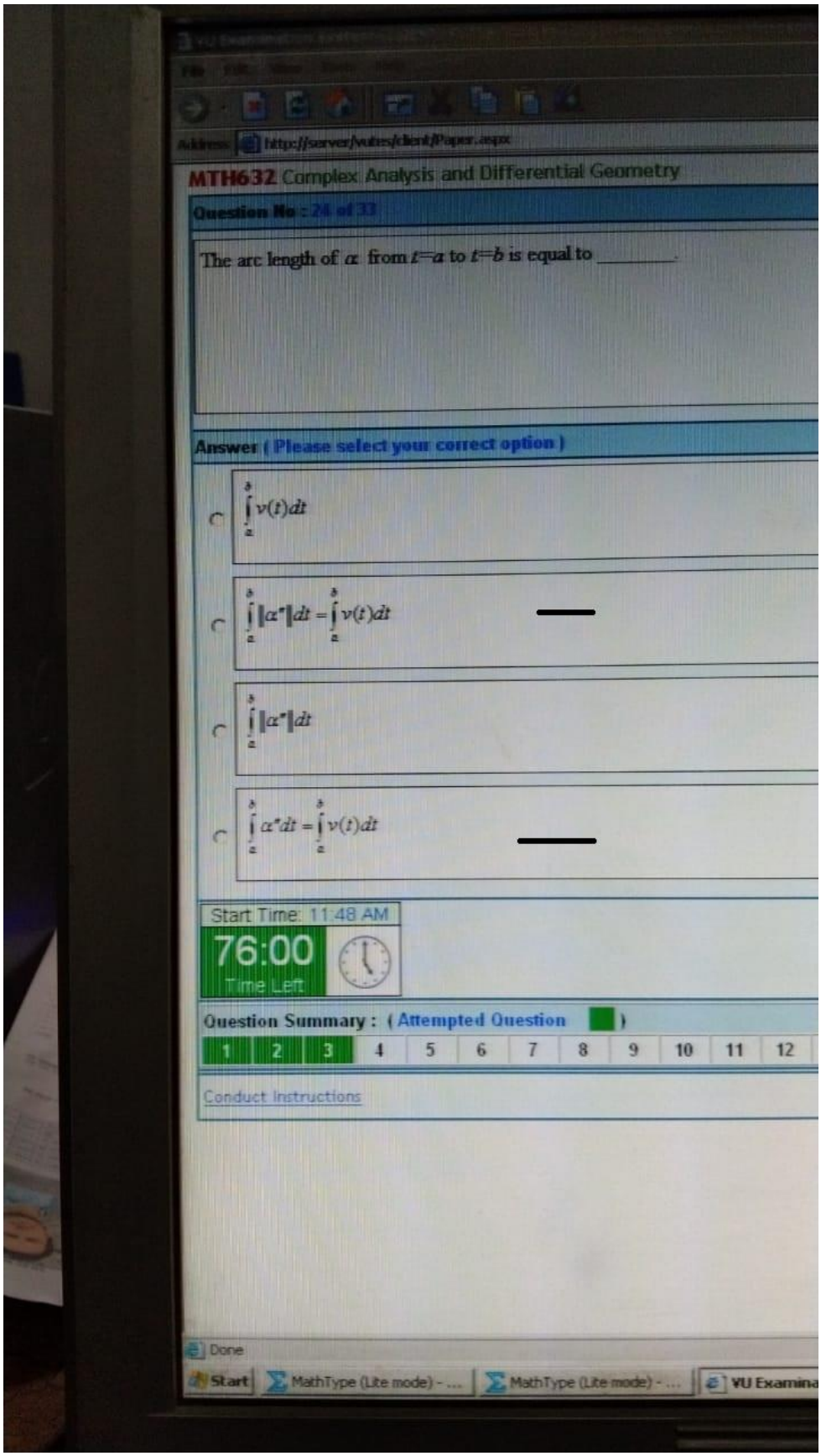

MTH632 Complex Analysis and Differential Geometry
Question No :
The poles of $\frac{1}{(z^2-1)^2}$ are _____.
Answer (Please select your correct option)
±i
±1
±2
±2i
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
20
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Click here to FINISH Your E
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to FIN
Conduct Instructions


MTH632: Complex Analysis and Differential Geometry
If p and q are points in R^n then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.q=q.p
p.p=q.q
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
Question No : 24 of 33
The arc length of α from $t=a$ to $t=b$ is equal to _____.
Answer (Please select your correct option)
$\int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt = \int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt$
$\int_a^b \alpha' dt = \int_a^b v(t)dt$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions
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MathType (Lite mode) - ...
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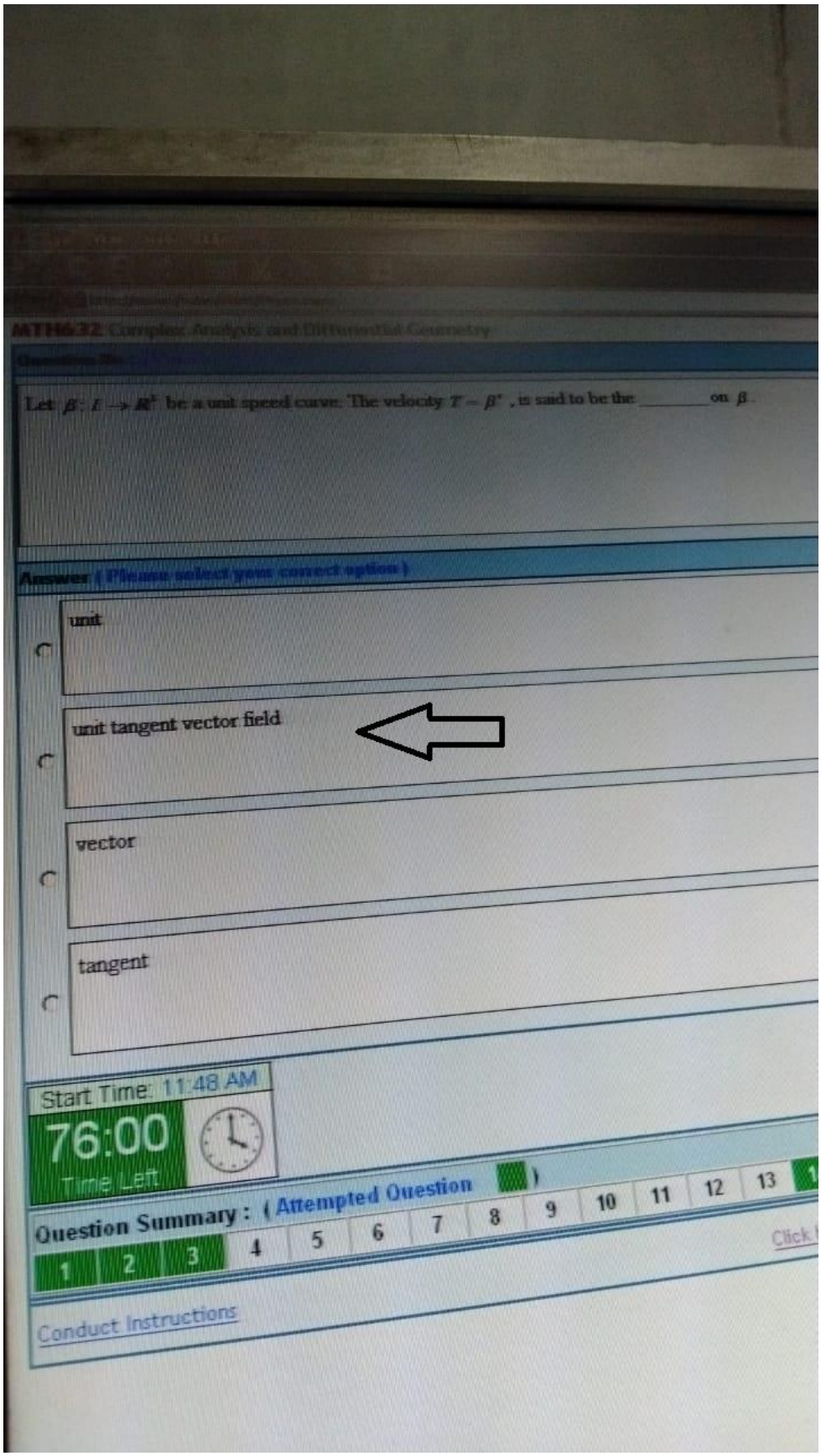
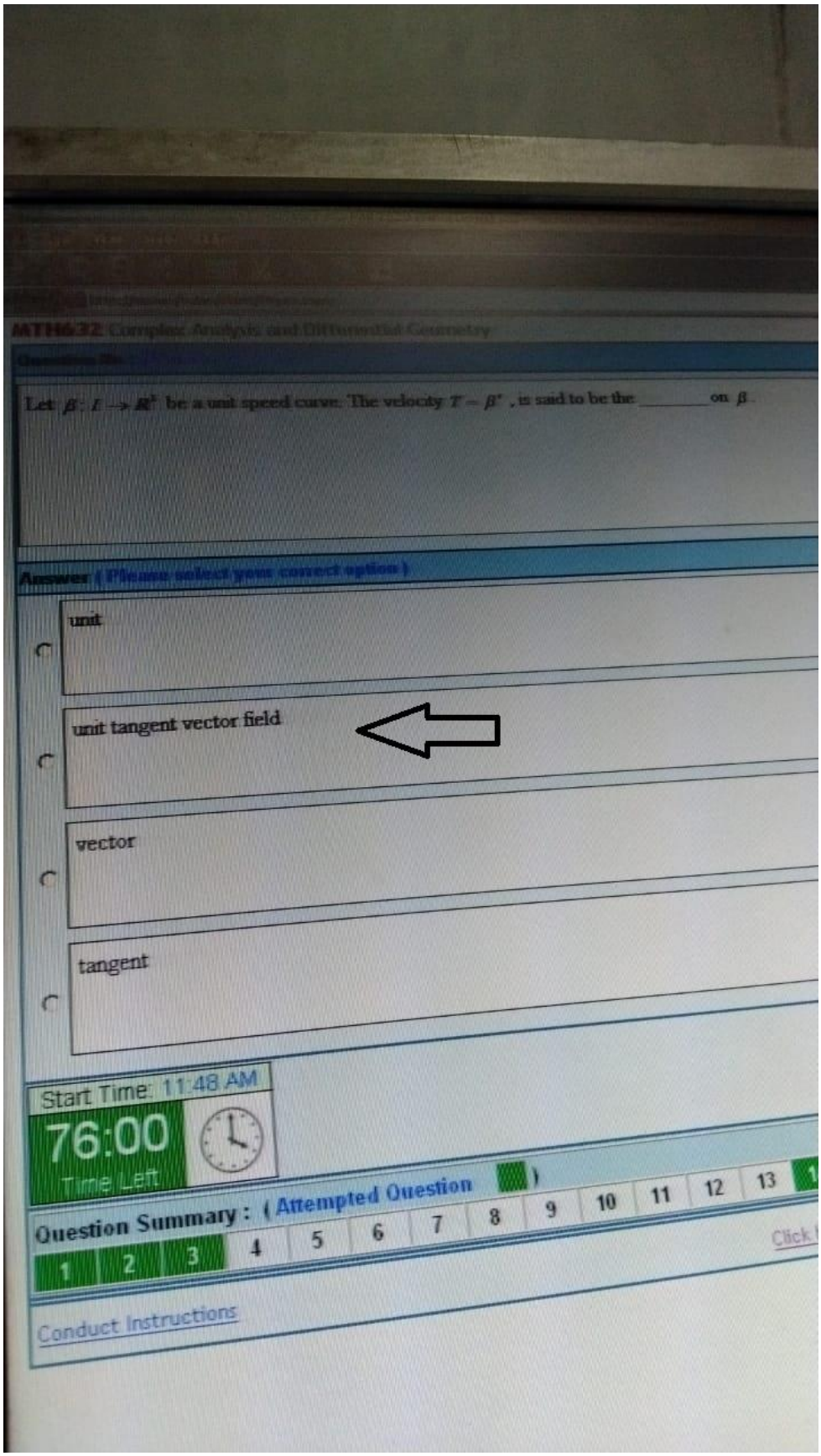
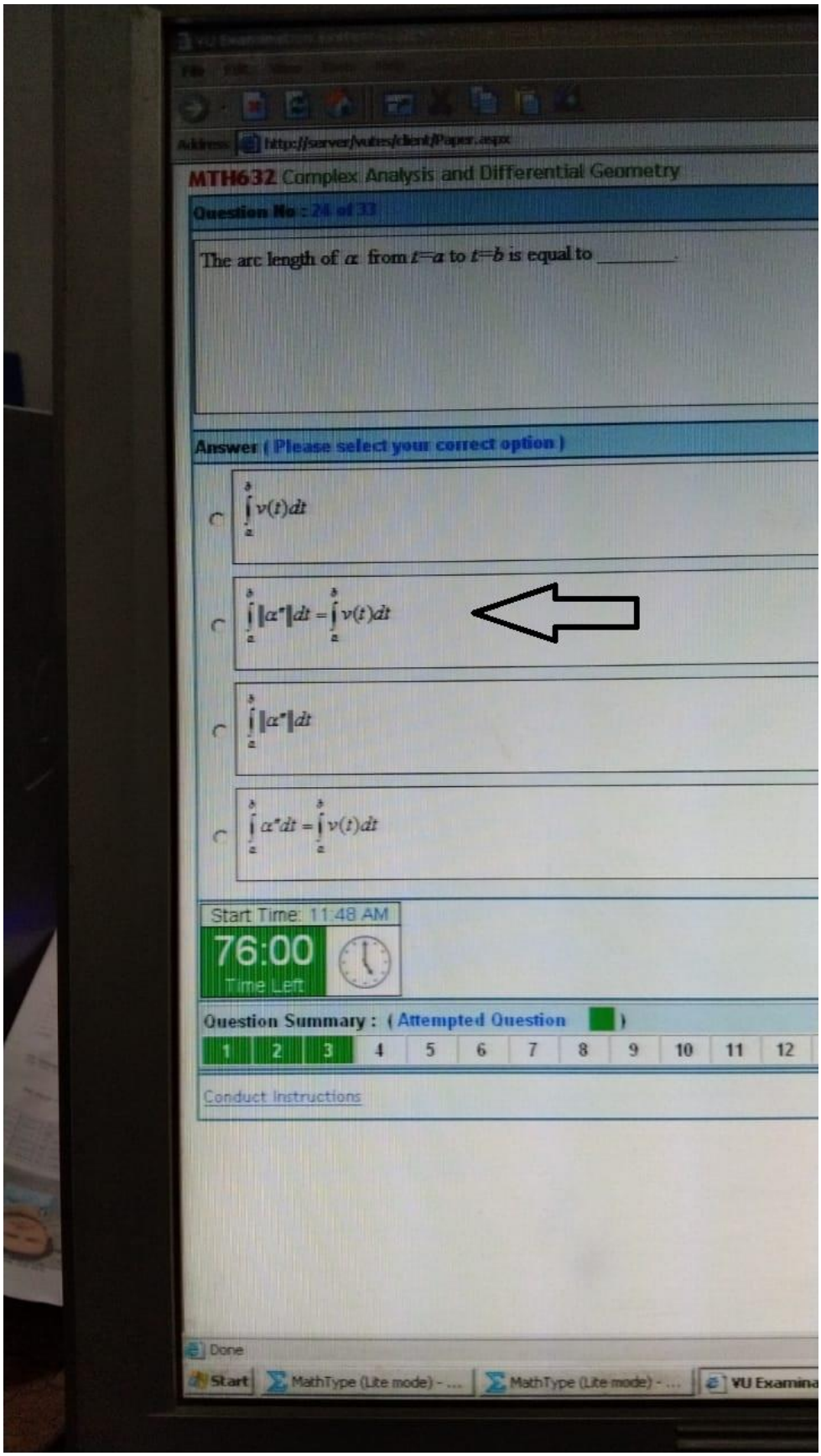
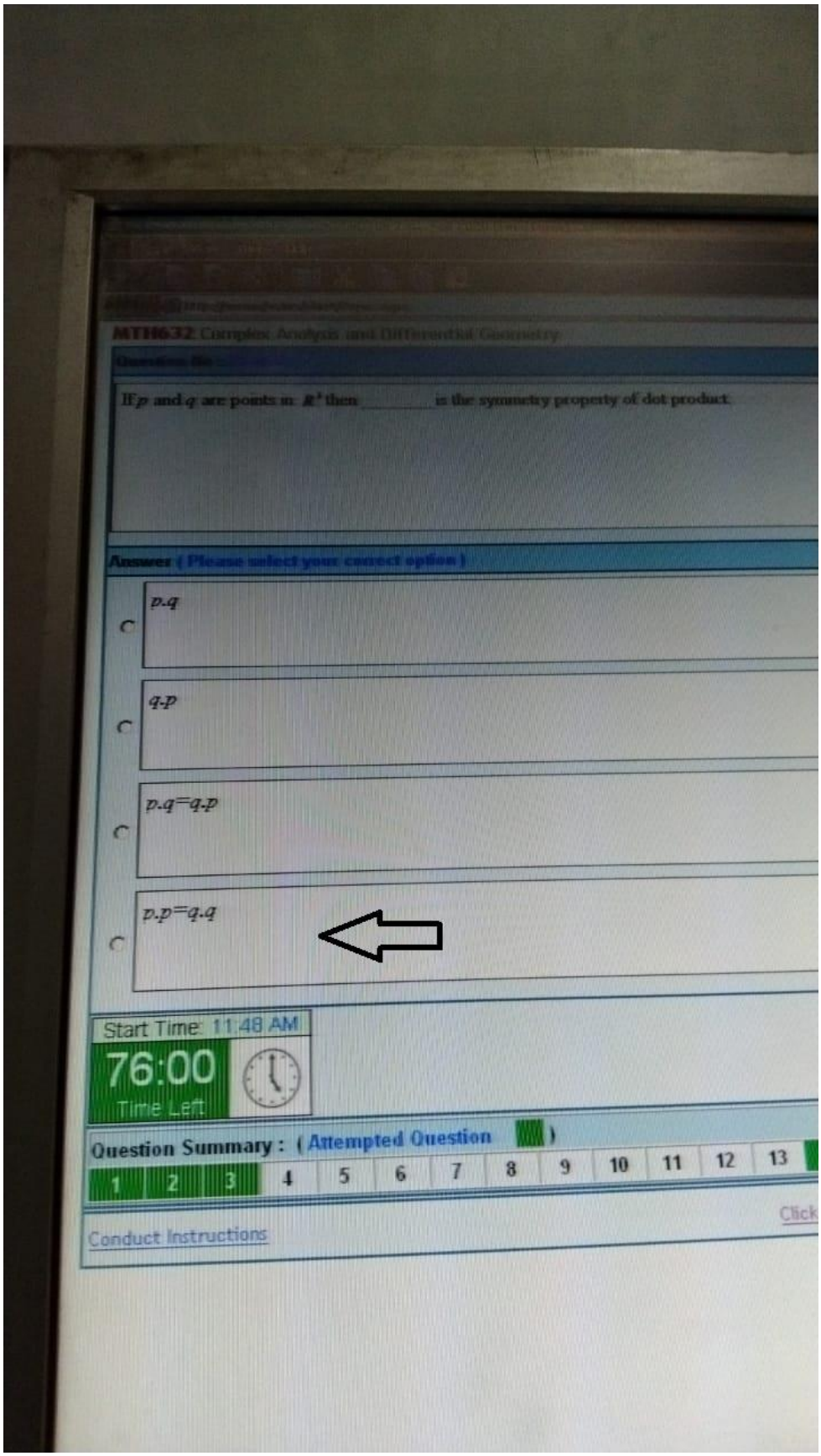
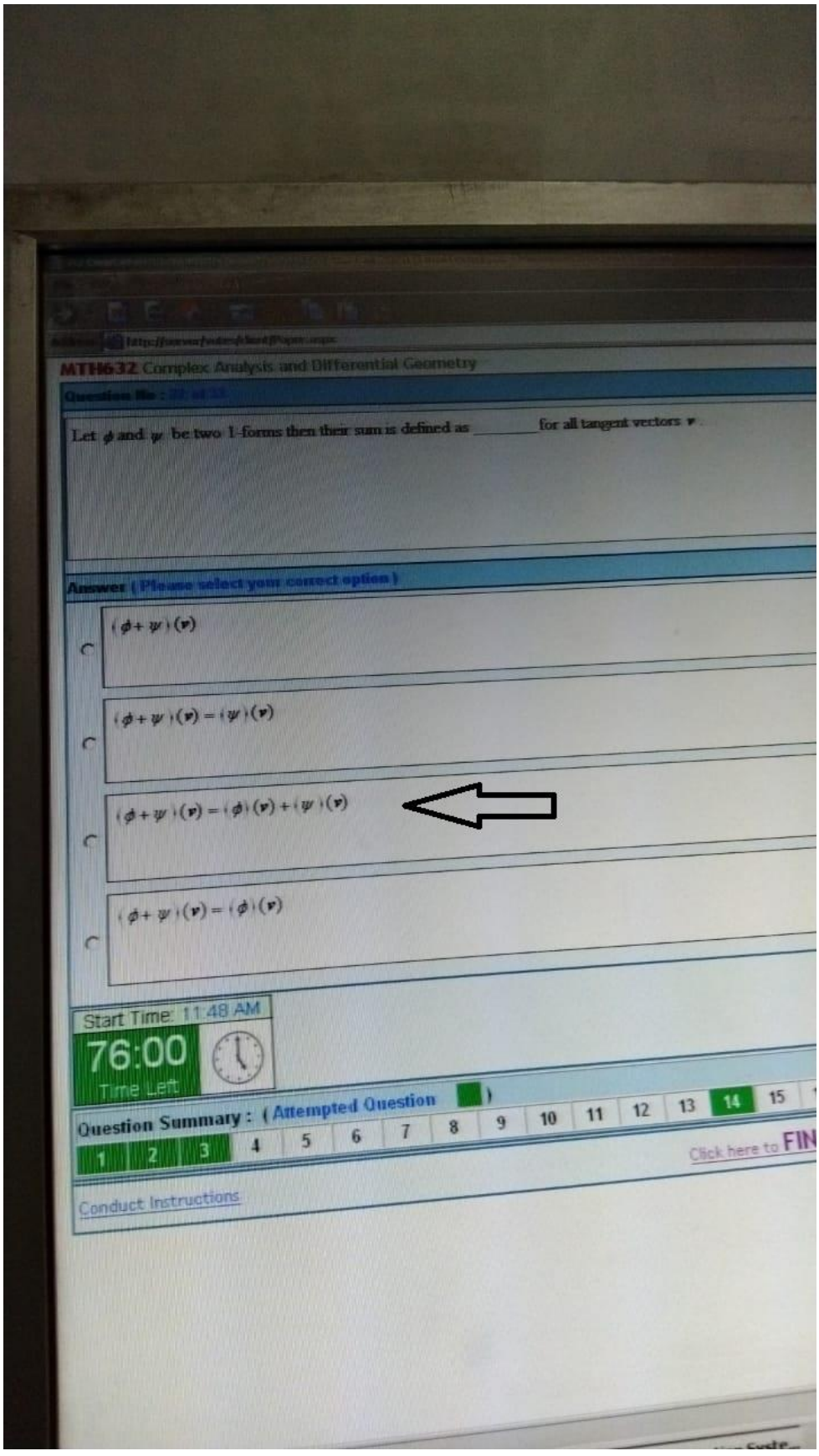
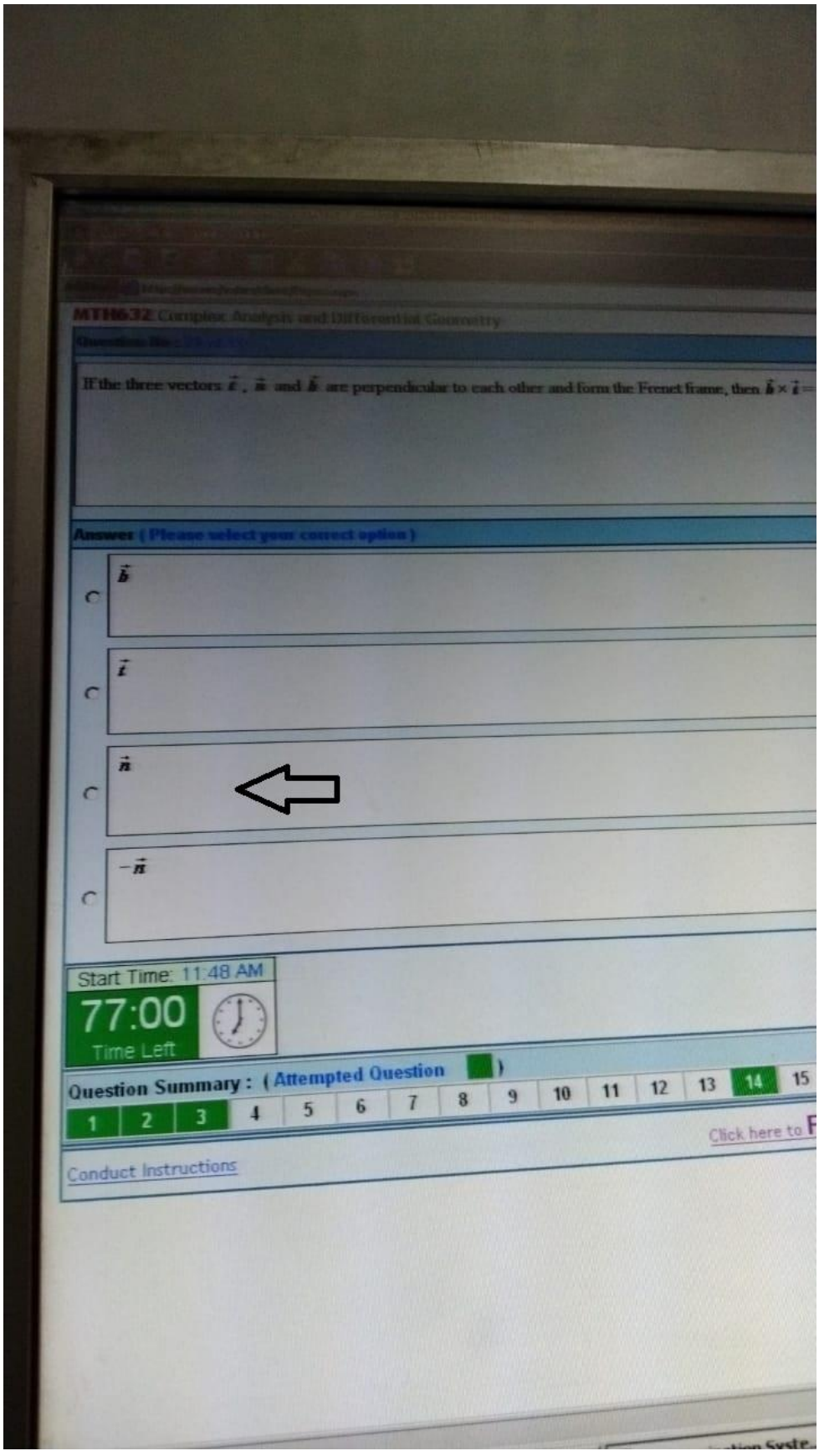

Let $\beta : I \to \mathbb{R}^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

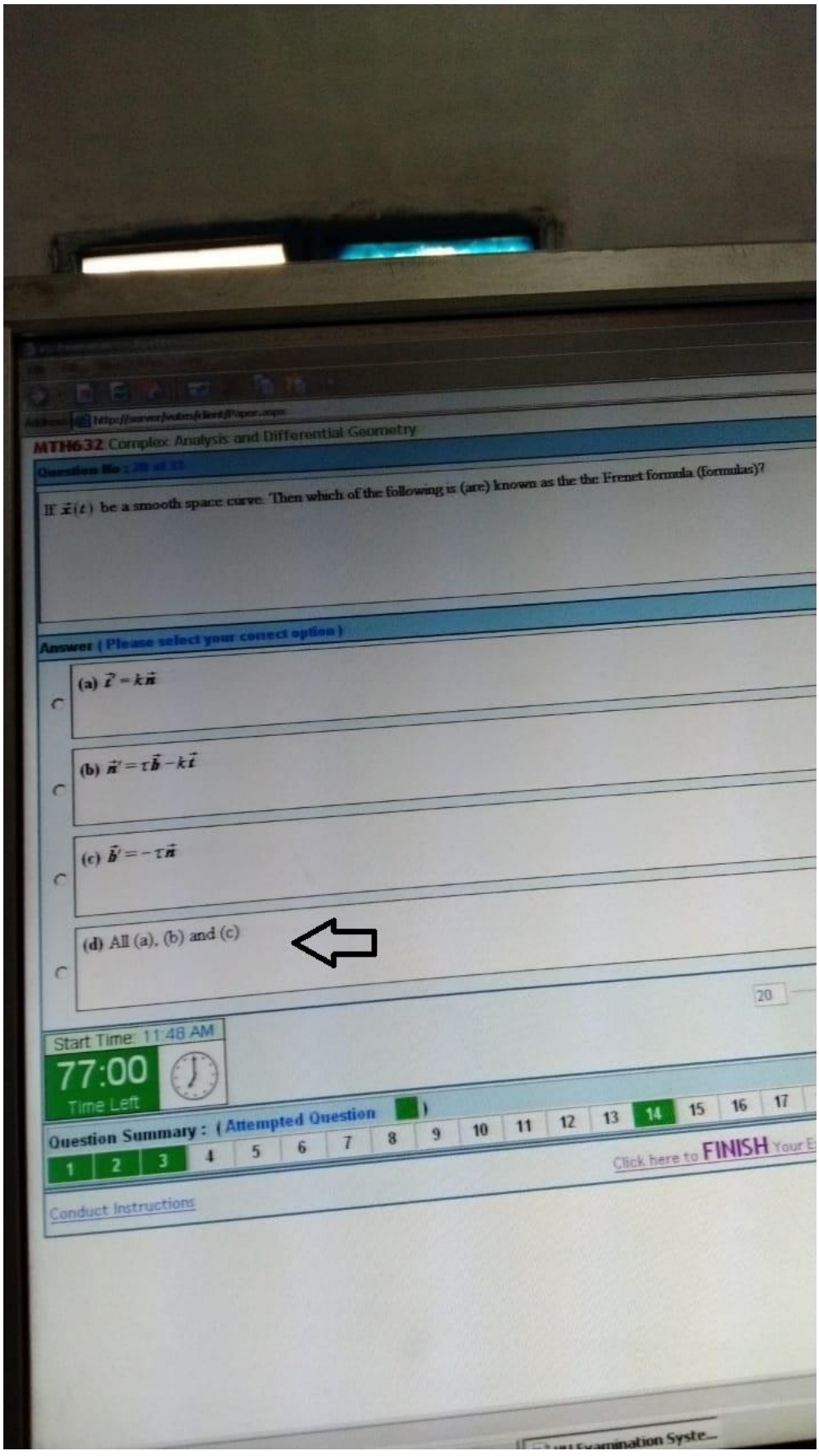

MTH632 Complex Analysis and Differential Geometry
Let $\beta : I \to \mathbb{R}^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

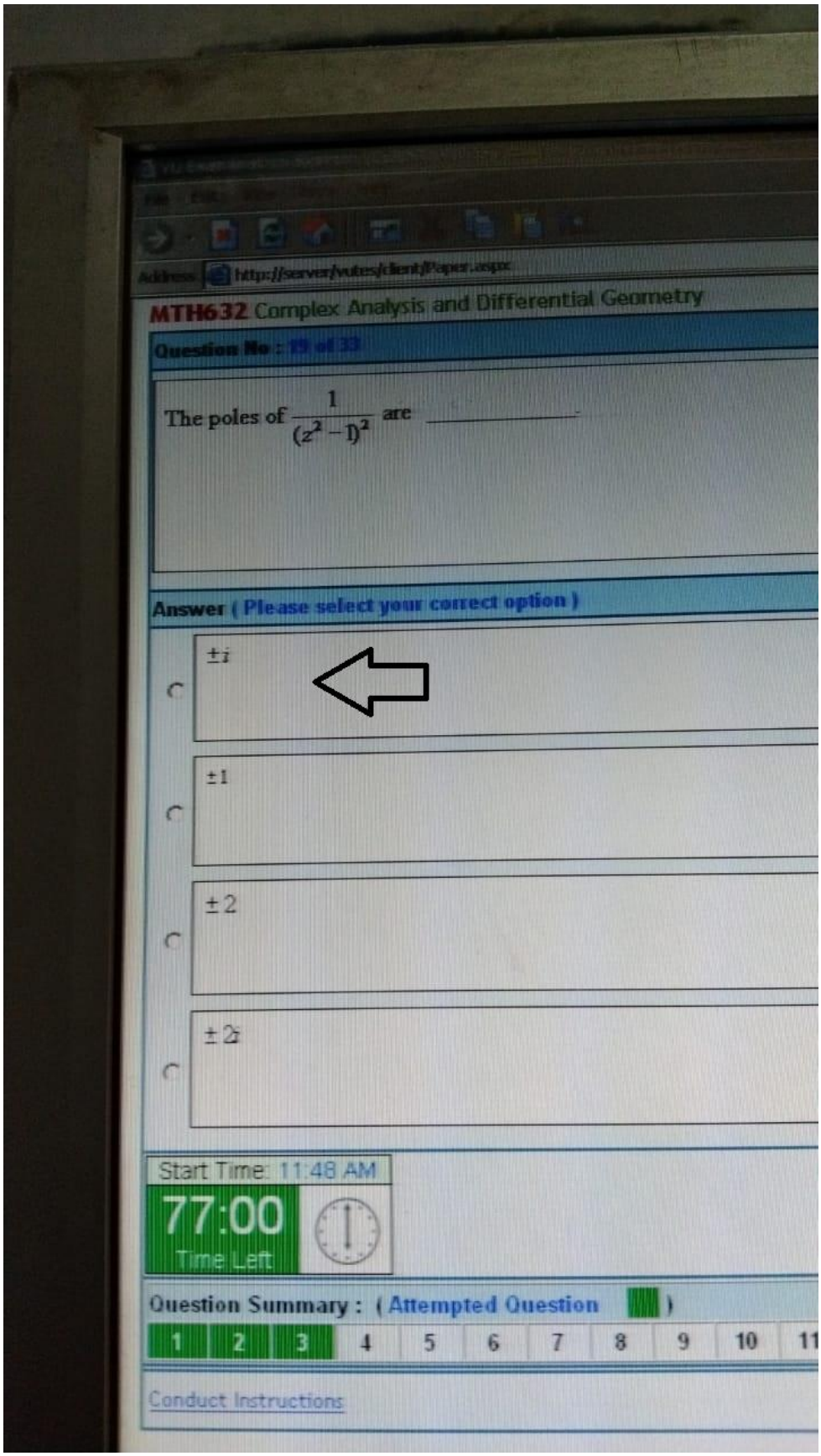

MTH632 Complex Analysis and Differential Geometry
Question No : 24 of 33
The arc length of α from $t=a$ to $t=b$ is equal to _____.
Answer (Please select your correct option)
$\int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt = \int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt$
$\int_a^b \alpha' dt = \int_a^b v(t)dt$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examina

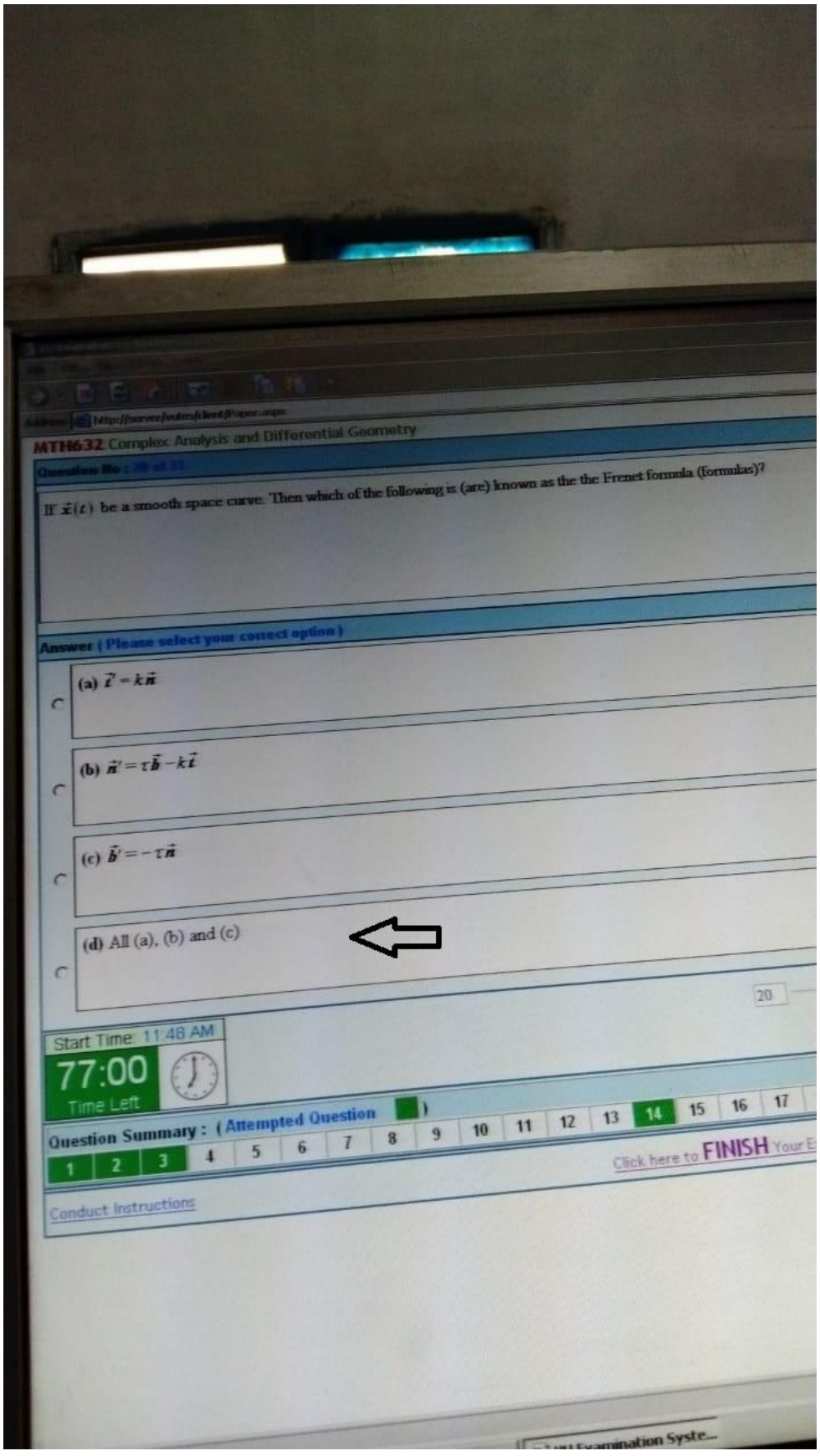

MTH632: Complex Analysis and Differential Geometry
If p and q are points in R^n then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.q=q.p
p.p=q.q
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

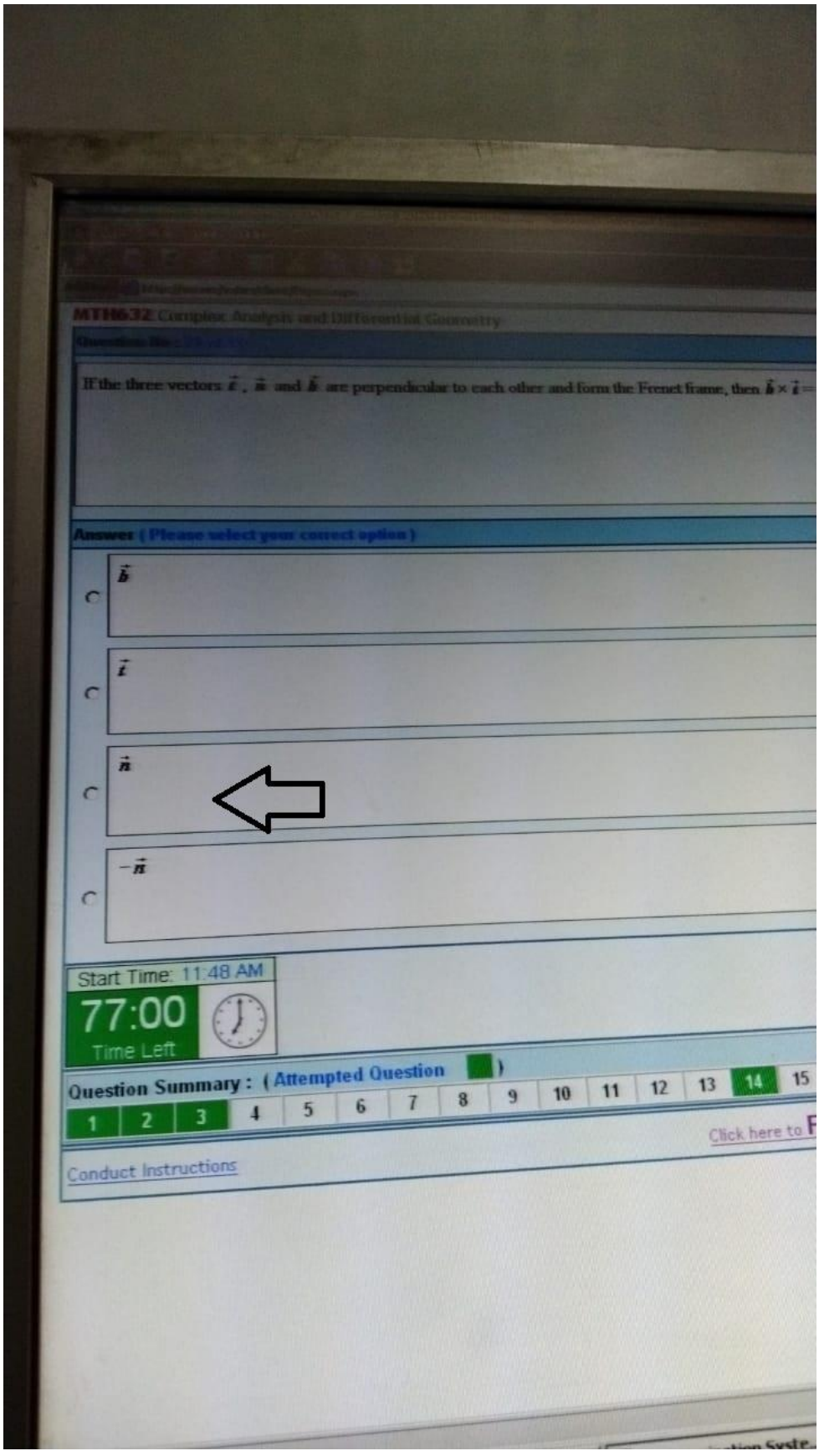

MTH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to FIN
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
20
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Click here to FINISH Your E
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
Question No :
The poles of $\frac{1}{(z^2-1)^2}$ are _____.
Answer (Please select your correct option)
±i
±1
±2
±2i
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
20
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Click here to FINISH Your E
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to
Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

Question No : 33 of 35

Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.

Answer (Please select your correct option)

- $(\phi+\psi)(v)$
- $(\phi+\psi)(v)=(\psi)(v)$
- $(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
- $(\phi+\psi)(v)=(\phi)(v)$

Start Time: 11:48 AM

76:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Click here to FIN

Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

Question No : 24 of 33

The arc length of α from $t=a$ to $t=b$ is equal to _____.

Answer (Please select your correct option)

- $\int_a^b v(t)dt$
- $\int_a^b \|\alpha'\|dt = \int_a^b v(t)dt$ ⇦
- $\int_a^b \|\alpha'\|dt$
- $\int_a^b \alpha' dt = \int_a^b v(t)dt$

Start Time: 11:48 AM

76:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12

Conduct Instructions

Done

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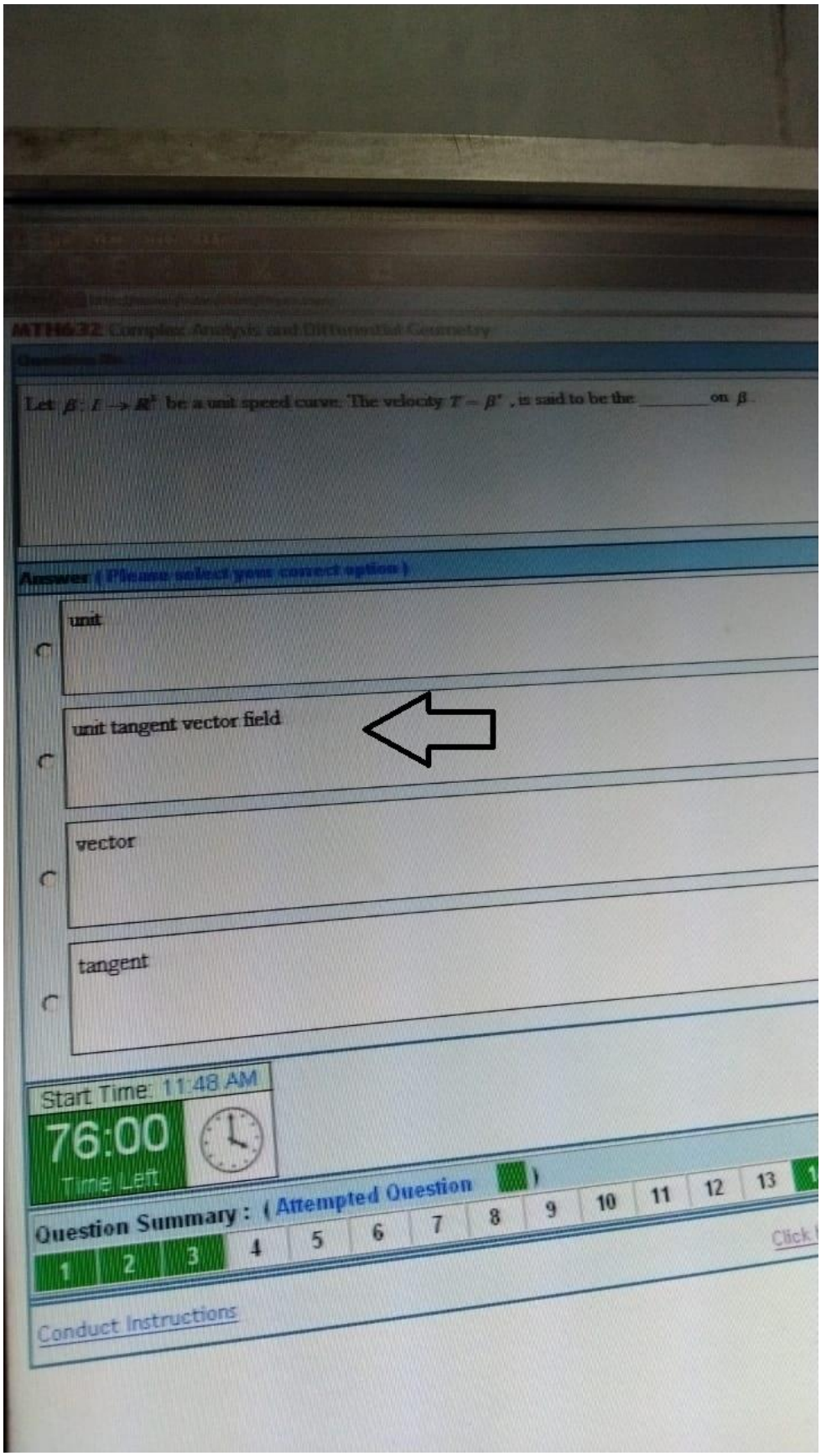

MTH632 Complex Analysis and Differential Geometry
Let $\beta : I \to \mathbb{R}^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

Question No : [illegible]

A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.

Answer (Please select your correct option)

line

point

Start Time: 11:48 AM

78:00

Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13

Conduct Instructions

Clic

MTH632 Complex Analysis and Differential Geometry

Question No : [illegible]

A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.

Answer (Please select your correct option)

line

point

Start Time: 11:48 AM

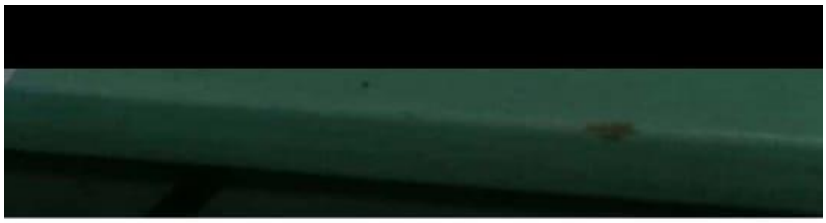
78:00

Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13

Conduct Instructions



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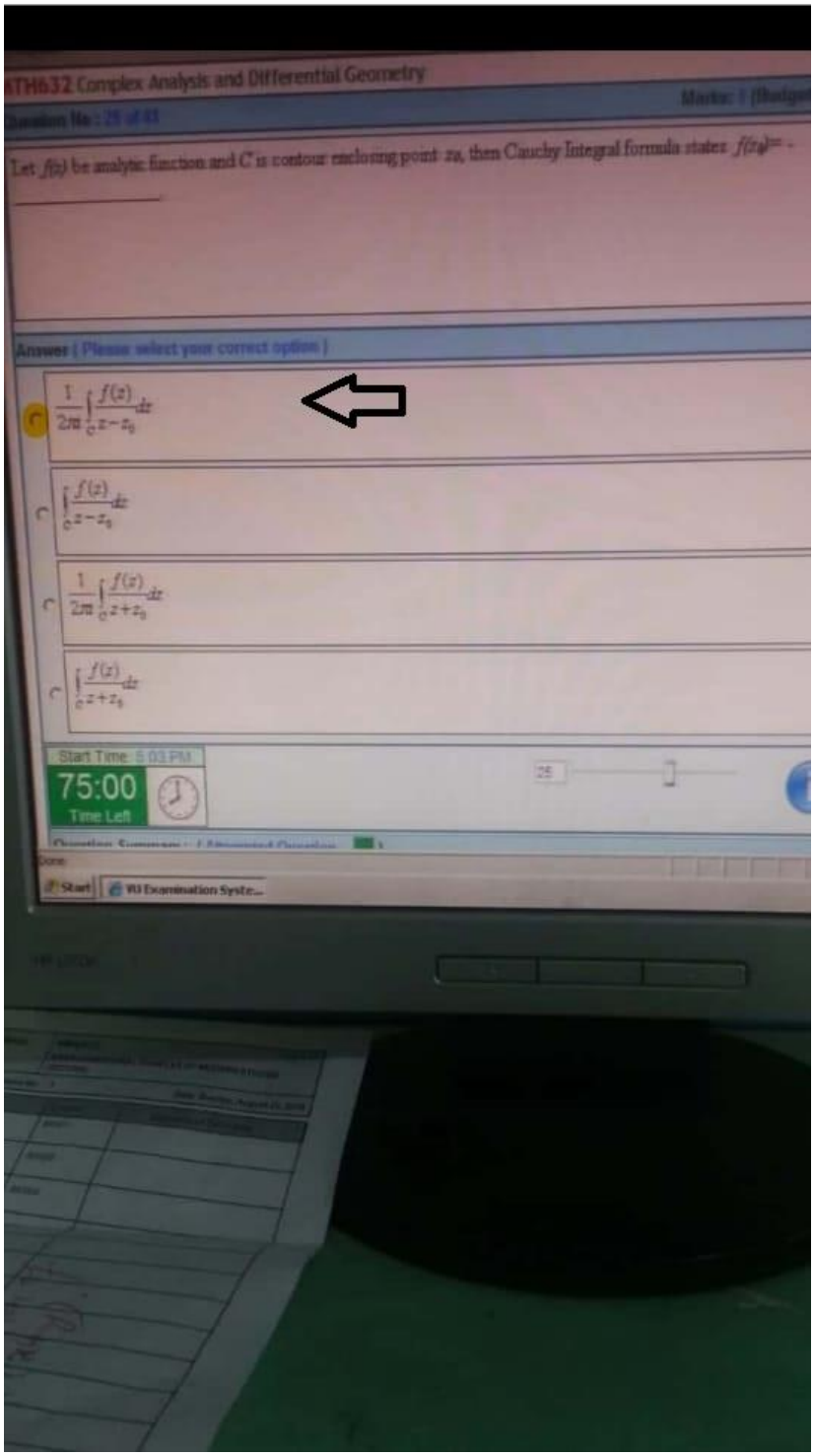

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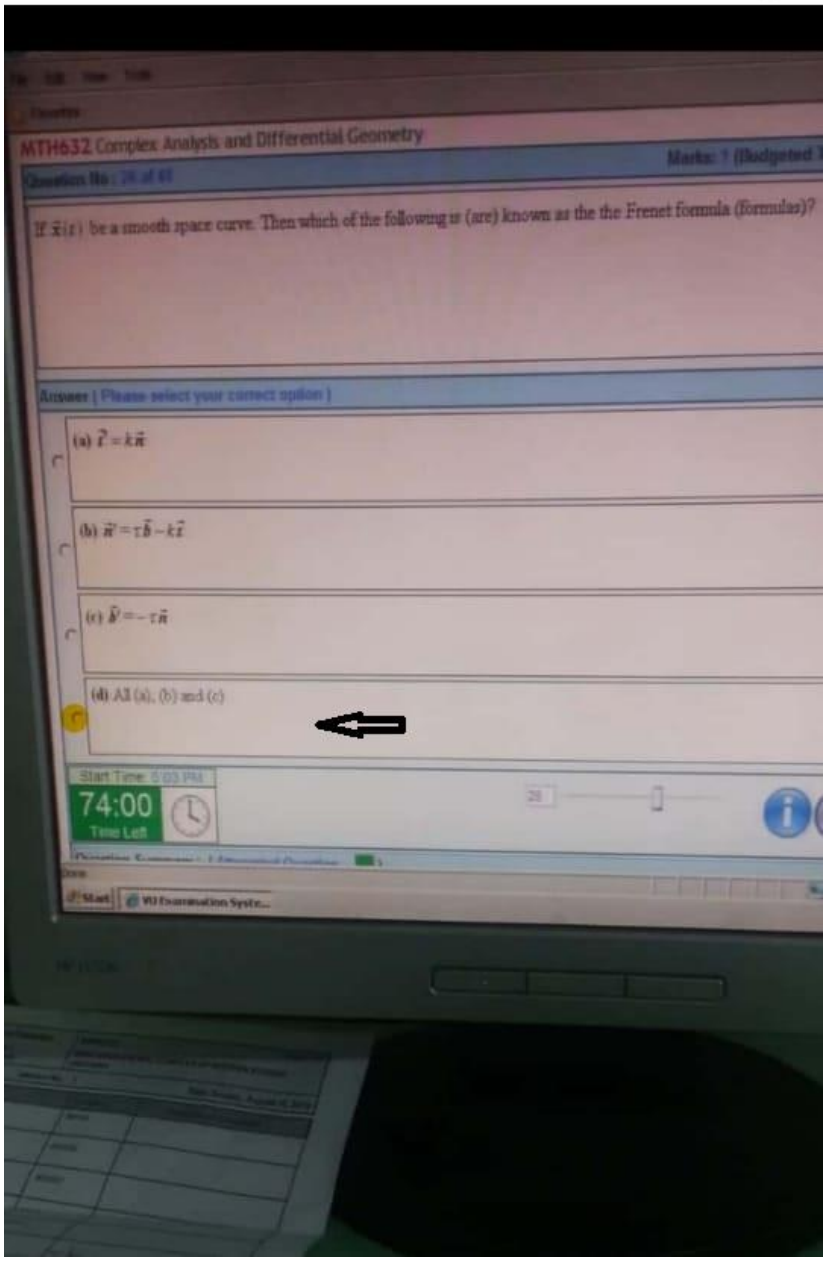

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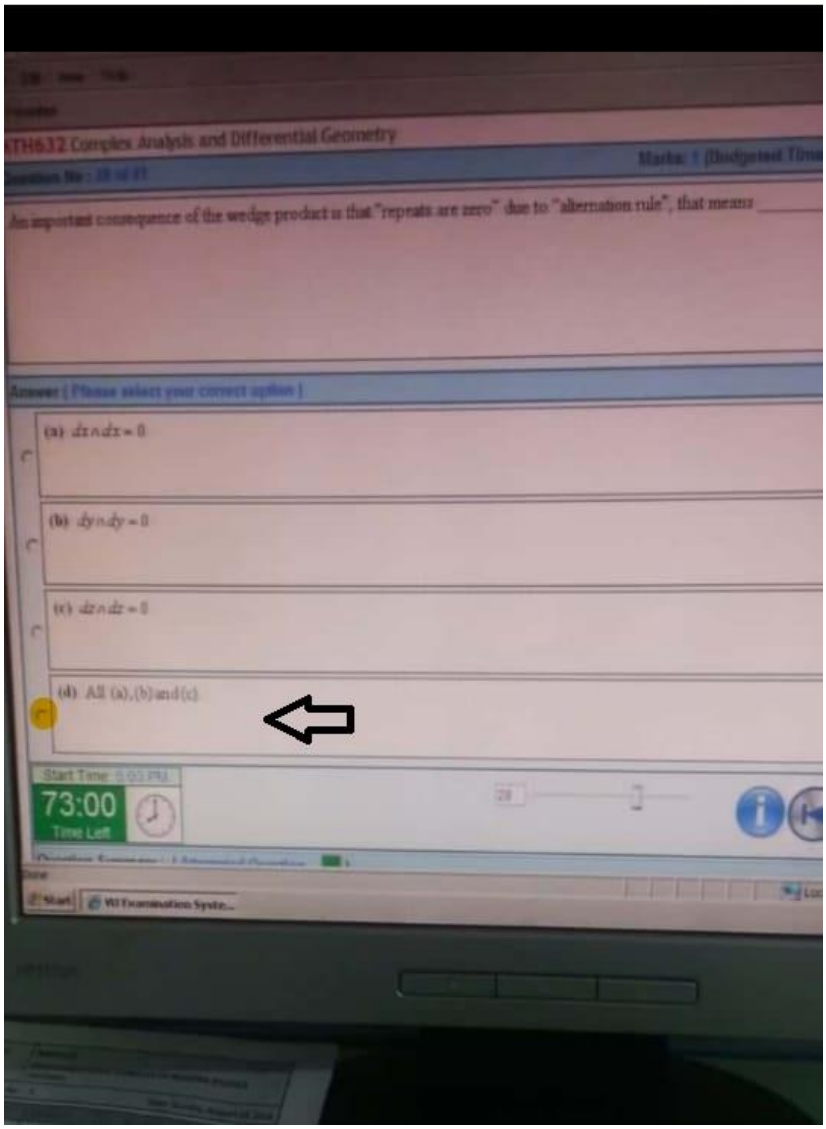

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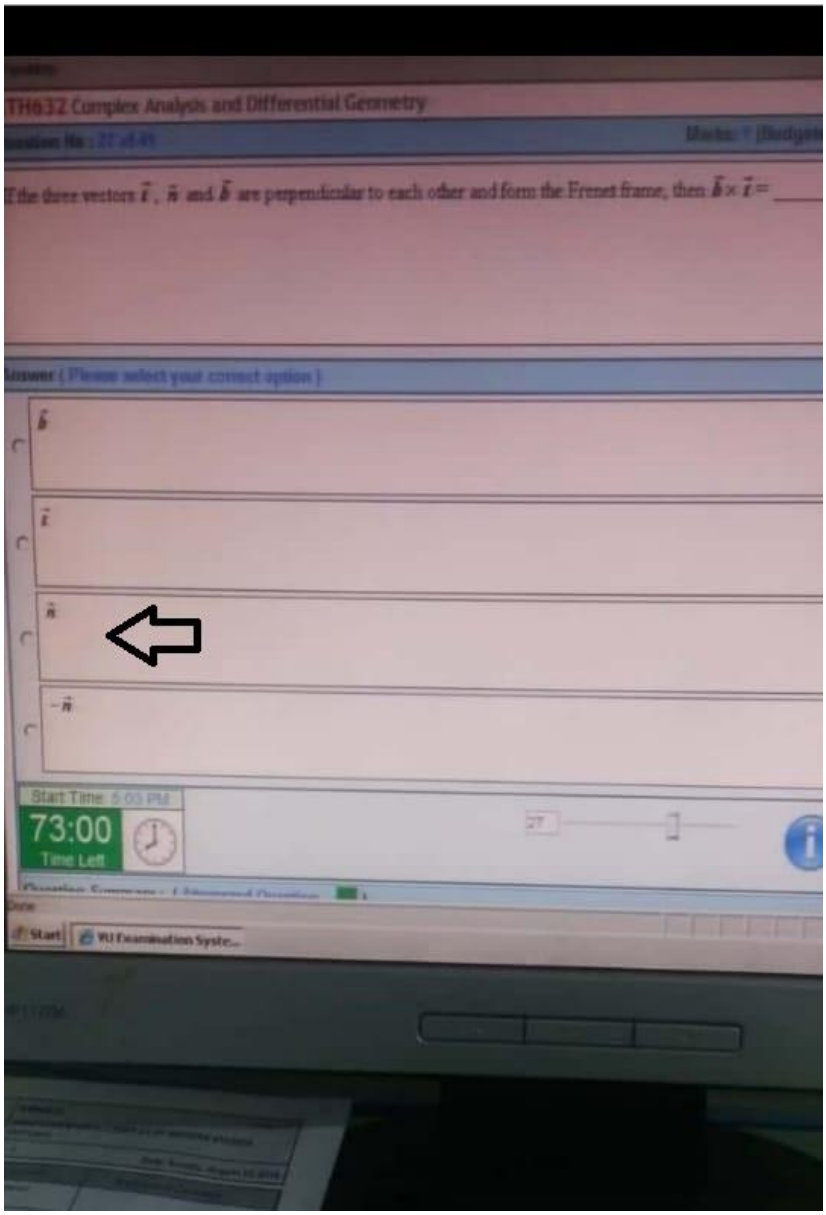
TH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 6:03 PM
73:00
Time Left

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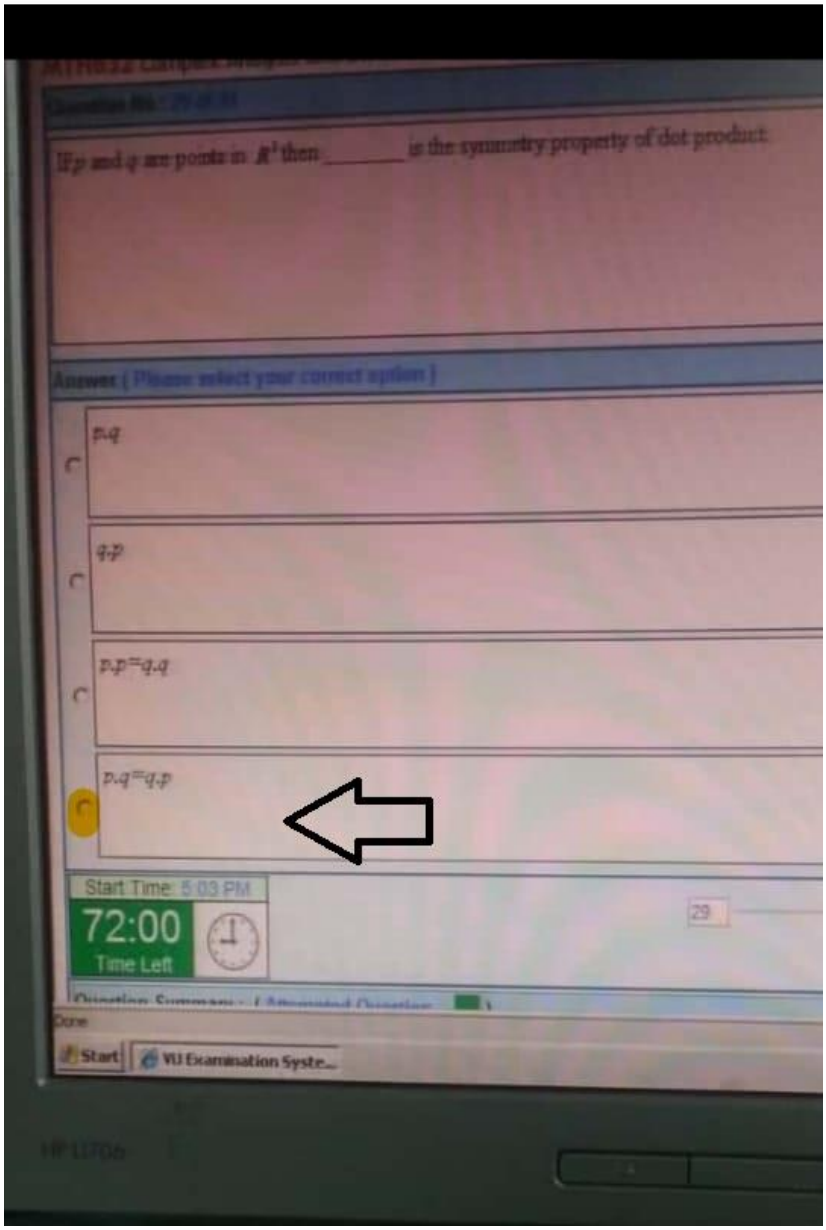
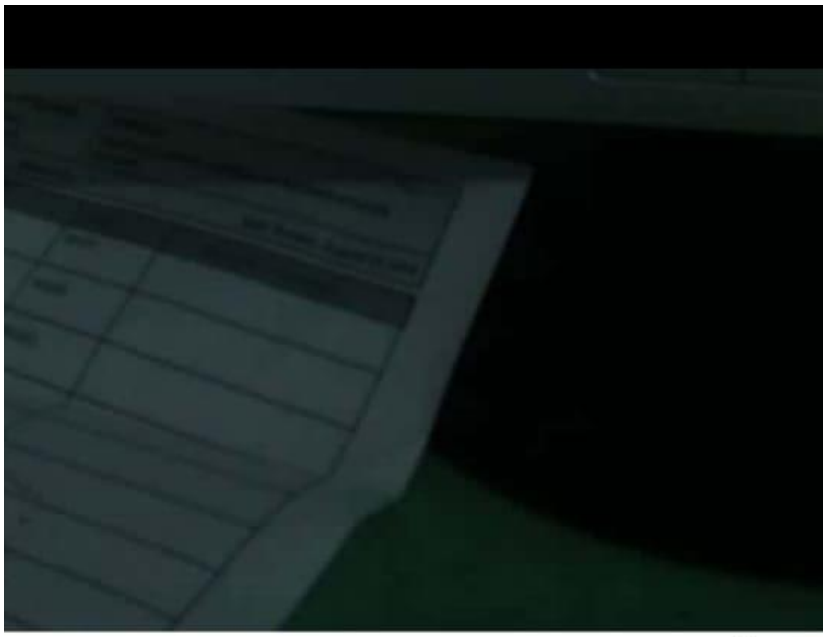
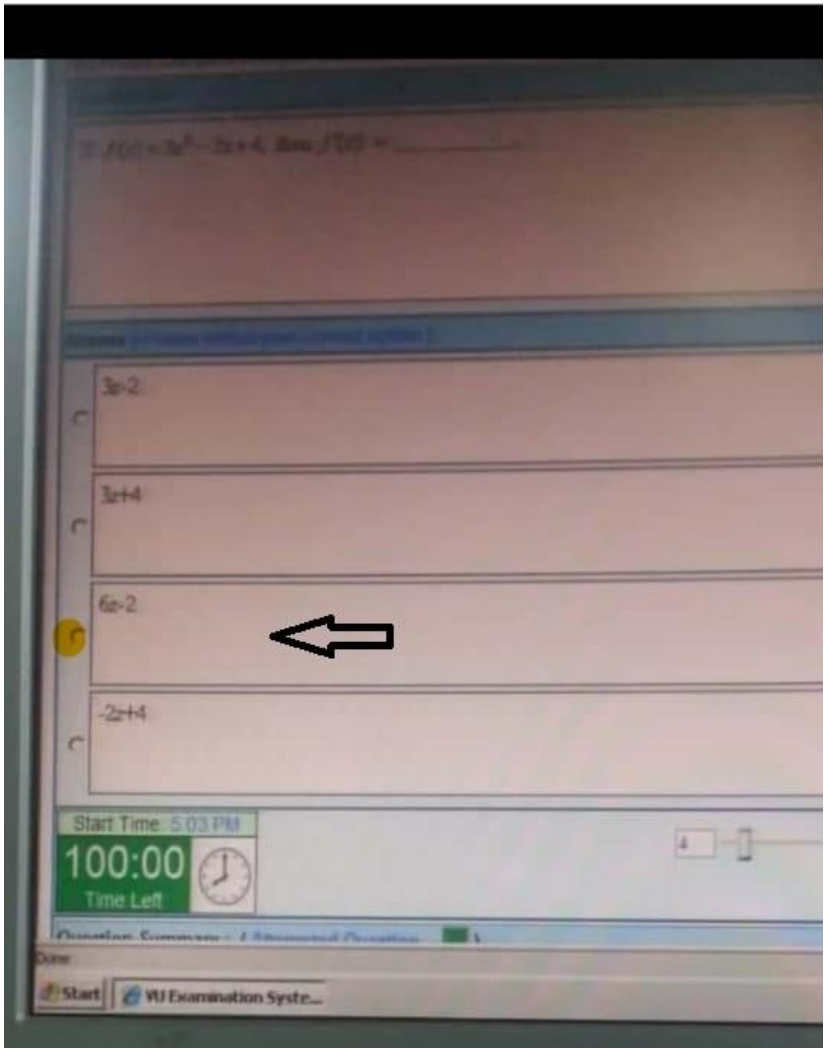
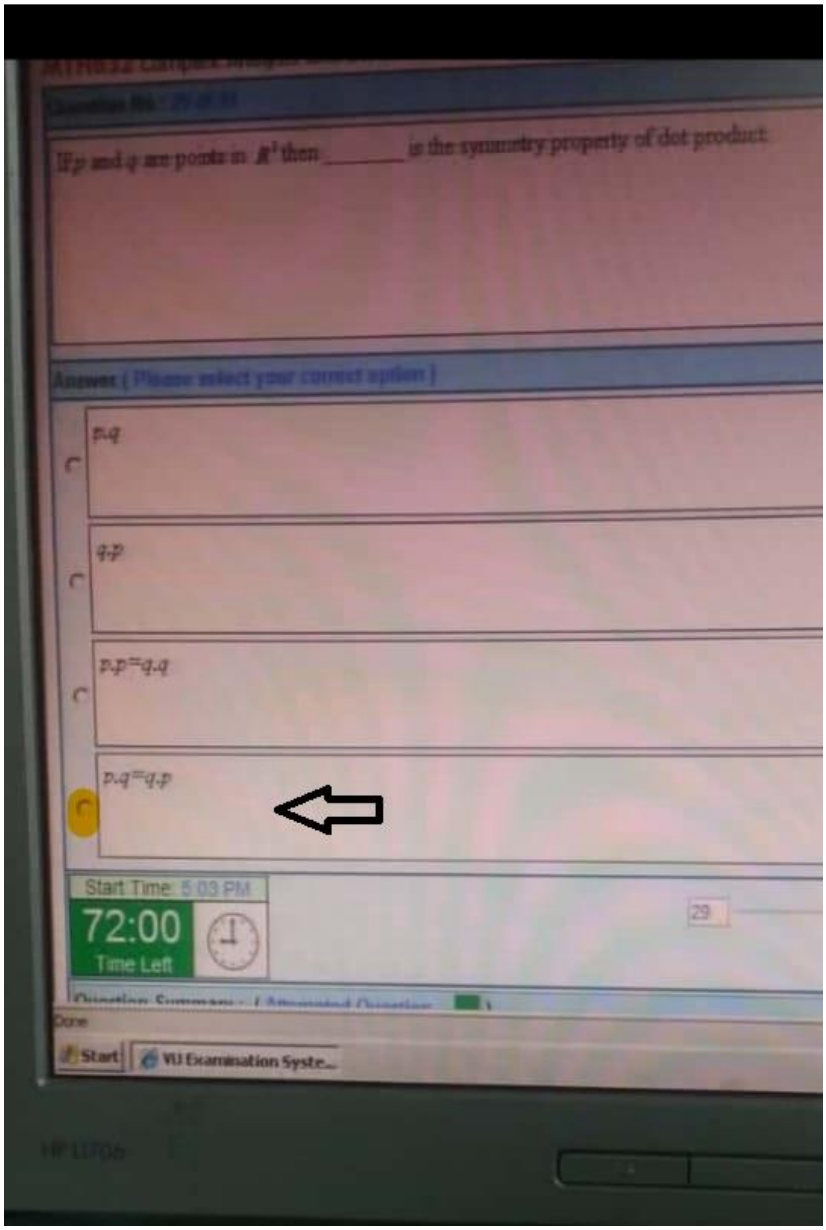

If p and q are points in R^3 then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.p=q.q
p.q=q.p
Start Time: 5:03 PM
72:00
Time Left
29
Start
VU Examination Syste...



Scanne

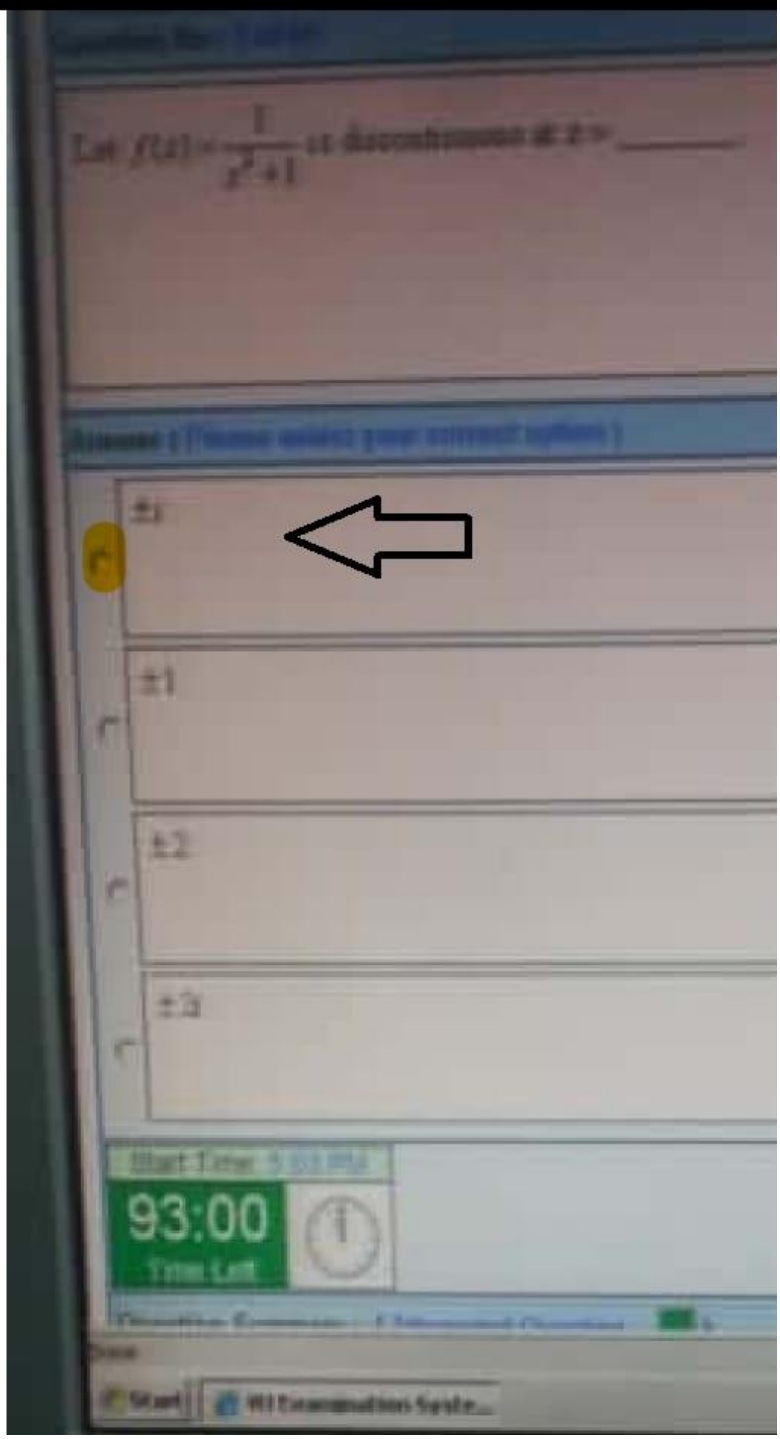

3x-2
3x+4
6x-2
-2x+4
Start Time: 5:03 PM
100:00
Time Left
Start
VU Examination Syste...

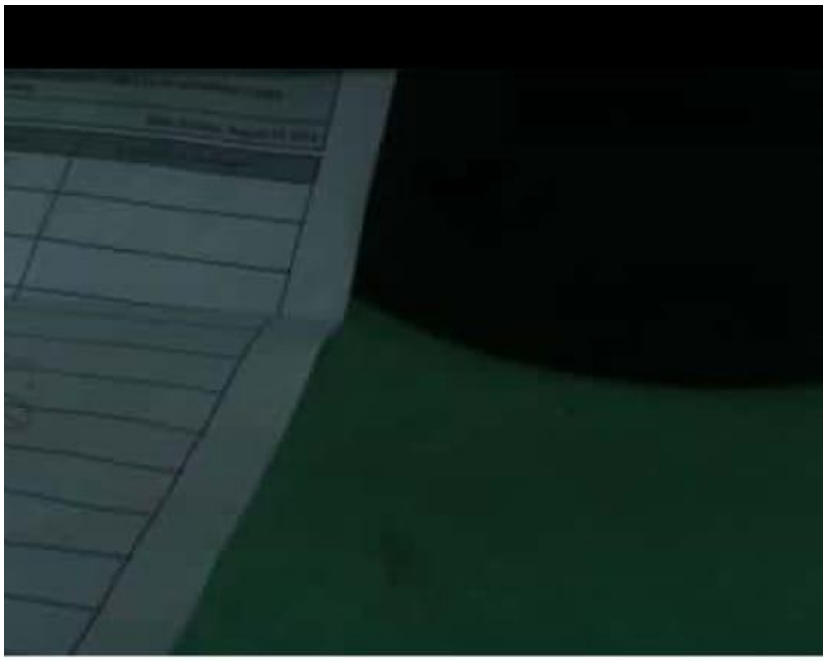
Scar


If p and q are points in R^3 then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.p=q.q
p.q=q.p
Start Time: 5:03 PM
72:00
Time Left
29
Start
VU Examination Syste...

[redacted]

[redacted]





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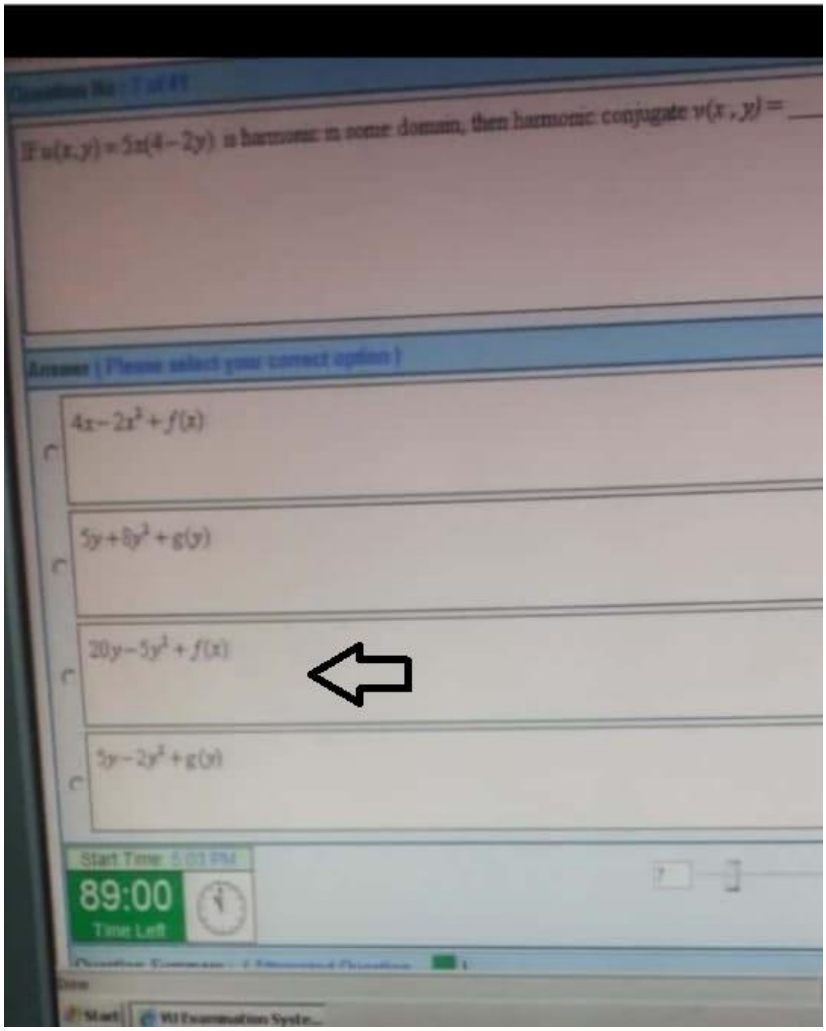
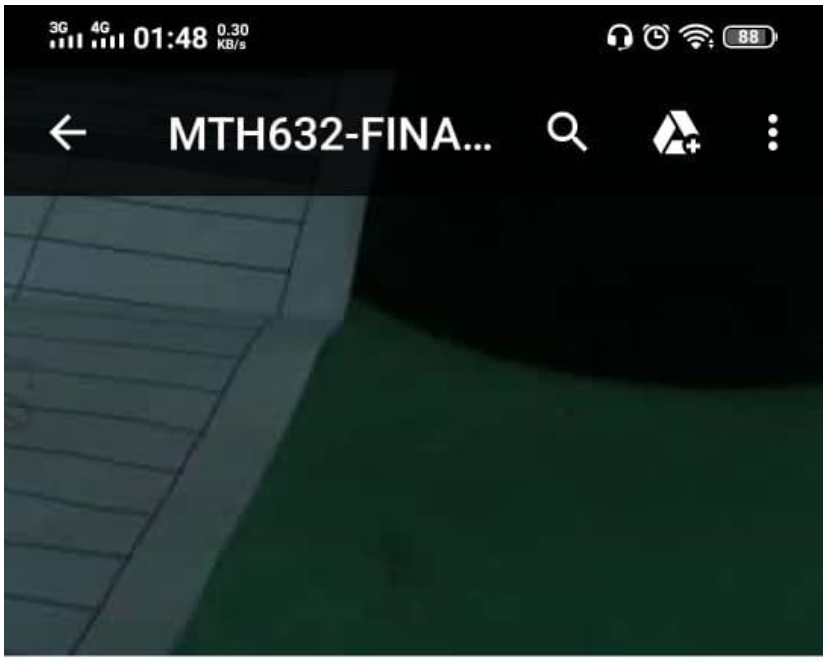

If u(x,y) = 5x(4−2y) is harmonic in some domain, then harmonic conjugate v(x , y) = ___
Answer (Please select your correct option)
4x−2x³ + f(x)
5y+8y³ + g(y)
20y−5y³ + f(x)
5y−2y³ + g(y)
Start Time: 5:03 PM
89:00
Time Left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MTH632-FINA...

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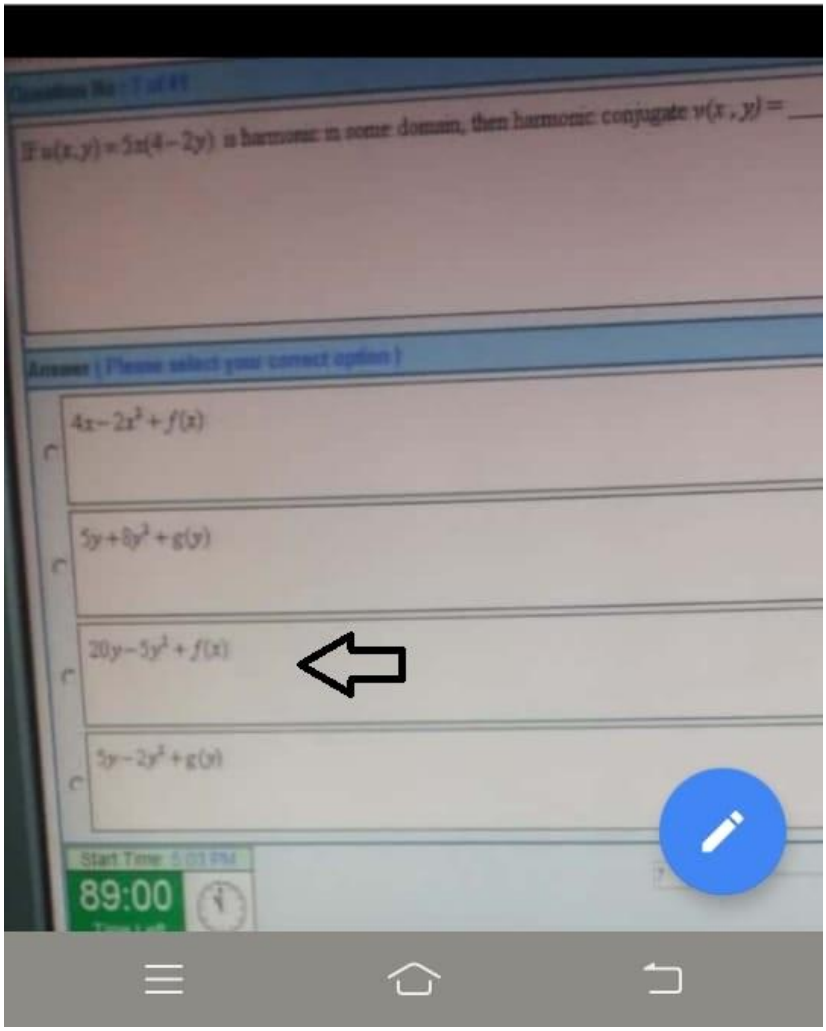
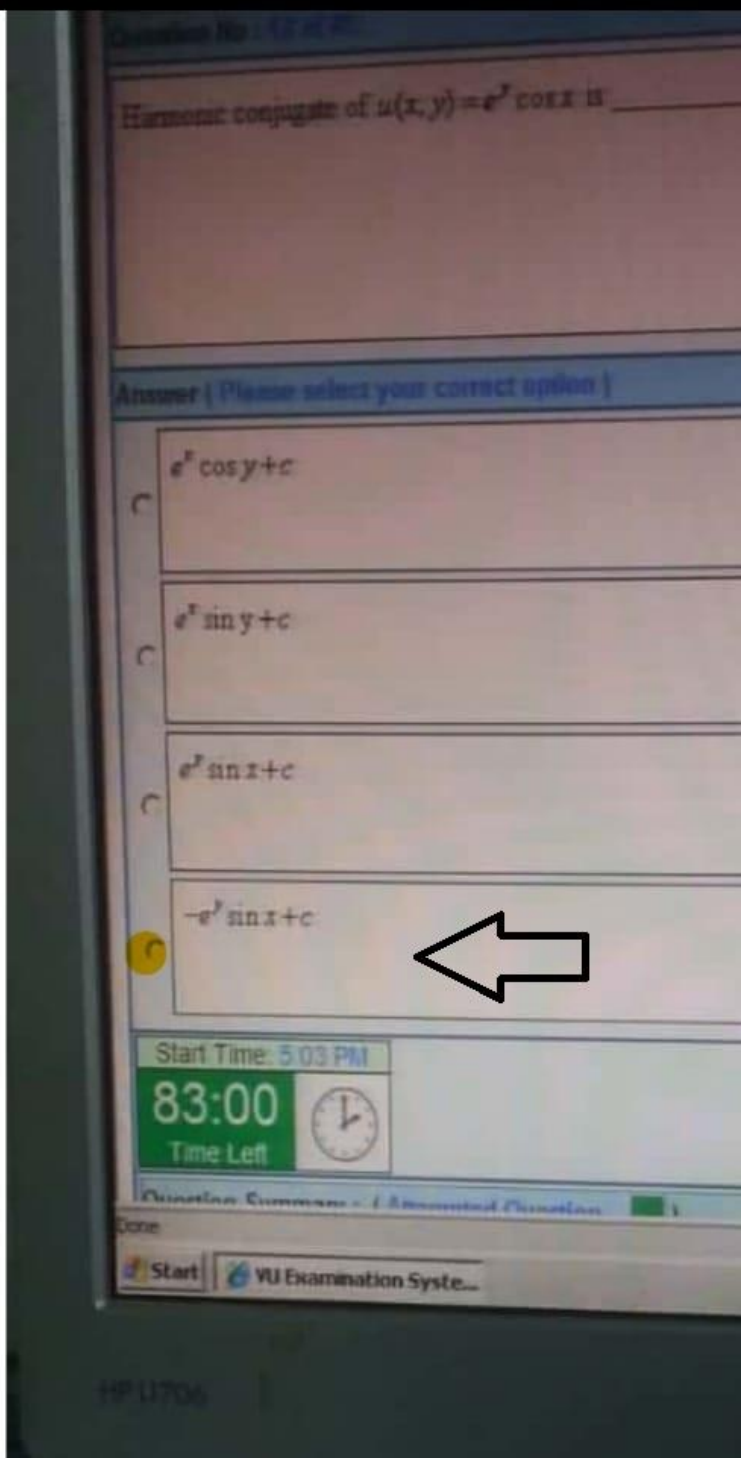
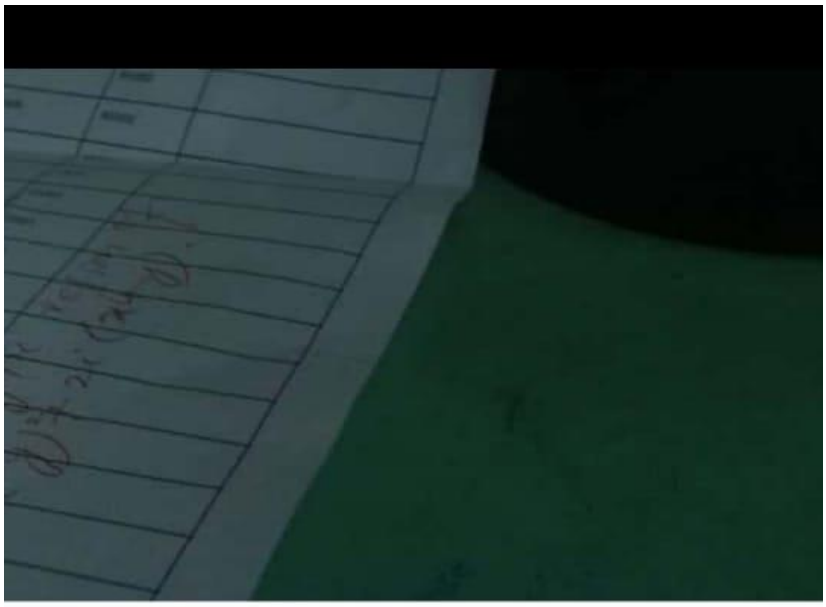

If u(x,y) = 5x(4−2y) is harmonic in some domain, then harmonic conjugate v(x, y) =
4x−2x³+f(x)
5y+8y³+g(y)
20y−5y³+f(x)
5y−2y³+g(y)
89:00


Harmonic conjugate of $u(x,y)=e^y \cos x$ is _____
Answer [Please select your correct option]
$e^y \cos y+c$
$e^y \sin y+c$
$e^y \sin x+c$
$-e^y \sin x+c$
Start Time: 5:03 PM
83:00
Time Left
Start
VU Examination Syste...



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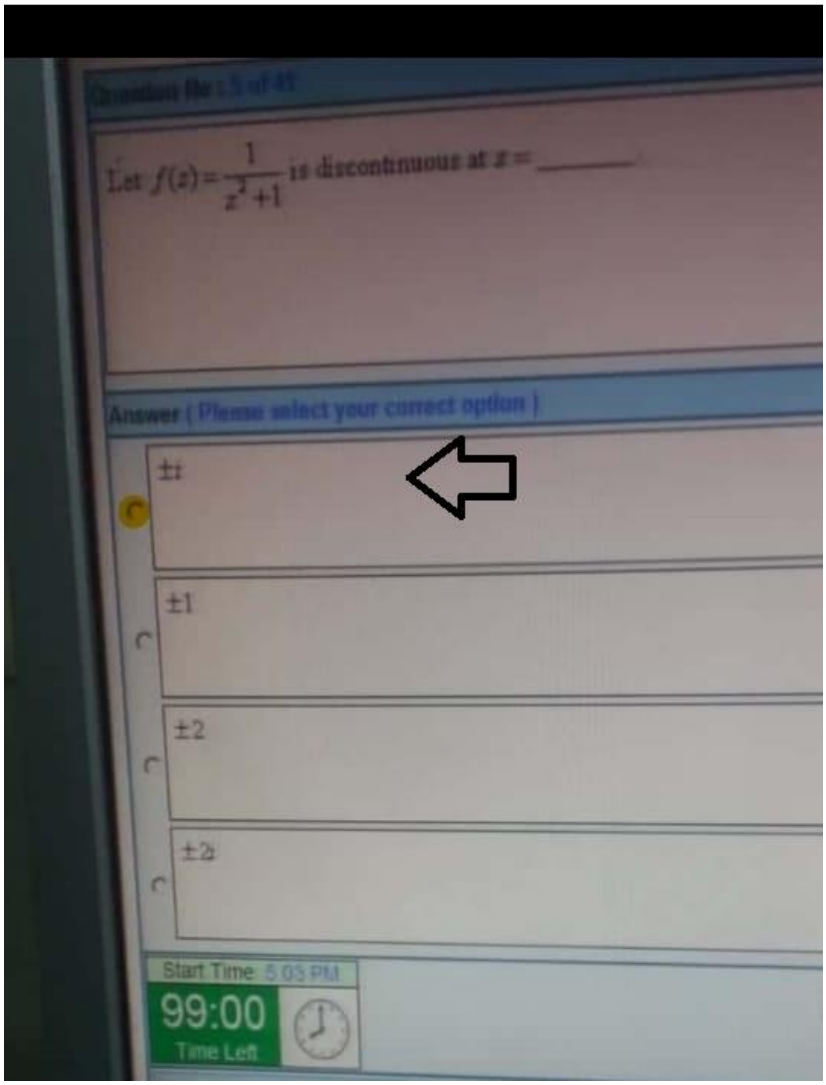
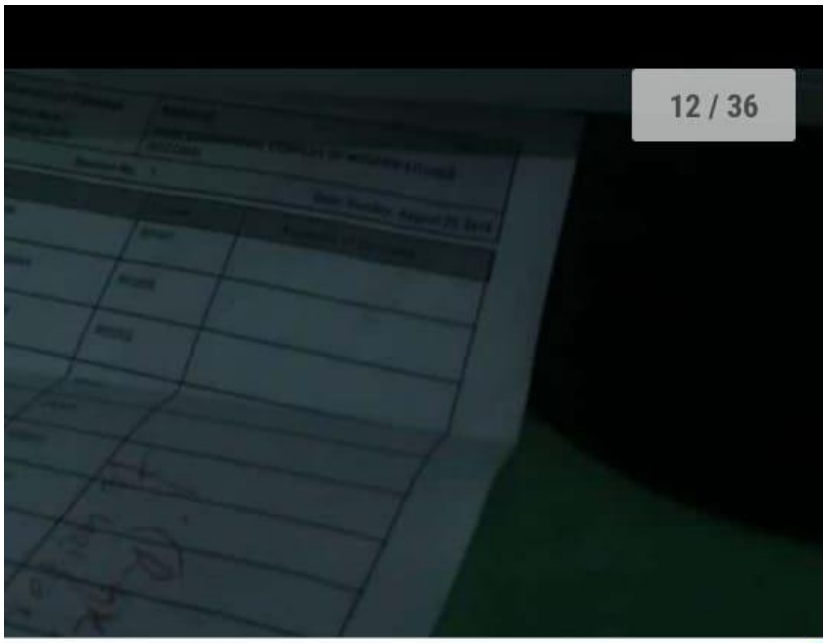
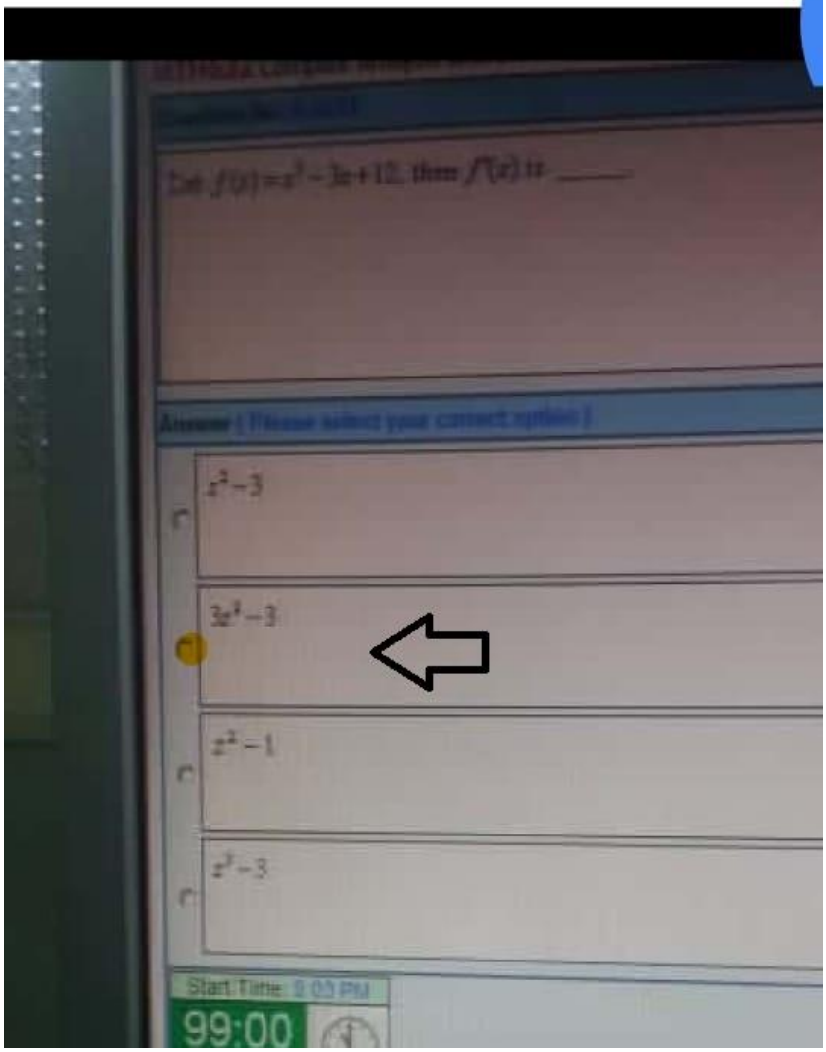
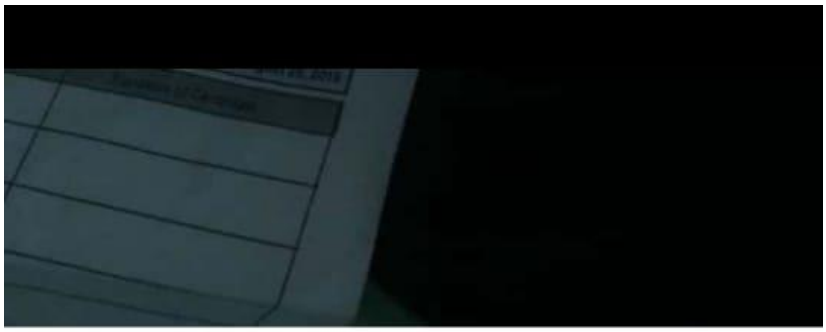

Let $f(x)=\frac{1}{x^2+1}$ is discontinuous at $x=$ _____
Answer (Please select your correct option)
±i
±1
±2
±2i
Start Time 5:03 PM
99:00
Time Left



Sca





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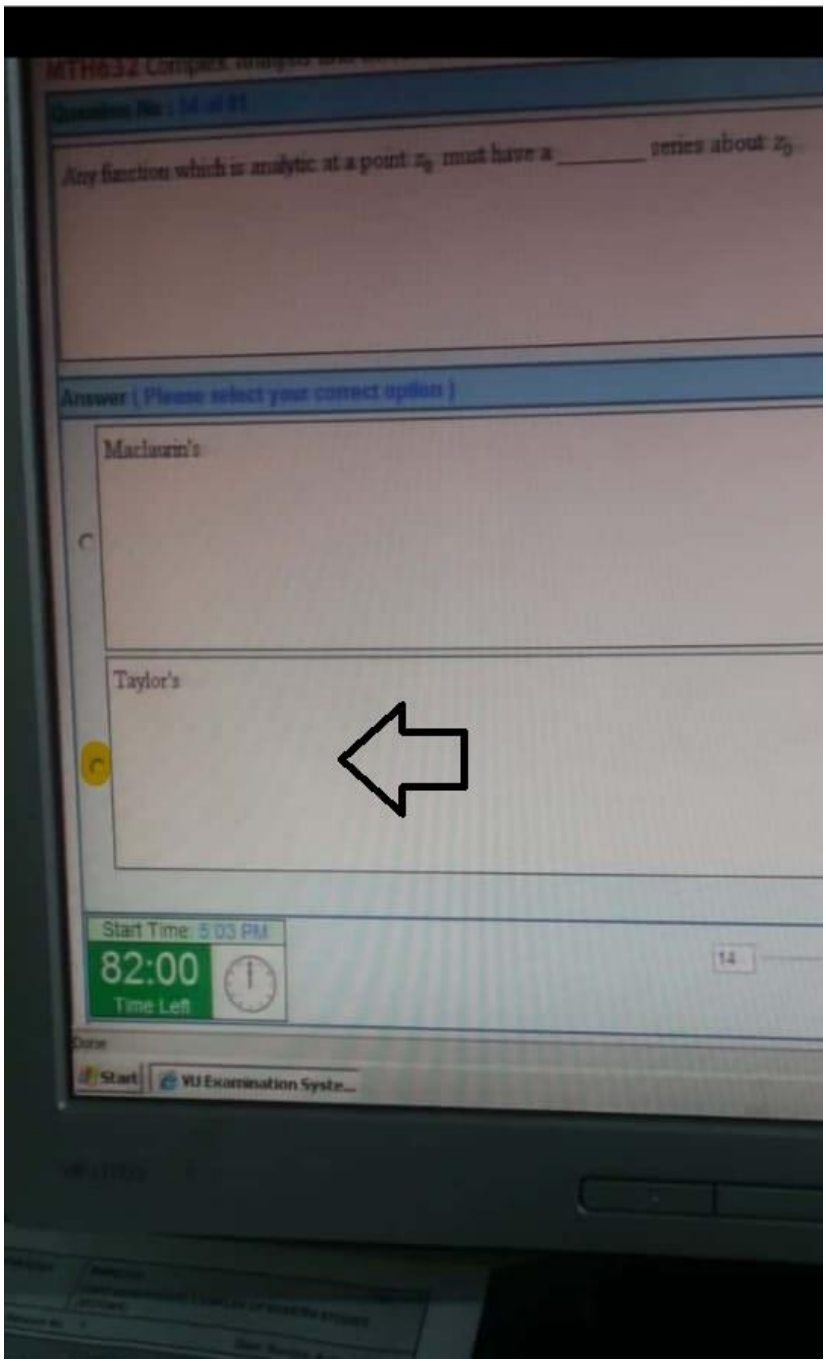
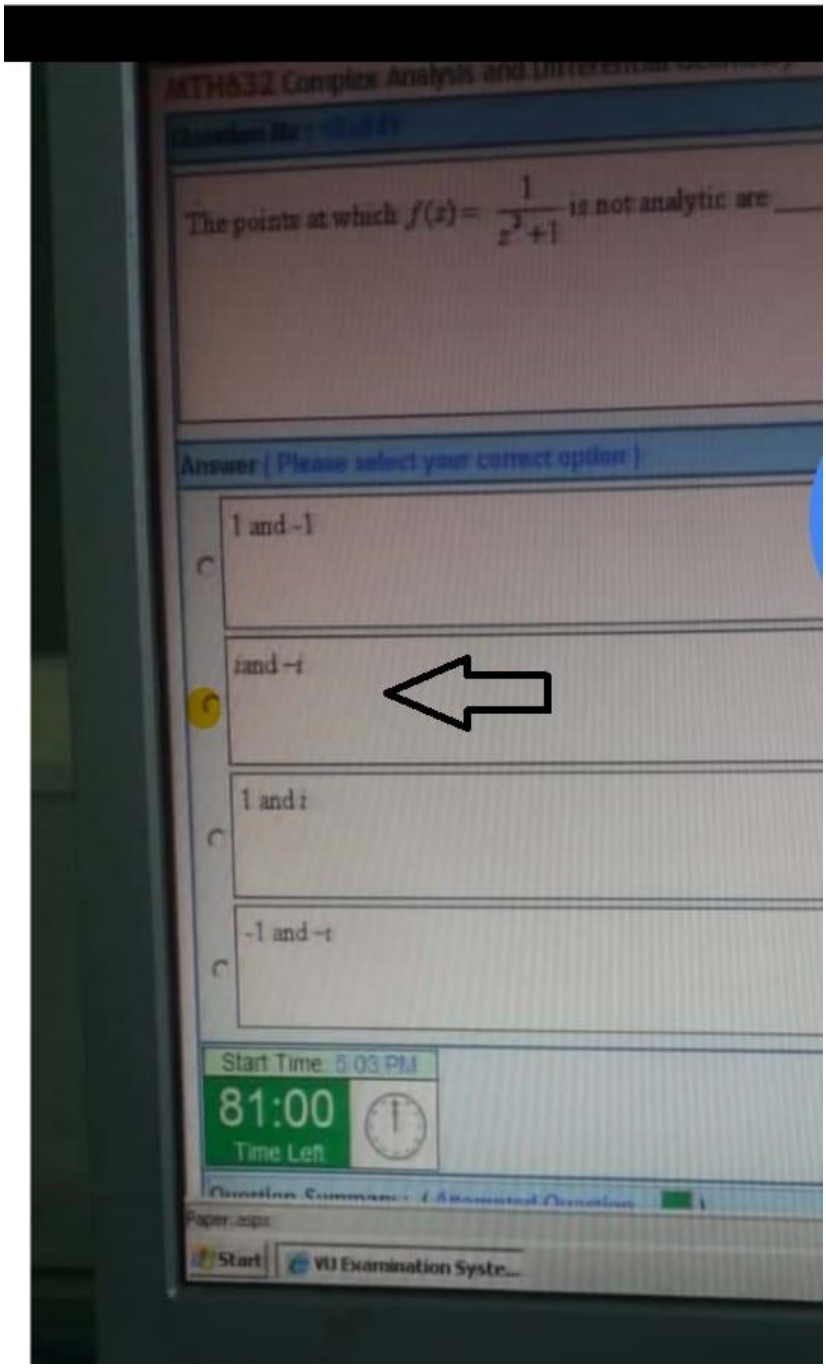
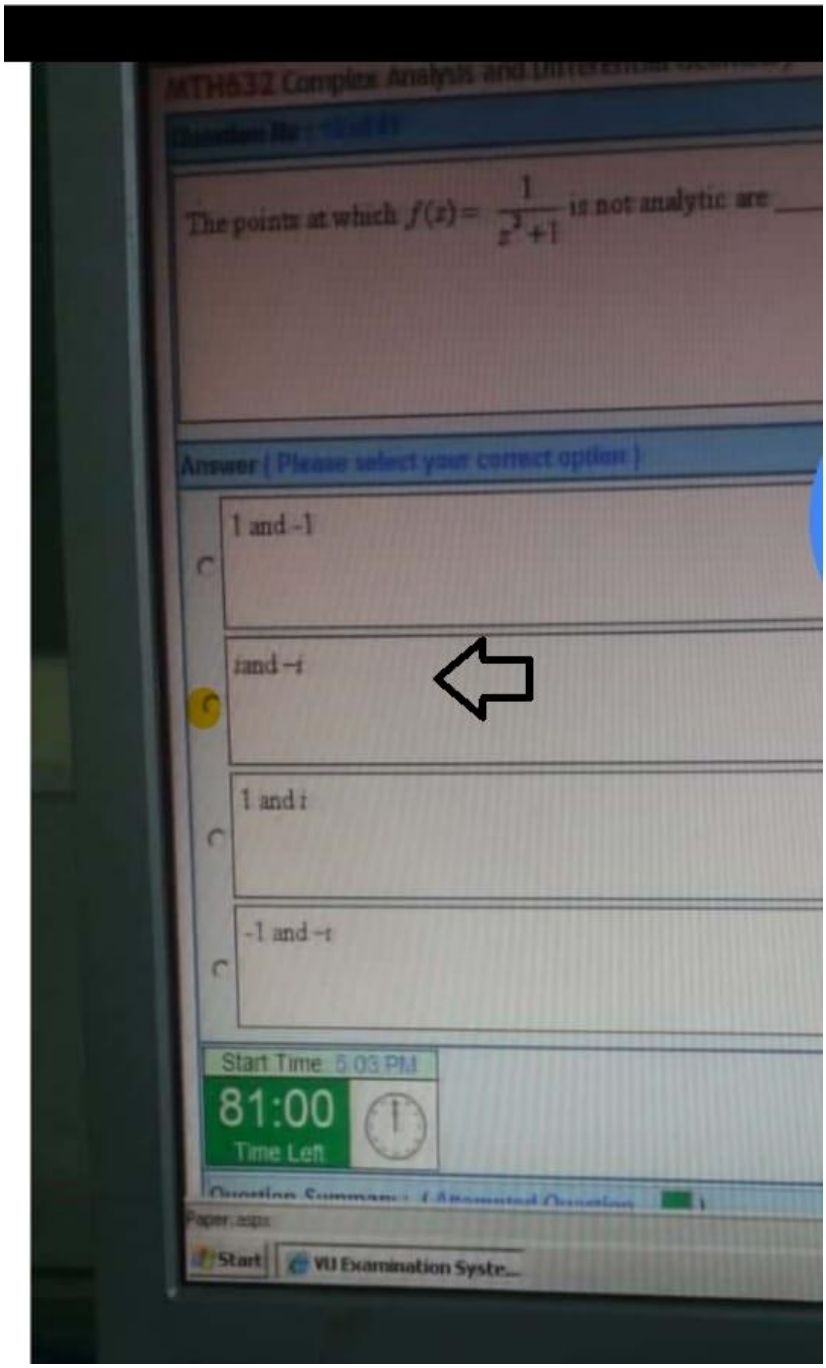
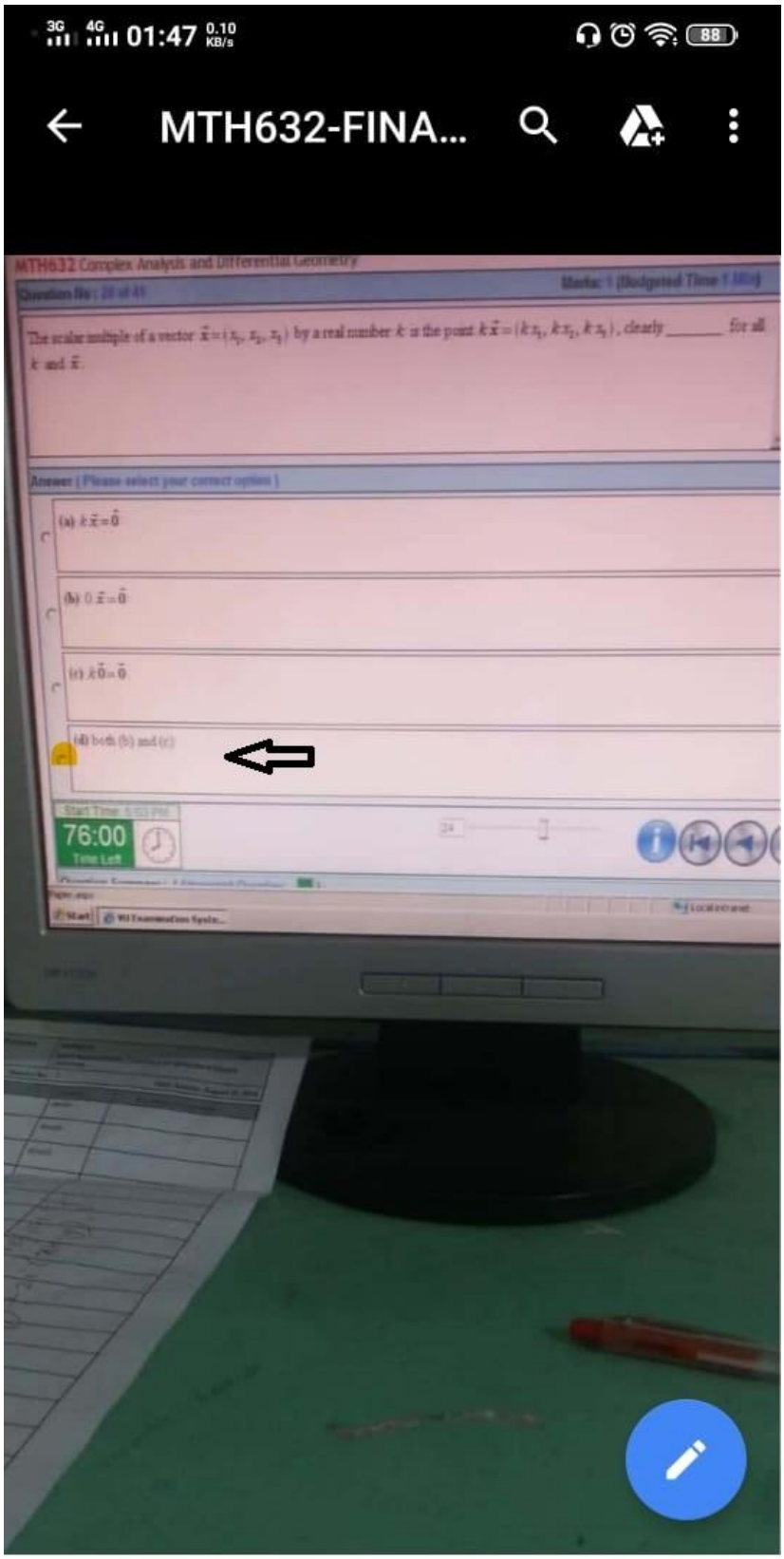
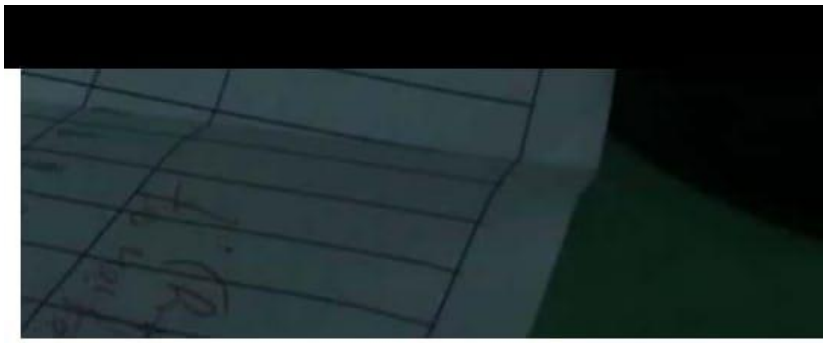
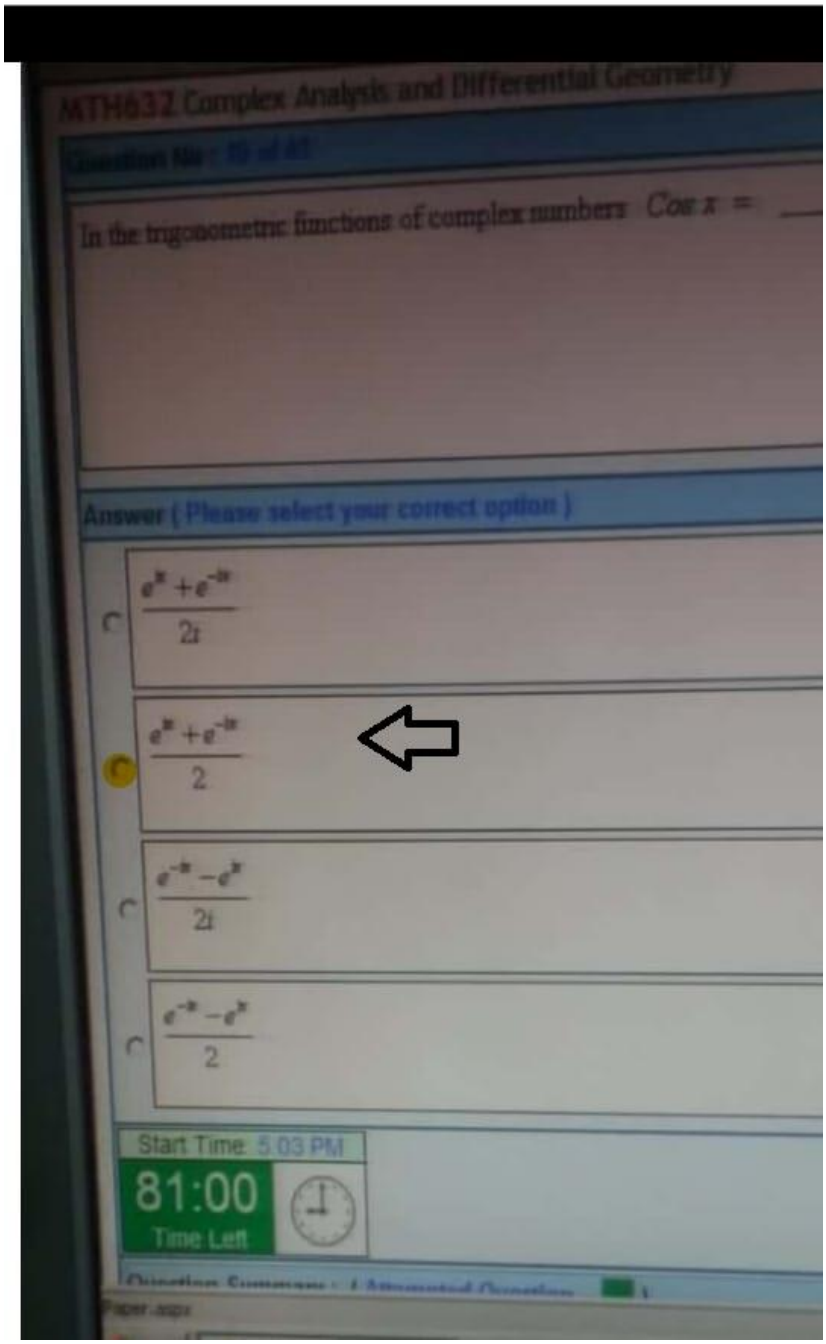

Any function which is analytic at a point z_0 must have a _____ series about z_0
Answer [Please select your correct option]
Maclaurin's
Taylor's
Start Time: 5:03 PM
82:00
Time Left
14
Start
VU Examination Syste...


MTH632 Complex Analysis and
The points at which $f(z) = \frac{1}{z^3+1}$ is not analytic are
Answer (Please select your correct option)
1 and -1
i and -i
1 and i
-1 and -i
Start Time 5:03 PM
81:00
Time Left
Start
VU Examination Syste...

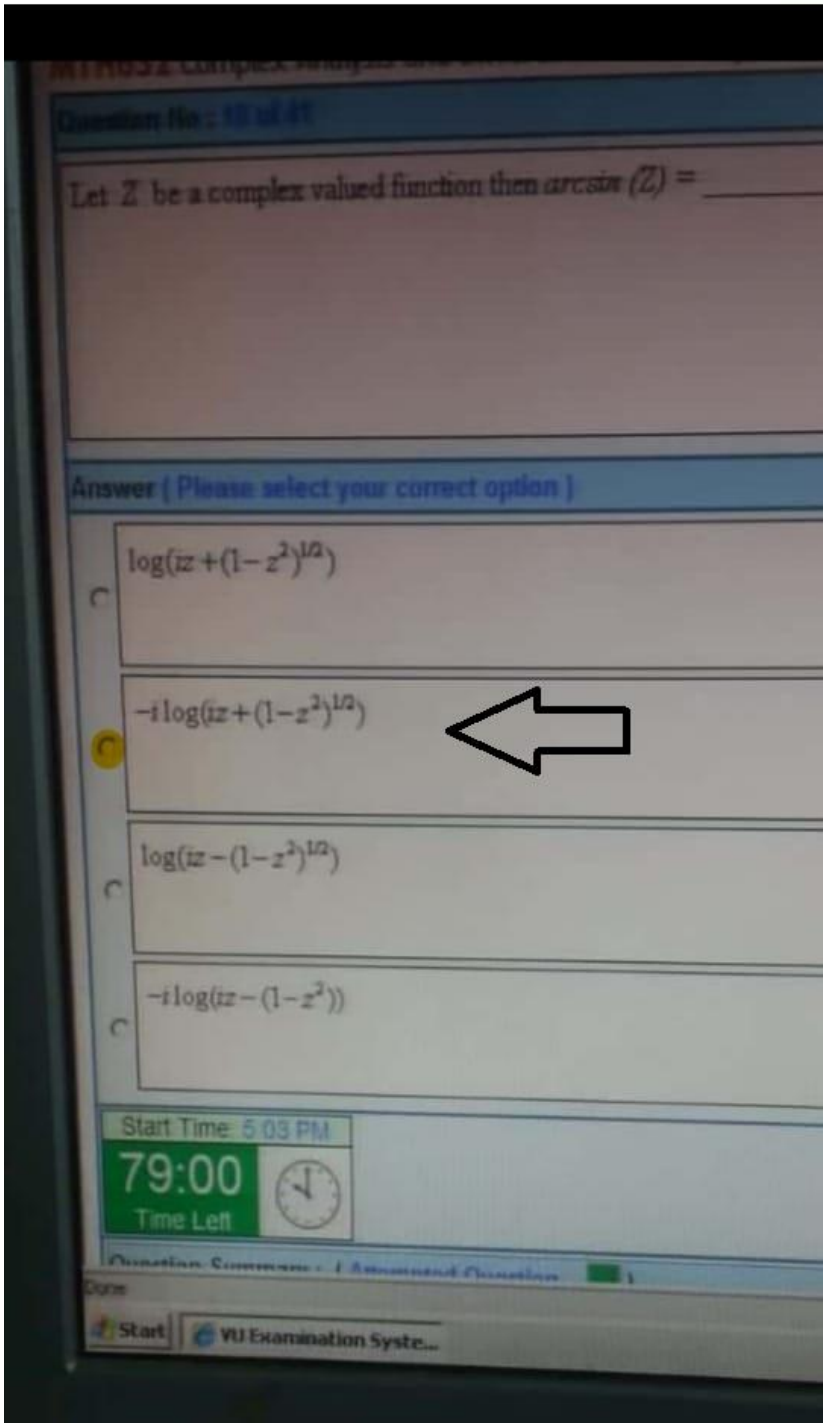

MTH632 Complex Analysis and Differential
The points at which $f(z) = \frac{1}{z^3+1}$ is not analytic are
Answer (Please select your correct option)
1 and -1
i and -i
1 and i
-1 and -i
Start Time 5:03 PM
81:00
Time Left
Start
VU Examination Syste...


01:47
MTH632-FINA...
MTH632 Complex Analysis and Differential Geometry
The scalar multiple of a vector $\vec{x}=(x_1, x_2, x_3)$ by a real number k is the point $k\vec{x}=(kx_1, kx_2, kx_3)$, clearly _____ for all k and $\vec{x}$.
Answer (Please select your correct option)
(a) $k\vec{x}=\vec{0}$
(b) $0\vec{x}=\vec{0}$
(c) $k\vec{0}=\vec{0}$
(d) both (b) and (c)
76:00
Time Left

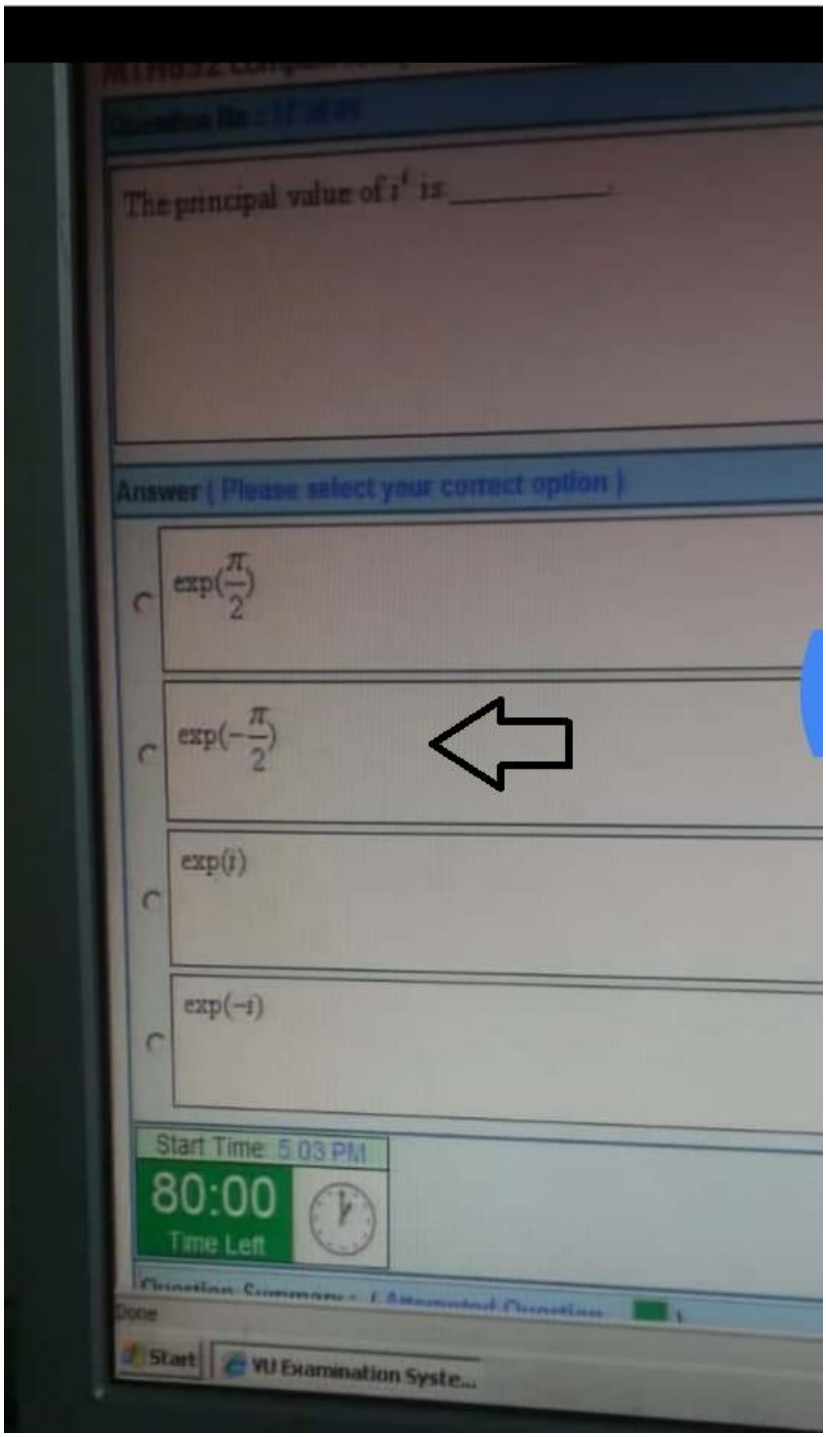



MTH632 Complex Analysis and Differential Geometry
In the trigonometric functions of complex numbers Cos x =
Answer (Please select your correct option)
$\frac{e^{ix}+e^{-ix}}{2i}$
$\frac{e^{ix}+e^{-ix}}{2}$
$\frac{e^{-ix}-e^{ix}}{2i}$
$\frac{e^{-ix}-e^{ix}}{2}$
Start Time: 5:03 PM
81:00
Time Left

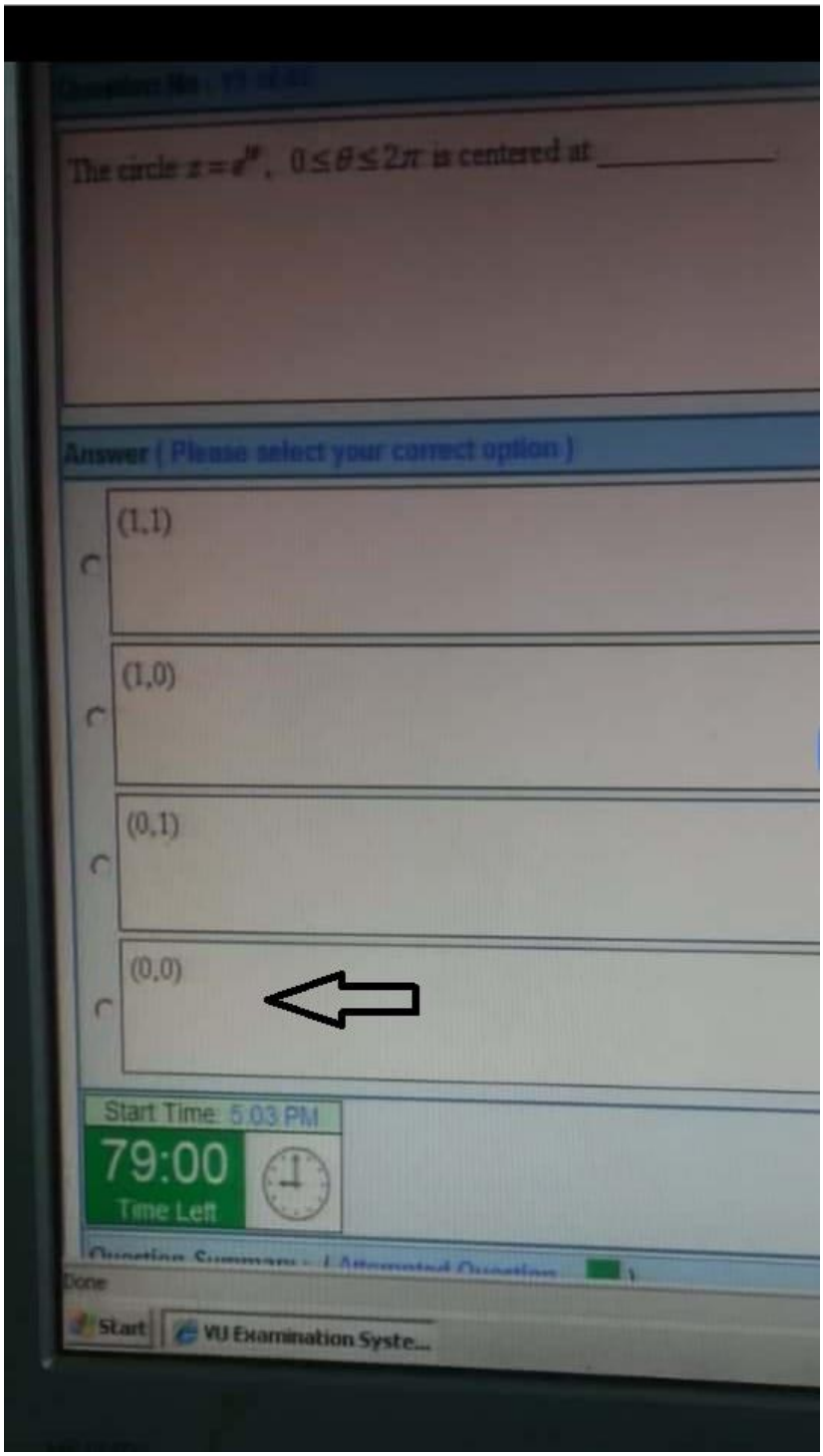
Sca


Let Z be a complex valued function then arcsin (Z) =
Answer (Please select your correct option)
log(iz +(1−z²)^1/2)
−i log(iz+(1−z²)^1/2)
log(iz −(1−z²)^1/2)
−i log(iz−(1−z²))
Start Time: 5:03 PM
79:00
Time Left
Start
VU Examination Syste...

Sca


The principal value of i^i is _____
Answer (Please select your correct option)
$\exp(\frac{\pi}{2})$
$\exp(-\frac{\pi}{2})$
$\exp(i)$
$\exp(-i)$
Start Time 5:03 PM
80:00
Time Left
Done
Start
VU Examination Syste...

Sca


The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 5:03 PM
79:00
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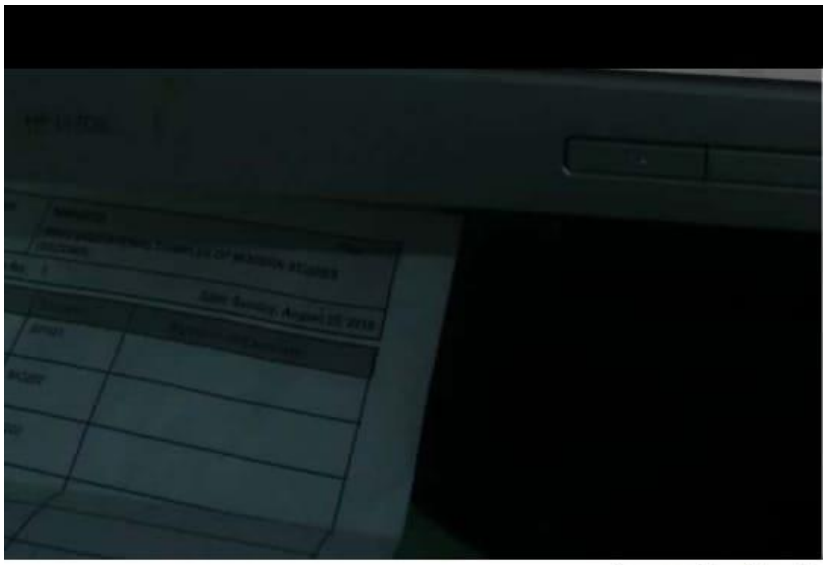
VU Examination Syste...



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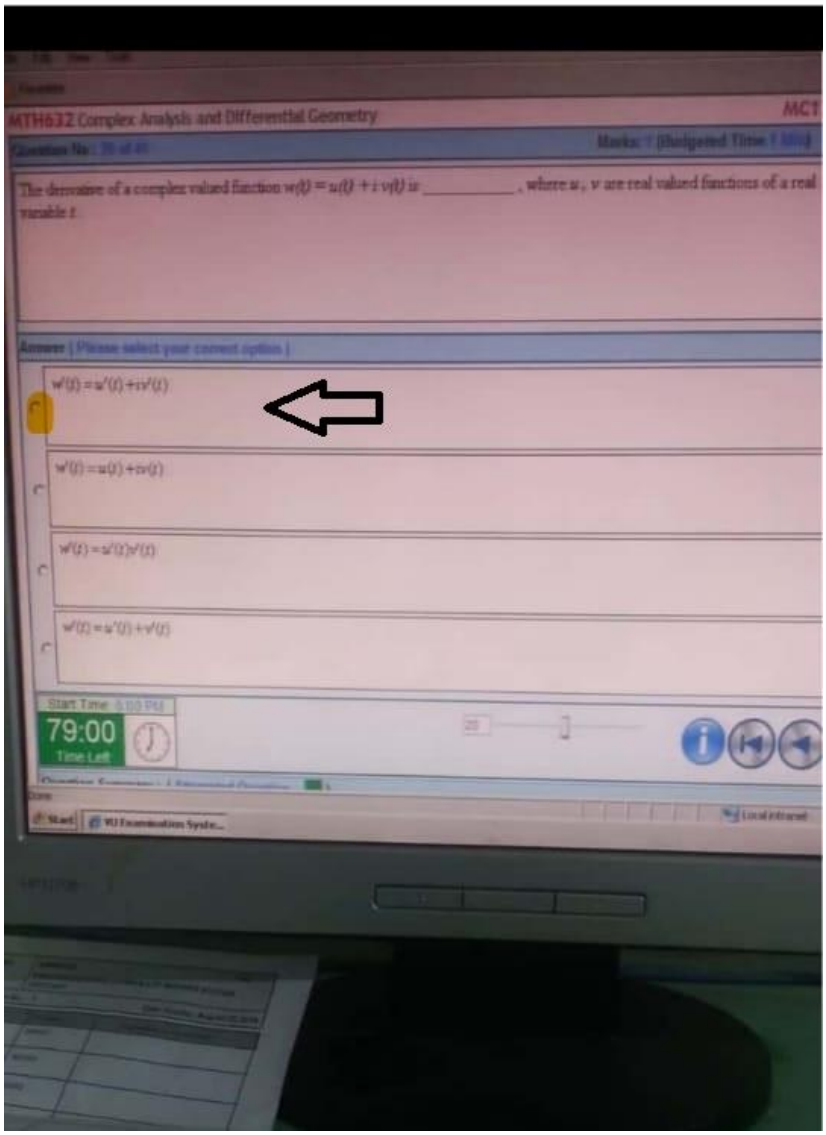

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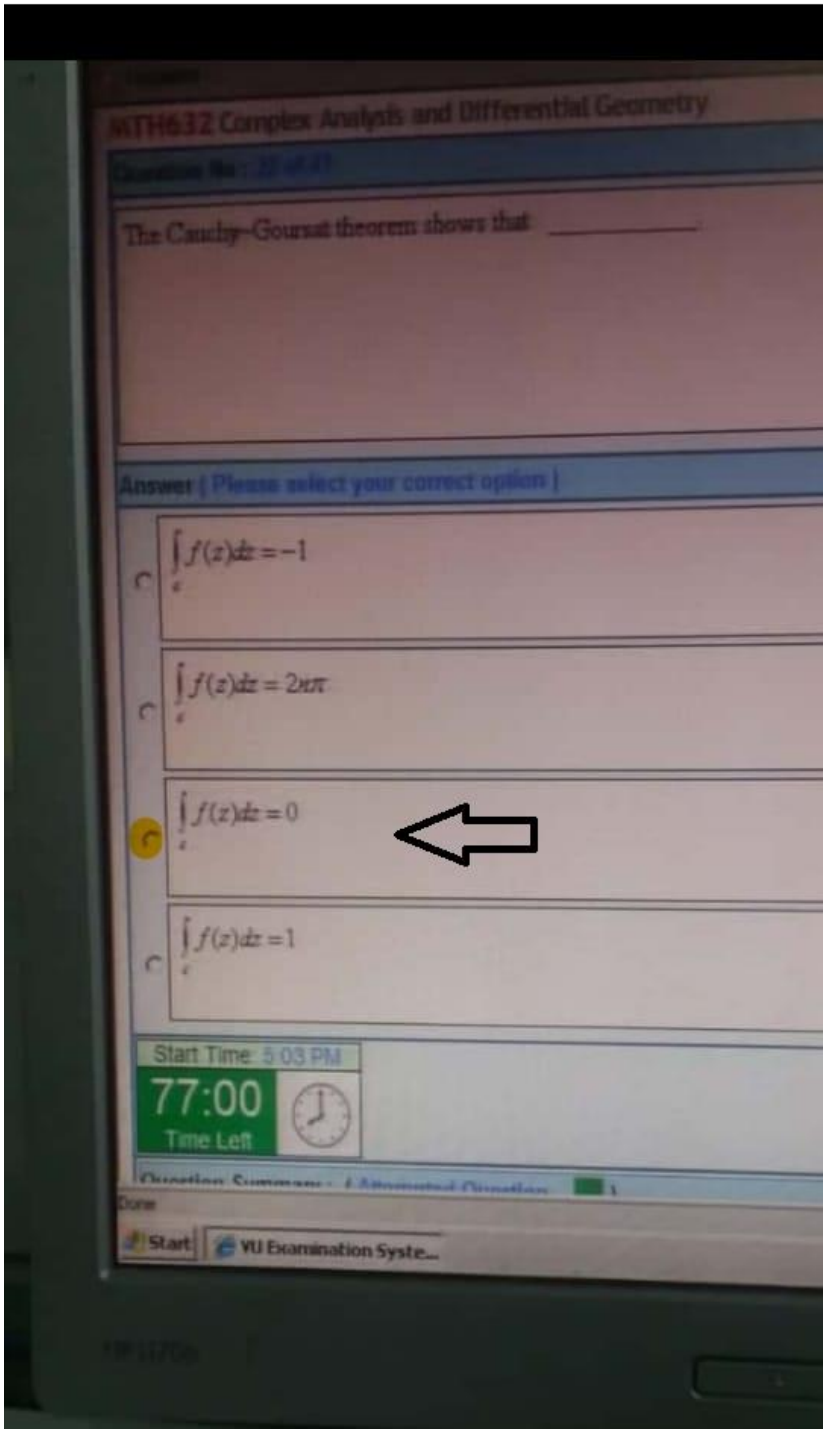
MTH632 Complex Analysis and Differential Geometry
The Cauchy-Gourst theorem shows that _____
Answer [Please select your correct option]
$\int_c f(z)dz = -1$
$\int_c f(z)dz = 2\pi\pi$
$\int_c f(z)dz = 0$
$\int_c f(z)dz = 1$
Start Time: 5:03 PM
77:00
Time Left
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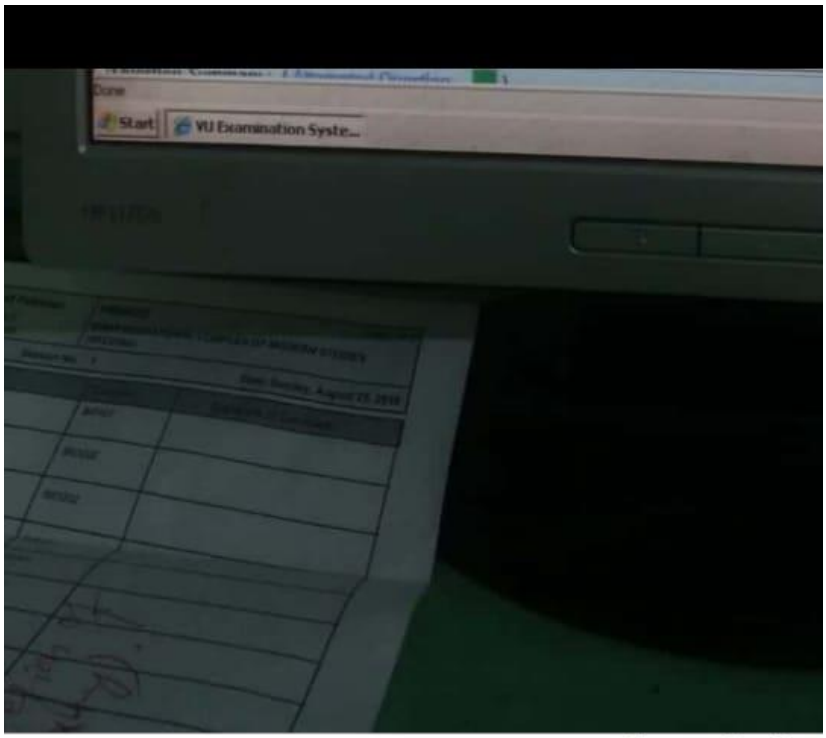
Start
VU Examination Syste...



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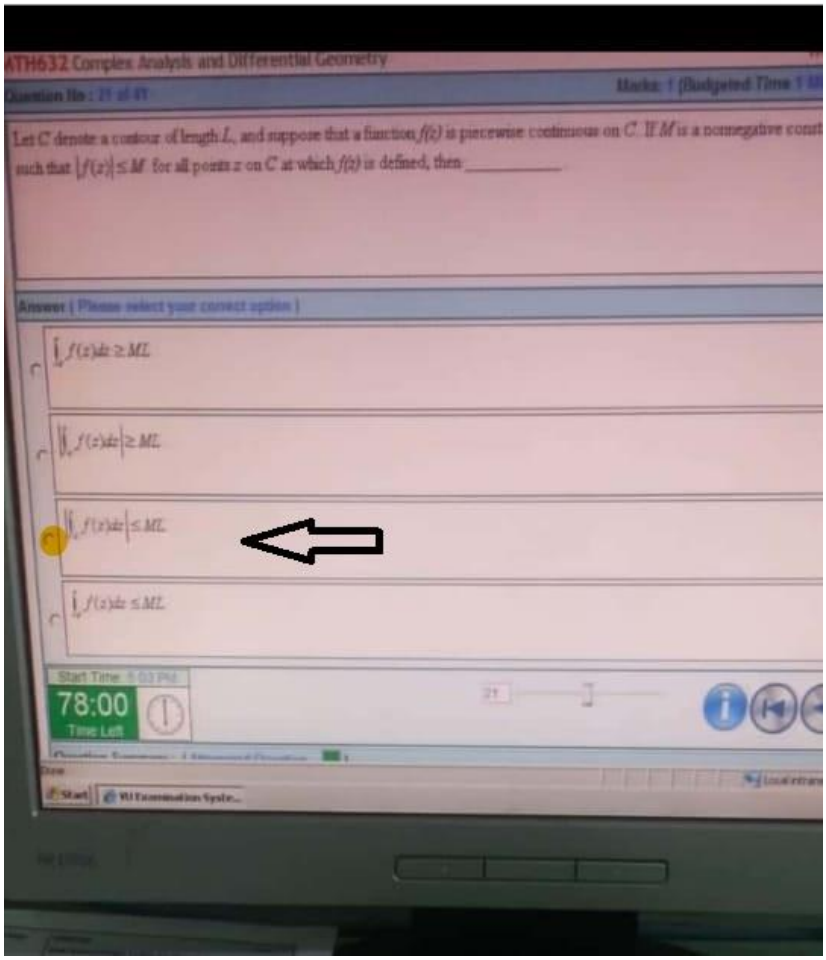

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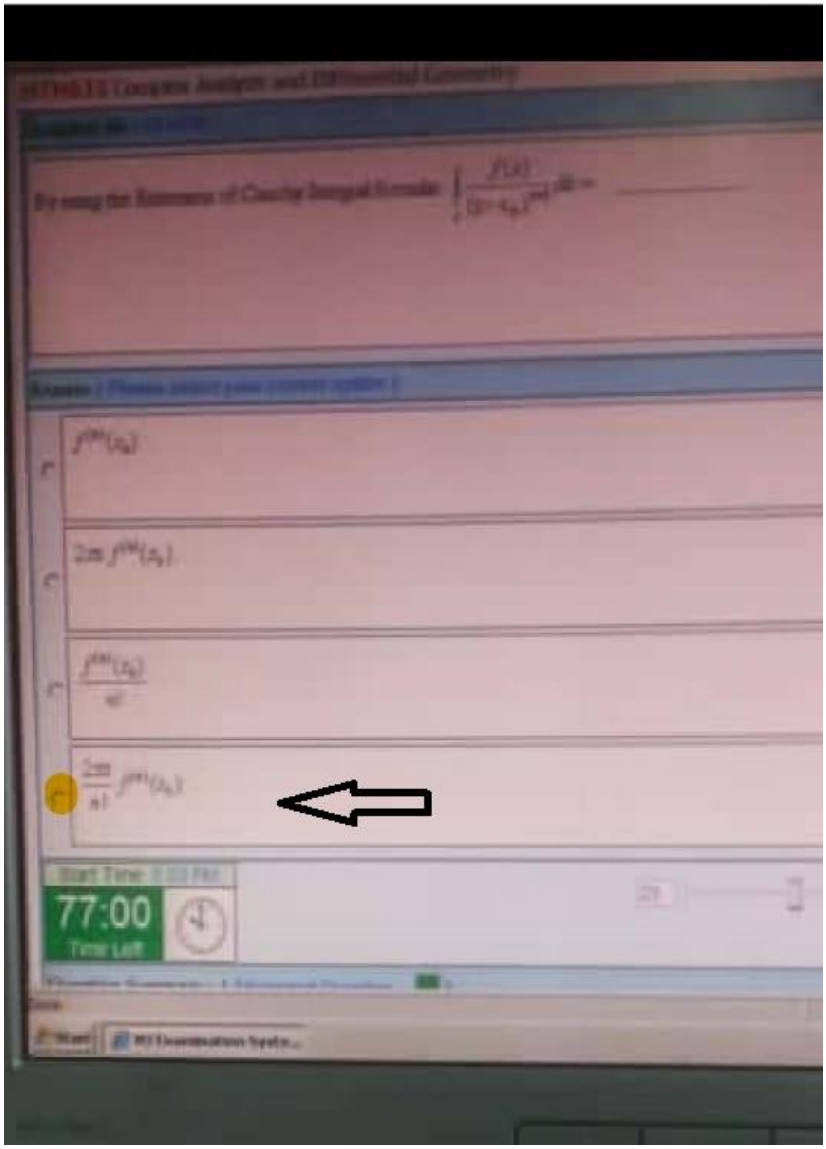

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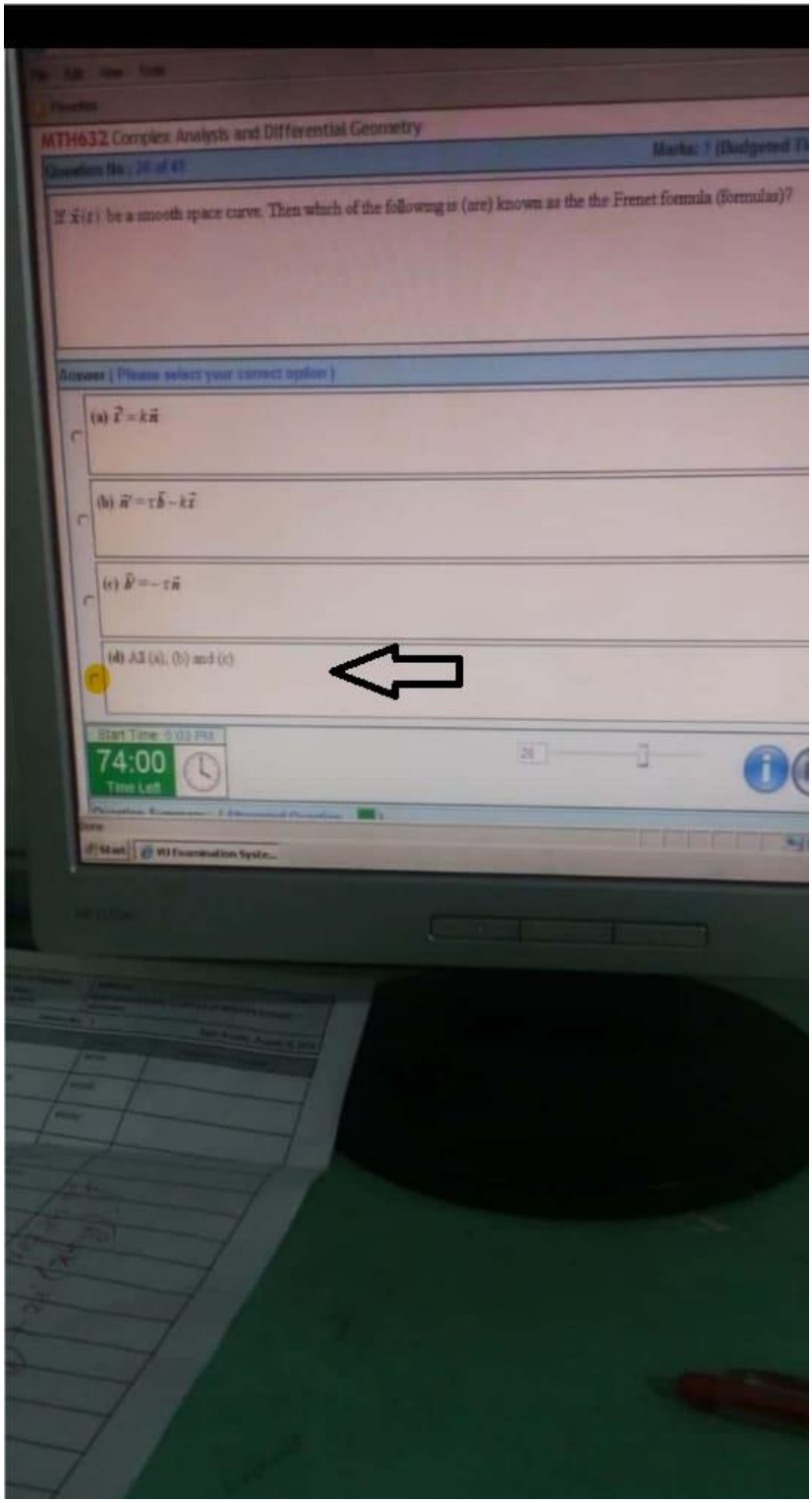

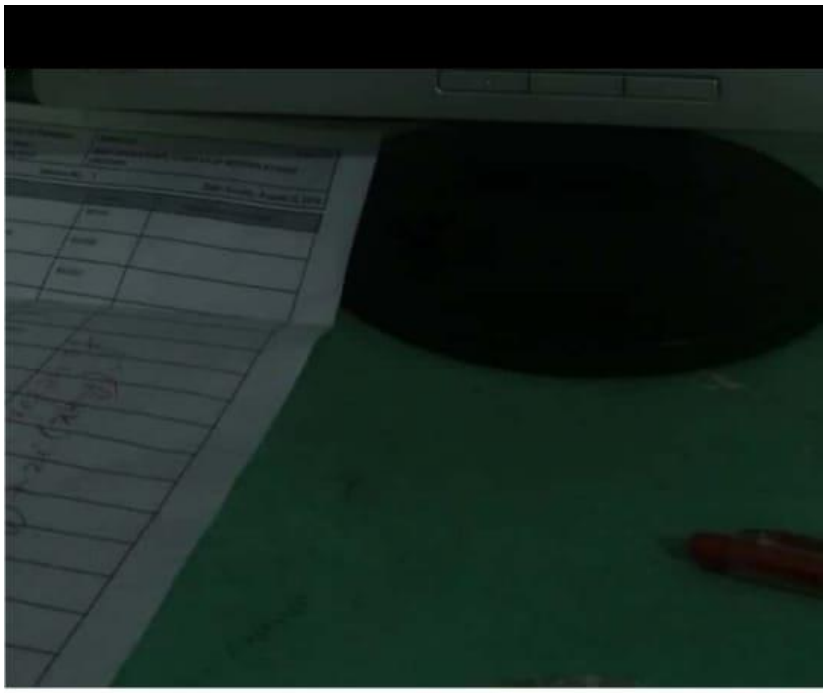

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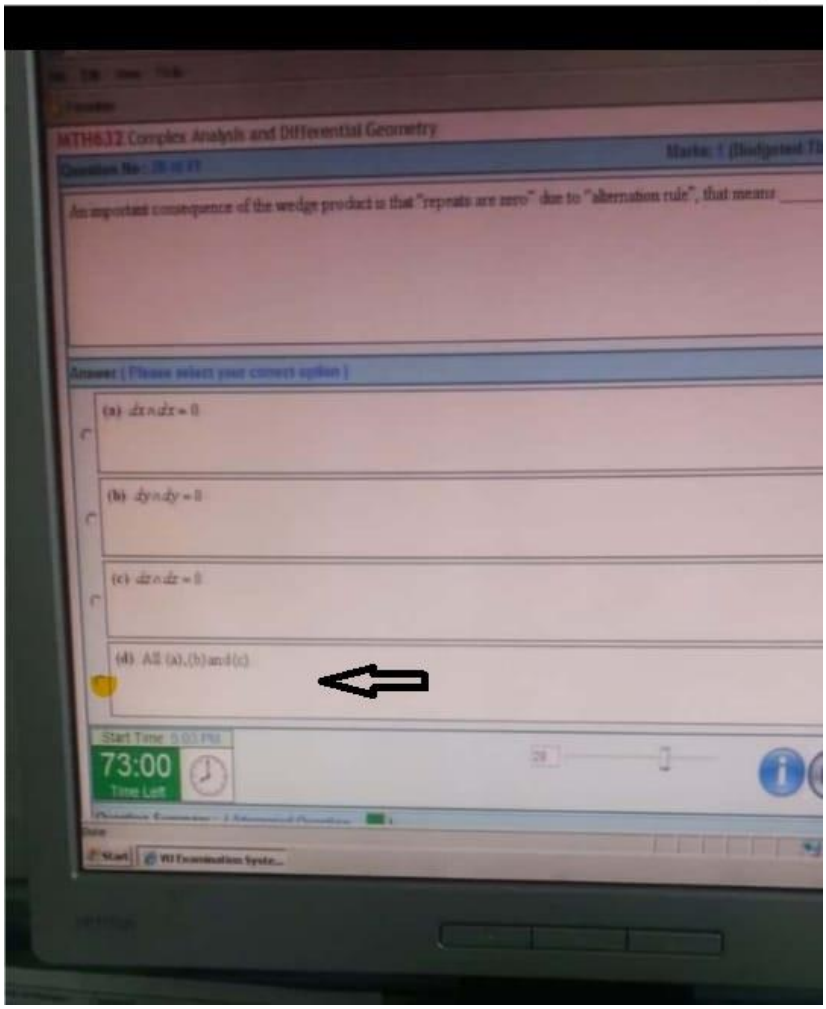

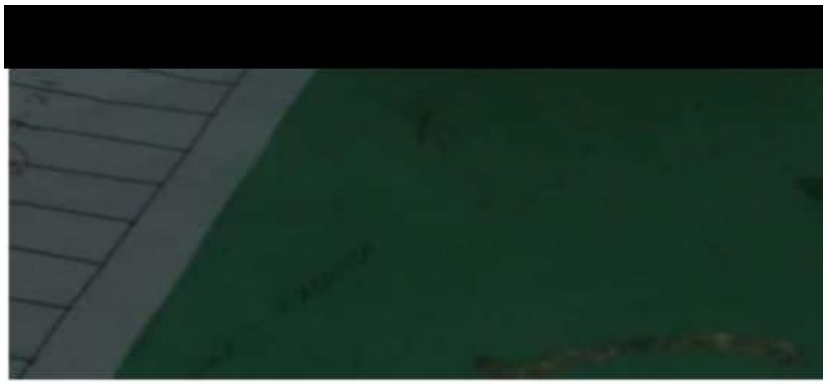

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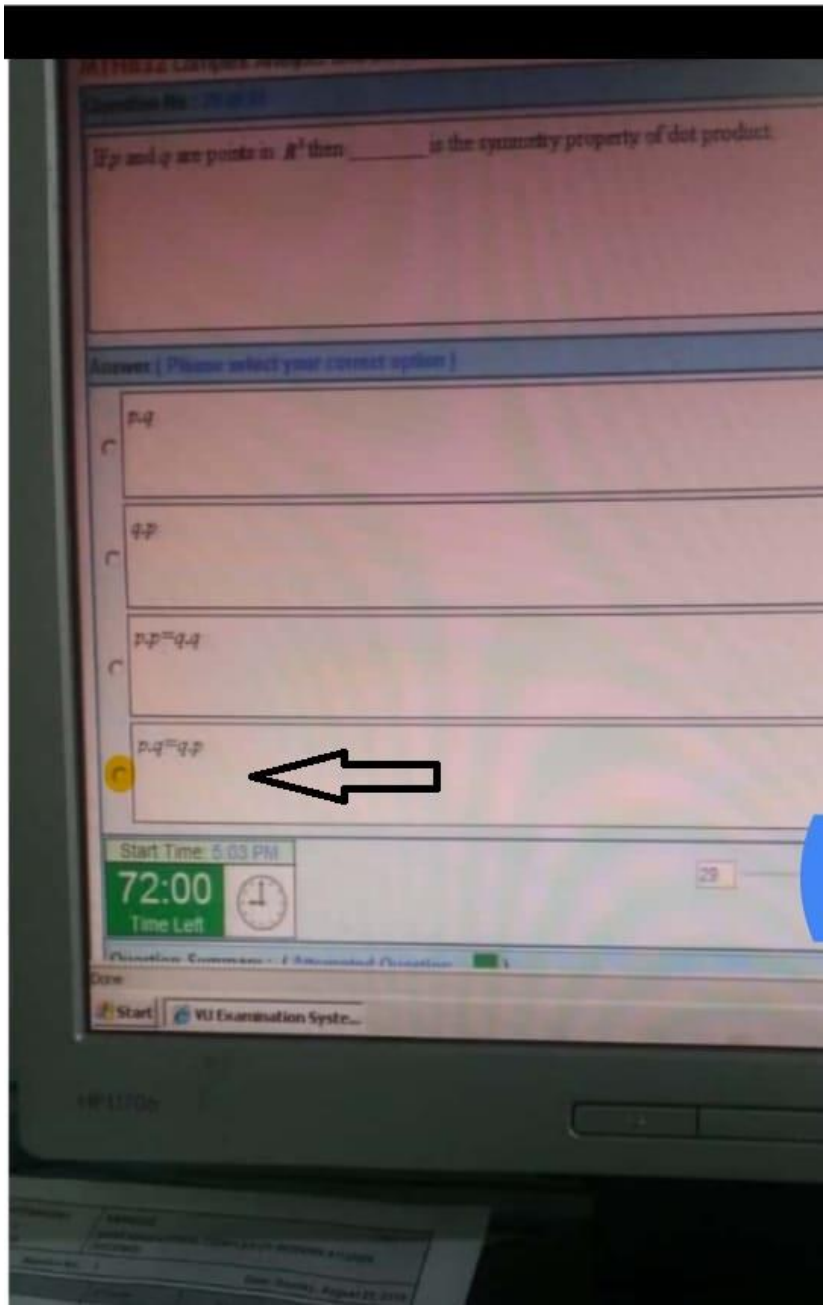
If p and q are points in R^3 then _____ is the symmetry property of dot product
Answer [Please select your correct option]
p.q
q.p
p.p=q.q
p.q=q.p
Start Time: 5:03 PM
72:00
Time Left
29
Start
VU Examination Syste...

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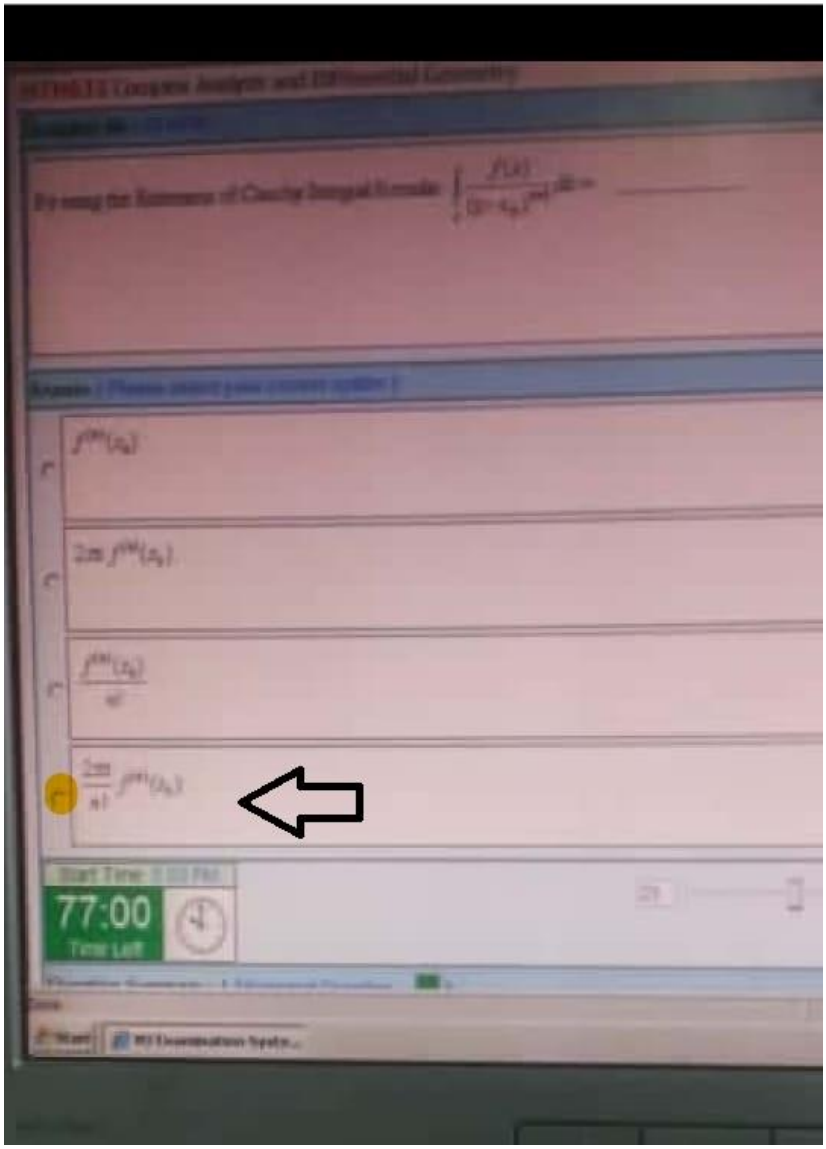

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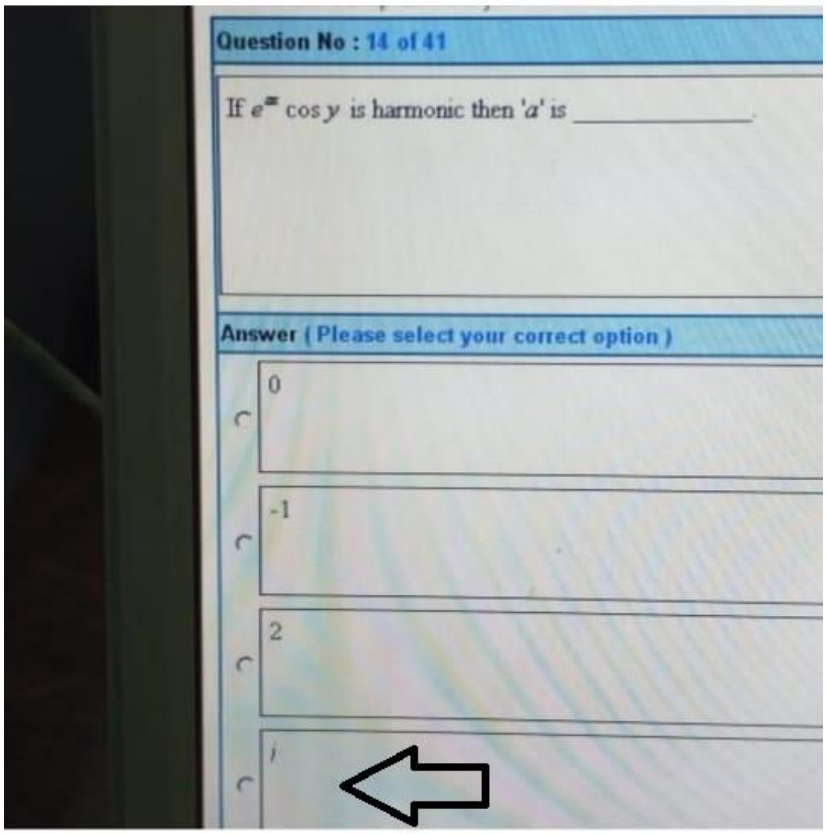
[redacted]



Question No : 14 of 41

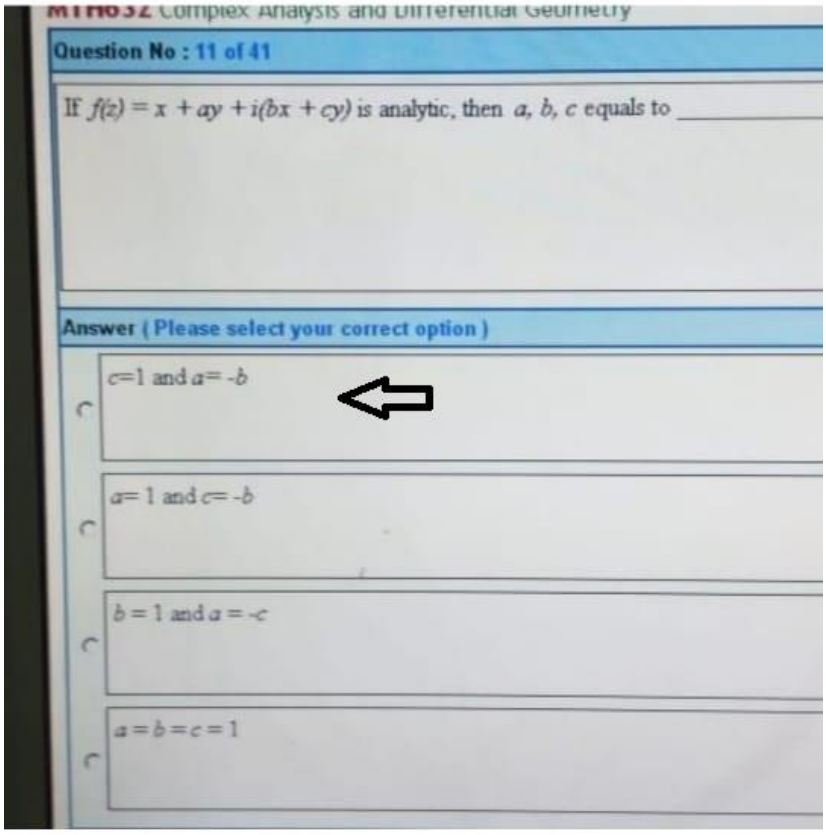
If $e^{ax} \cos y$ is harmonic then 'a' is _____.

Answer (Please select your correct option)

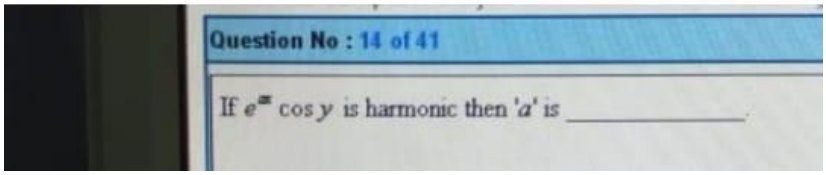
- 0
- -1
- 2
- i

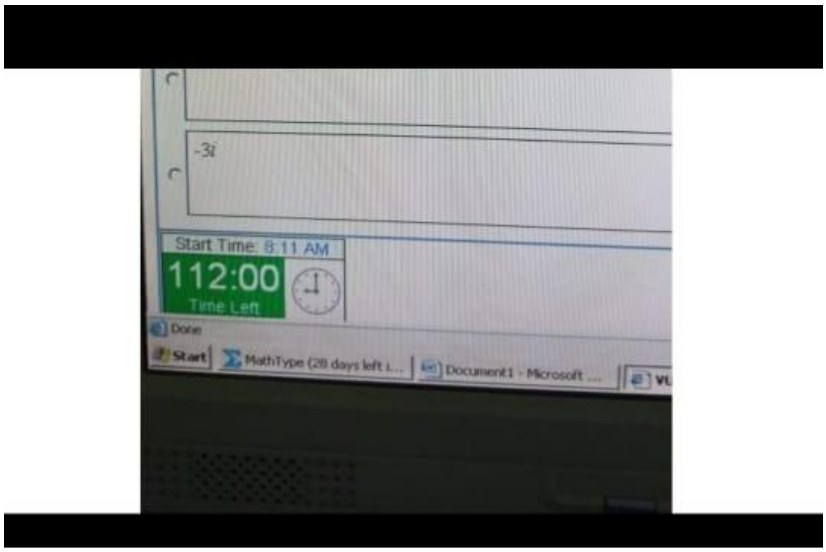
[redacted]

[redacted]



[redacted]




-3i
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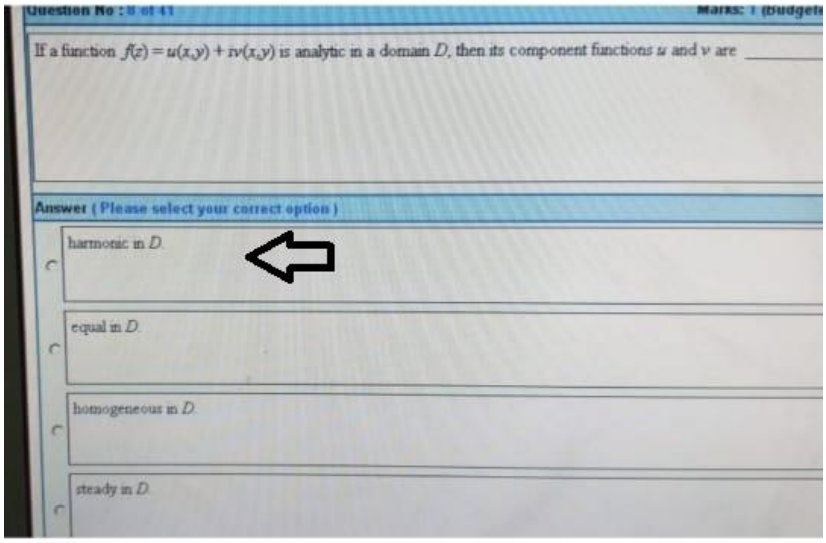
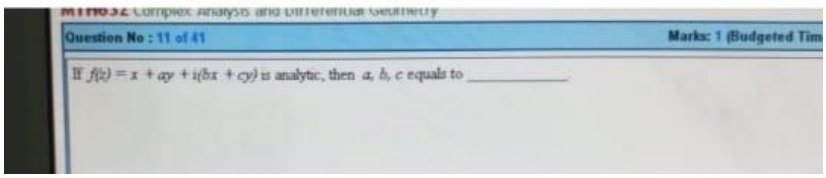
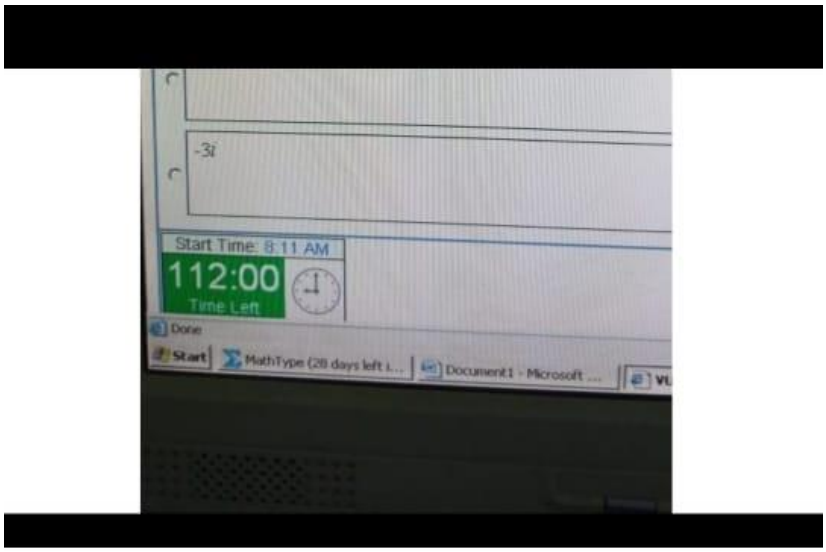

If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____
Answer (Please select your correct option)
harmonic in D.
equal in D.
homogeneous in D.
steady in D.


Question No : 11 of 41
Marks: 1 (Budgeted Tim
If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____.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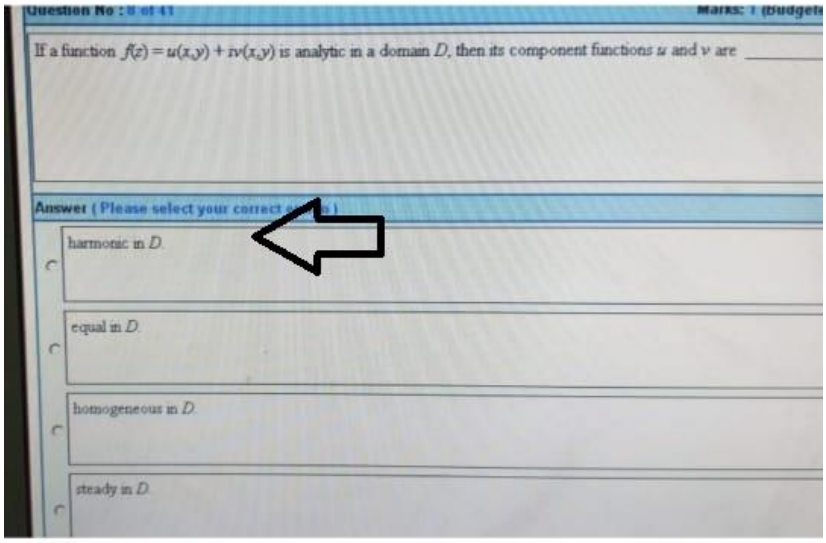
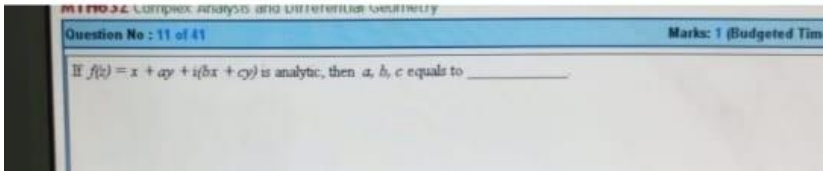
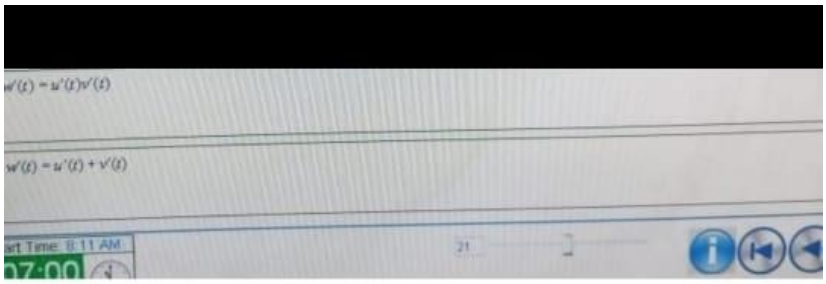
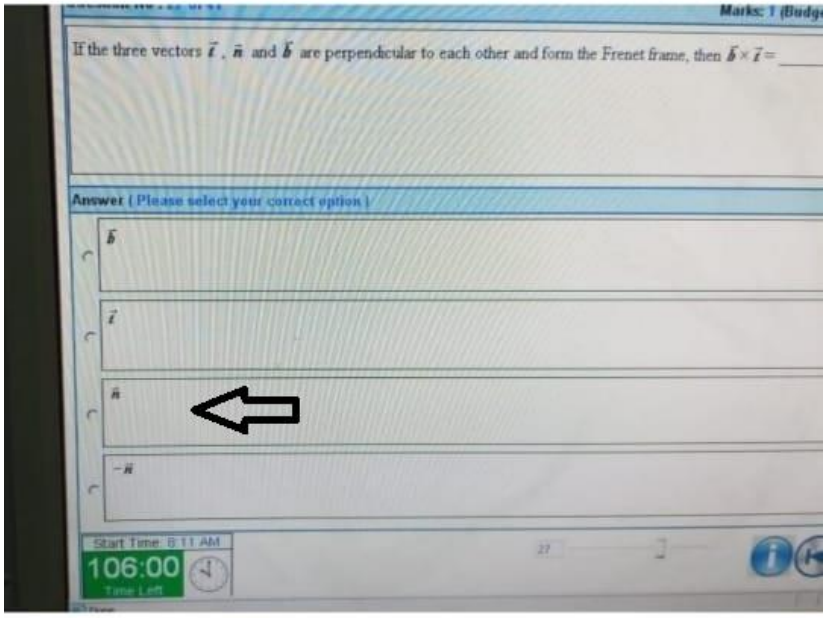

If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____
Answer (Please select your correct
harmonic in D.
equal in D.
homogeneous in D.
steady in D.


Question No : 11 of 41
If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____.


$w'(t) = u'(t)v'(t)$
$w'(t) = u'(t) + v'(t)$
Start Time: 8:11 AM
07:00
21


Marks: 1 (Budg
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ ____
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 8:11 AM
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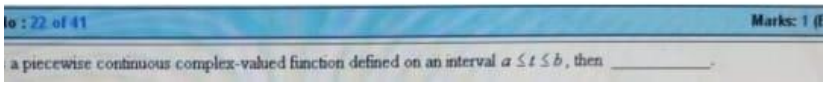
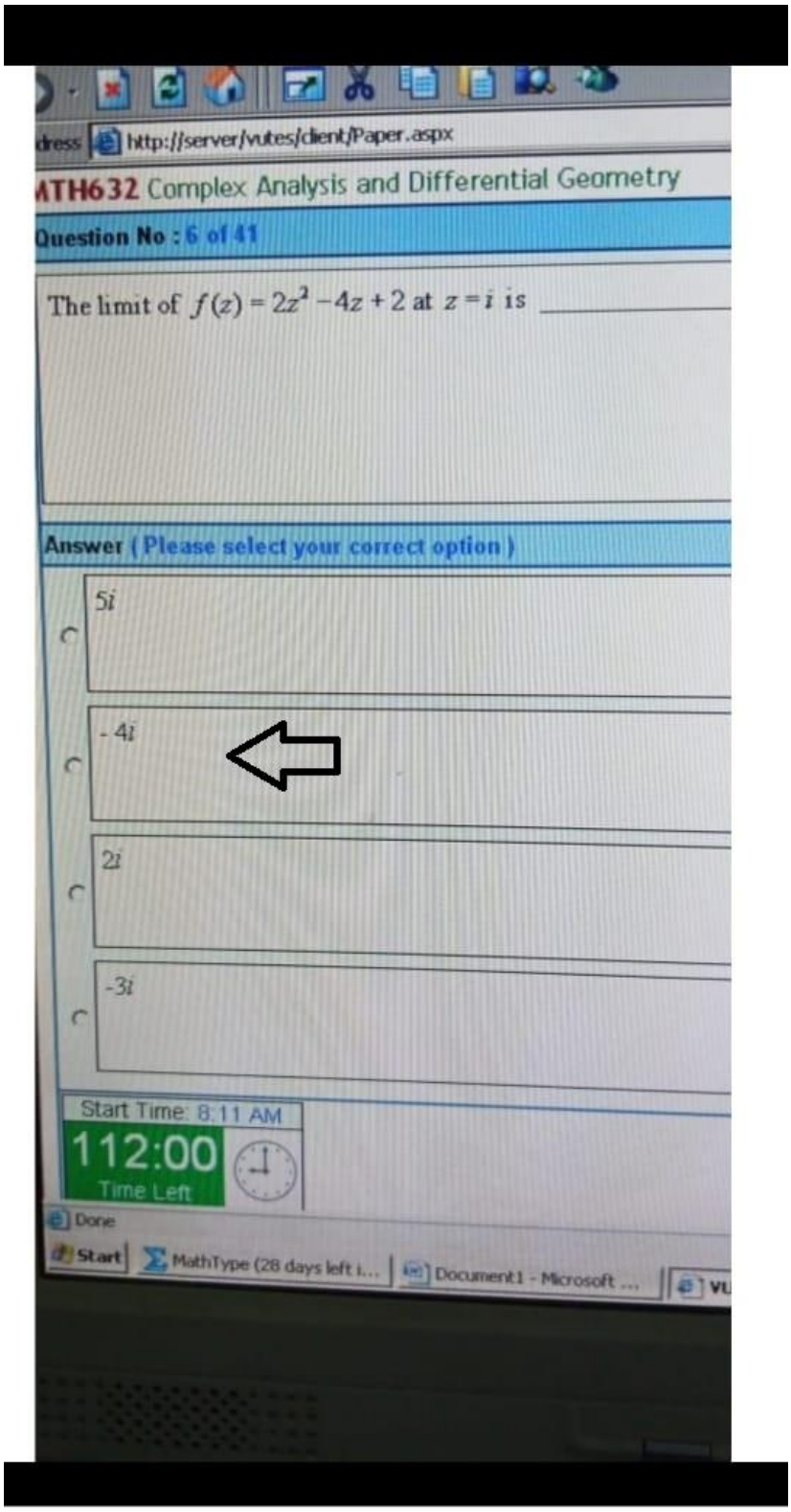
Time Left
27


lo : 22 of 41
Marks: 1 (
a piecewise continuous complex-valued function defined on an interval $a \le t \le b$, then _____.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ATH632 Complex Analysis and Differential Geometry
Question No : 6 of 41
The limit of $f(z) = 2z^2 - 4z + 2$ at $z = i$ is _____
Answer (Please select your correct option)
5i
- 4i
2i
-3i
Start Time: 8:11 AM
112:00
Time Left
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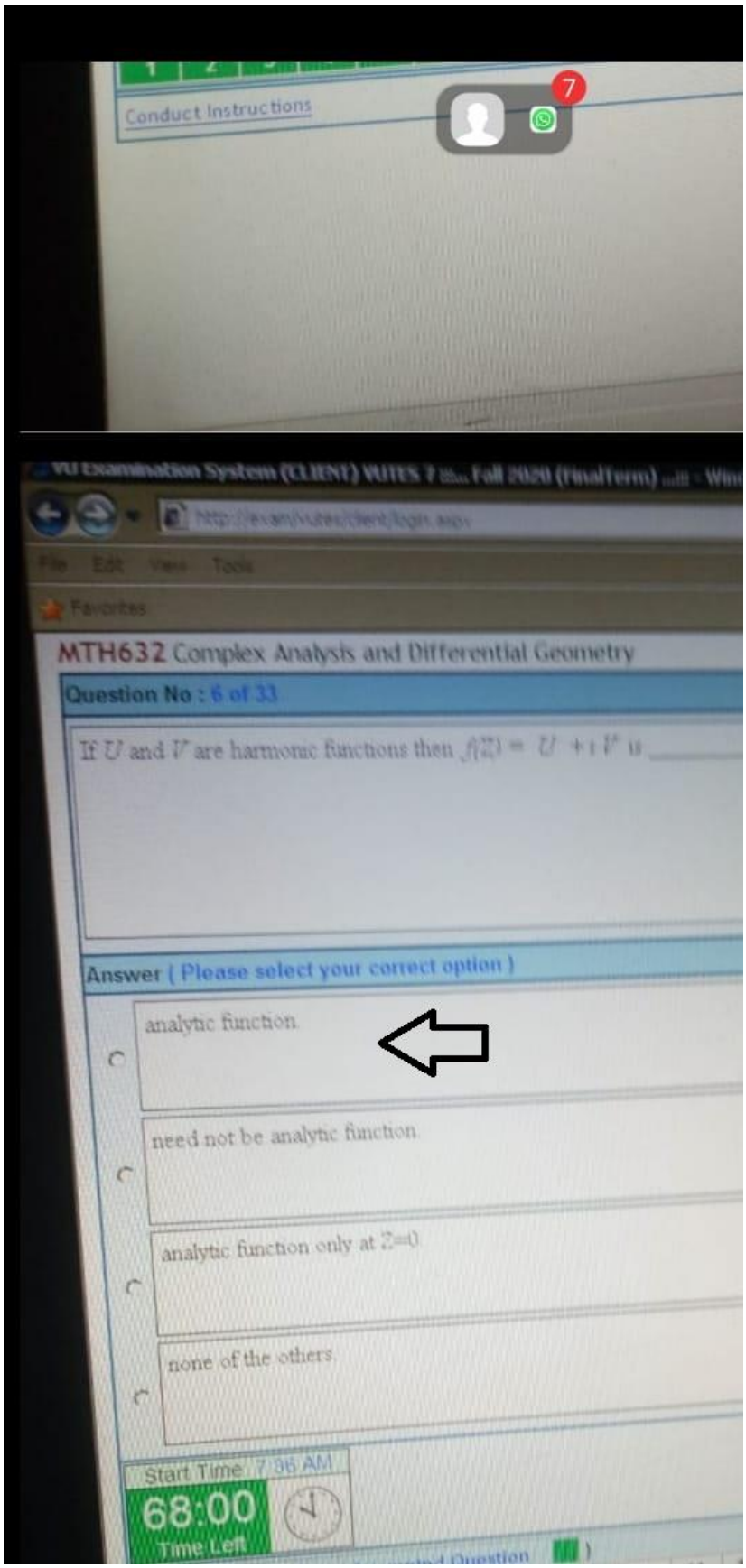
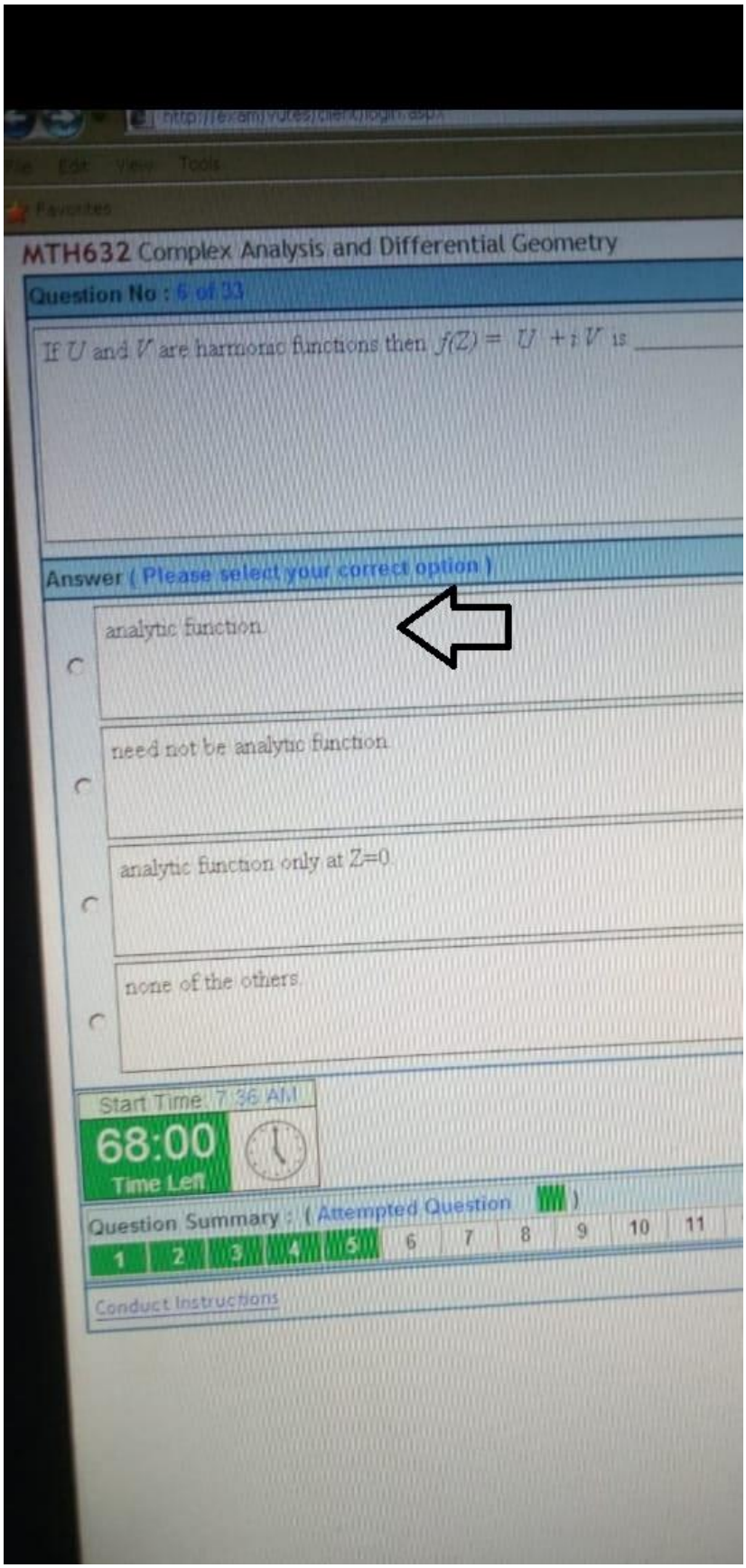
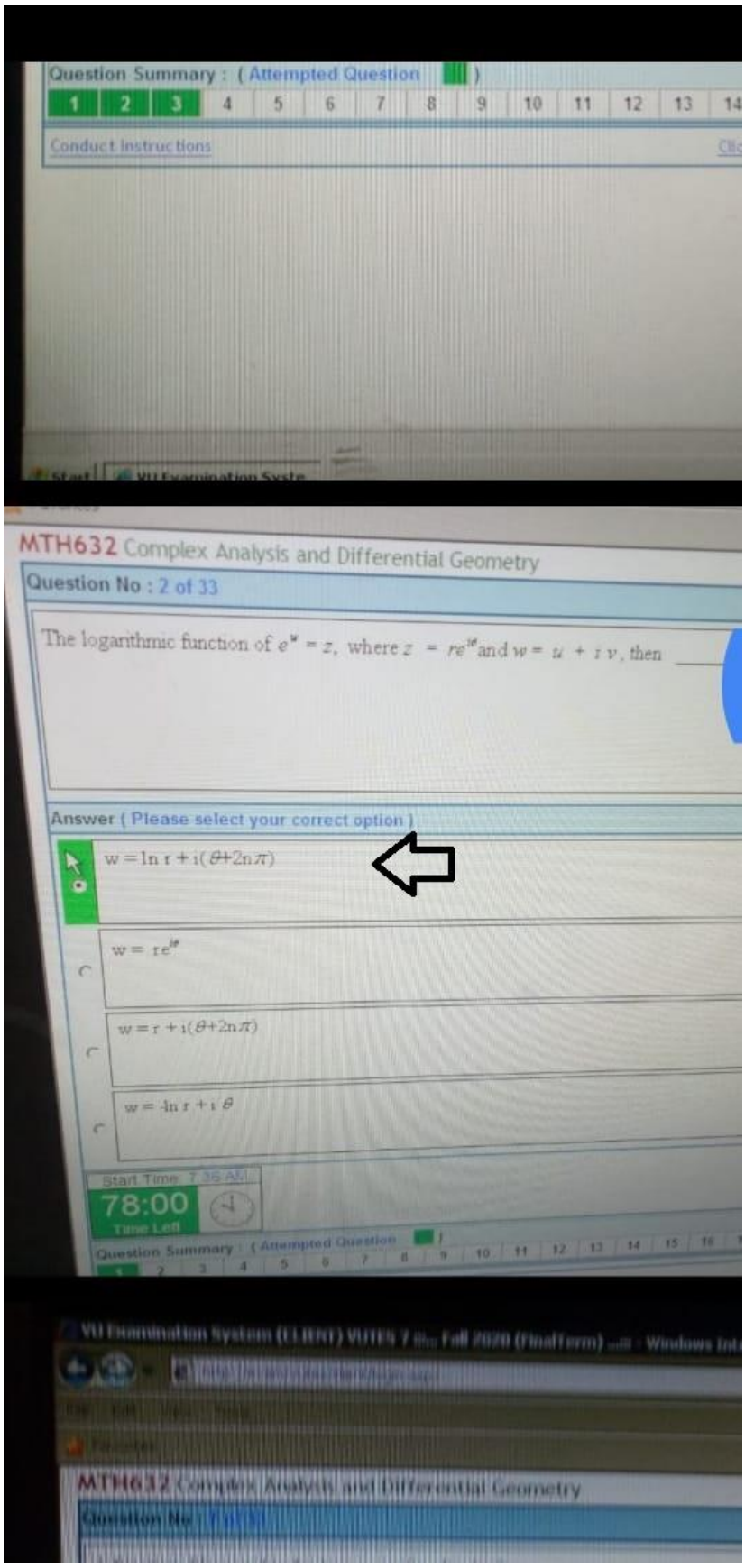
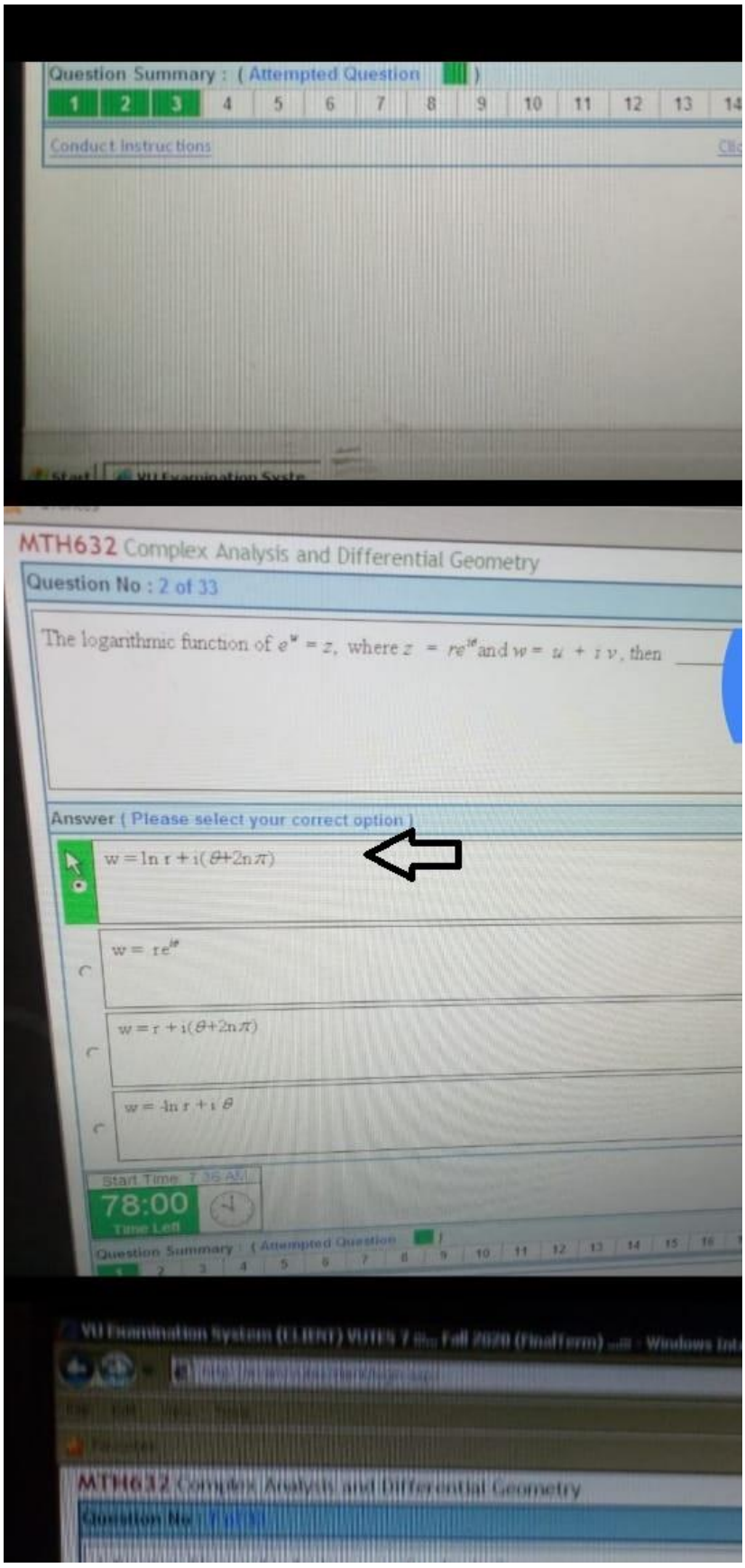
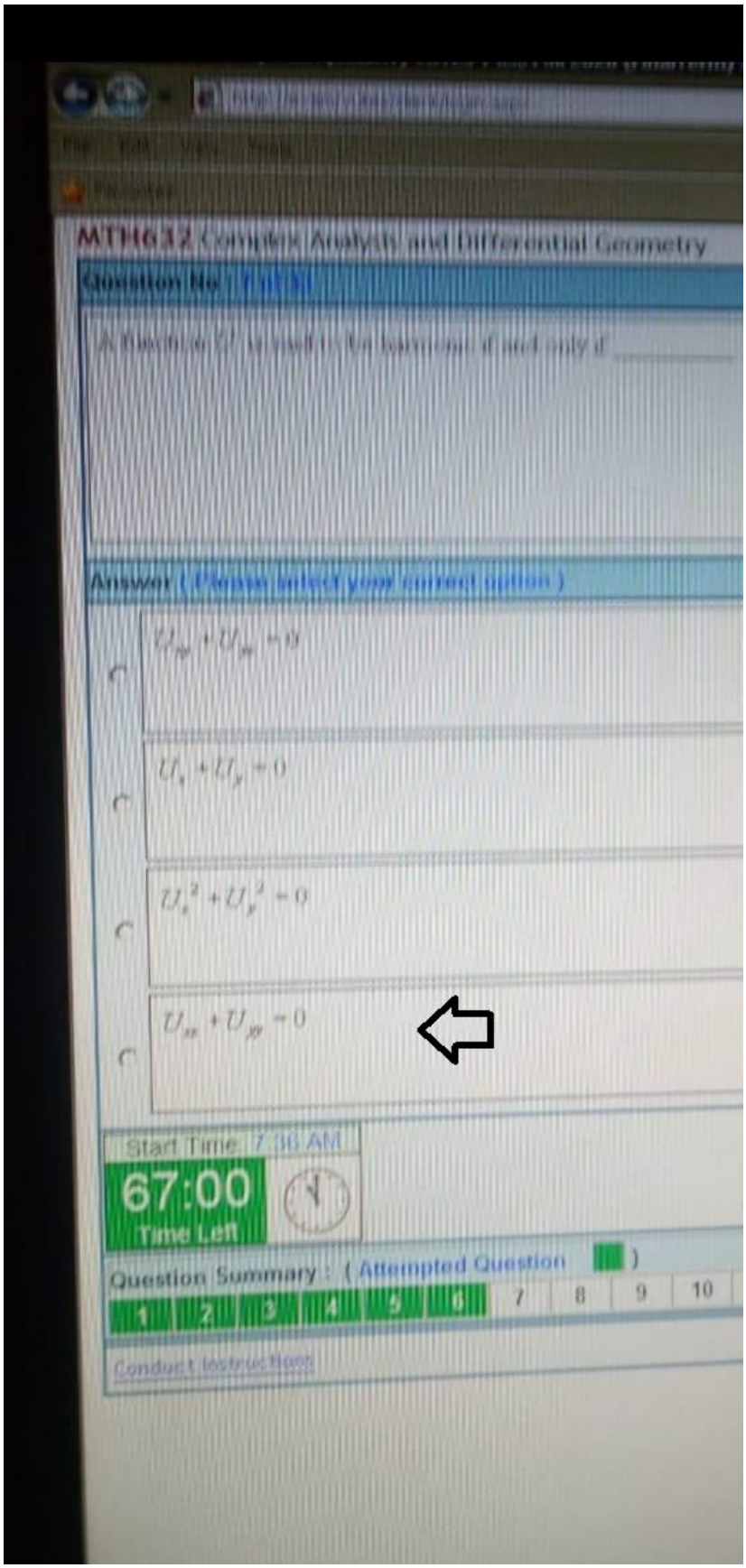
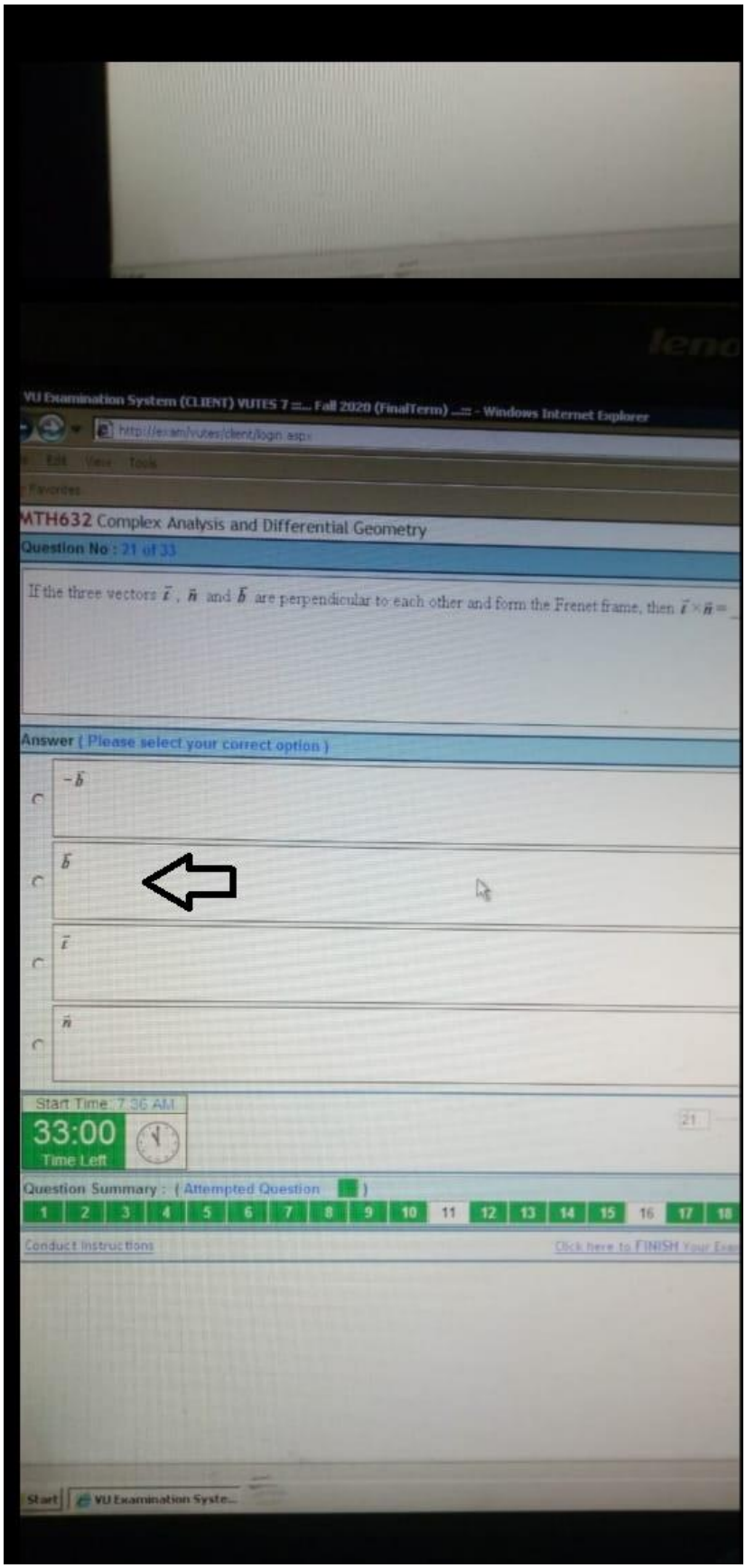

Conduct Instructions
7
VU Examination System (CLIENT) VUTES ? ... Fall 2020 (FinalTerm)
File Edit View Tools
Favorites
MTH632 Complex Analysis and Differential Geometry
Question No : 6 of 33
If U and V are harmonic functions then f(z) = U + i V is
Answer (Please select your correct option)
analytic function
need not be analytic function
analytic function only at z=0
none of the others
Start Time: 7:36 AM
68:00
Time Left

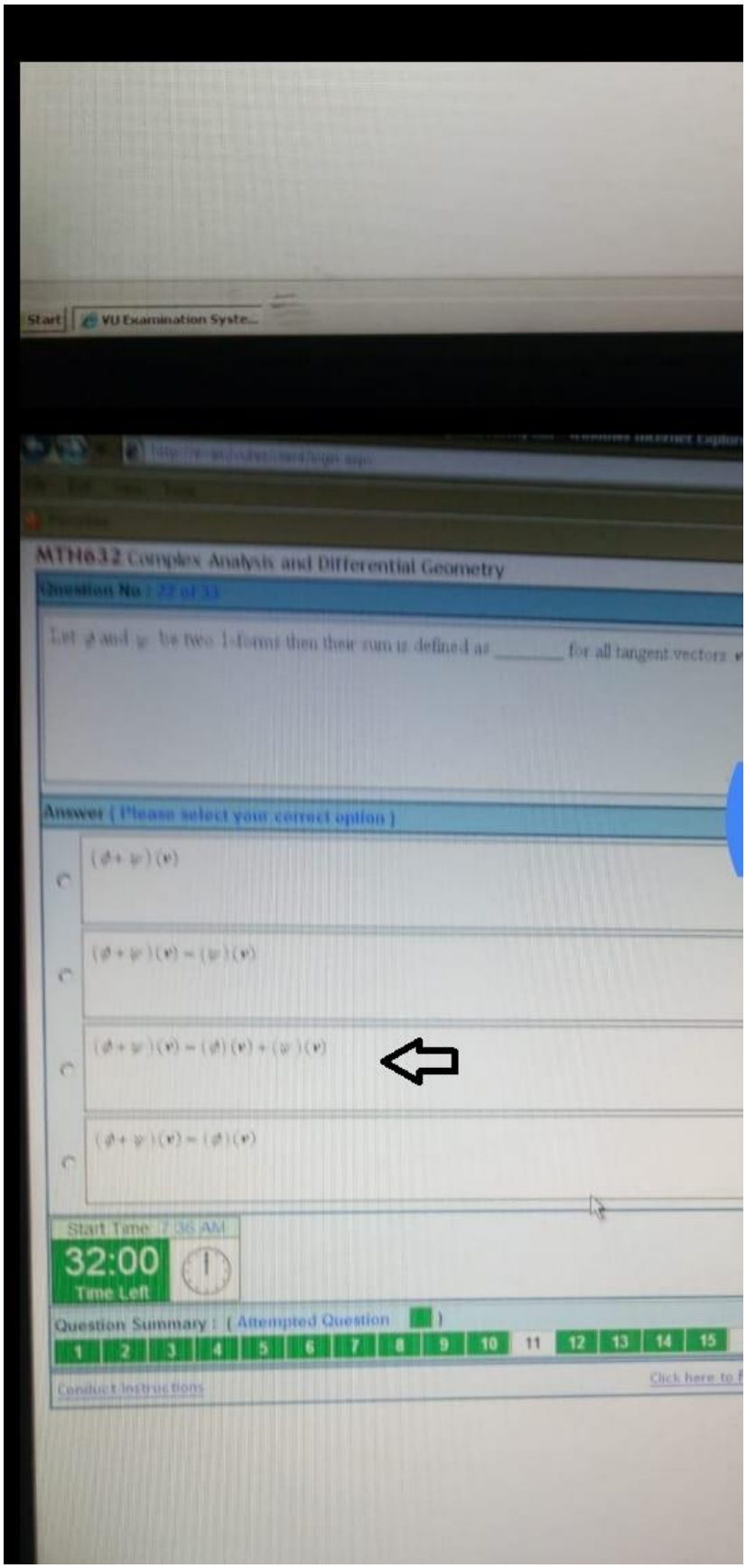

MTH632 Complex Analysis and Differential Geometry
Question No : 6 of 33
If U and V are harmonic functions then $f(Z) = U + iV$ is _____
Answer (Please select your correct option)
analytic function
need not be analytic function
analytic function only at Z=0
none of the others
Start Time: 7:36 AM
68:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions

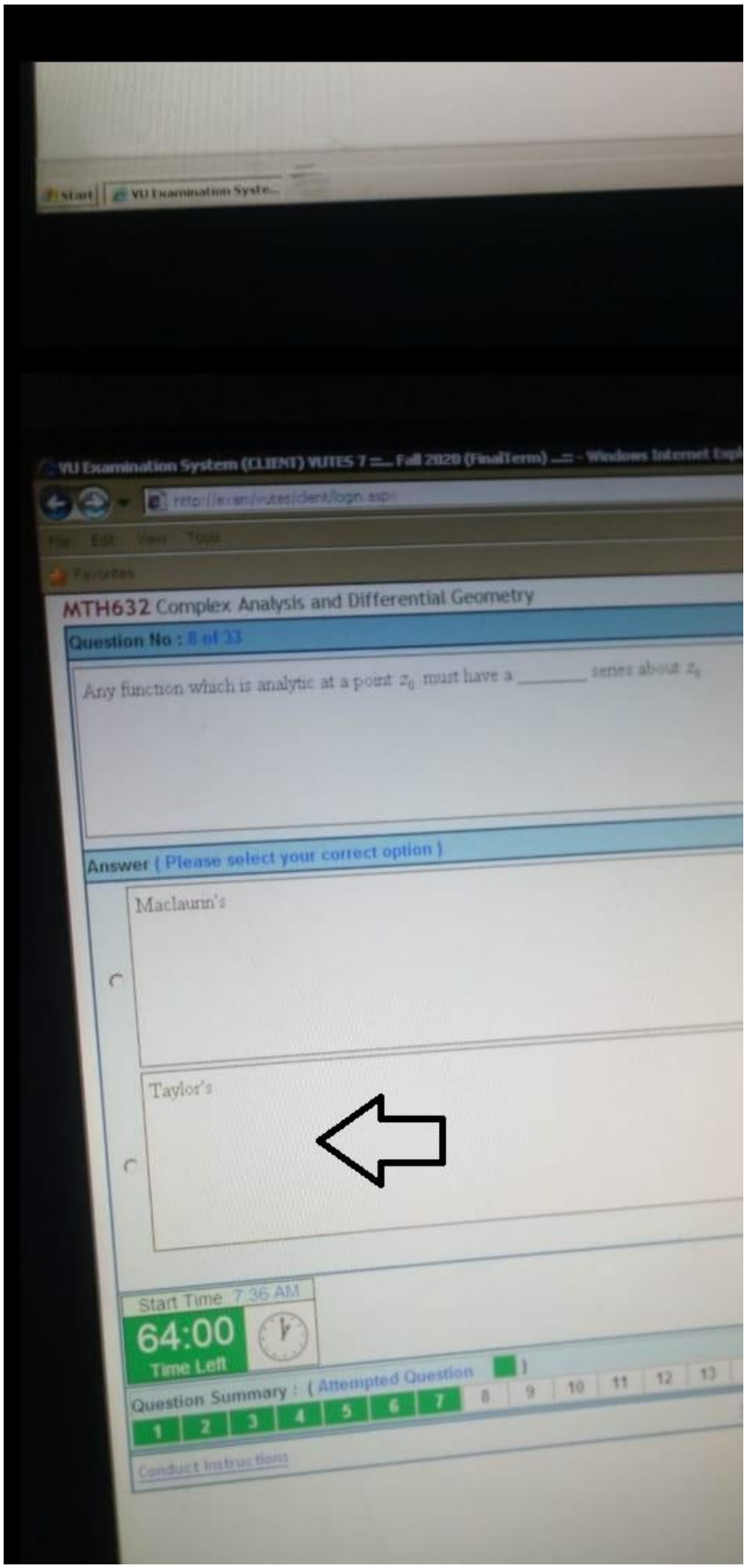

Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{i\theta}$
$w = r + i(\theta + 2n\pi)$
$w = \ln r + i\theta$
Start Time: 7:36 AM
78:00
Time Left
Question Summary (Attempted Question)
MTH632 Complex Analysis and Differential Geometry

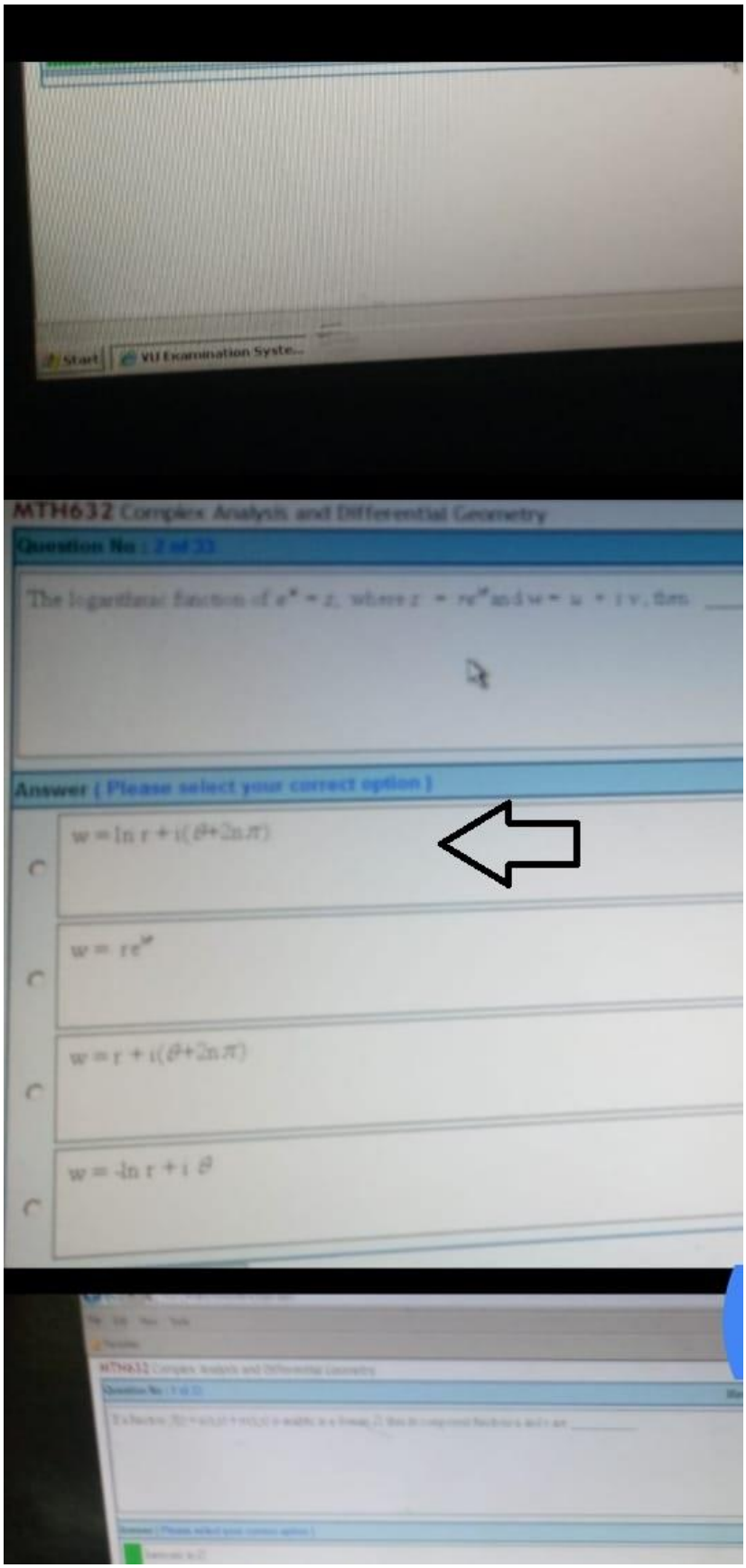

Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{i\theta}$
$w = r + i(\theta + 2n\pi)$
$w = \ln r + i\theta$
Start Time: 7:36 AM
78:00
Time Left
Question Summary (Attempted Question)
MTH632 Complex Analysis and Differential Geometry

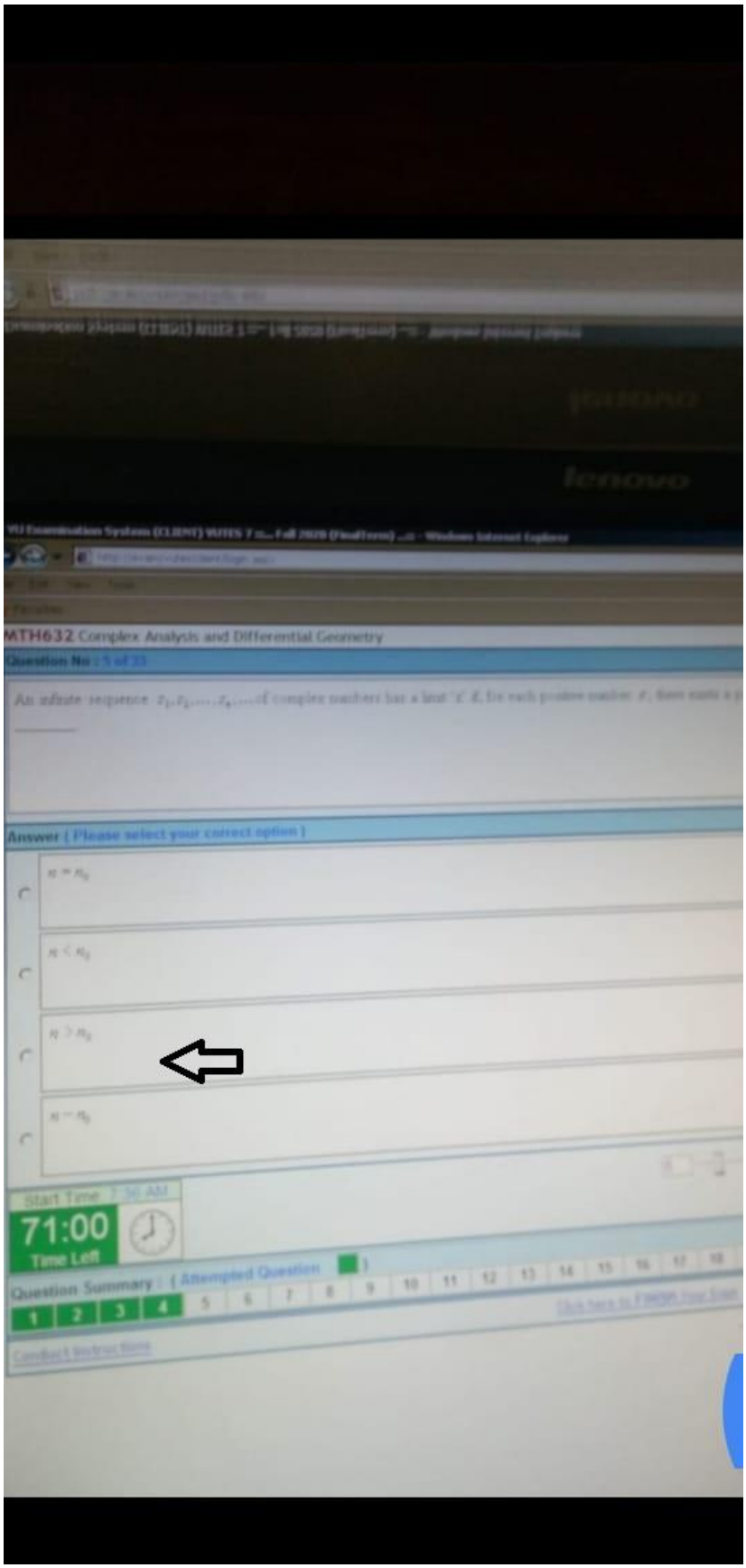

MTH632 Complex Analysis and Differential Geometry
Question No
Answer (Please select your correct option)
$U_{xx} + U_{yy} = 0$
$U_x + U_y = 0$
$U_x^2 + U_y^2 = 0$
$U_{xx} + U_{yy} = 0$
Start Time: 7:36 AM
67:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10
Conduct Instructions

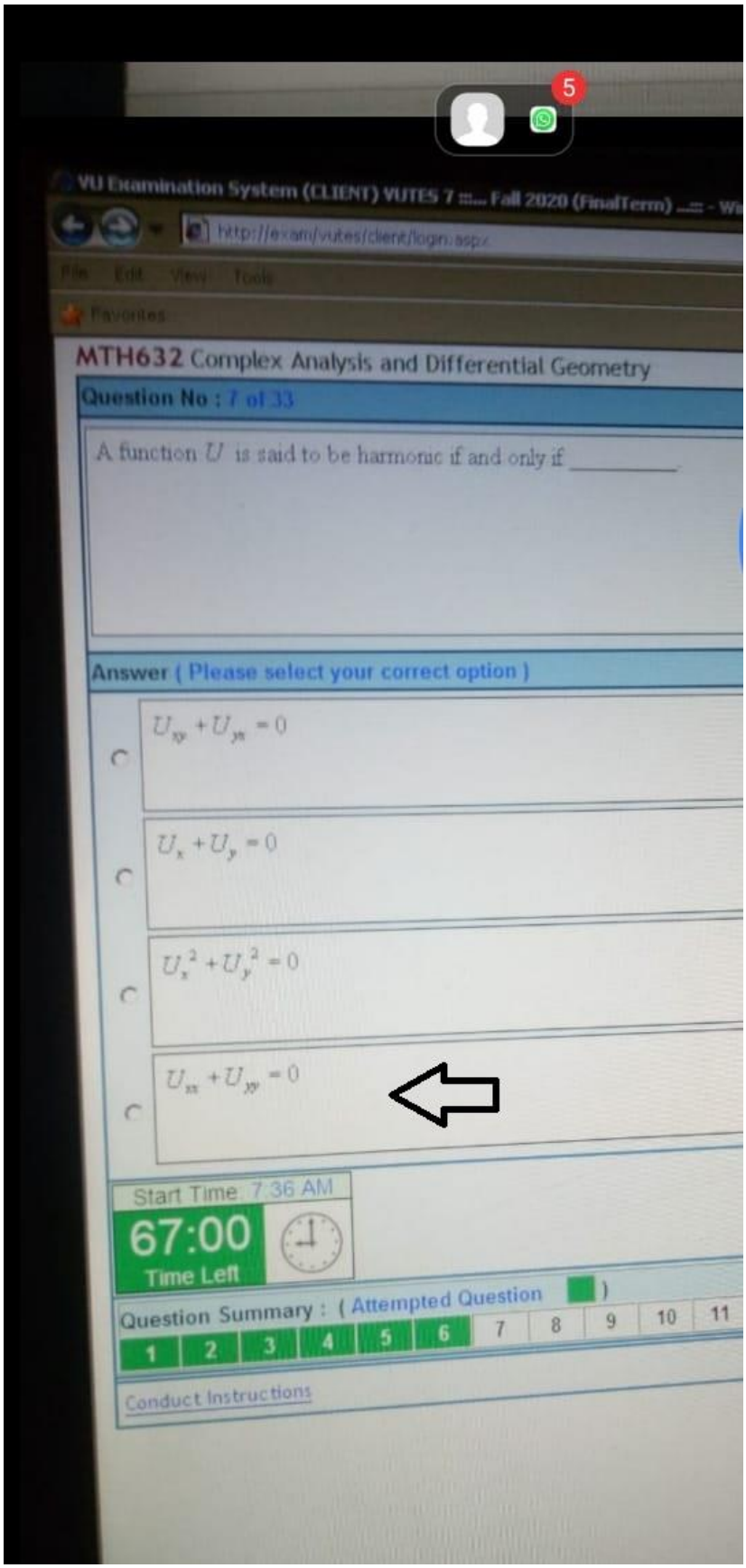

VU Examination System (CLIENT) VUTES 7 ... Fall 2020 (FinalTerm) ... - Windows Internet Explorer
MTH632 Complex Analysis and Differential Geometry
Question No : 21 of 33
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{t} \times \vec{n} =$
Answer (Please select your correct option)
$-\vec{b}$
$\vec{b}$
$\vec{t}$
$\vec{n}$
Start Time: 7:36 AM
33:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Conduct Instructions
Click here to FINISH Your Exam
Start
VU Examination Syste...

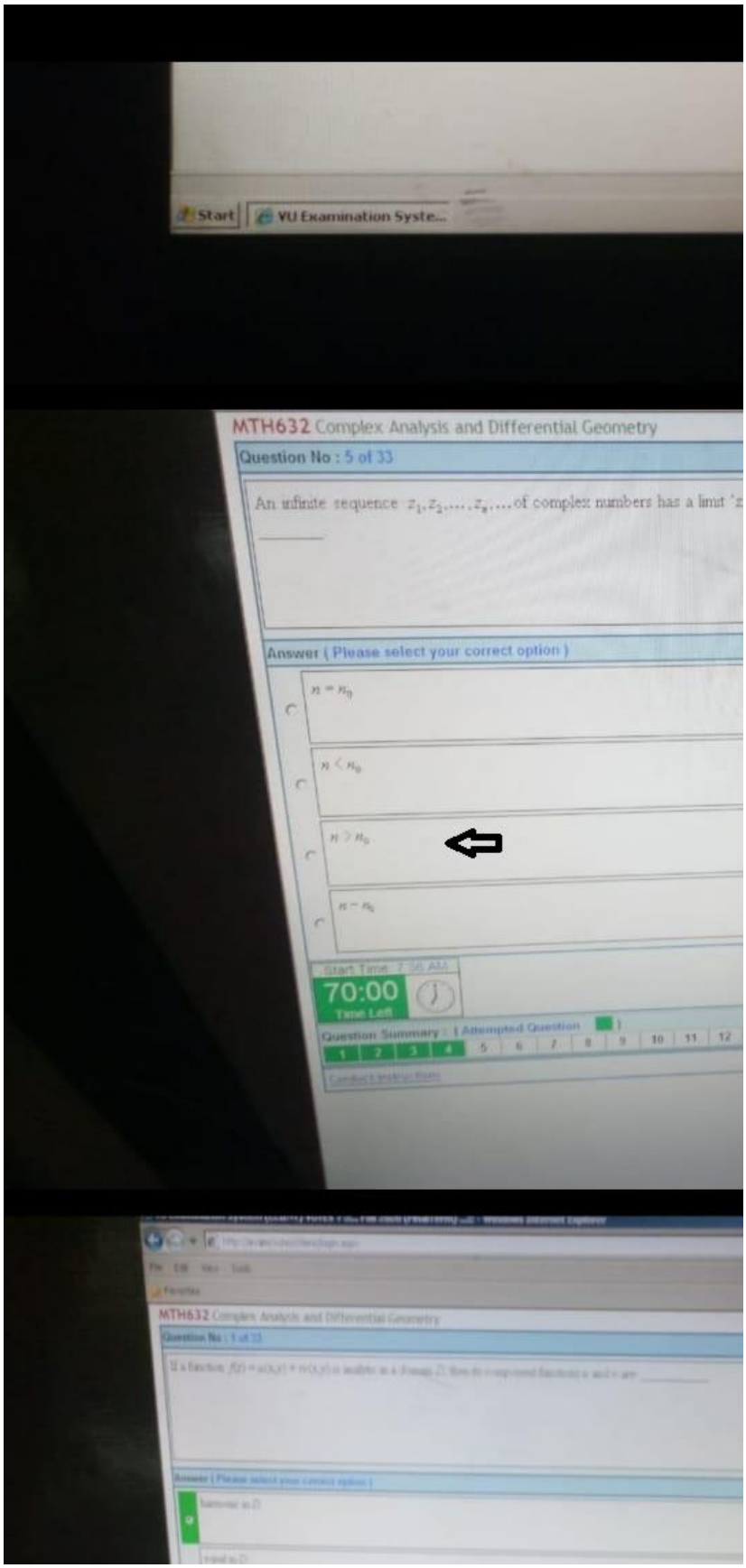

Start
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No : 22 of 33
Answer (Please select your correct option)
Start Time : 7:36 AM
32:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions

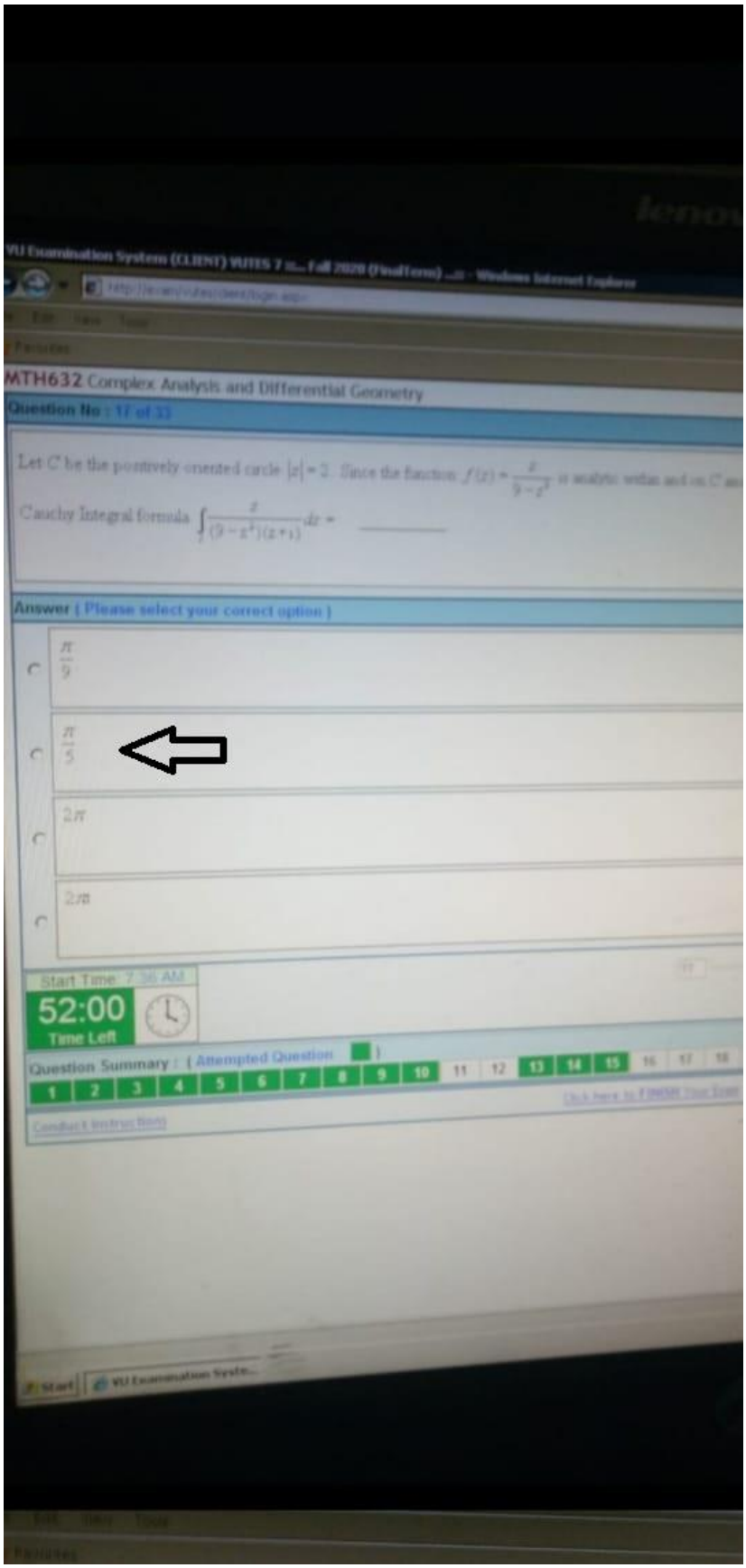

VU Examination System (CLIENT) VUTES 7 — Fall 2020 (FinalTerm)
MTH632 Complex Analysis and Differential Geometry
Question No : 8 of 33
Any function which is analytic at a point z_0 must have a _____ series about z_0
Answer (Please select your correct option)
Maclaurin's
Taylor's
Start Time 7:36 AM
64:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

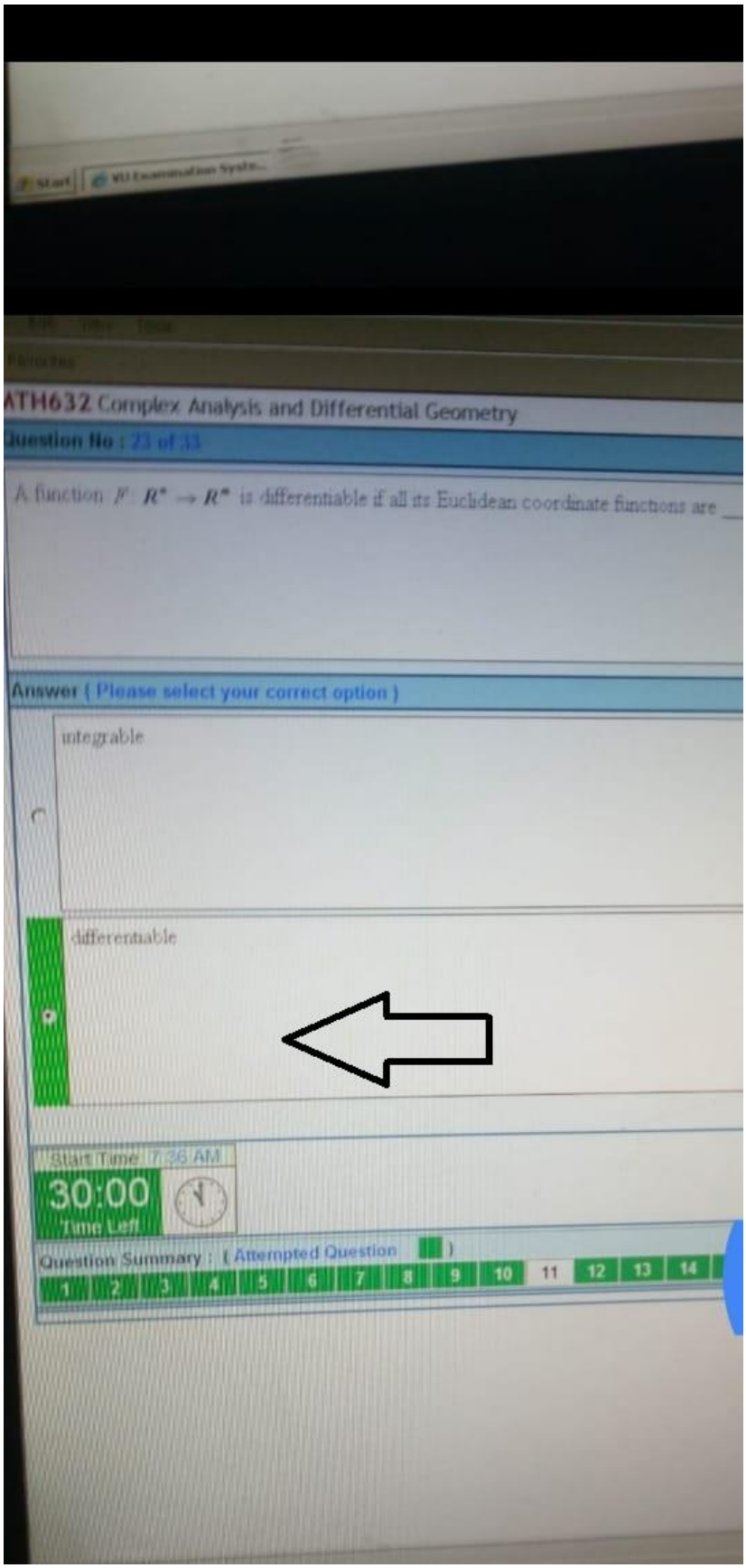

Start
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No :
Answer (Please select your correct option)
w = ln r + i(θ+2nπ)
w = re^θ
w = r + i(θ+2nπ)
w = -ln r + i θ

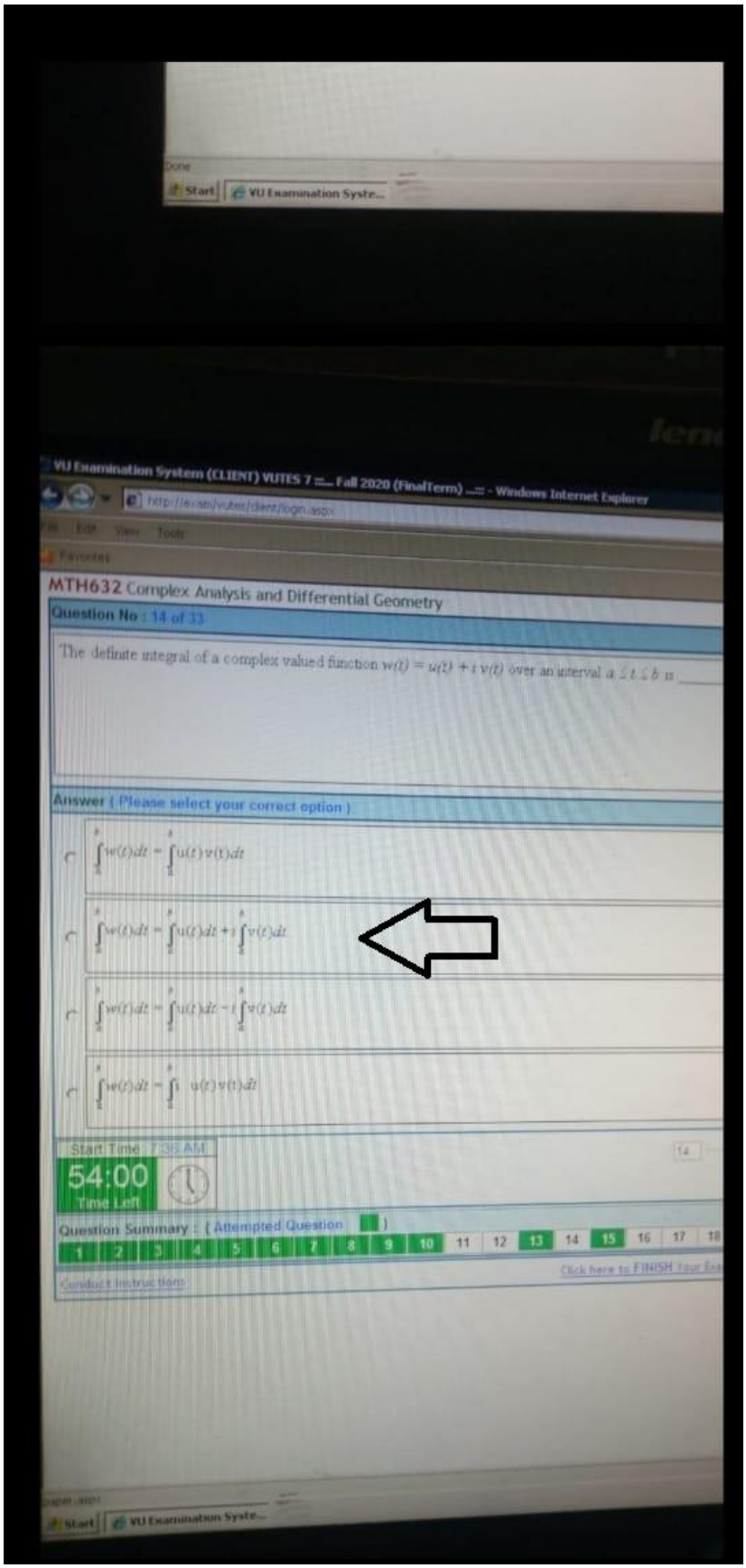

MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
Answer [Please select your correct option]
71:00
Time Left
Question Summary : [Attempted Question]


VU Examination System (CLIENT) VUTES 7 ::... Fall 2020 (FinalTerm) ...::
MTH632 Complex Analysis and Differential Geometry
Question No : 7 of 33
A function U is said to be harmonic if and only if _____.
Answer (Please select your correct option)
$U_{xy} + U_{yx} = 0$
$U_x + U_y = 0$
$U_x^2 + U_y^2 = 0$
$U_{xx} + U_{yy} = 0$
Start Time 7:36 AM
67:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions


Start
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
An infinte sequence $z_1, z_2, \ldots, z_n, \ldots$ of complex numbers has a limit
Answer (Please select your correct option)
$n = n_0$
$n < n_0$
$n > n_0$
$n = n_0$
70:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12


MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
Let C be the positively oriented circle |z| = 2. Since the function
Cauchy Integral formula
Answer (Please select your correct option)
Start Time: 7:36 AM
52:00
Time Left
Question Summary : (Attempted Question)


ATH632 Complex Analysis and Differential Geometry
Question No : 23 of 33
A function $F: R^n \to R^m$ is differentiable if all its Euclidean coordinate functions are
Answer (Please select your correct option)
integrable
differentiable
Start Time: 7:36 AM
30:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14


Start
VU Examination Syste...
VU Examination System (CLIENT) VUTES 7 ... Fall 2020 (FinalTerm) ... - Windows Internet Explorer
MTH632 Complex Analysis and Differential Geometry
Question No : 14 of 33
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is
Answer (Please select your correct option)
$\int_a^b w(t)\,dt = \int_a^b u(t)\,v(t)\,dt$
$\int_a^b w(t)\,dt = \int_a^b u(t)\,dt + i\int_a^b v(t)\,dt$
$\int_a^b w(t)\,dt = \int_a^b u(t)\,dt - i\int_a^b v(t)\,dt$
$\int_a^b w(t)\,dt = \int_a^b u(t)\,v(t)\,dt$
Start Time: 7:35 AM
54:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
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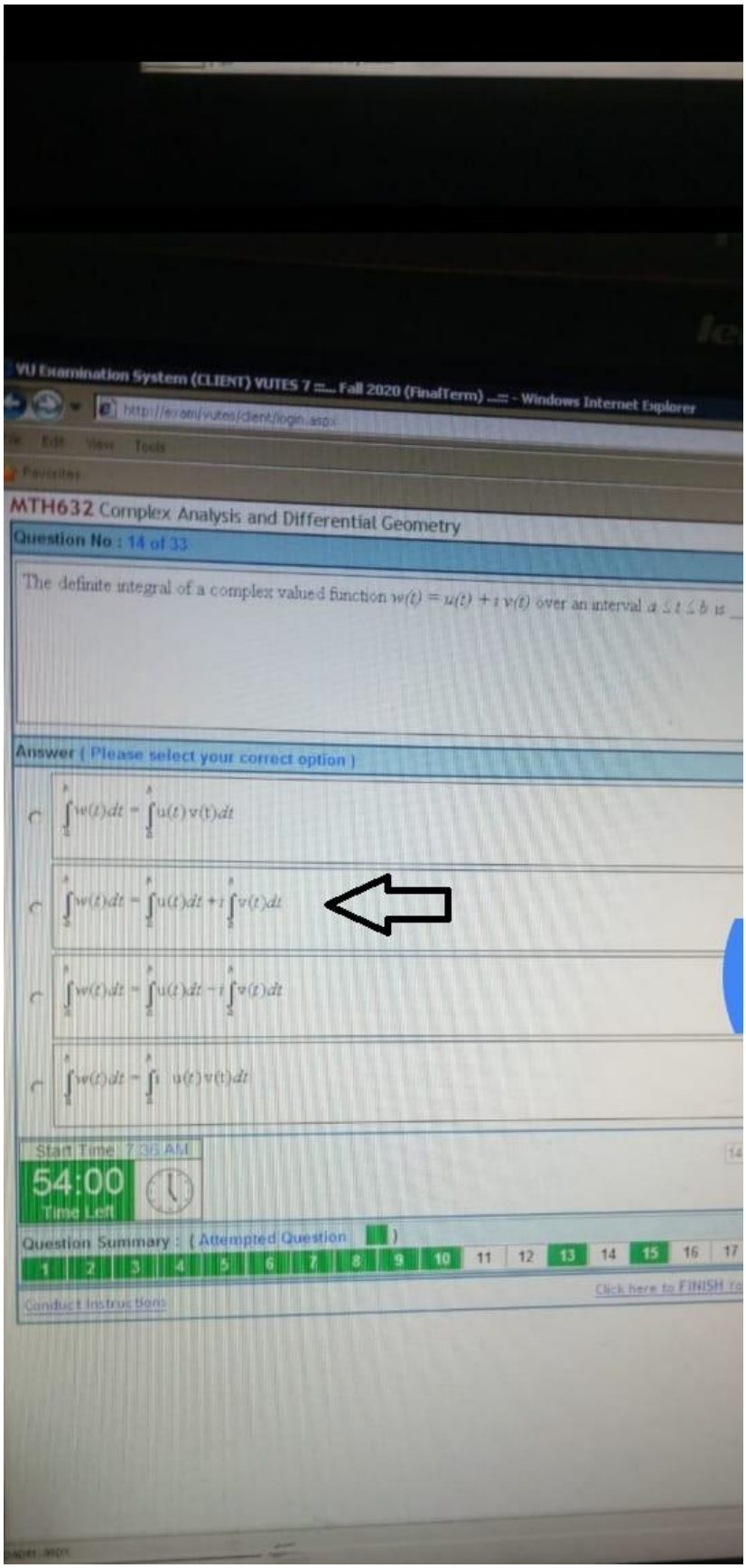
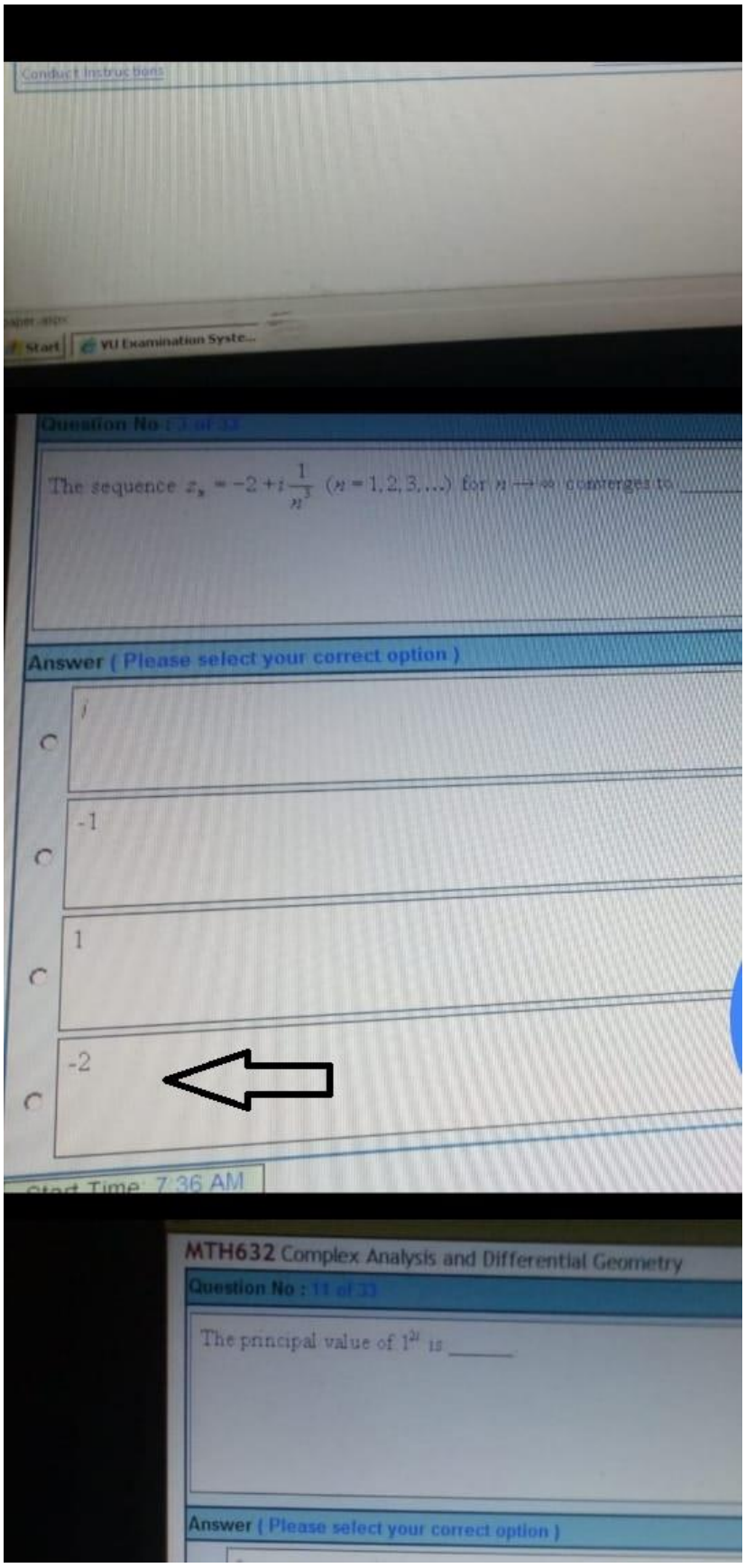
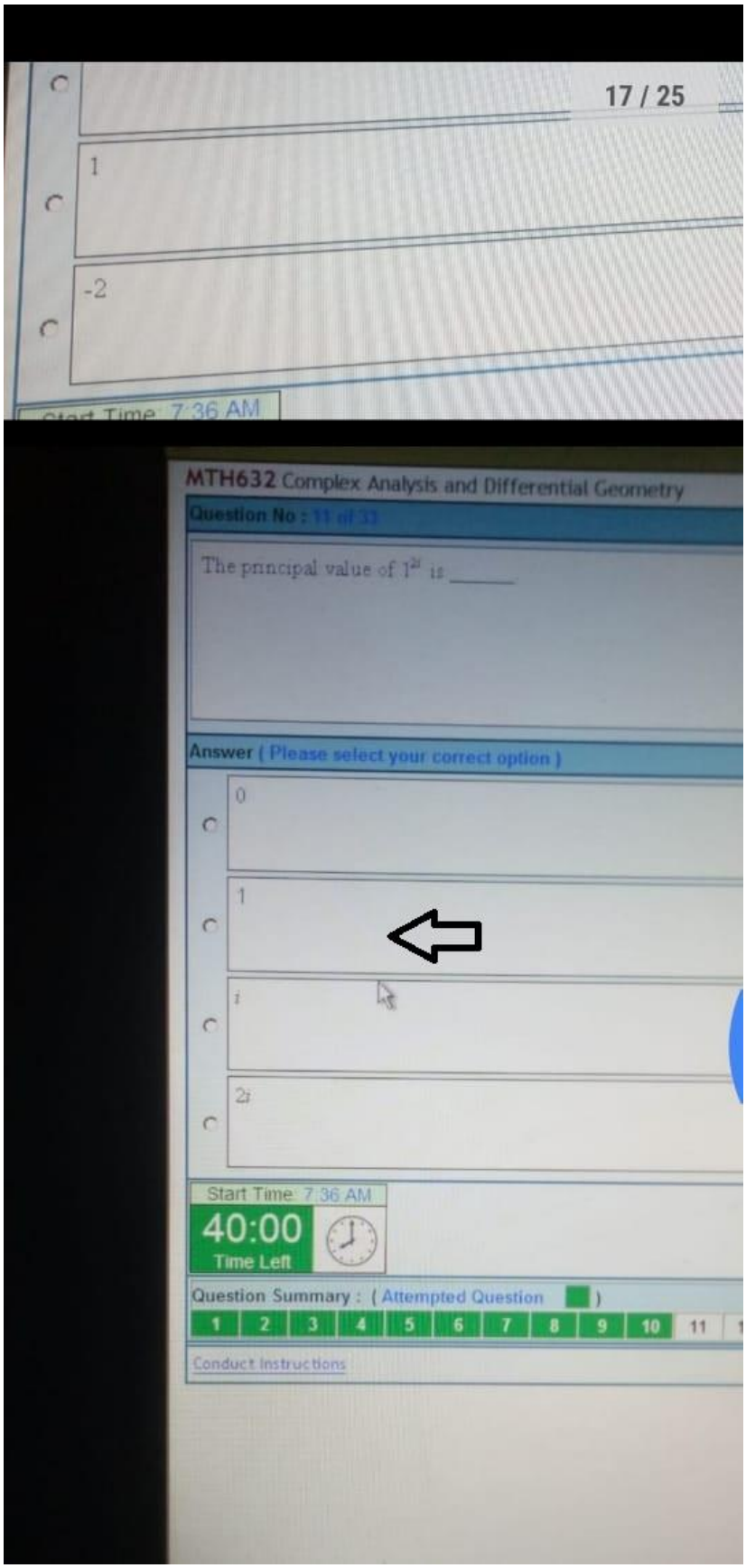
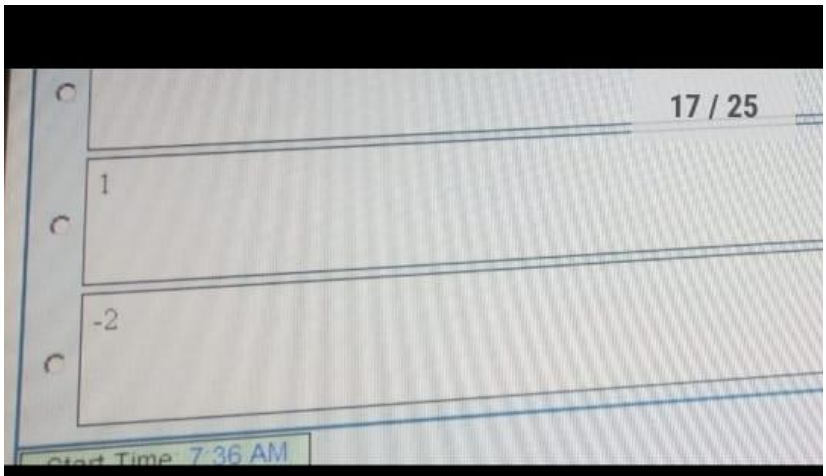
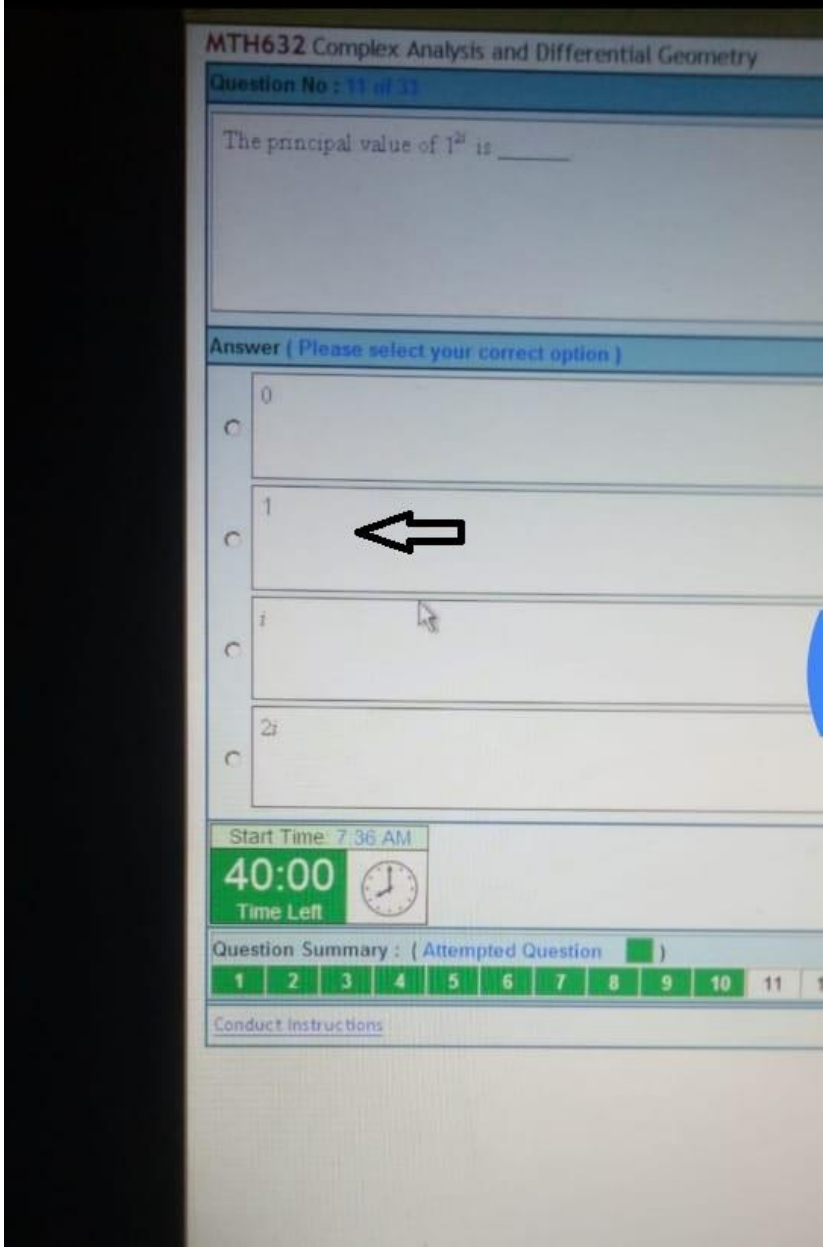
VU Examination Syste...

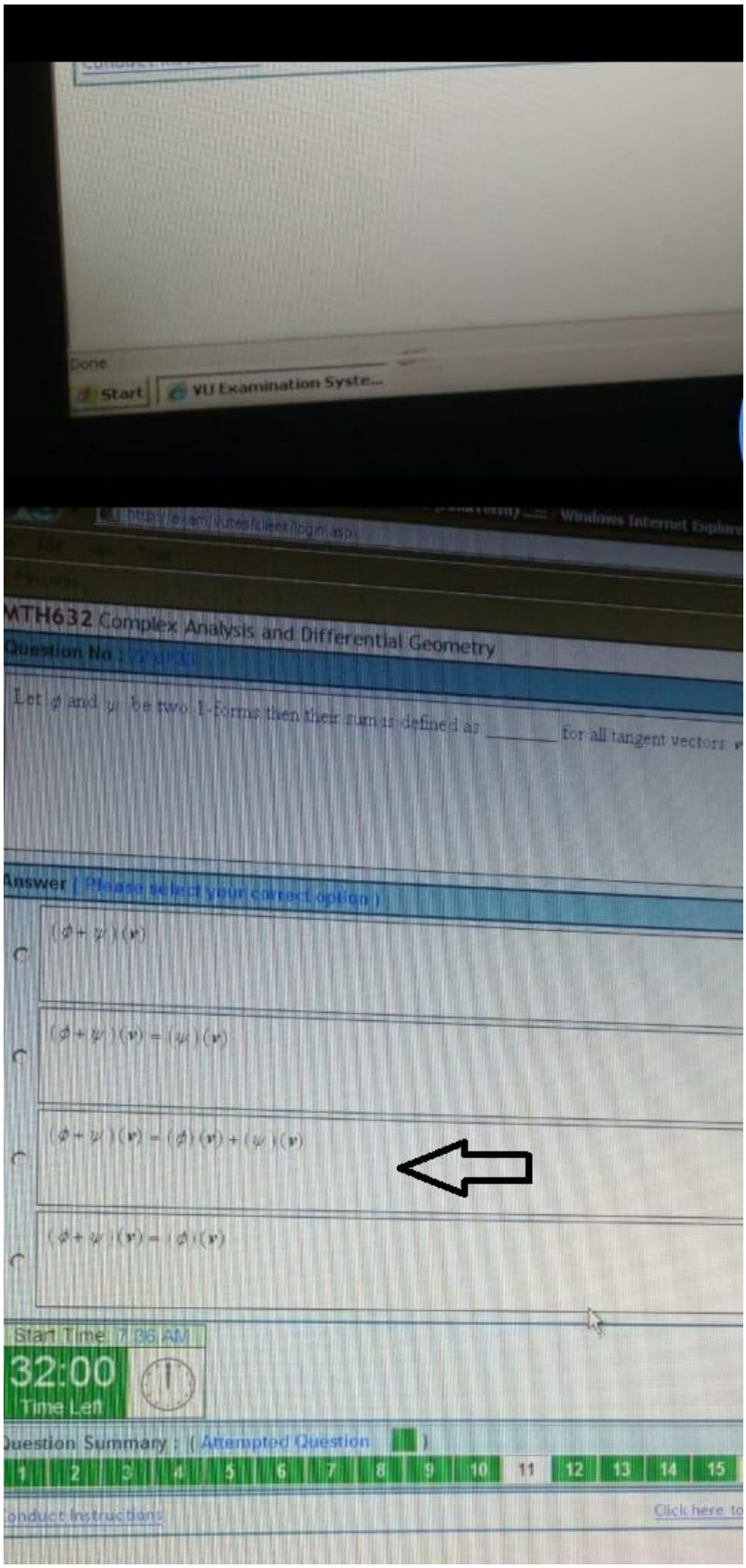

VU Examination System (CLIENT) VUTES 7 ::... Fall 2020 (FinalTerm) ...:: - Windows Internet Explorer
MTH632 Complex Analysis and Differential Geometry
Question No : 14 of 33
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
Start Time 7:35 AM
54:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Conduct Instructions


Conduct Instructions
Start
VU Examination Syste...
Question No : 3 of 33
The sequence $z_n = -2 + i\frac{1}{n^3}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____
Answer (Please select your correct option)
i
-1
1
-2
Start Time: 7:36 AM
MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 33
The principal value of 1^{2i} is _____
Answer (Please select your correct option)


17 / 25
1
-2
Start Time: 7:36 AM
MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 33
The principal value of 1^{2i} is _____
Answer (Please select your correct option)
0
1
i
2i
Start Time: 7:36 AM
40:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions


17 / 25
1
-2
Start Time: 7:36 AM


MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 33
The principal value of 1^{2i} is _____
Answer (Please select your correct option)
0
1
i
2i
Start Time: 7:36 AM
40:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions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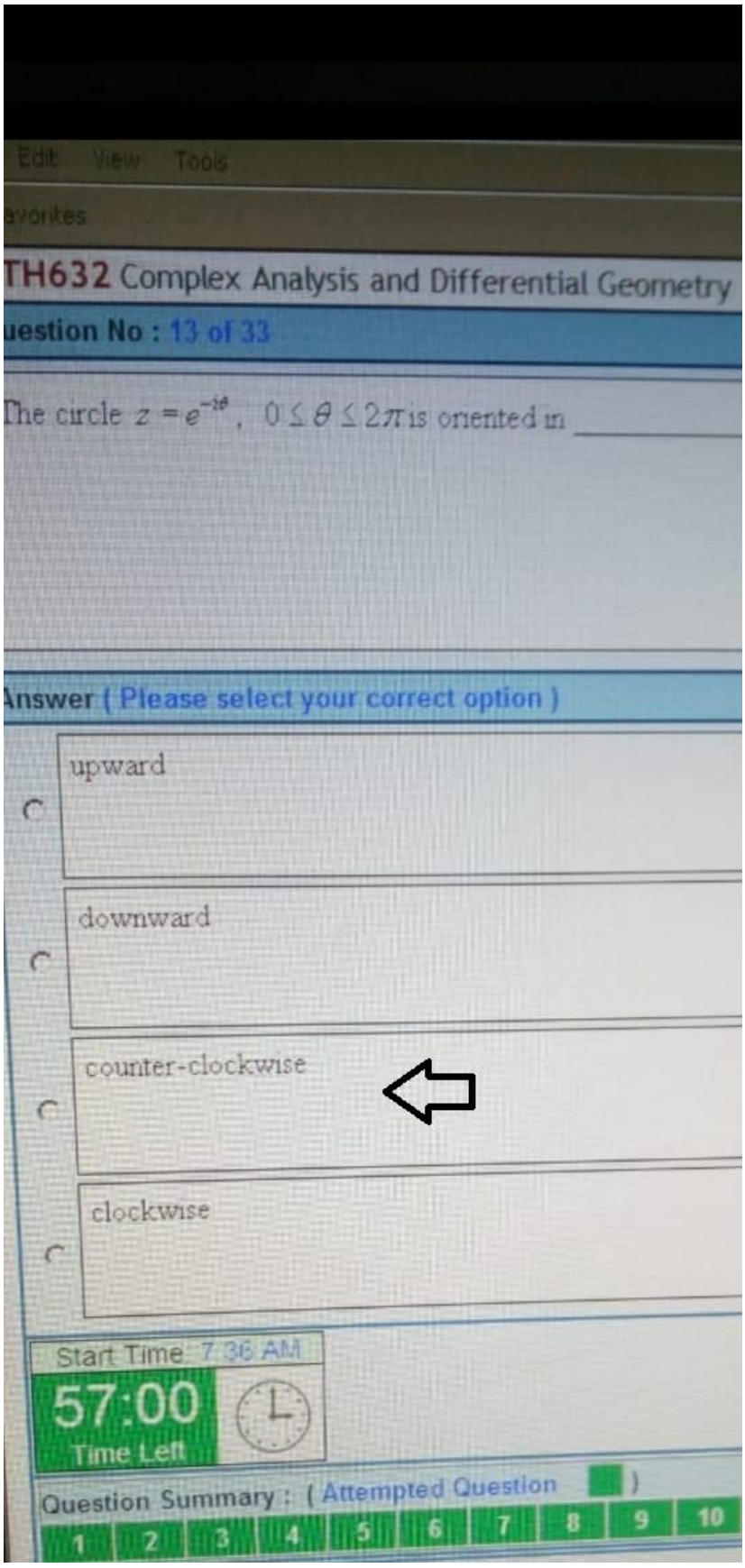
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No :
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 7:06 AM
32:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to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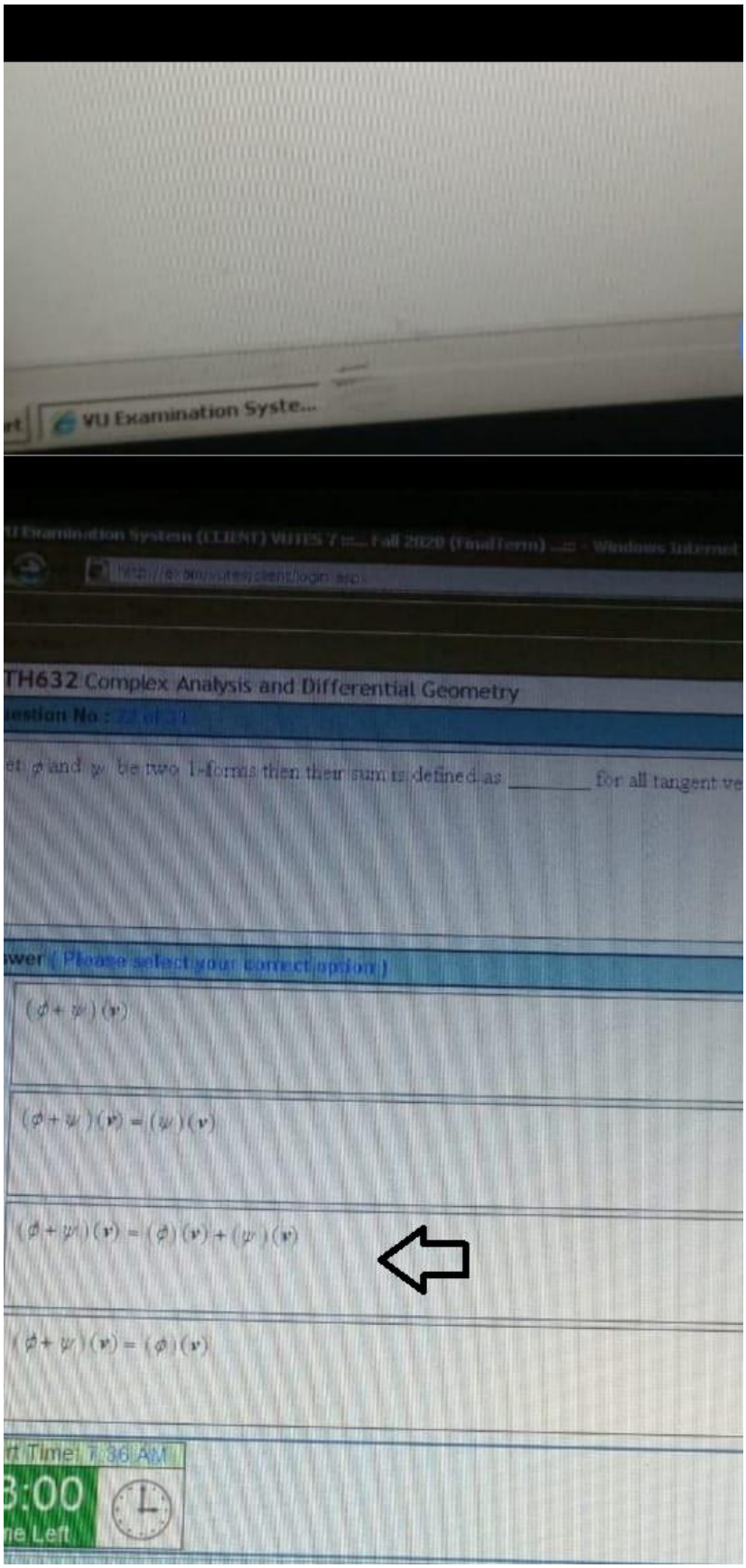
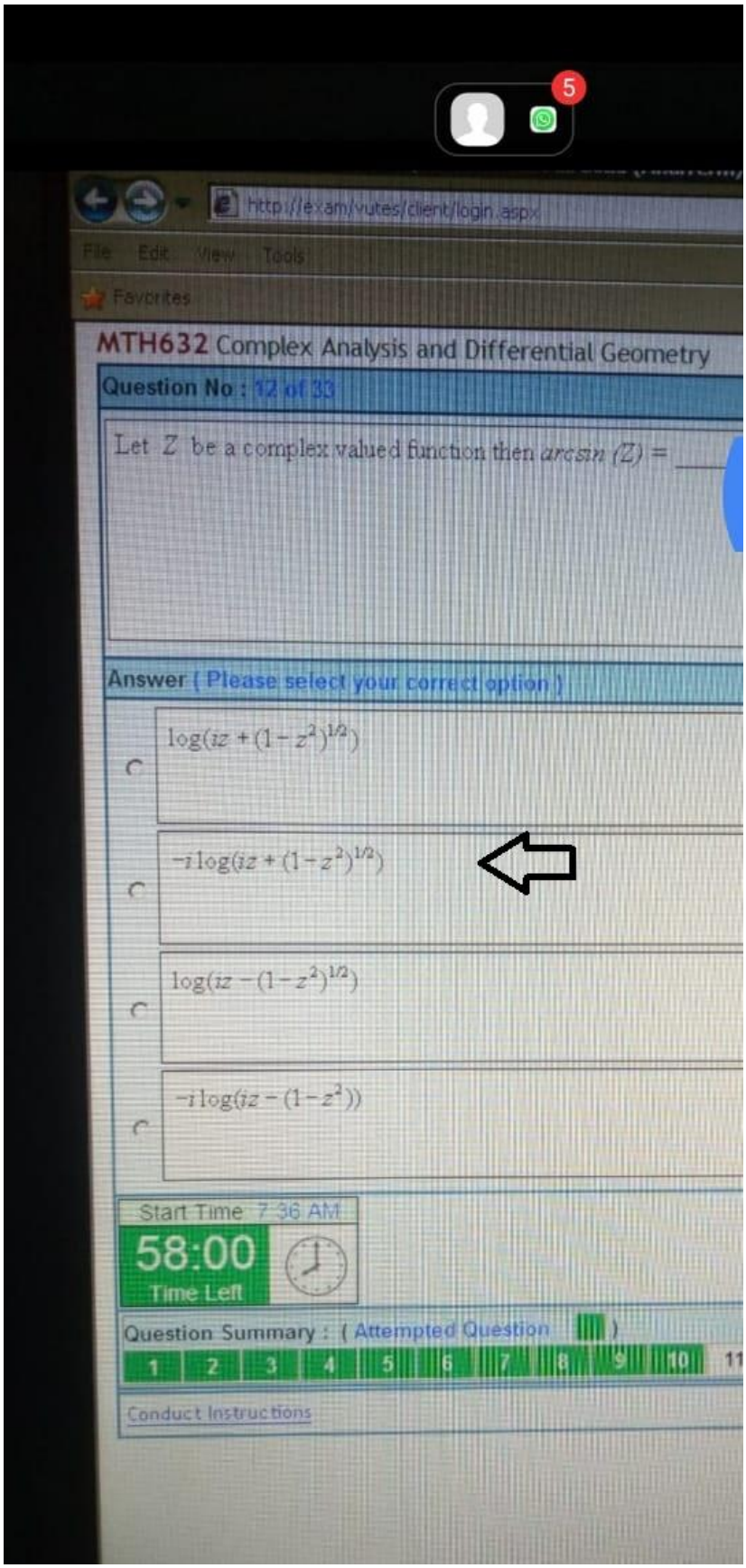
TH632 Complex Analysis and Differential Geometry
uestion No : 13 of 33
The circle $z = e^{-i\theta}$, $0 \le \theta \le 2\pi$ is oriented in _____
Answer (Please select your correct option)
upward
downward
counter-clockwise
clockwise
Start Time: 7:36 AM
57:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10


VU Examination Syste...
TH632 Complex Analysis and Differential Geometry
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$


5
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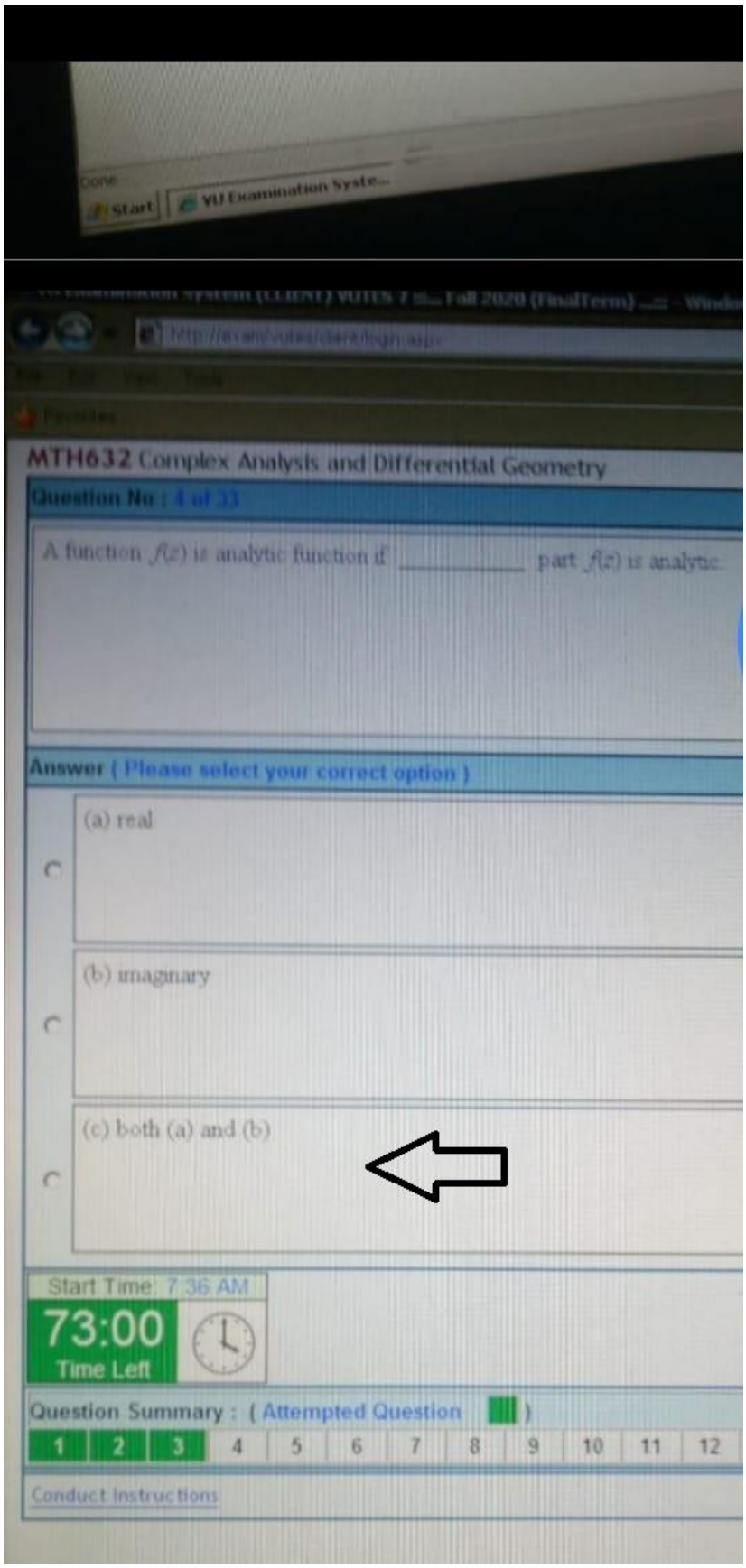
MTH632 Complex Analysis and Differential Geometry
Question No : 12 of 30
Let Z be a complex valued function then arcsin (Z) = ____
Answer (Please select your correct option)
log(iz + (1 − z²)^1/2)
−i log(iz + (1 − z²)^1/2)
log(iz − (1 − z²)^1/2)
−i log(iz − (1 − z²))
Start Time 7:36 AM
58:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions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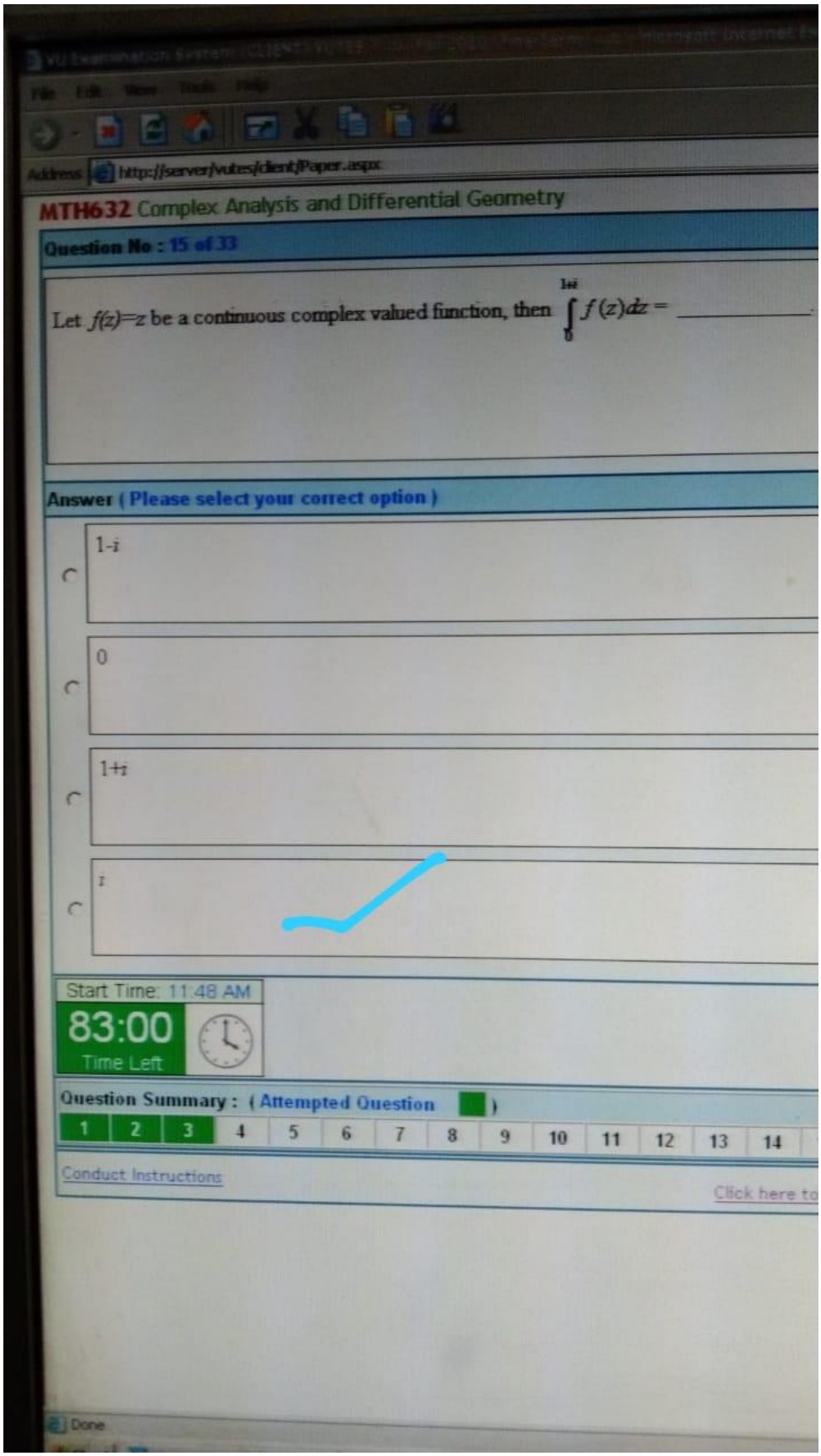
Start
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No : 4 of 33
A function f(z) is analytic function if _____ part f(z) is analytic
Answer (Please select your correct option)
(a) real
(b) imaginary
(c) both (a) and (b)
Start Time: 7:36 AM
73:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let $f(z)=z$ be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
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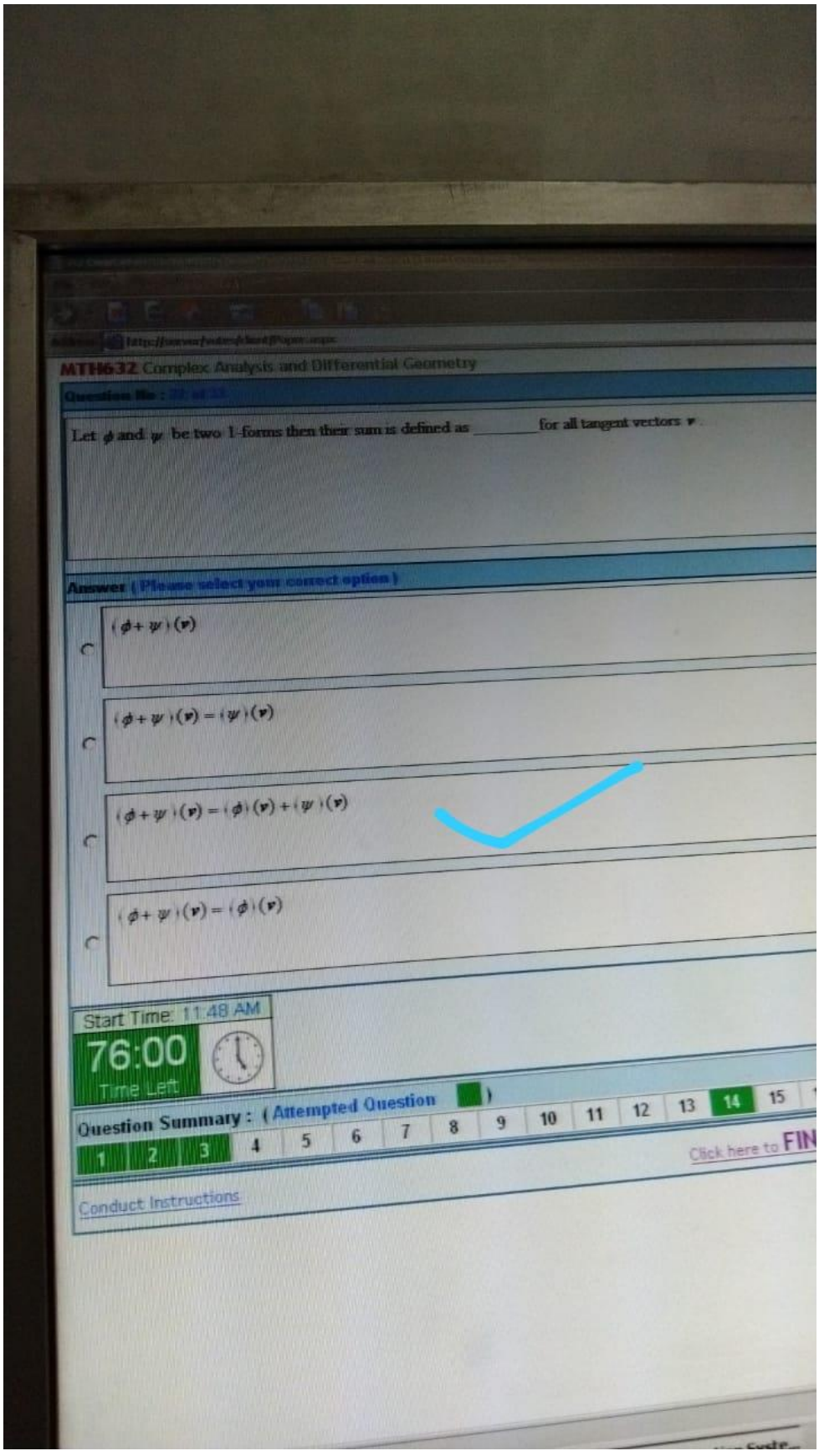
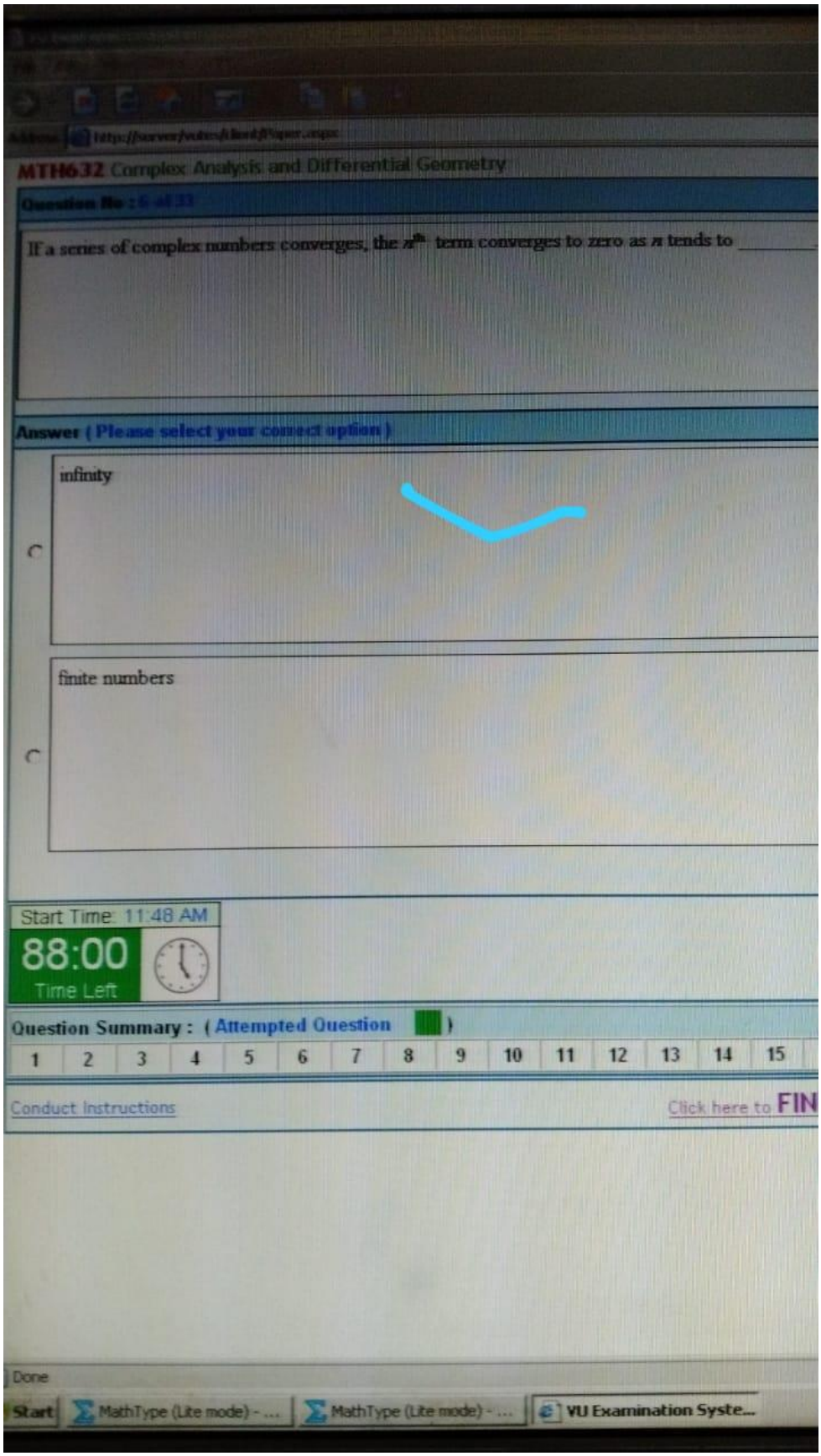
Done


MTH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to FIN
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the n^{th} term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
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Done
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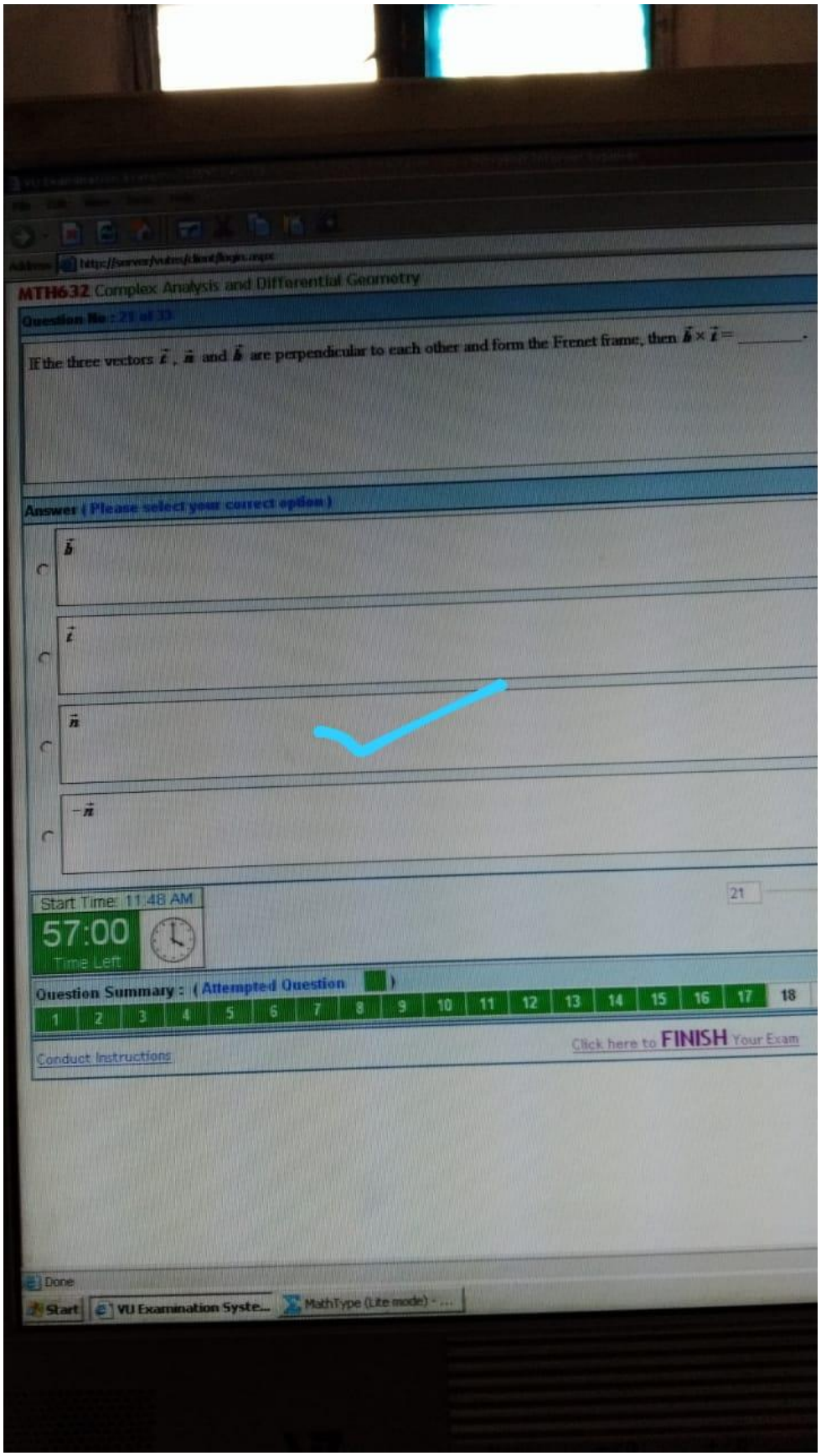
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


MTH632 Complex Analysis and Differential Geometry
Question No : 21 of 33
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ _____.
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
57:00
Time Left
21
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Conduct Instructions
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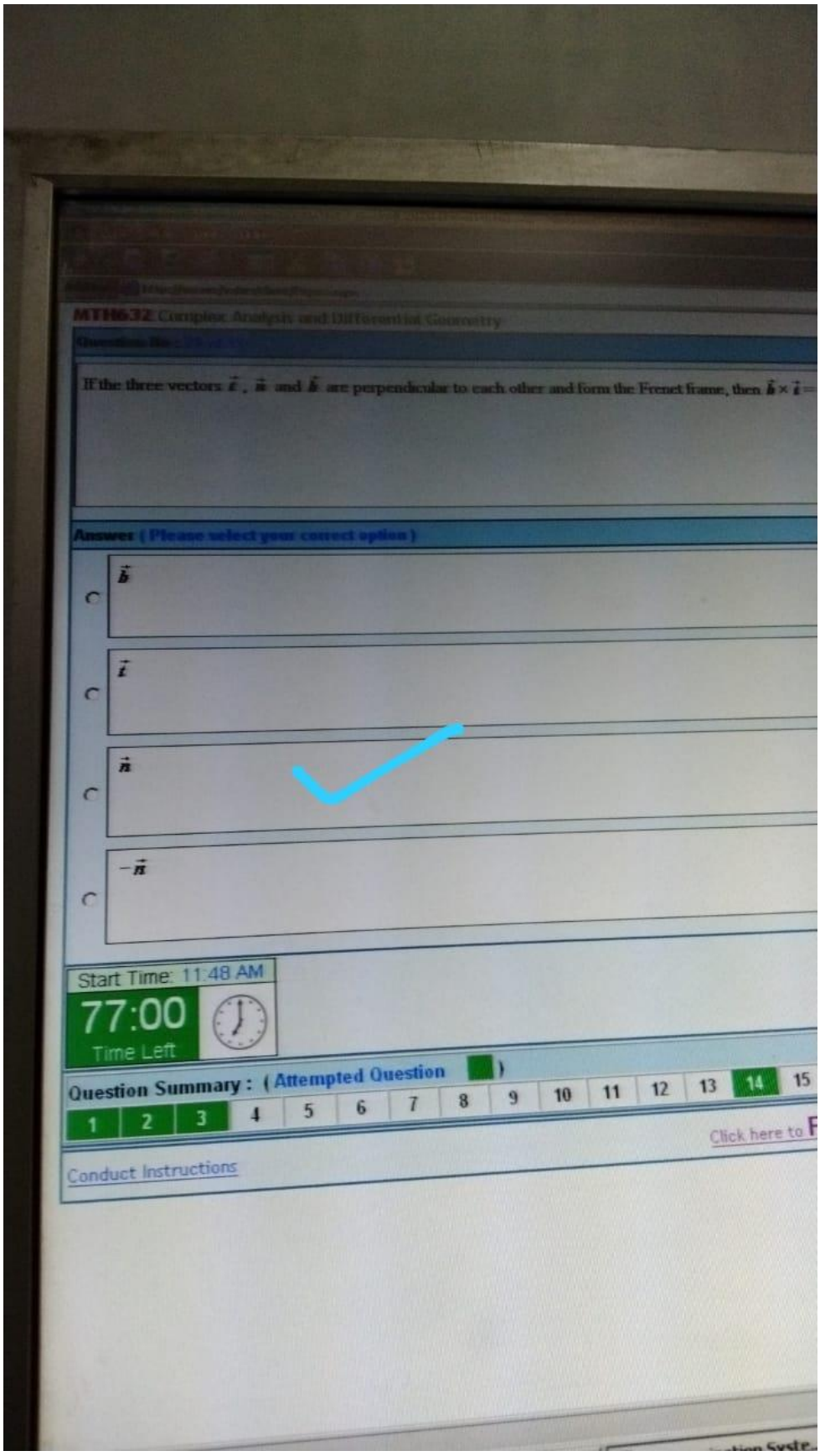
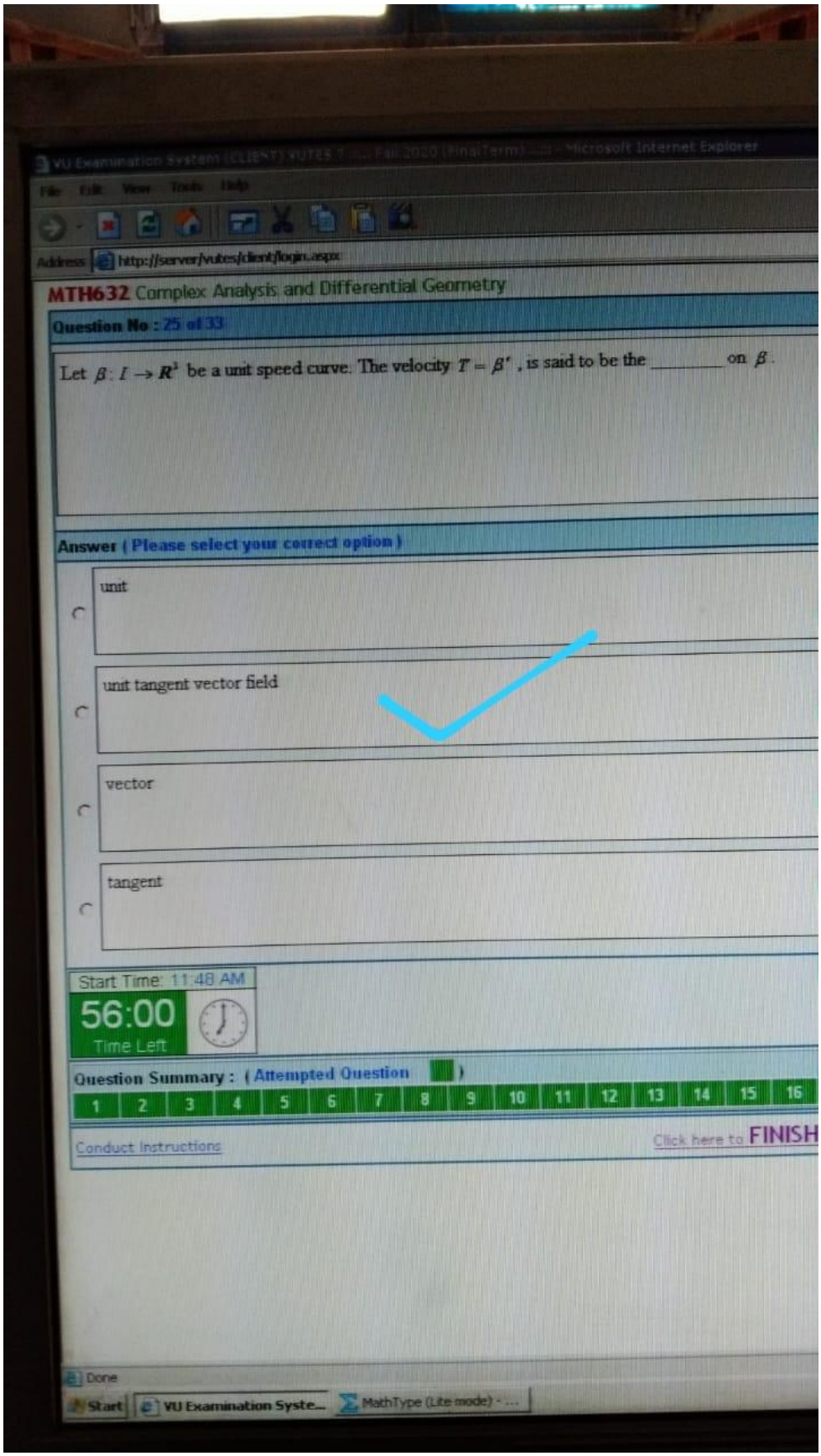
Done
Start
VU Examination Syste...
MathType (Lite mode) - ...


MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to
Conduct Instructions


Microsoft Internet Explorer
Address http://server/vutes/client/login.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 25 of 33
Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
56:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Conduct Instructions
Click here to FINISH
Done
Start
VU Examination Syste...
MathType (Lite mode) - ...

MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 20

The principal value of 1^{2i} is _____.

Answer (Please select your correct option)

- 0
- 1
- i
- 2i

Start Time: 11:48 AM

88:00

Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10

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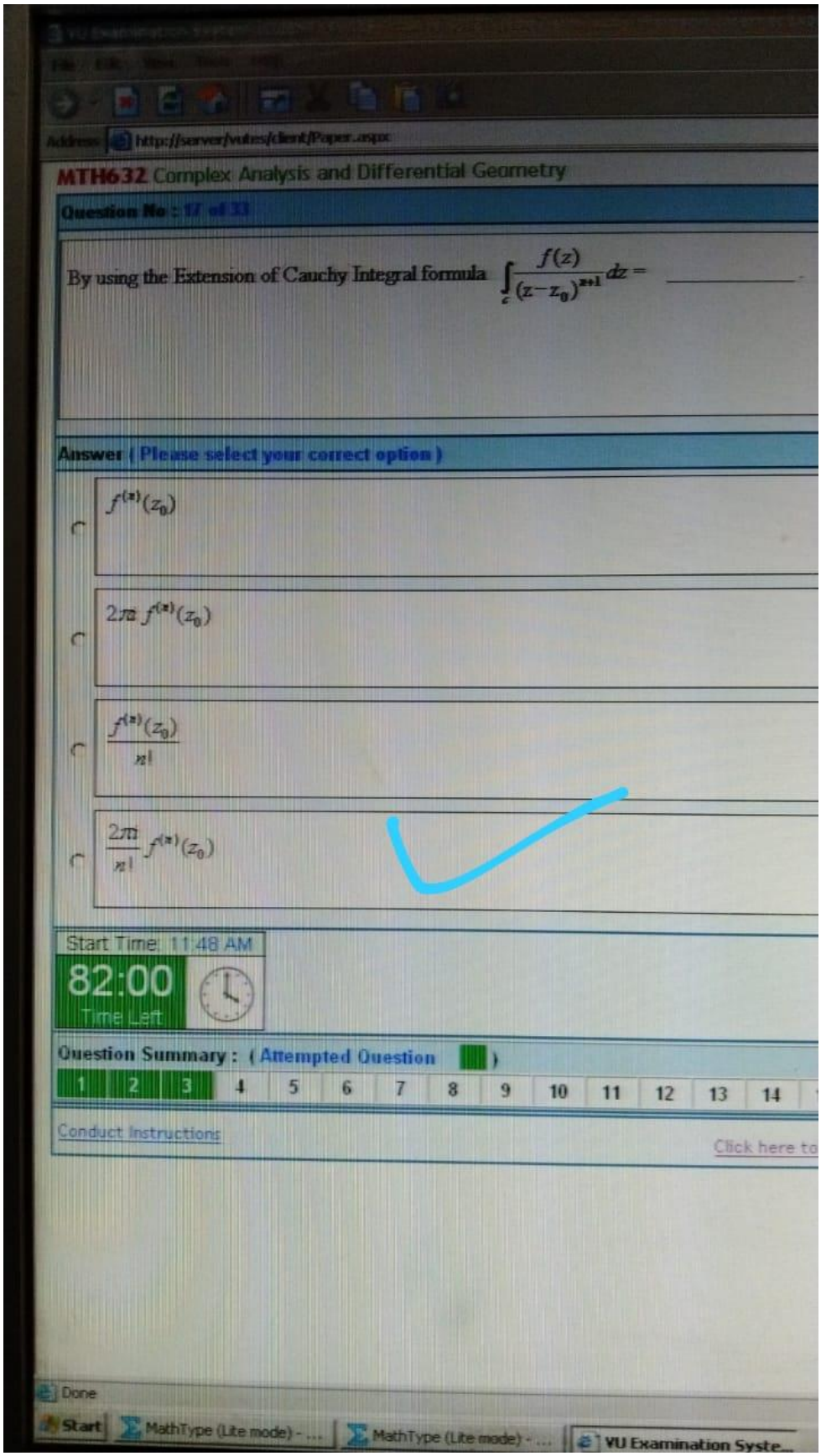
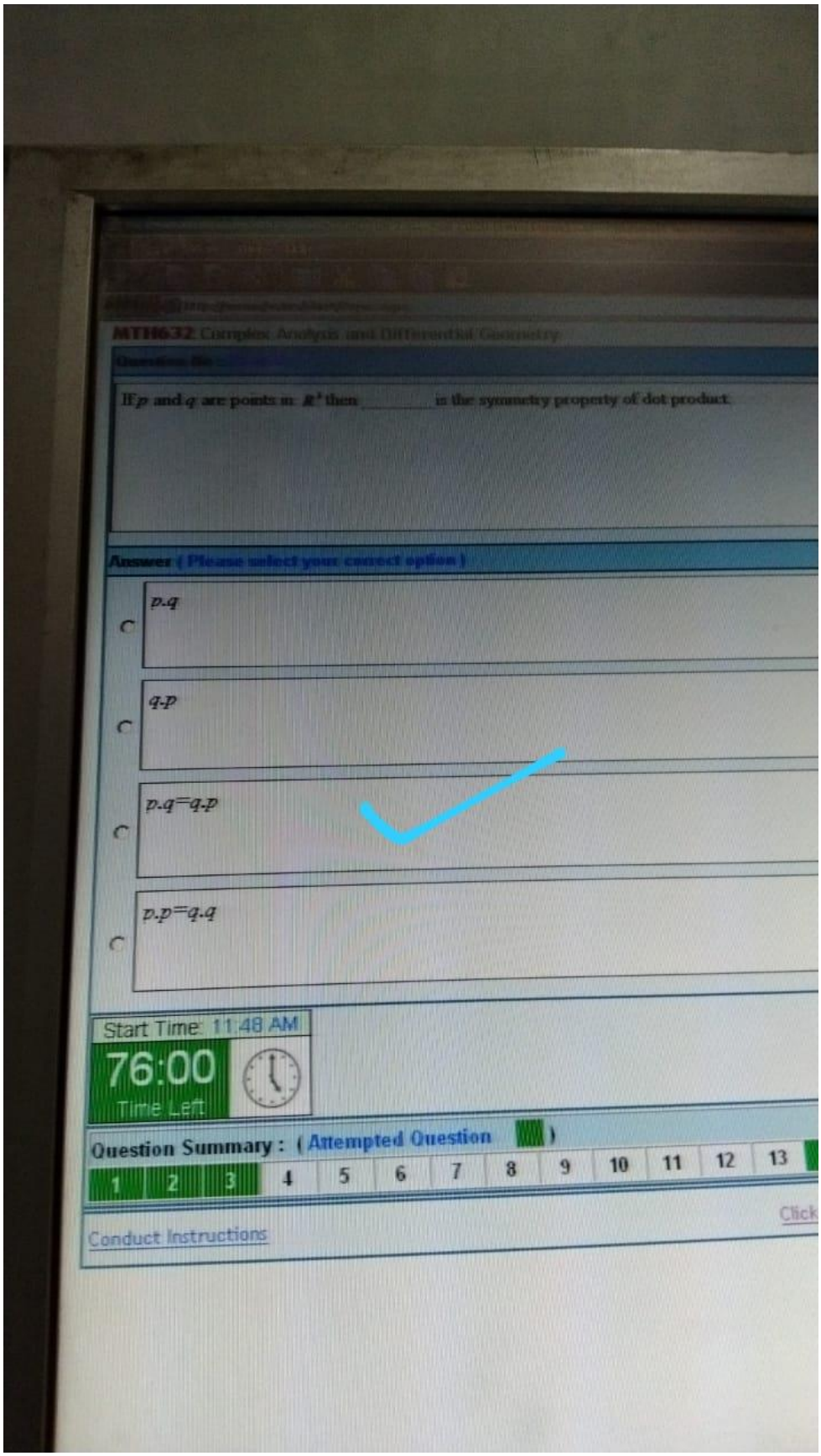
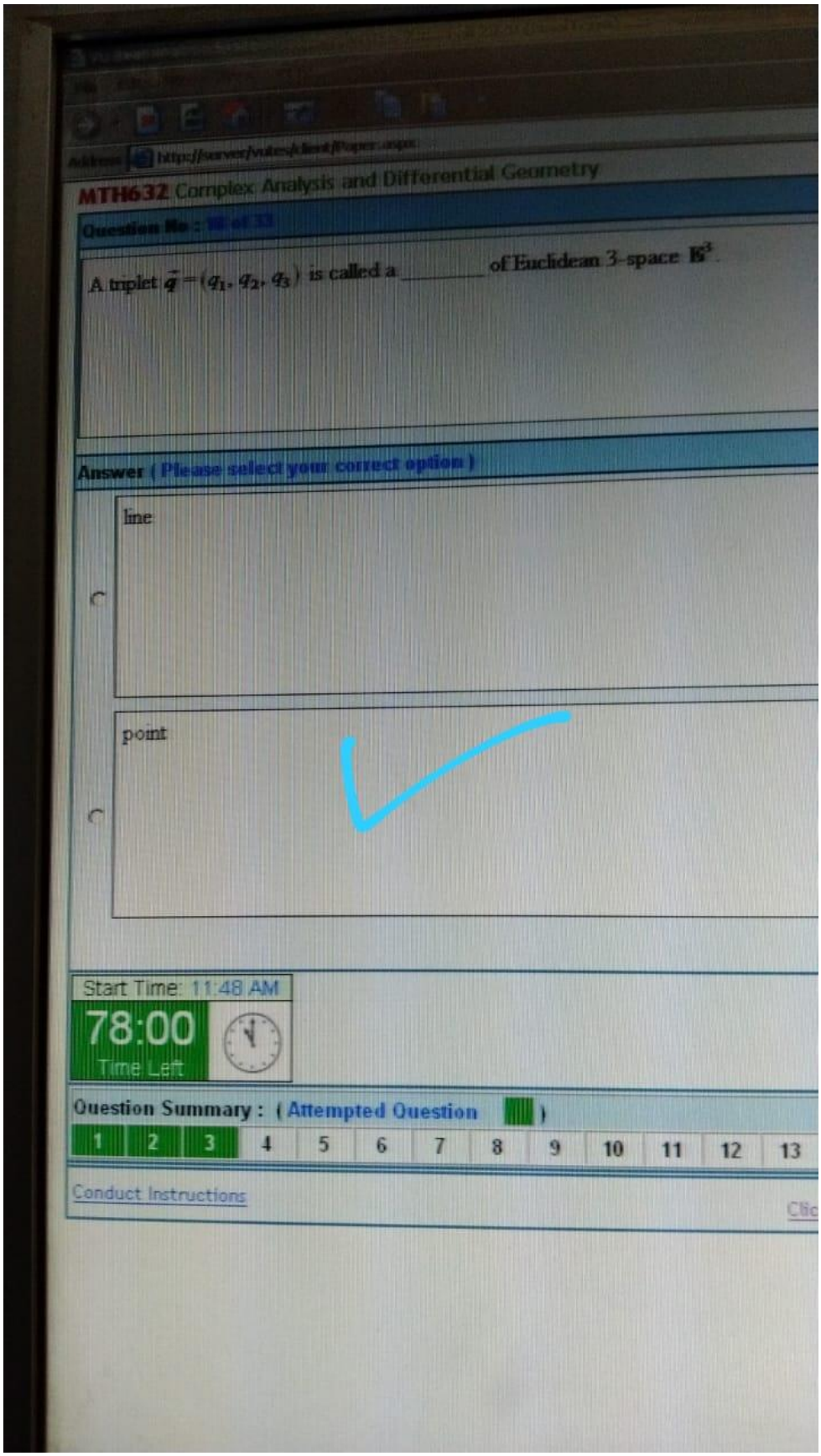
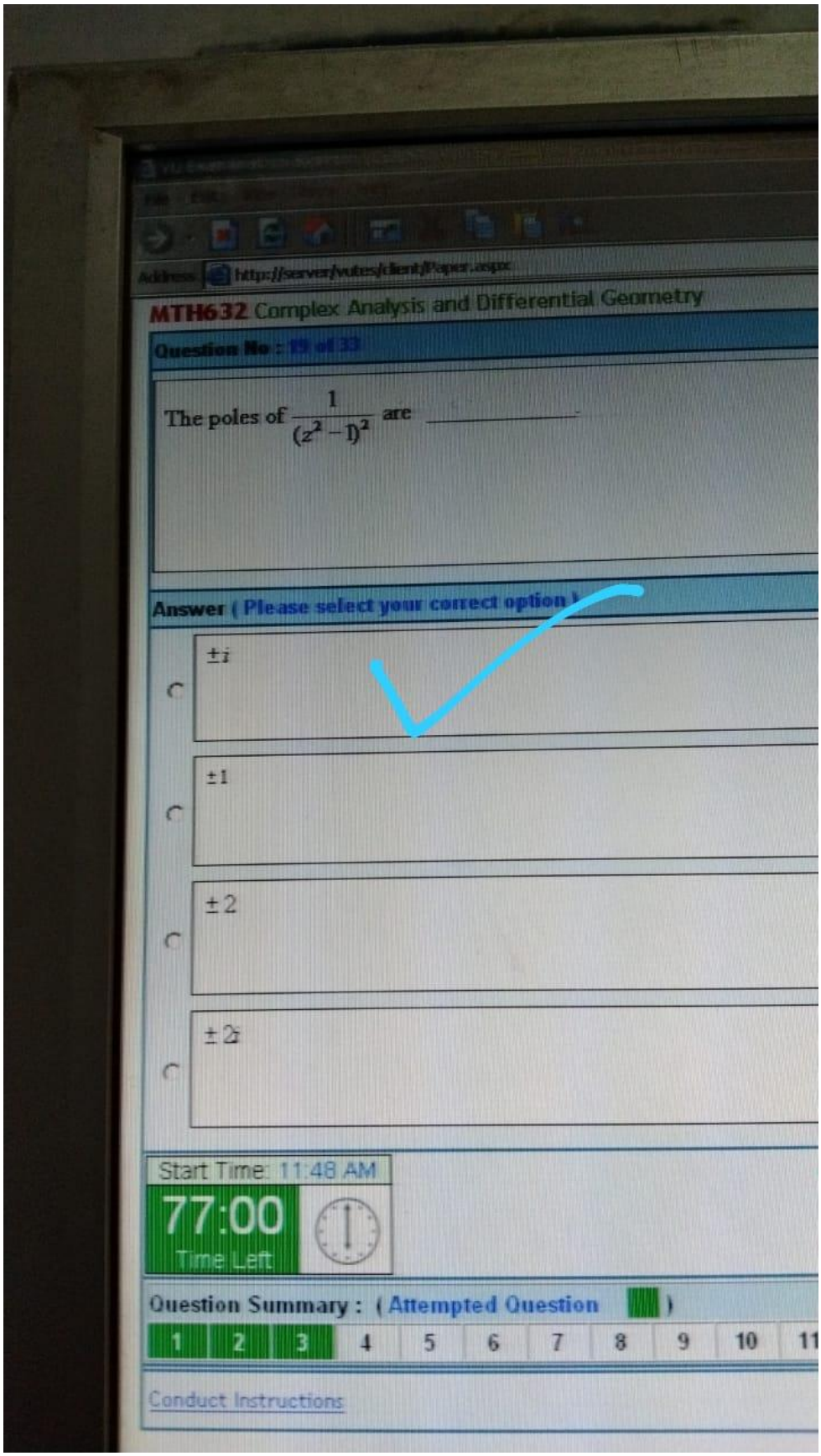
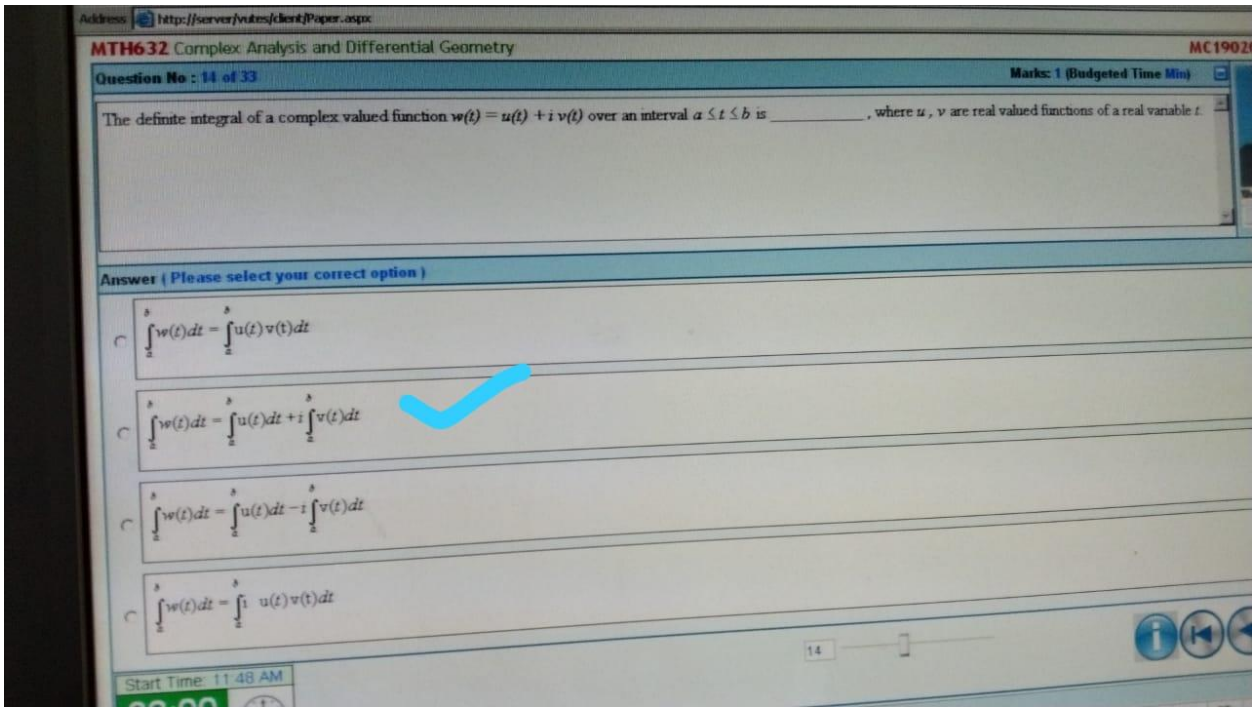

MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____.
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


MTH632 Complex Analysis and Differential Geometry
If p and q are points in R³ then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.q=q.p
p.p=q.q
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time: 11:48 AM
78:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
Question No :
The poles of $\frac{1}{(z^2-1)^2}$ are _____.
Answer (Please select your correct option)
±i
±1
±2
±2i
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions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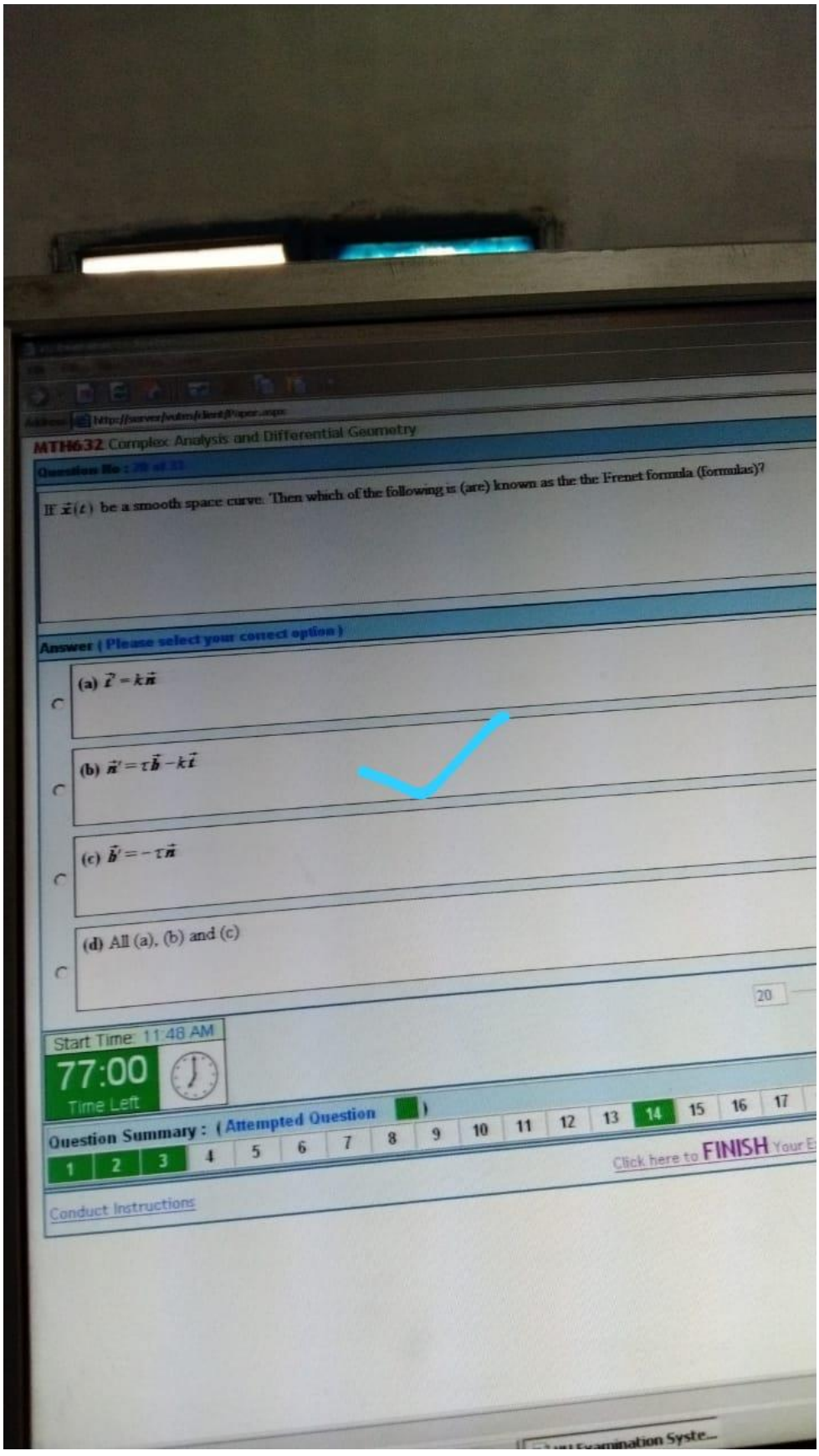
MTH632 Complex Analysis and Differential Geometry
MC1902
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functions of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b i\ u(t)v(t)dt$
14
Start Time: 11:48 AM


MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
20
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
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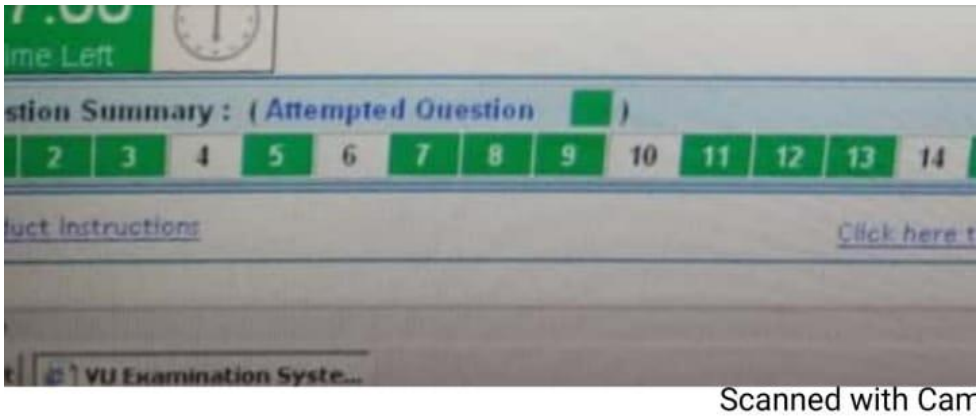
Conduct Instructions



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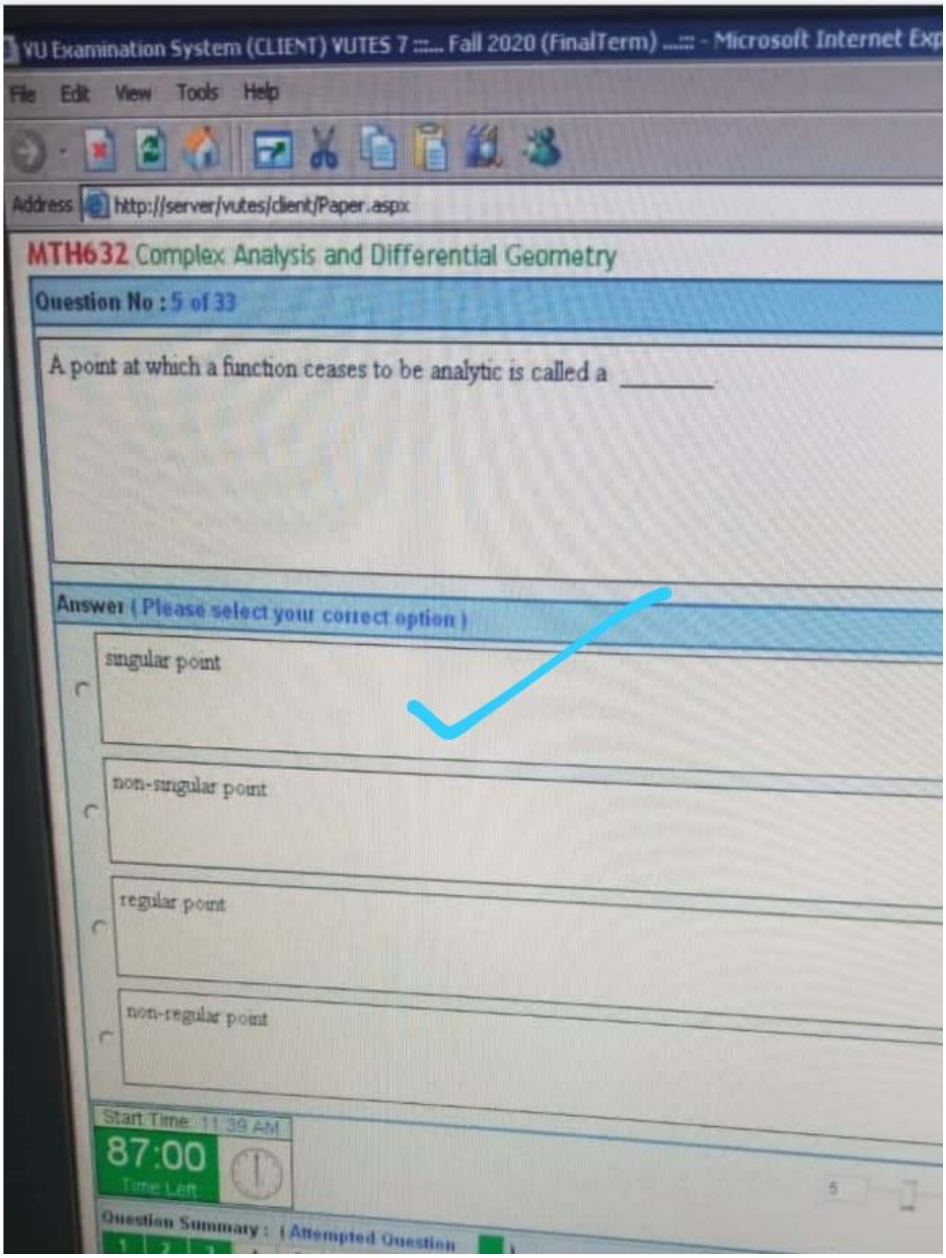

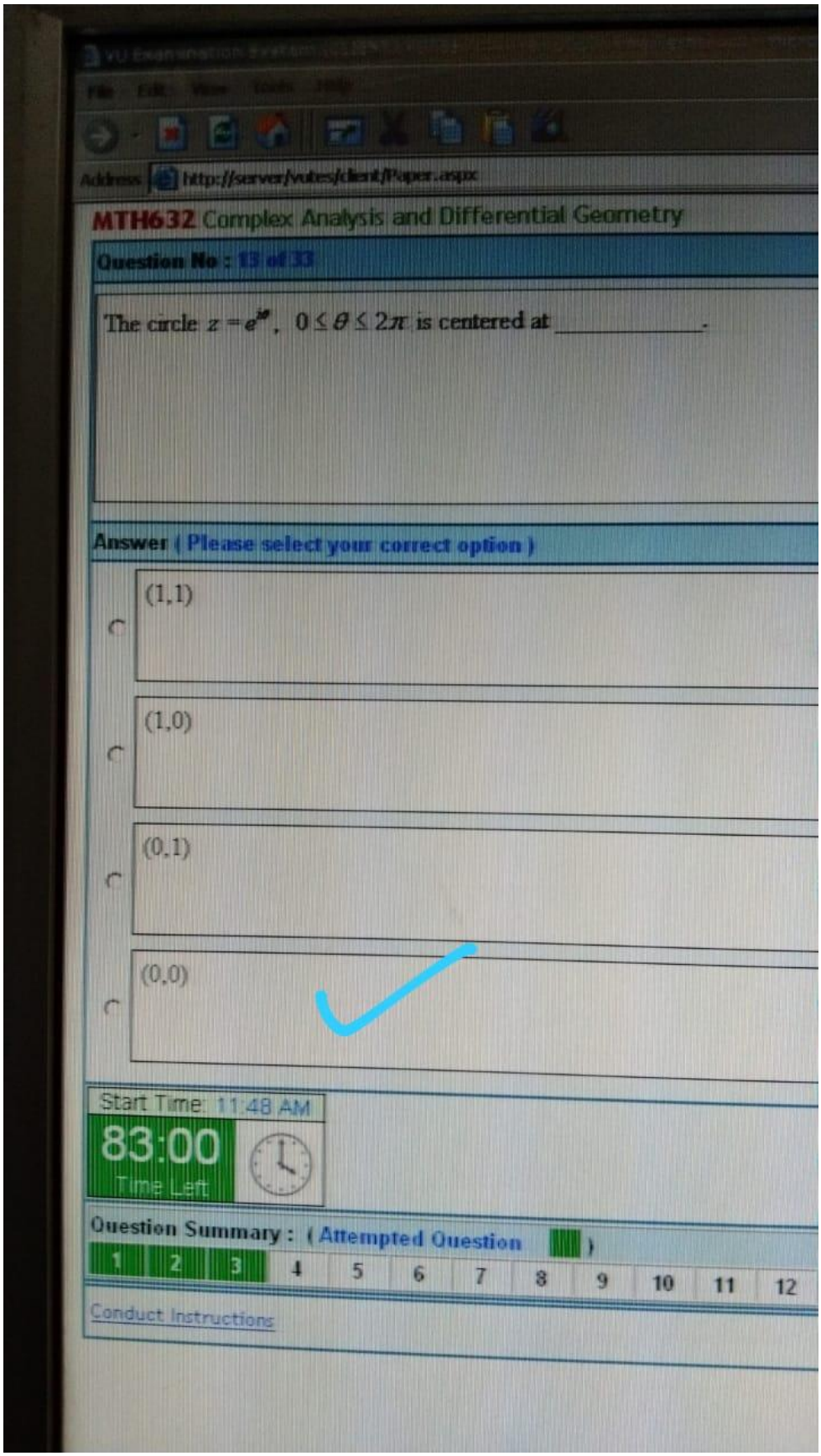
Address http://server/votes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

Conduct Instructions
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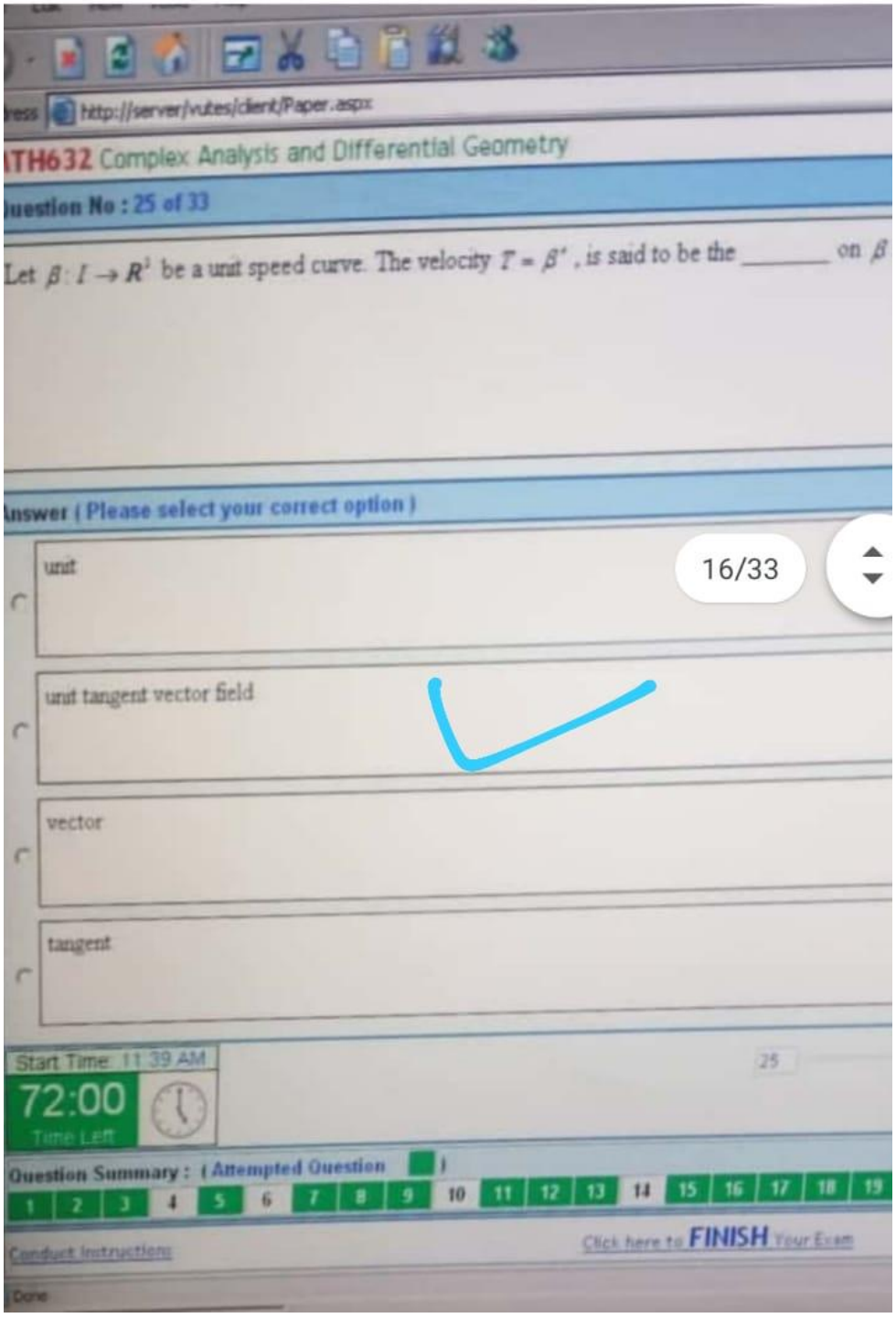
http://server/vutes/client/Paper.aspx
ATH632 Complex Analysis and Differential Geometry
Question No : 25 of 33
Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β
Answer (Please select your correct option)
unit
16/33
unit tangent vector field
vector
tangent
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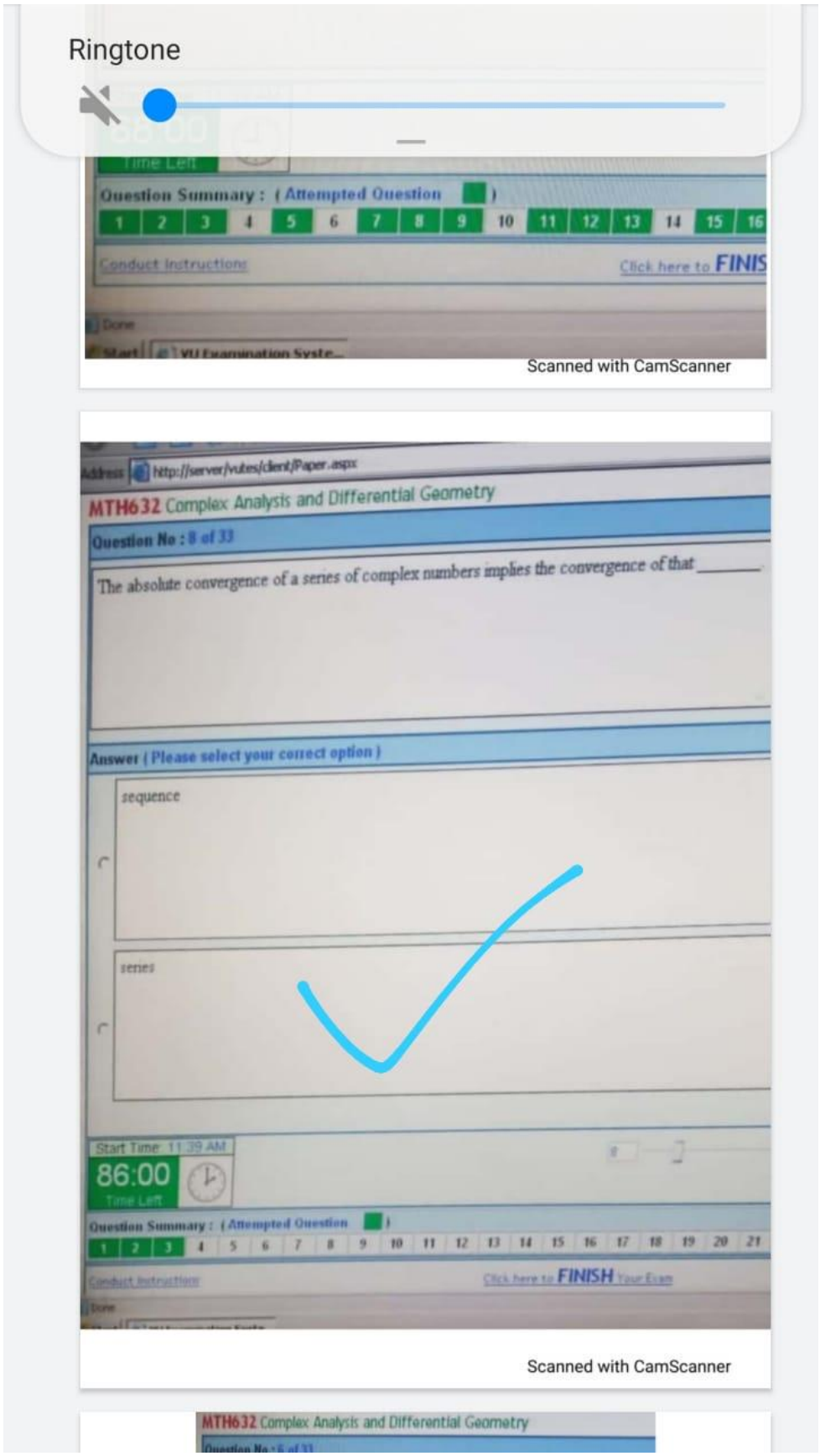
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MTH632 Complex Analysis and Differential Geometry
Question No : 8 of 33
The absolute convergence of a series of complex numbers implies the convergence of that _____.
Answer (Please select your correct option)
sequence
series
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Question Summary : (Attempted Question)
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MTH632 Complex Analysis and Differential Geometry

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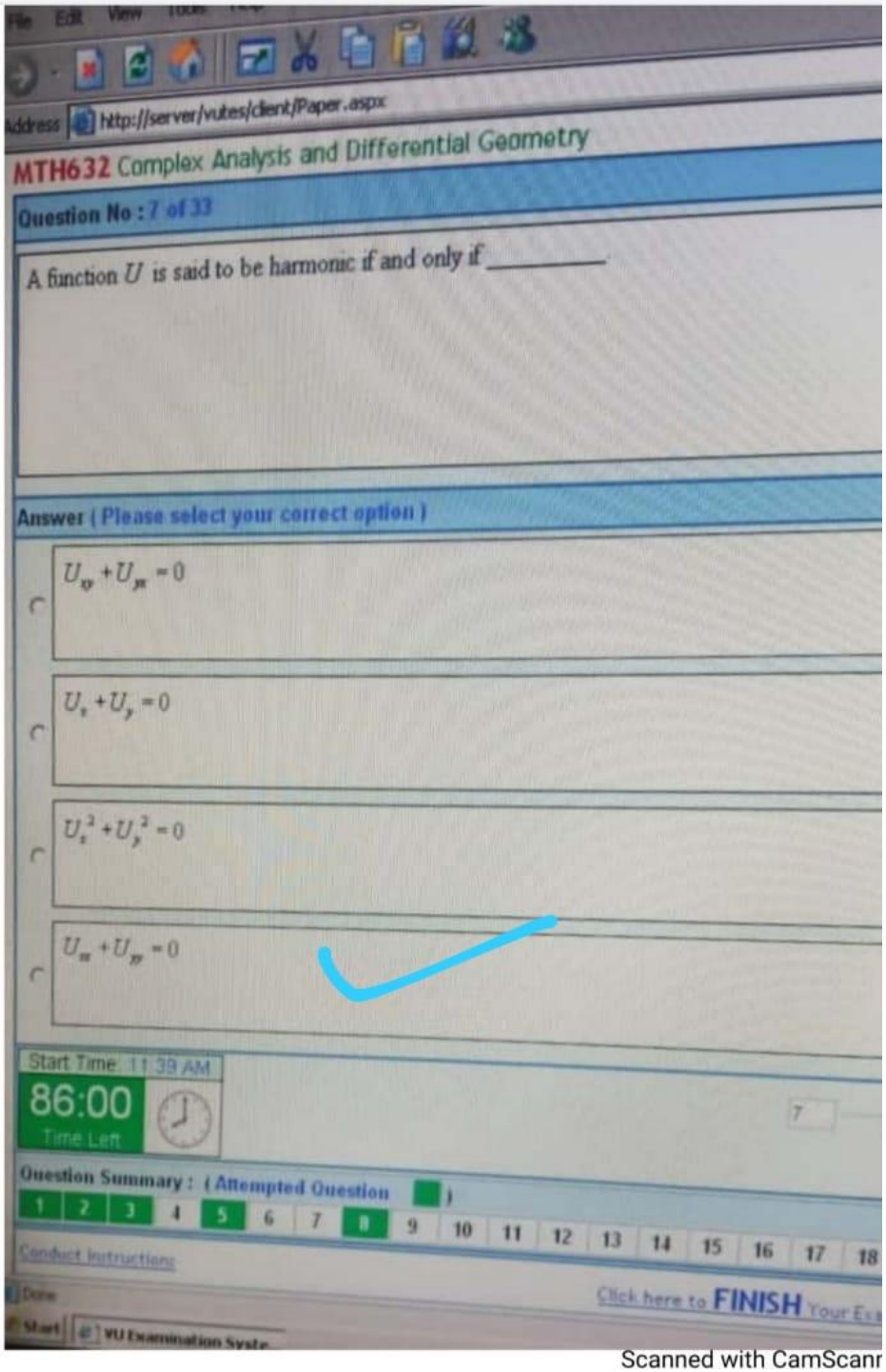

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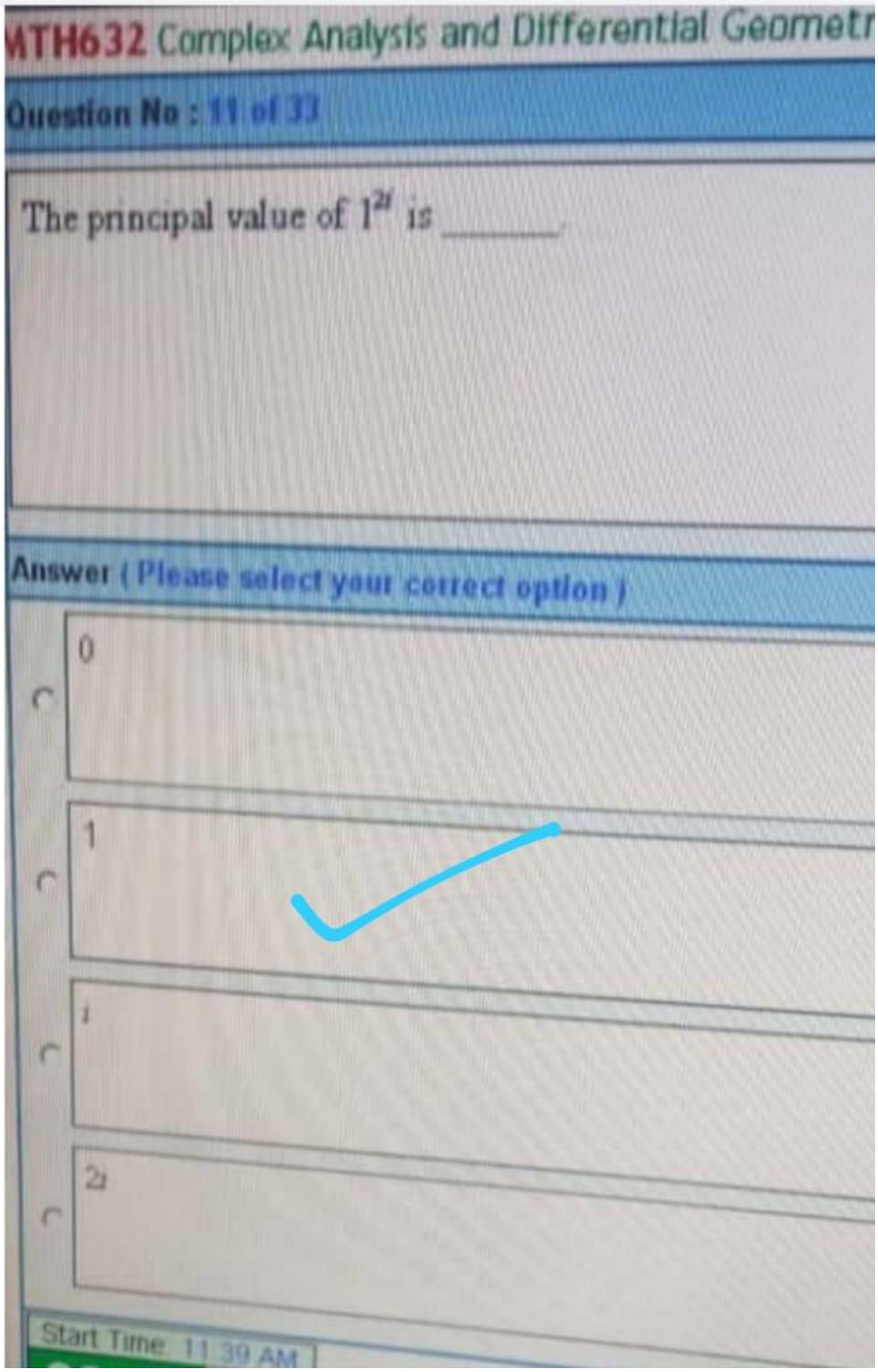
MTH632 Complex Analysis and Differential Geometr
Question No : 11 of 33
The principal value of 1^{2i} is _____.
Answer (Please select your correct option)
0
1
i
2i
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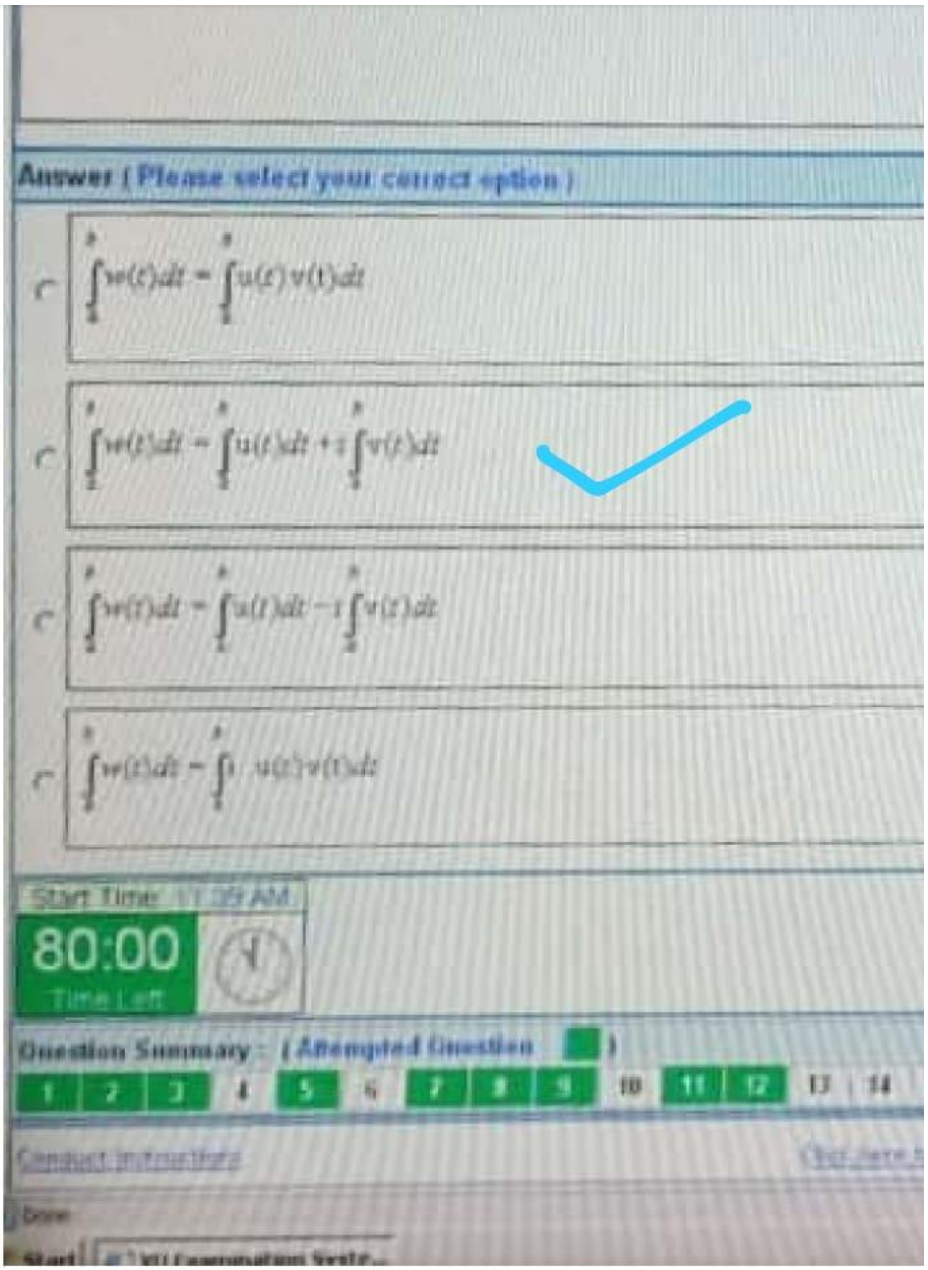
Answer (Please select your correct option)
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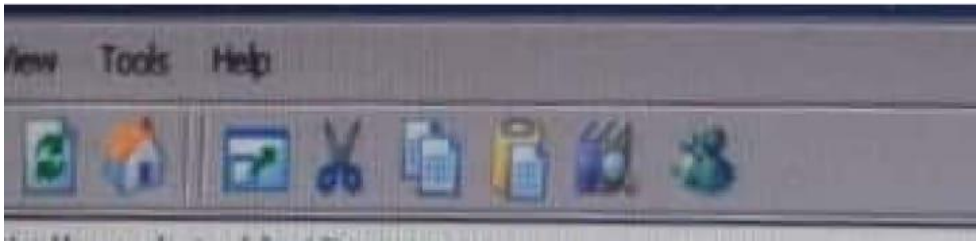
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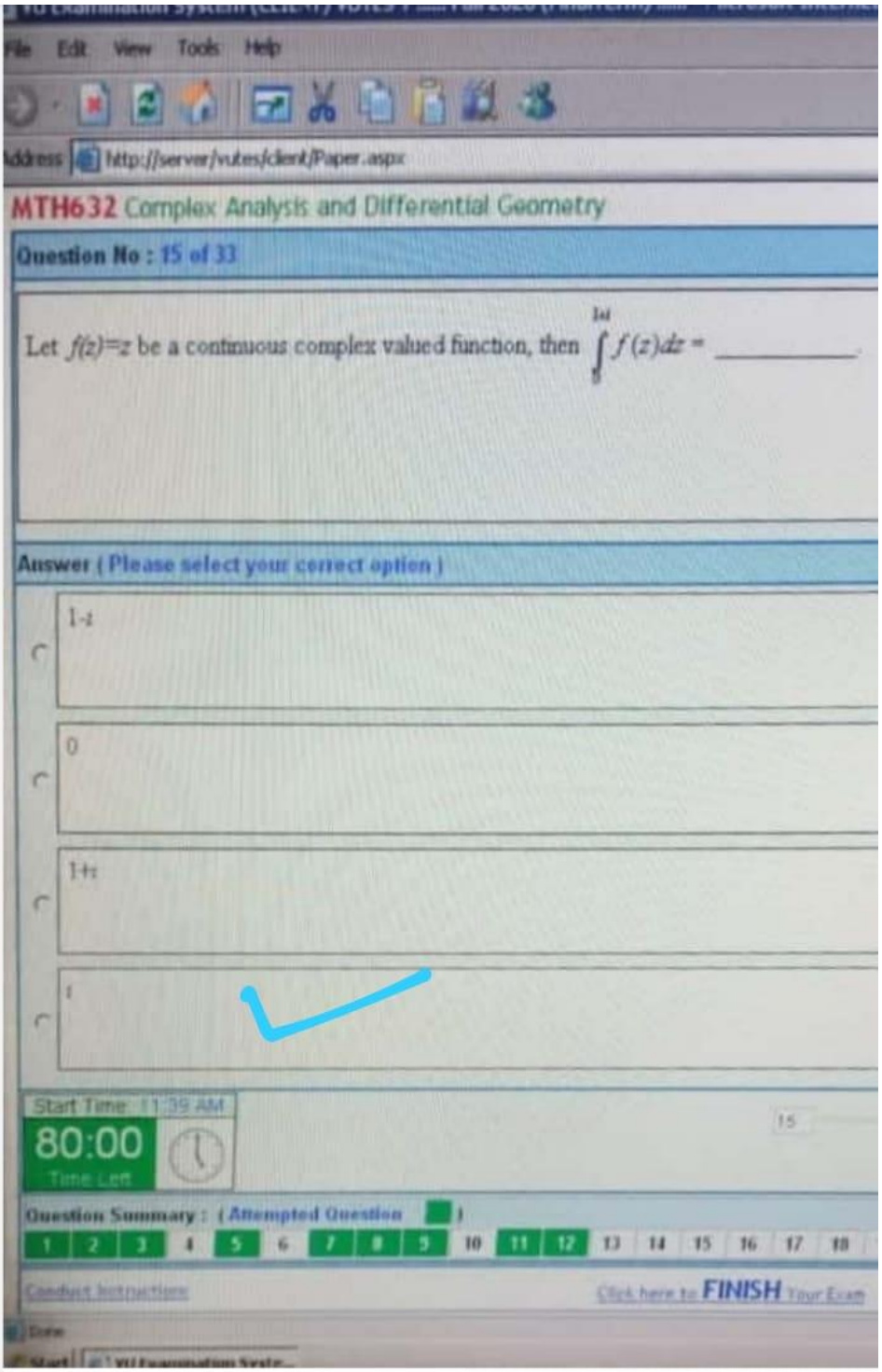
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MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let f(z)=z be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____.
Answer (Please select your correct option)
1-i
0
1+i
i
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Question Summary : (Attempted Question)
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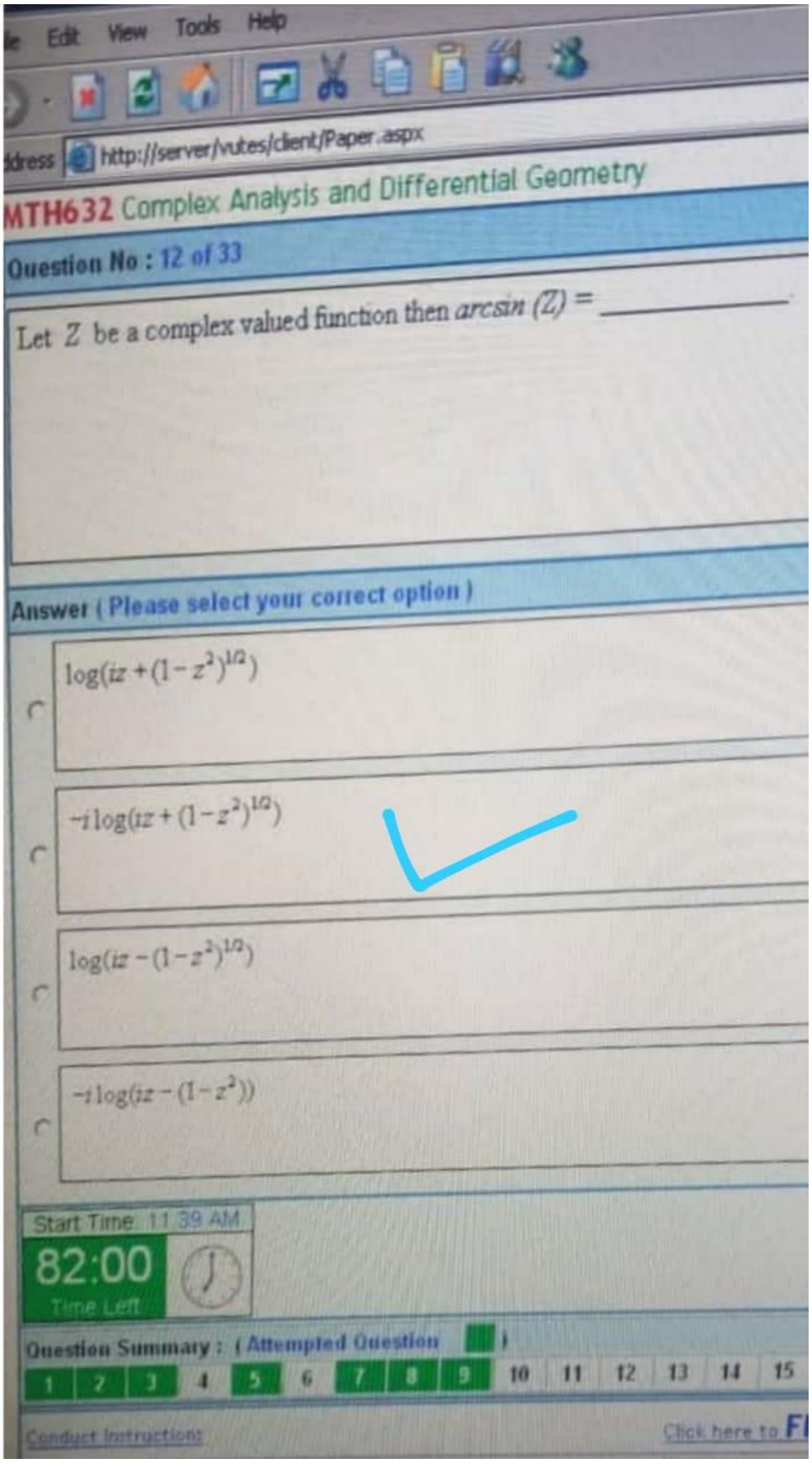
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MTH632 Complex Analysis and Differential Geometry
Question No : 12 of 33
Let Z be a complex valued function then arcsin (Z) = _____
Answer (Please select your correct option)
log(iz + (1 − z²)^1/2)
−i log(iz + (1 − z²)^1/2)
log(iz − (1 − z²)^1/2)
−i log(iz − (1 − z²))
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MTH632 Complex Analysis and Differential Geometry

Question No : 39 of 41

Show that

(a) $Log(1+i)^2 = 2Log(1+i)$ and (b) $Log(-1+i)^2 \neq 2Log(-1+i)$

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry **MC1804**

Question No : 34 of 41 **Marks: 2 (Budgeted Time Min)**

If tangent vector along the helix $\vec{x}(t) = a(\cos t)\hat{e}_1 + a(\sin t)\hat{e}_2 + b(t)\hat{e}_3$, $a, b \neq 0$ is $\vec{x}'(t) = -a(\sin t)\hat{e}_1 + a(\cos t)\hat{e}_2 + b\hat{e}_3$.

Then evaluate the unit tangent vector $\vec{t}(s) = \dot{x} = \dfrac{\vec{x}'(t)}{|\vec{x}'(t)|}$.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 32 of 41

Test whether the function $w = 2xy + i(x^2 - y^2)$ is analytic or not.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 6 of 41

The limit of $f(z) = 2z^2 - 4z + 2$ at $z = i$ is _____

Answer (Please select your correct option)

- $5i$
- $-4i$
- $2i$
- $-3i$

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Question No : 8 of 41 Marks: 1 (Budgeted Time

If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____.

Answer (Please select your correct option)

- harmonic in D.
- equal in D.
- homogeneous in D.
- steady in D.

Question No : 11 of 41 **Marks: 1 (Budgeted Time Min)**

If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____.

Answer (Please select your correct option)

- $c=1$ and $a=-b$
- $a=1$ and $c=-b$
- $b=1$ and $a=-c$
- $a=b=c=1$

Question No : 14 of 41 **Marks: 1**

If $e^{ax}\cos y$ is harmonic then 'a' is _____.

Answer (Please select your correct option)

- 0
- -1
- 2
- i

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MTH632 Complex Analysis and Differential Geometry

Question No : 16 of 41

If $f(z) = z(2-z)$, then $f(1+i) =$ _____.

Answer (Please select your correct option)

- 0
- 2
- i
- $-i$

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Question No : 21 of 41

Marks: 1 (Budgeted Time Min)

The derivative of a complex valued function $w(t) = u(t) + i\,v(t)$ is _____ , where u , v are real valued functions of a real variable t.

Answer (Please select your correct option)

- $w'(t) = u'(t) + iv'(t)$
- $w'(t) = u(t) + iv(t)$
- $w'(t) = u'(t)v'(t)$
- $w'(t) = u'(t) + v'(t)$

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21

Marks: 1 (Budgeted Time Min)

If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ _____.

Answer (Please select your correct option)

- $\vec{b}$
- $\vec{t}$
- $\vec{n}$
- $-\vec{n}$

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Question No : 22 of 41 Marks: 1 (Budgeted Time

If $w(t)$ is a piecewise continuous complex-valued function defined on an interval $a \le t \le b$, then _____.

Answer (Please select your correct option)

- $\left|\int_a^b w(t)dt\right| \le \int_a^b |w(t)|dt$
- $\left|\int_a^b w(t)dt\right| \ge \int_a^b |w(t)|dt$
- $\left|\int_a^b w(t)dt\right| = \int_a^b |w(t)|dt$
- $\int_a^b w(t)dt \ge \left|\int_a^b |w(t)|dt\right|$

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MTH632 Complex Analysis and Differential Geome

Question No : 18 of 41

The principal value of i^i is _____.

Answer (Please select your correct option)

- $\exp(\frac{\pi}{2})$
- $\exp(-\frac{\pi}{2})$
- $\exp(i)$
- $\exp(-i)$

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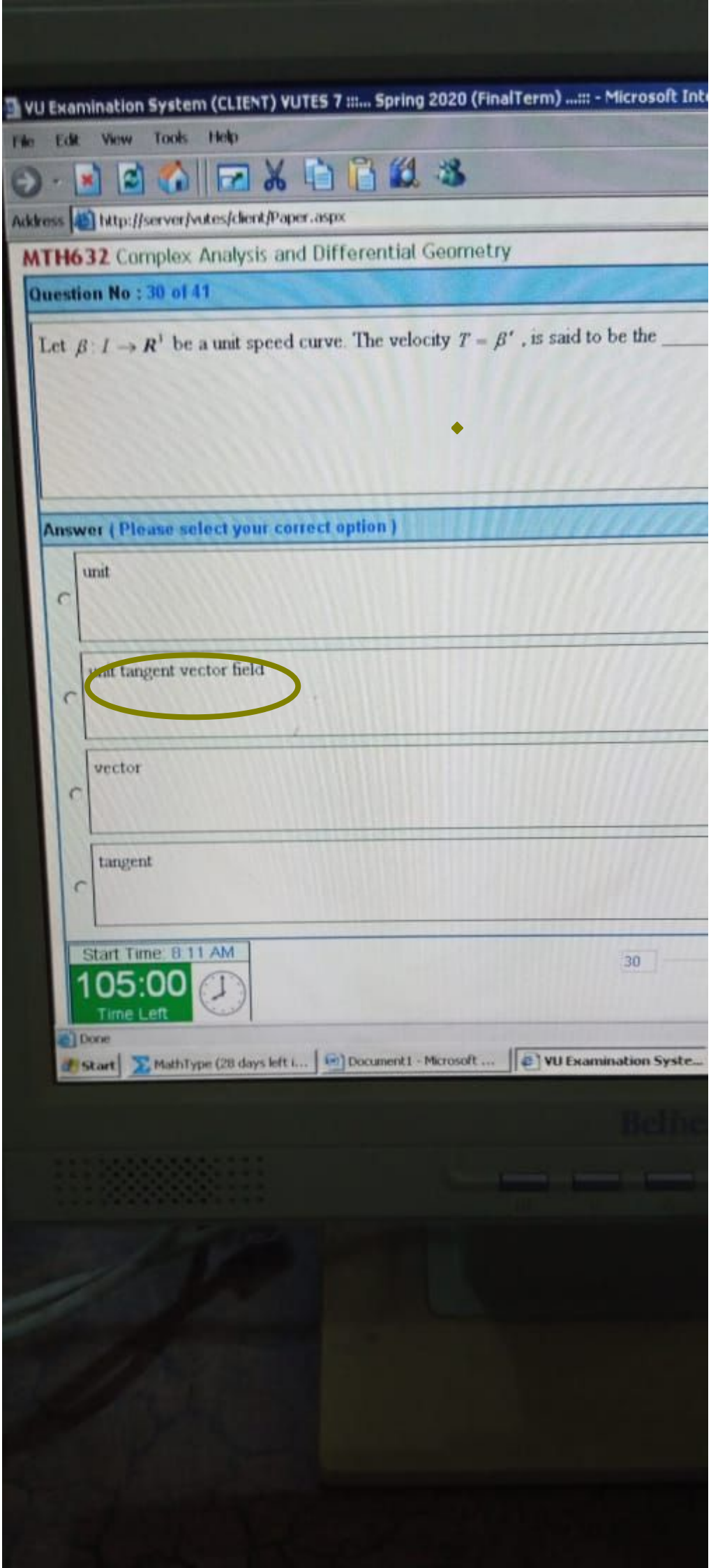
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MTH632 Complex Analysis and Differential Geometry
Question No : 30 of 41
Let $\beta : I \to R^1$ be a unit speed curve. The velocity $T = \beta'$, is said to be the ____
Answer (Please select your correct option)
unit
tangent vector field
vector
tangent
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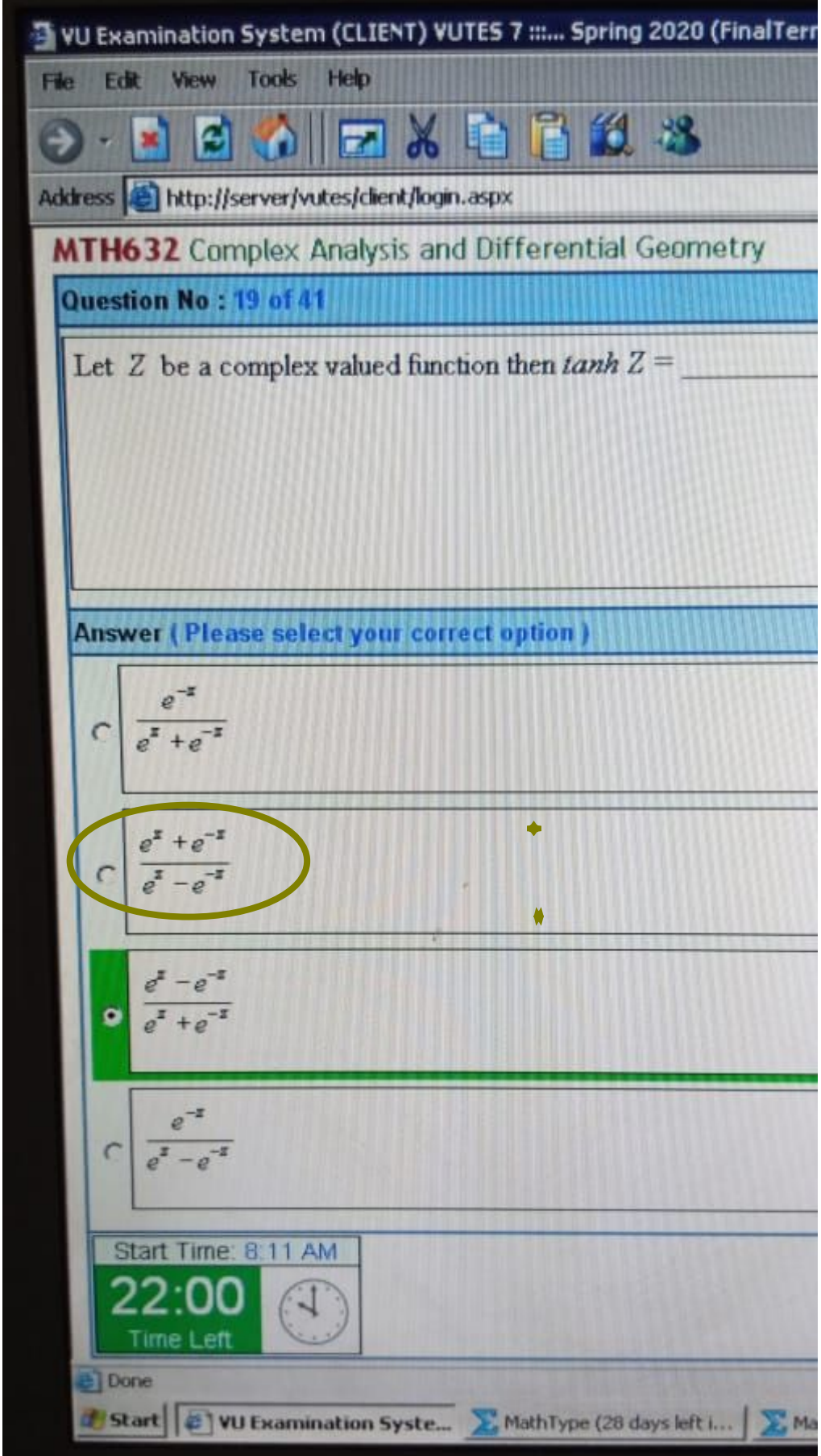
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MTH632 Complex Analysis and Differential Geometry
Question No : 19 of 41
Let Z be a complex valued function then tanh Z = _____
Answer (Please select your correct option)
$\frac{e^{-z}}{e^{z}+e^{-z}}$
$\frac{e^{z}+e^{-z}}{e^{z}-e^{-z}}$
$\frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$
$\frac{e^{-z}}{e^{z}-e^{-z}}$
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MTH632 Complex Analysis and Differential Geometry

Question No : 28 of 41

Marks: 1 (Budgeted Time Min)

The differential of f is, $df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$, where df calculates the small changes in the value of f when there is small change in _____.

Answer (Please select your correct option)

(a) x

(b) y

(c) z

(d) All (a), (b) and (c).

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Question No : 41 of 41 **Marks: 5 (Budgeted Time Min)**

Let C_1 denote the positively oriented boundary of the square whose sides lie along the lines $x = \pm 1, y = \pm 1$ and let C_2 be the positively oriented circle $|z| = 4$ (given in Fig. below). Explain why, $\int_{C_1} f(z)dz = \int_{C_2} f(z)dz$ where $f(z) = \dfrac{z+2}{\sin(z/2)}$?

y

Answer (Please click here to Add Answer)

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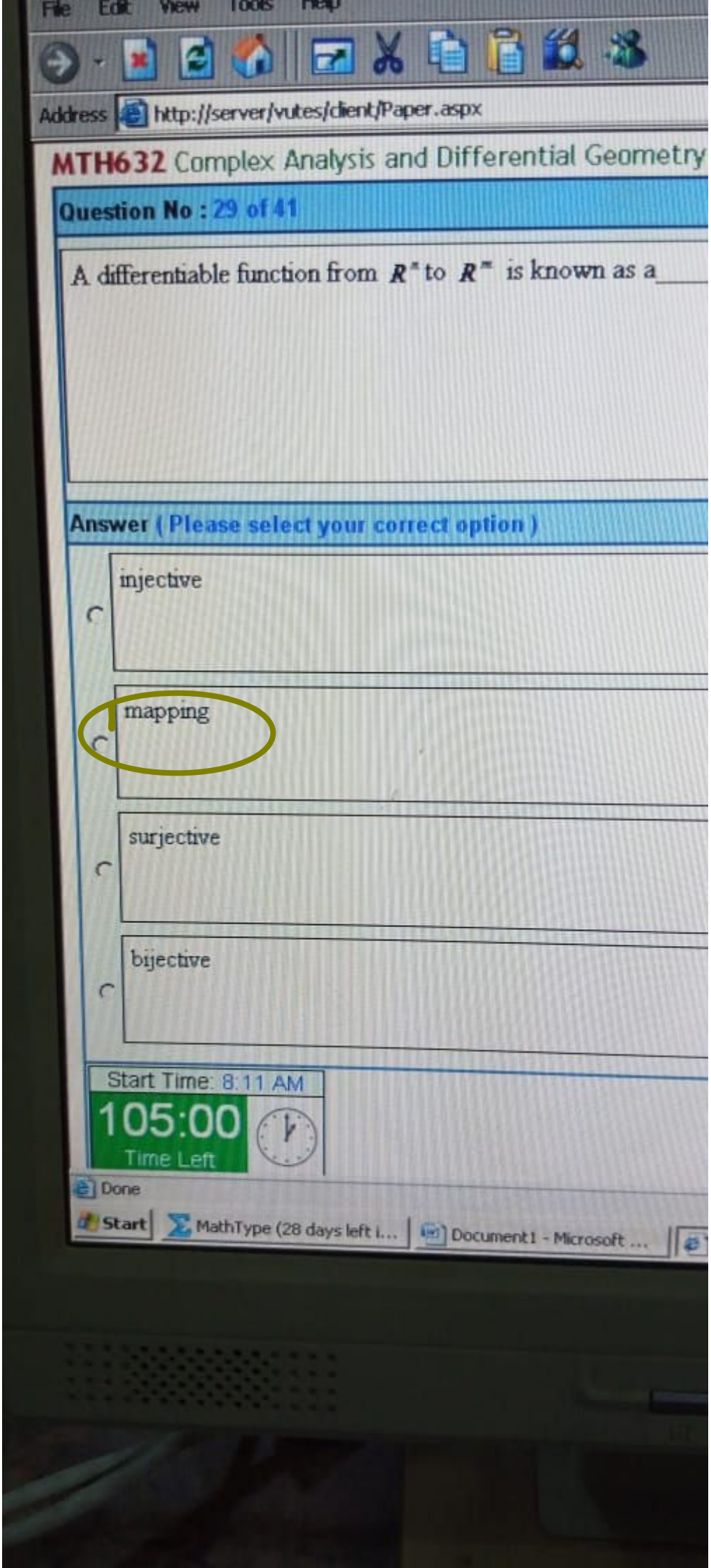
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MTH632 Complex Analysis and Differential Geometry
Question No : 29 of 41
A differentiable function from R^n to R^m is known as a____
Answer (Please select your correct option)
injective
mapping
surjective
bijective
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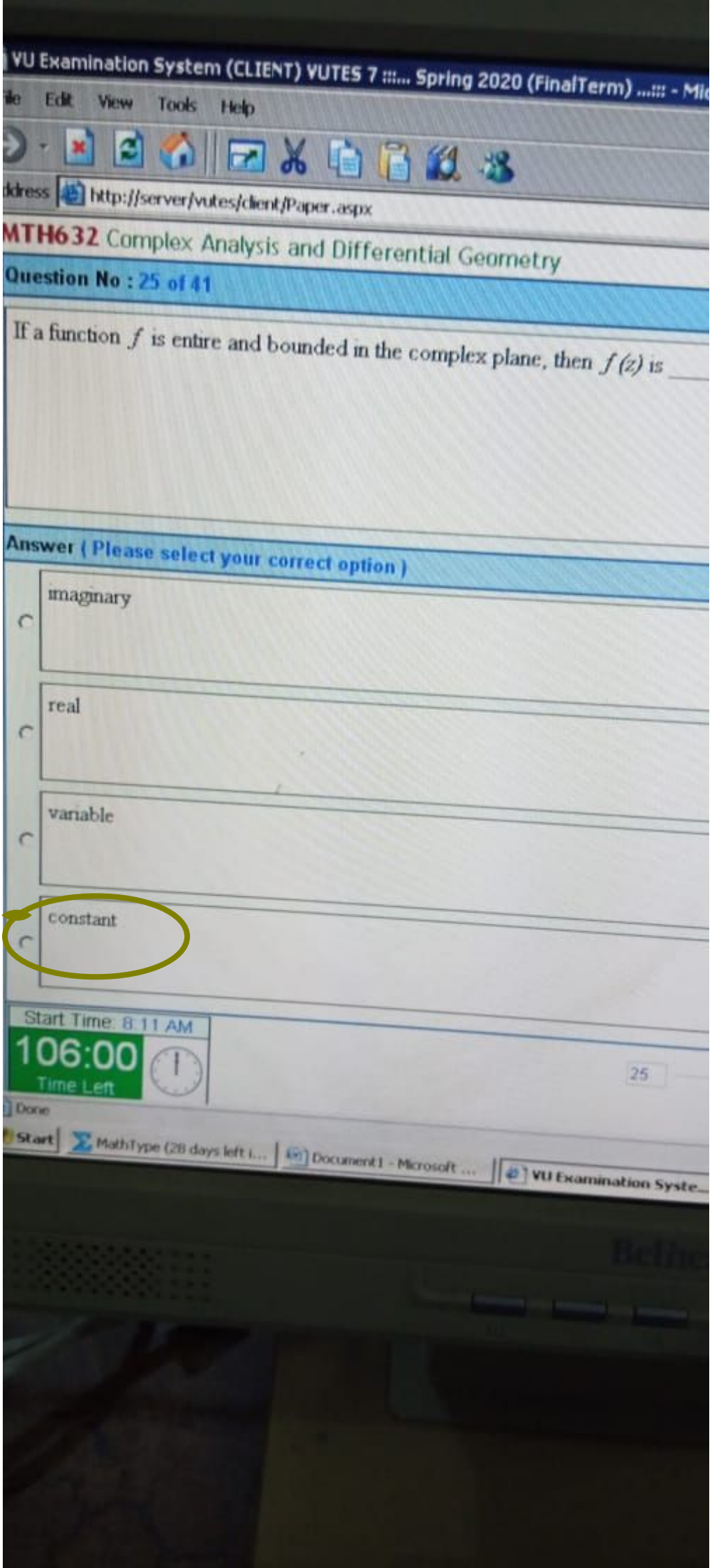
MTH632 Complex Analysis and Differential Geometry

Question No : 26 of 41 **Marks: 1 (Budgeted Time Min)**

The scalar multiple of a vector $\vec{x} = (x_1, x_2, x_3)$ by a real number k is the point $k\vec{x} = (kx_1, kx_2, kx_3)$, clearly _____ for all k and $\vec{x}$.

Answer (Please select your correct option)

- (a) $k\vec{x} = \vec{0}$
- (b) $0\vec{x} = \vec{0}$
- (c) $k\vec{0} = \vec{0}$
- (d) both (b) and (c)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MTH632 Complex Analysis and Differential Geometry
Question No : 25 of 41
If a function f is entire and bounded in the complex plane, then f (z) is
Answer (Please select your correct option)
imaginary
real
variable
constant
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MTH632 Complex Analysis and Differential Geometry

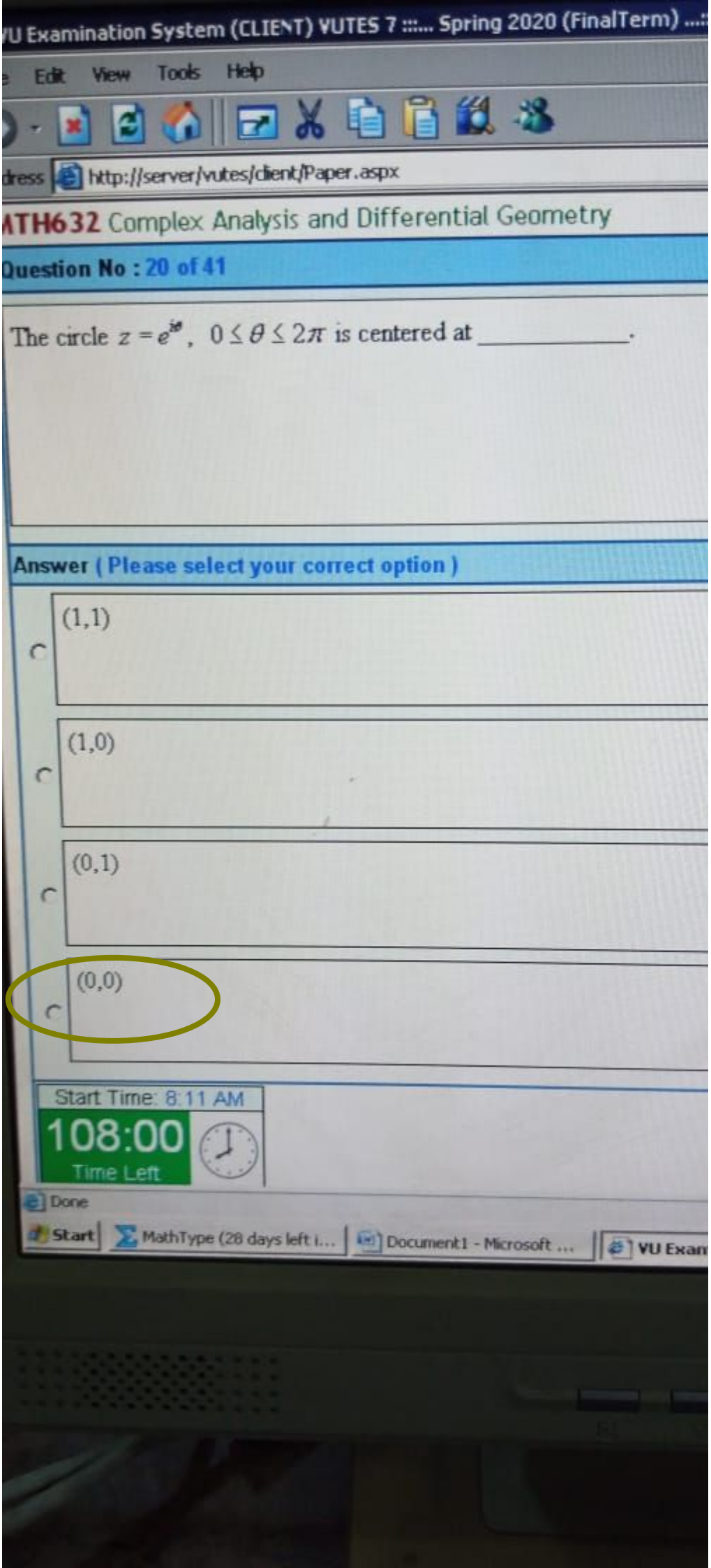
Marks: 1 (Budgeted Time Min)

Question No : 24 of 41

Let C be the positively oriented circle $|z|=2$. Since the function $f(z)=\dfrac{z}{9-z^2}$ is analytic within and on C and since the point $z_0=-i$ is interior to C, then using the Cauchy Integral formula $\int_C \dfrac{z}{(9-z^2)(z+i)}dz =$ _____.

Answer (Please select your correct option)

- $\dfrac{\pi}{9}$
- $\dfrac{\pi}{5}$
- 2π
- $2\pi i$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TH632 Complex Analysis and Differential Geometry
Question No : 20 of 41
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
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MTH632 Complex Analysis and Differential Geometry

Question No : 17 of 41

In the trigonometric functions of complex numbers $Sin\ x =$ ___

Answer (Please select your correct option)

- $\frac{e^{ix} - e^{-ix}}{2i}$
- $\frac{e^{ix} - e^{-ix}}{2}$
- $\frac{e^{-ix} - e^{ix}}{2i}$
- $\frac{e^{-ix} - e^{ix}}{2}$

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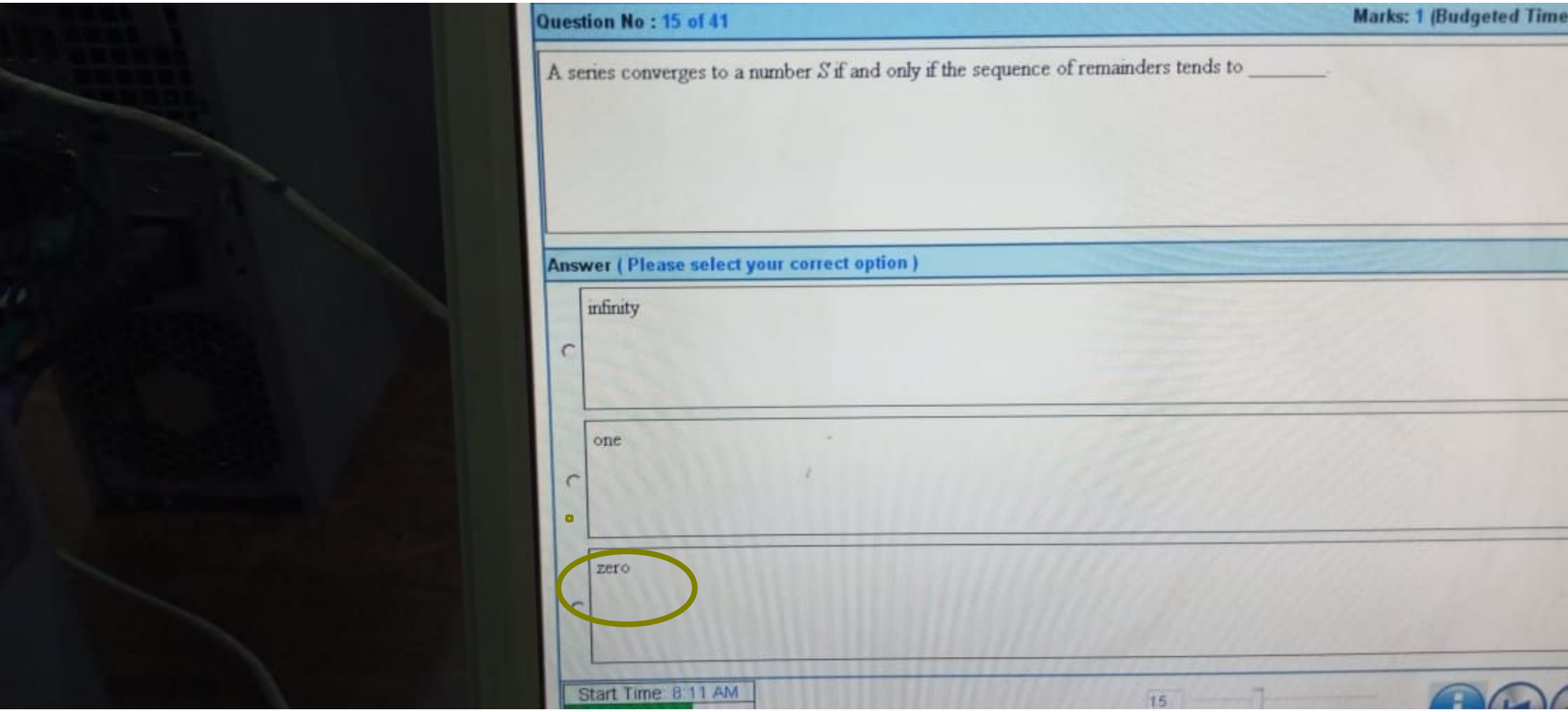
Question No : 15 of 41
Marks: 1 (Budgeted Time
A series converges to a number S if and only if the sequence of remainders tends to _____.
Answer (Please select your correct option)
infinity
one
zero
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Question No : 13 of 41 Marks: 1 (Budg

If U and V are harmonic functions then $f(Z) = U + iV$ is _____.

Answer (Please select your correct option)

- analytic function
- need not be analytic function
- analytic function only at Z=0
- none of the others

MTH032 Complex Analysis and Differential Geometry

Question No : 12 of 41 **Marks: 1 (Budgeted Tim**

If a series of complex numbers converges, the n^{th} term converges to zero as n tends to _____.

Answer (Please select your correct option)

- infinity
- finite numbers

MTH632 Complex Analysis and Differential Geometry

Question No : 9 of 41 **Marks: 1 (Bu**

If Z_1 and Z_2 are two nonzero complex numbers then $Log(z_1 z_2) =$ _____.

Answer (Please select your correct option)

- $Log\, z_1\, Log\, z_2$
- $Log\, z_1 / Log\, z_2$
- $Log\, z_1 - Log\, z_2$
- $Log\, z_1 + Log\, z_2$

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MTH632 Complex Analysis and Differentia

Question No : 7 of 41

Let $f(z) = z^3 - 3z + 12$, then $f'(z)$ is _____

Answer (Please select your correct option)

- $z^2 - 3$
- $3z^2 - 3$
- $z^2 - 1$
- $z^3 - 3$

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MTH632 Complex Analysis and Differential Geometry

Question No : 3 of 41

The complex numbers $f(z) = 3iz/5$ then $\lim_{z \to 10} f(z) =$ ____

Answer (Please select your correct option)

- $6i$
- $2i$
- $3i$
- $10i$

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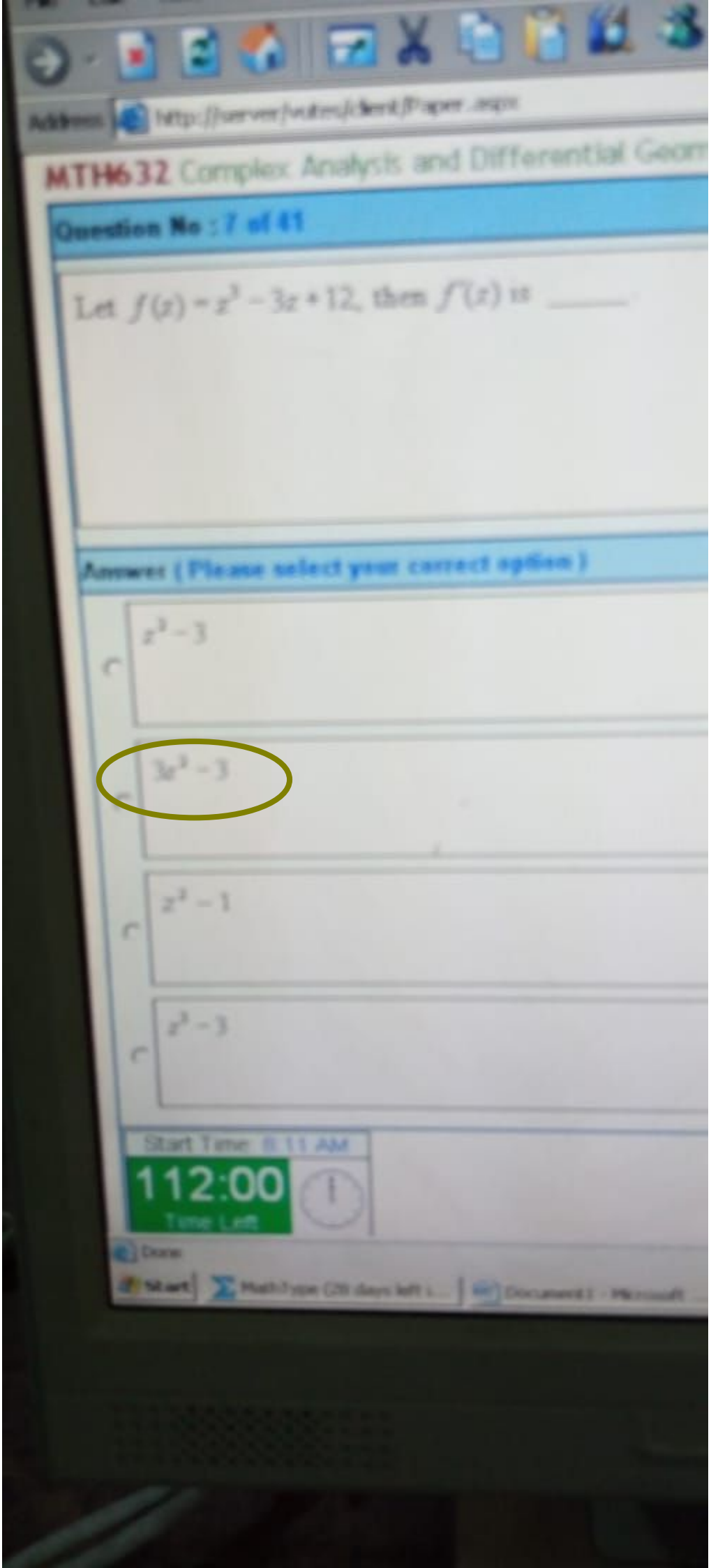
MTH632 Complex Analysis and Differential Geom
Question No : 7 of 41
Let $f(z) = z^3 - 3z + 12$, then $f'(z)$ is _____
Answer (Please select your correct option)
$z^3 - 3$
$3z^3 - 3$
$z^3 - 1$
$z^3 - 3$
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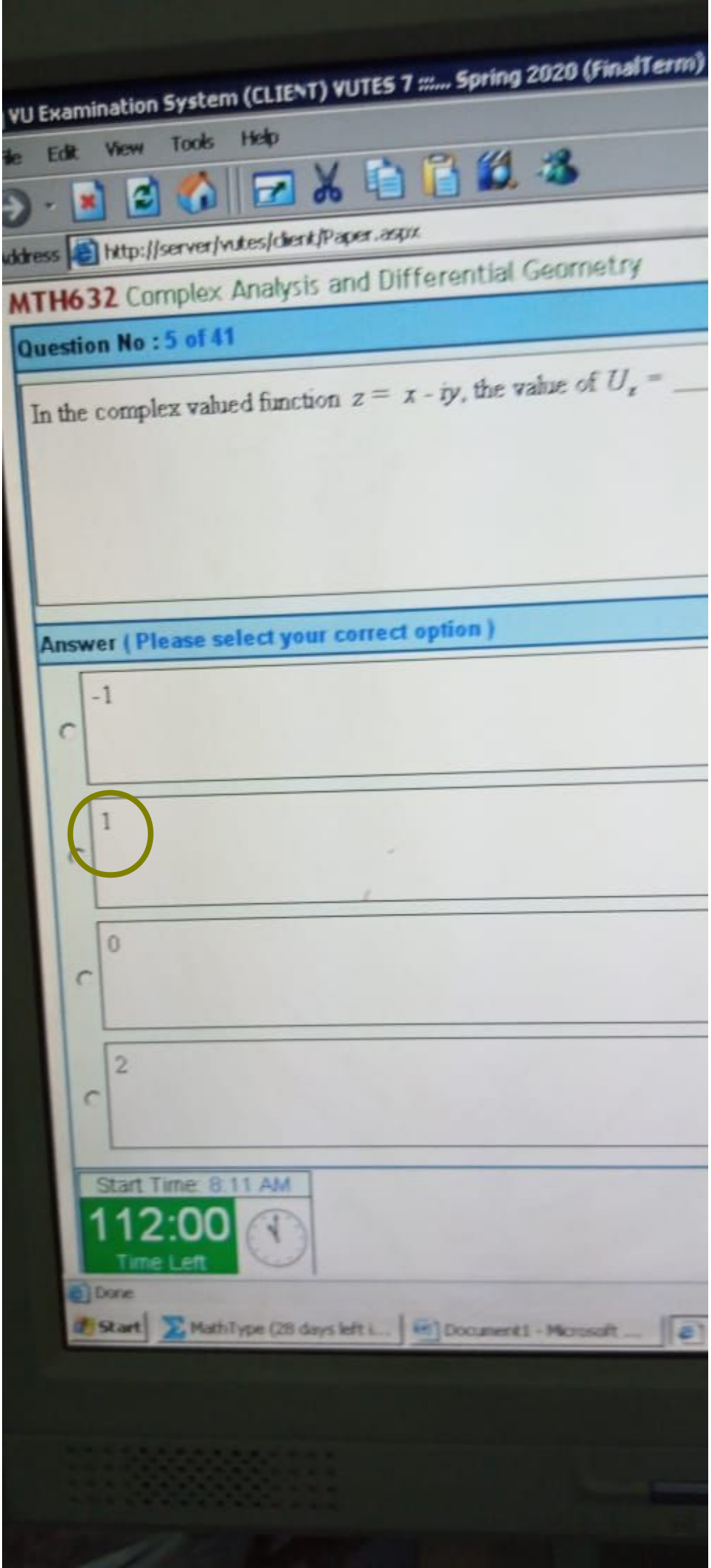
Time Left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MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 41
In the complex valued function $z = x - iy$, the value of $U_x =$ ___
Answer (Please select your correct option)
-1
1
0
2
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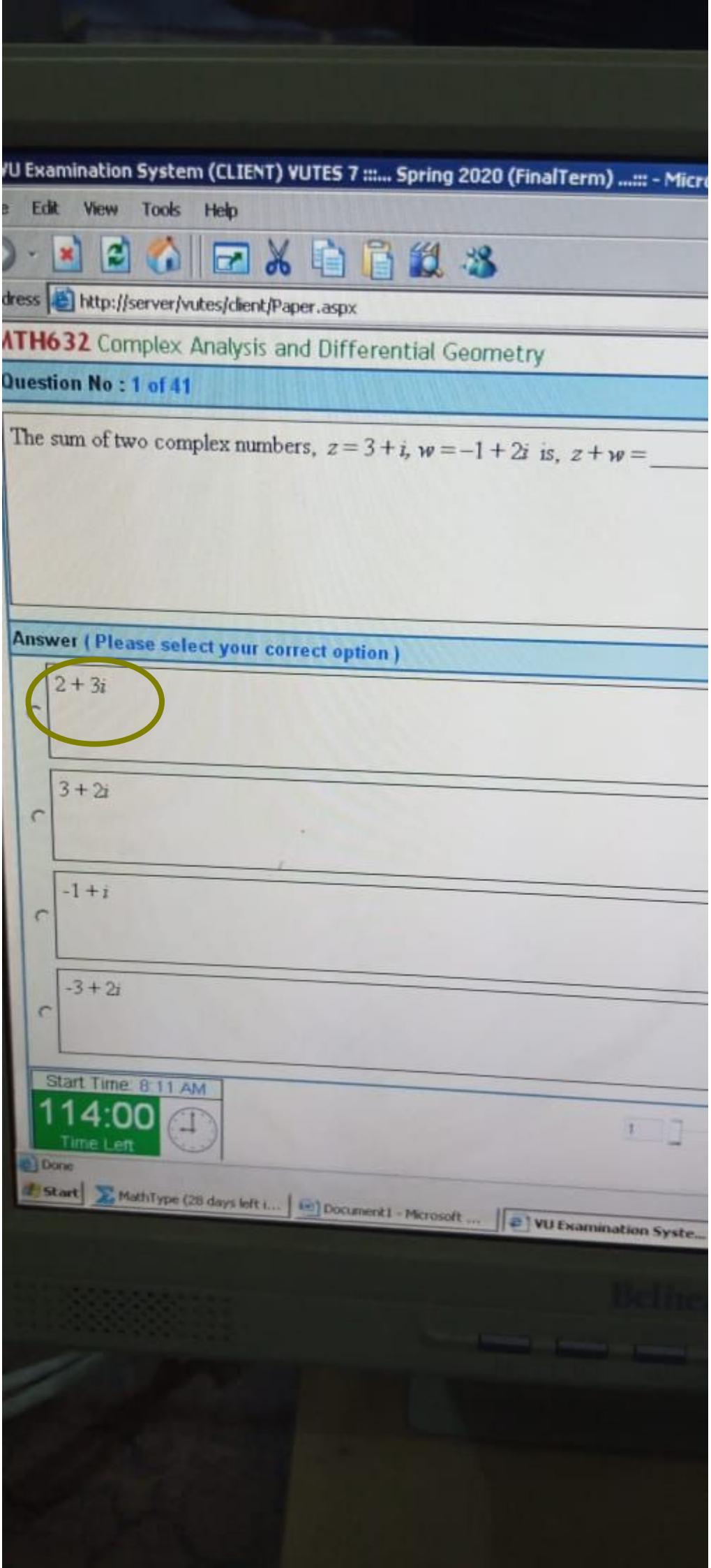
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TH632 Complex Analysis and Differential Geometry
Question No : 1 of 41
The sum of two complex numbers, $z = 3 + i$, $w = -1 + 2i$ is, $z + w =$ ____
Answer (Please select your correct option)
2 + 3i
3 + 2i
-1 + i
-3 + 2i
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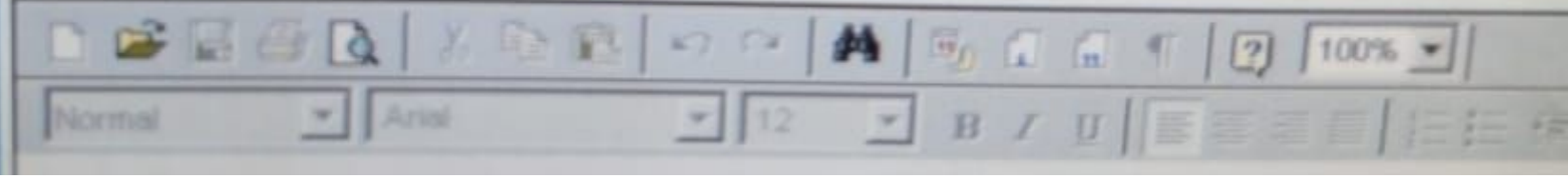
MTH632 Complex Analysis and Differential Geometry

Question No : 31 of 41

Find $f'(z)$, when $f(z)=(1-4z^2)^3$.

Answer (Please click here to Add Answer)



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MTH632 Complex Analysis and Differential Geometry

Question No : 36 of 41 Mar

If $\lim_{n\to\infty} z_n = z$ then by using the inequality, $||z_n|-|z|| \le |z_n - z|$, show that $\lim_{n\to\infty} |z_n| = |z|$.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis Differential Geometry

Question No : 32 of 41 **Marks**

Test whether the function $w = 2xy + i(x^2 - y^2)$ is analytic or not.

Answer (Please click here to Add Answer)

100%

MTH632 Complex Analysis and Differential Geometry

Question No : 33 of 41

Evaluate the integral: $\int_1^2 \left(\frac{1}{t} - i\right)^2 dt$.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 38 of 41 Mar

Let f and g be functions, ϕ and ψ be 1-forms. Then show that $d(fg) = df\,g + f\,dg$.

MTH632 Complex Analysis and Differential Geometry

Question No : 37 of 41

Show that $\frac{d}{dz}(\operatorname{arcsinh}(z)) = \frac{1}{\sqrt{1+z^2}}$.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 40 of 41

If $u = \log(x^2 + y^2)$ find v and $f(z)$ such that $f(z) = u + iv$ is analytic.

Answer (Please click here to Add Answer)

100%

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Question No : 23 of 41 **Marks: 1 (Budgeted Tim**

Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C \frac{z^2}{z-3} dz =$ _____

Answer (Please select your correct option)

- 3
- 0
- 1
- -3

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MTH632 Complex Analysis and Differential Geometry

Question No : 41 of 41 MC1

Marks: 5 (Budgeted Time Min)

Let C_1 denote the positively oriented boundary of the square whose sides lie along the lines $x = \pm 1, y = \pm 1$ and let C_2 be the positively oriented circle $|z| = 4$ (given in Fig. below). Explain why, $\int_C f(z)dz = \int_{C_1} f(z)dz$ where $f(z) = \dfrac{z+2}{\sin(z/2)}$?

Answer (Please click here to Add Answer)

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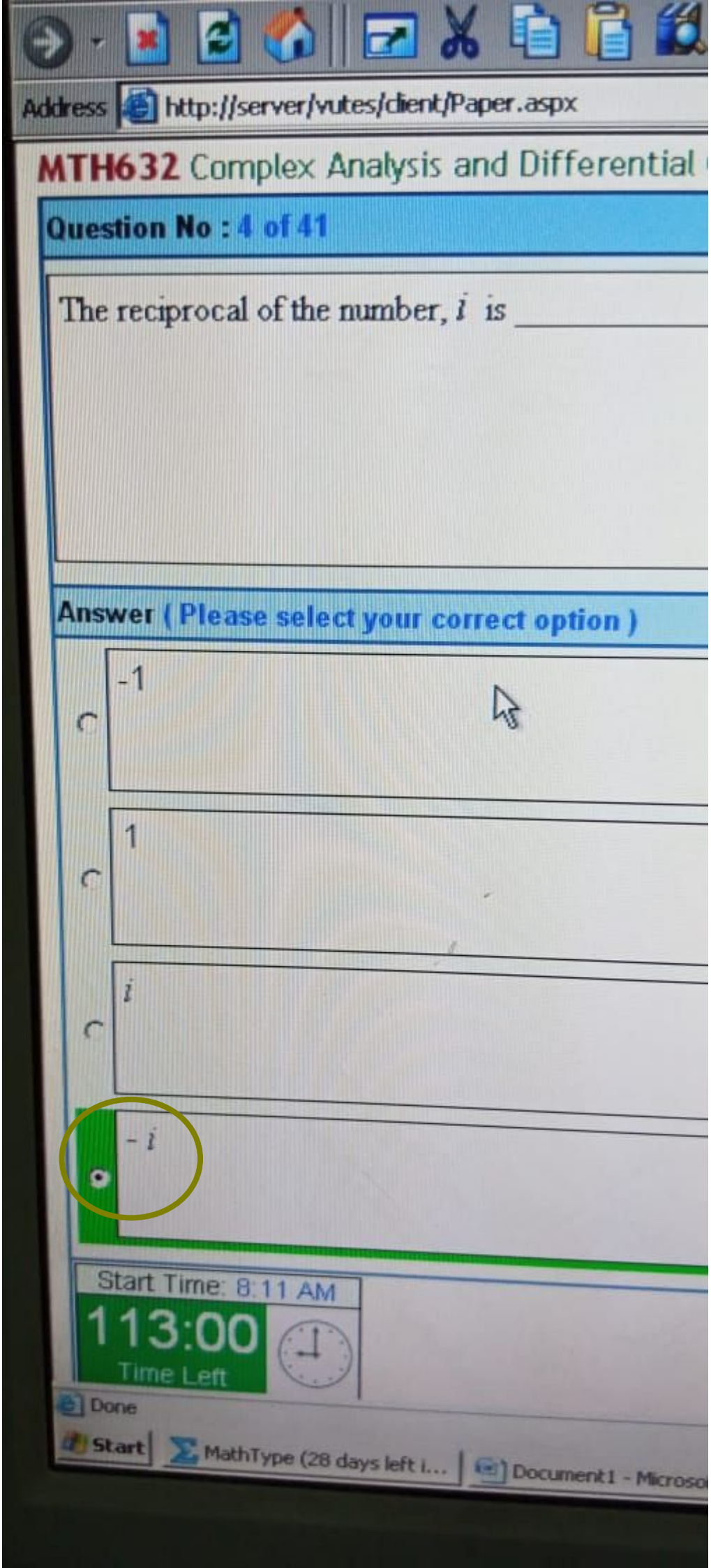
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MTH632 Complex Analysis and Differential
Question No : 4 of 41
The reciprocal of the number, i is _____
Answer (Please select your correct option)
-1
1
i
$-i$
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MTH632 Complex Analysis and Differential Geometry

Question No : 35 of 41

Evaluate $\lim_{z \to i} \frac{(z^2 + 4z + 2)}{z + 1}$.

Answer (Please click here to Add Answer)

MTH632 Complex Analysis and Differential Geometry

Question No : 10 of 41

Marks: 1 (Budgeted Time Min)

The sequence $z_n = \frac{i}{n^2}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____.

Answer (Please select your correct option)

- 1
- 0
- infinity
- 1/2

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Question No : 23 of 41 Marks: 1 (Budgeted Time Min)

Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C \frac{z^2}{z-3} dz =$ _____.

Answer (Please select your correct option)

- 3
- 0
- 1
- -3

23

Address http://server/vutes/client/login.aspx

Question No : 24 of 41 Marks: 1 (Budgeted Time

By using the Extension of Cauchy Integral formula $\int_z \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____

Answer (Please select your correct option)

- $f^{(n)}(z_0)$
- $2\pi i\, f^{(n)}(z_0)$
- $\frac{f^{(n)}(z_0)}{n!}$
- $\frac{2\pi i}{n!} f^{(n)}(z_0)$

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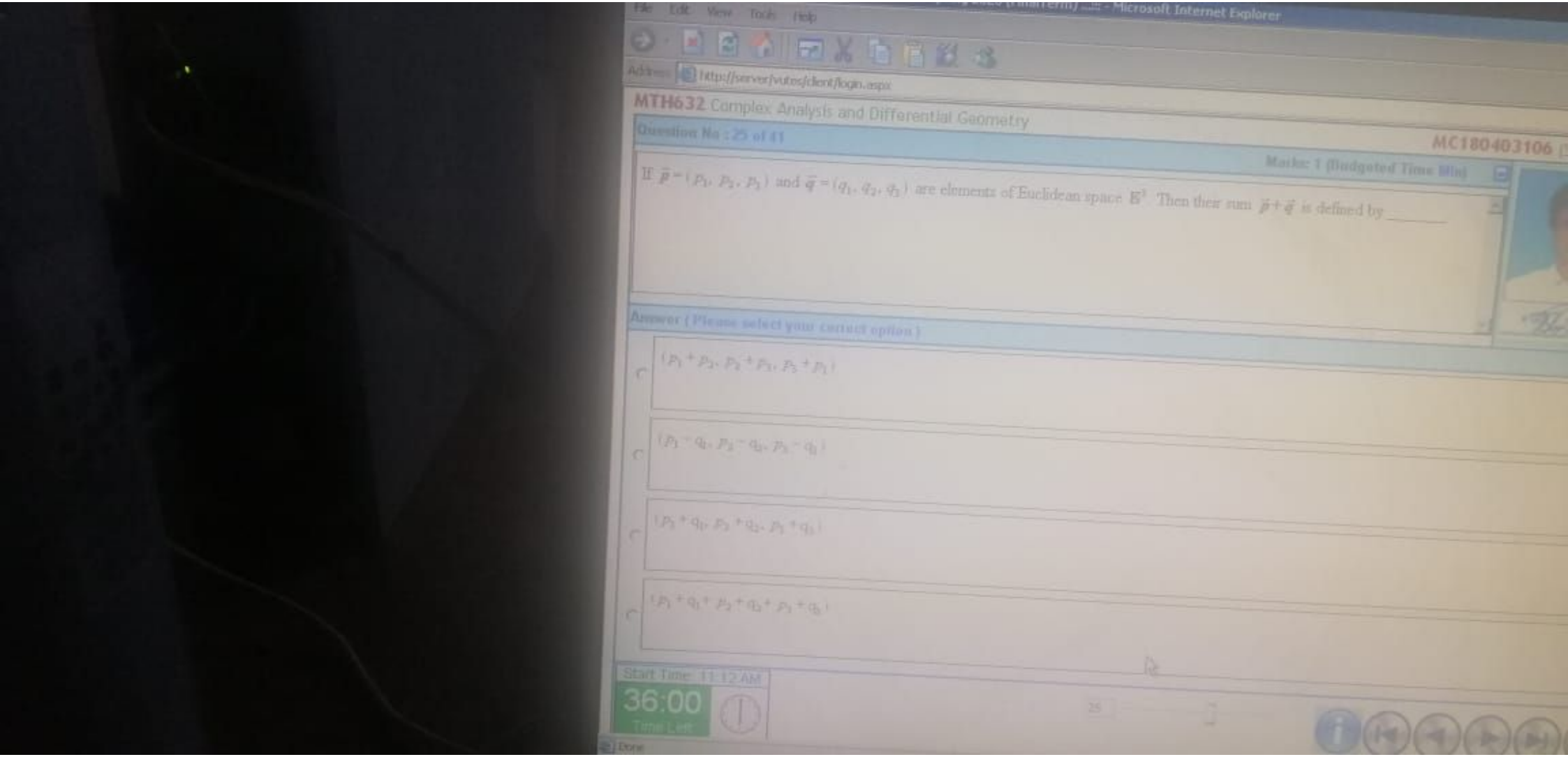
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MTH632 Complex Analysis and Differential Geometry
Question No : 25 of 41
MC180403106
Marks: 1 (Budgeted Time: 1Min)
If $\vec{p} = (p_1, p_2, p_3)$ and $\vec{q} = (q_1, q_2, q_3)$ are elements of Euclidean space E^3. Then their sum $\vec{p} + \vec{q}$ is defined by _____
Answer (Please select your correct option)
C $(p_1 + p_2, p_2 + p_3, p_3 + p_1)$
C $(p_1 - q_1, p_2 - q_2, p_3 - q_3)$
C $(p_1 + q_1, p_2 + q_2, p_3 + q_3)$
C $(p_1 + q_1 + p_2 + q_2 + p_3 + q_3)$
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Question No : 30 of 41 **Marks: 1 (Budgeted Time Min)**

The restriction of a curve $\alpha: I \to R^1$ to a close interval $[a, b]$ is known as a _____.

Answer (Please select your correct option)

- curve
- segment
- line
- curve segment

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Address http://server/vutes/client/Paper.aspx

Question No : 29 of 41 **Marks: 1 (Budgeted Time Min)**

If p and q are points in R^3 then _____ is the symmetry property of dot product.

Answer (Please select your correct option)

- $p.q$
- $q.p$
- $p.q=q.p$
- $p.p=q.q$

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MTH632 Complex Analysis and Differential Geometry

Question No : 18 of 41 Marks: 1 (Budgeted Time

The principal value of i^i is _____.

Answer (Please select your correct option)

- $\exp(\frac{\pi}{2})$
- $\exp(-\frac{\pi}{2})$
- $\exp(i)$
- $\exp(-i)$

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Address http://server/vutes/client/Paper.aspx

Question No : 26 of 41 **Marks: 1 (Budgeted Time Min)**

Let $f(z)$ be analytic function and C is contour enclosing point z_0, then Cauchy Integral formula states $f(z_0)$= -

_____.

Answer (Please select your correct option)

- $\frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}dz$
- $\int_C \frac{f(z)}{z-z_0}dz$
- $\frac{1}{2\pi i}\int_C \frac{f(z)}{z+z_0}dz$
- $\int_C \frac{f(z)}{z+z_0}dz$

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Question No : 28 of 41 **Marks: 1 (Budgeted Time Min)**

The differential of f is, $df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$, where df calculates the small changes in the value of f when there is small change in _____

Answer (Please select your correct option)

- (a) x
- (b) y
- (c) z
- (d) All (a), (b) and (c)

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Marks: 1 (Budgeted Time

Question No : 23 of 41

Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C \frac{z^2}{z-3} dz =$ _____

Answer (Please select your correct option)

- 3
- 0
- 1
- -3

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23

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Marks: 1 (Budgeted Time Min)

Question No : 21 of 41

The answer of definite integral of a complex valued function $w(t) = \int_1^2 i dt$ is _____

Answer (Please select your correct option)

- 1
- 2
- i
- 0

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Question No : 22 of 41 Marks: 1 (Budgeted Time Min)

Let $f(z)=z$ be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____

Answer (Please select your correct option)

- 1-i
- 0
- 1+i
- i

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Marks: 1 (Budgeted

Question No : 20 of 41

The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is oriented in _____ direction.

Answer (Please select your correct option)

- upward
- downward
- counter-clockwise
- clockwise

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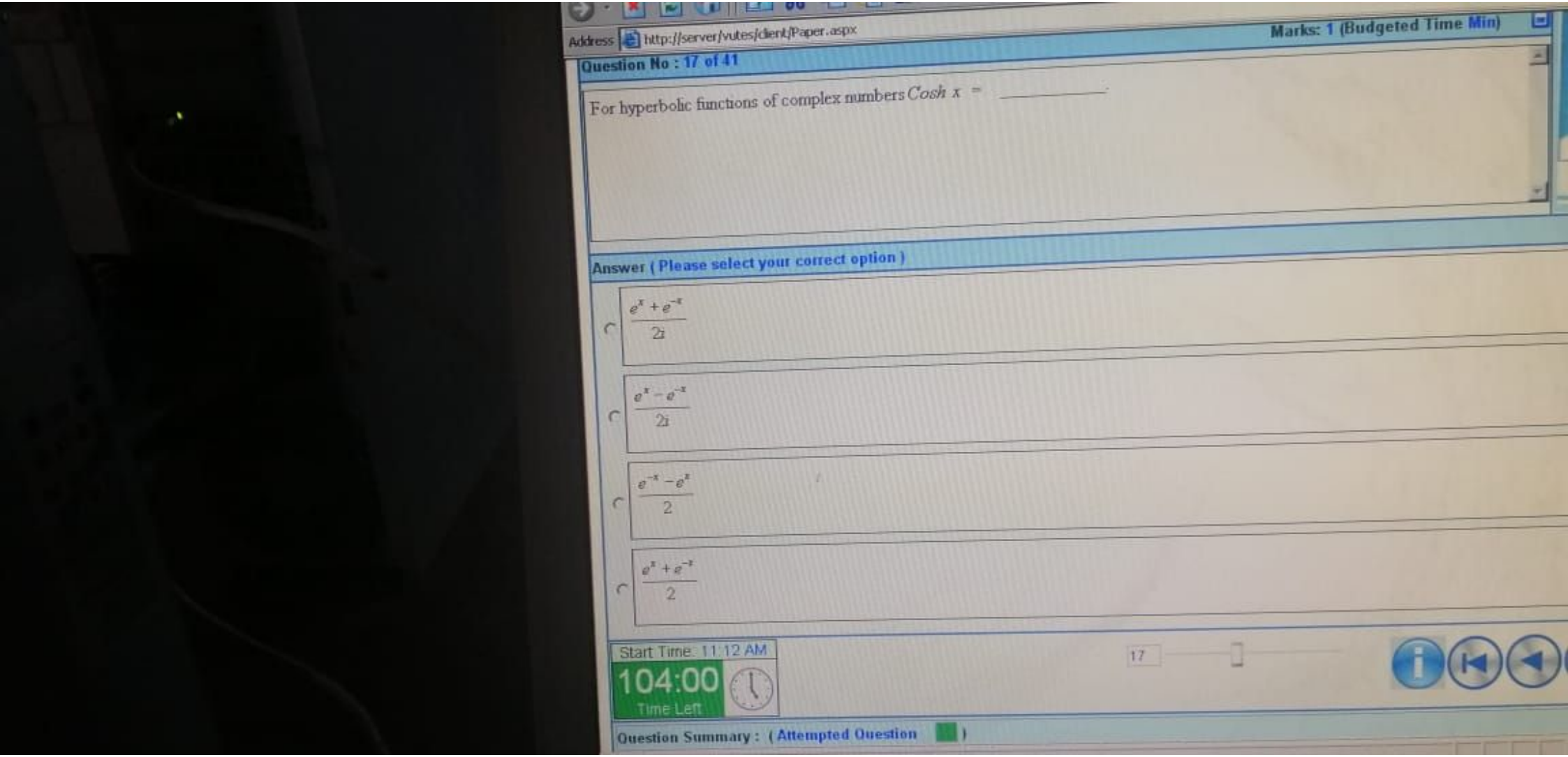
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Marks: 1 (Budgeted Time Min)
Question No : 17 of 41
For hyperbolic functions of complex numbers Cosh x = _____
Answer (Please select your correct option)
$\frac{e^x + e^{-x}}{2i}$
$\frac{e^x - e^{-x}}{2i}$
$\frac{e^{-x} - e^x}{2}$
$\frac{e^x + e^{-x}}{2}$
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Question Summary : (Attempted Question)

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Question No : 19 of 41 Marks: 1 (Budgeted Time Min)

Let Z be a complex valued function then $\tanh Z =$ _____.

Answer (Please select your correct option)

- $\frac{e^{-z}}{e^{z}+e^{-z}}$
- $\frac{e^{z}+e^{-z}}{e^{z}-e^{-z}}$
- $\frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$
- $\frac{e^{-z}}{e^{z}-e^{-z}}$

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Question No : 16 of 41 **Marks: 1 (Budgeted Time Min)**

The points at which $f(z) = \frac{1}{z^2+1}$ is not analytic are _____

Answer (Please select your correct option)

- 1 and -1
- i and $-i$
- 1 and i
- -1 and $-i$

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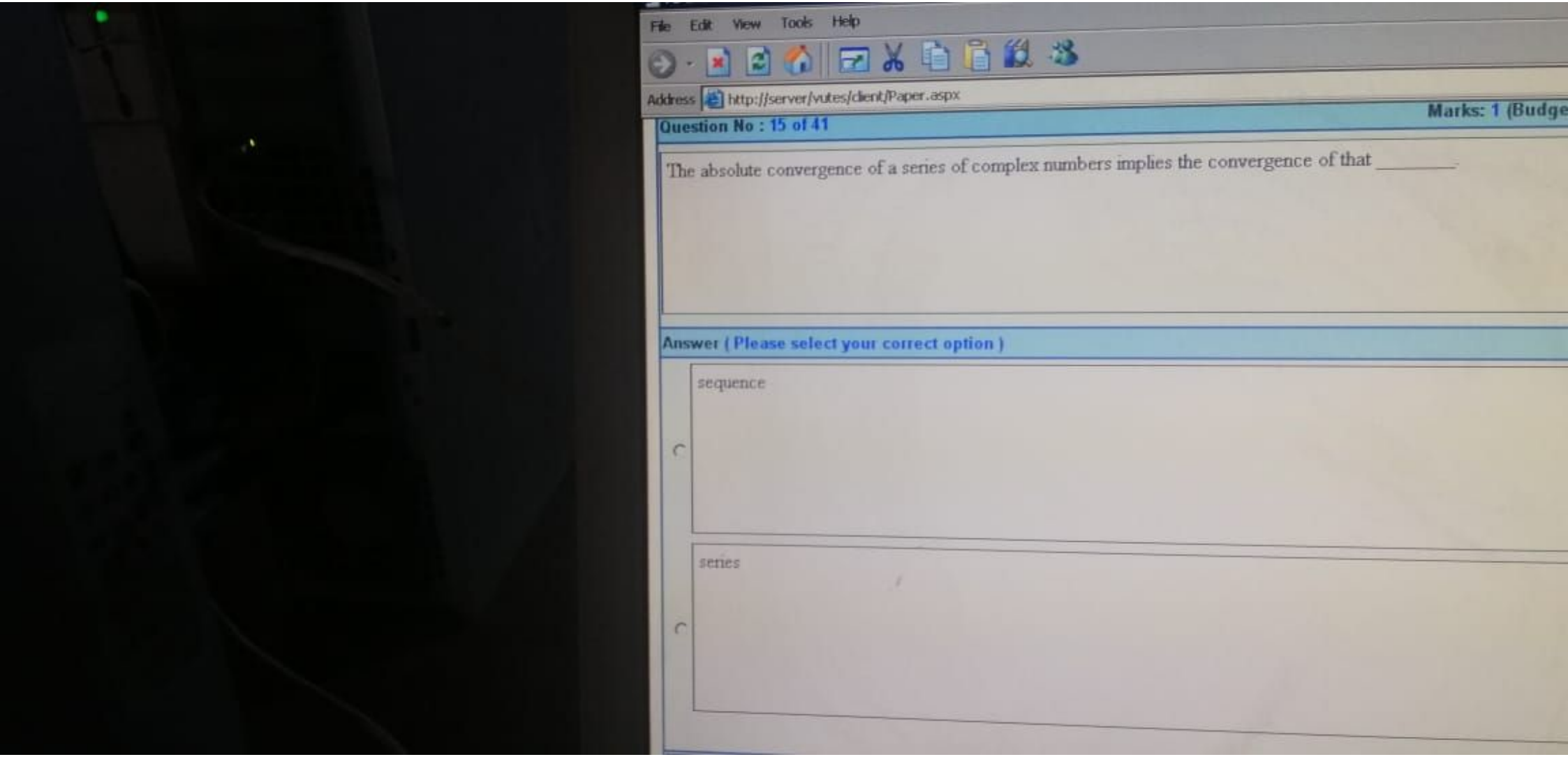
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Question No : 15 of 41
Marks: 1 (Budge
The absolute convergence of a series of complex numbers implies the convergence of that _____.
Answer (Please select your correct option)
sequence
series

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Question No : 14 of 41 **Marks: 1 (Budgeted Time Mi**

Harmonic conjugate of $u(x,y) = e^{y}\cos x$ is _____.

Answer (Please select your correct option)

- $e^{x}\cos y + c$
- $e^{x}\sin y + c$
- $e^{y}\sin x + c$
- $-e^{y}\sin x + c$

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Question No : 13 of 41 **Marks: 1 (Budgeted Time Min)**

If U and V are harmonic functions then $f(Z) = U + iV$ is _____.

Answer (Please select your correct option)

- analytic function.
- need not be analytic function.
- analytic function only at Z=0.
- none of the others.

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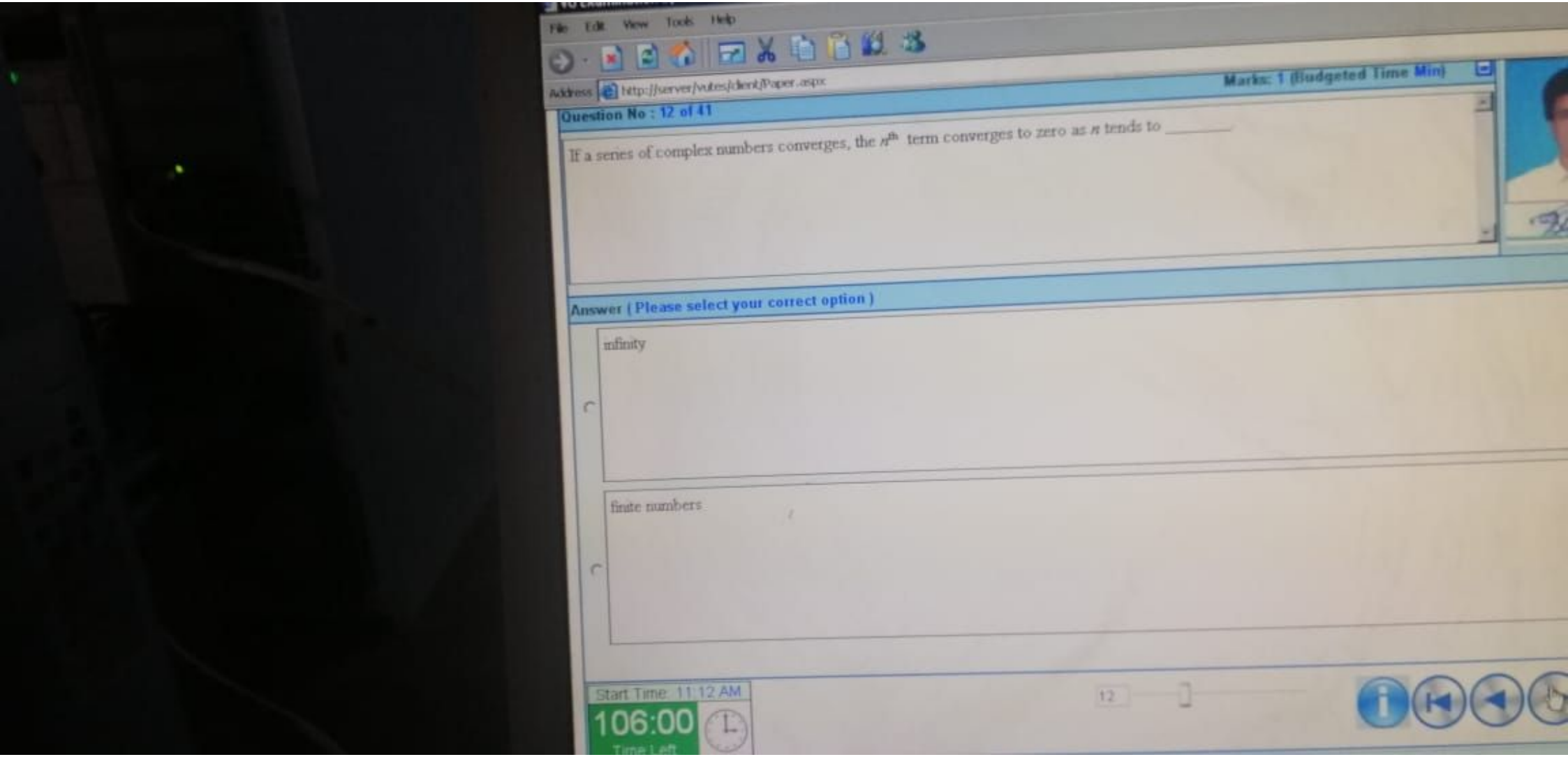
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Marks: 1 (Budgeted Time Min)
Question No : 12 of 41
If a series of complex numbers converges, the n^{th} term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
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Question No : 11 of 41 **Marks: 1 (Budgeted Time Min)**

A point at which a function ceases to be analytic is called a _____.

Answer (Please select your correct option)

- singular point
- non-singular point
- regular point
- non-regular point

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Question No : 10 of 41 Marks: 1 (Budgeted Time Min)

Any polynomial $p(z) = a_0 + a_1 z + a_2 z^2 + \ldots + a_n z^n \quad (a_n \neq 0)$ of degree n $(n \geq 1)$ has at least one zero. That is, there exists at least _____ z_0, such that $P(z_0) = 0$.

Answer (Please select your correct option)

- one point
- n points
- no points
- 10 points

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Marks: 1 (Budgeted Time Min)

Question No : 8 of 41

If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____.

Answer (Please select your correct option)

- harmonic in D.
- equal in D.
- homogeneous in D.
- steady in D.

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Question No : 9 of 41 **Marks: 1 (Budgeted Time Mi**

The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then _____.

Answer (Please select your correct option)

- $w = \ln r + i(\theta + 2n\pi)$
- $w = re^{i\theta}$
- $w = r + i(\theta + 2n\pi)$
- $w = -\ln r + i\,\theta$

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MTH632 Complex Analysis and Differential Geometry

MC18040310

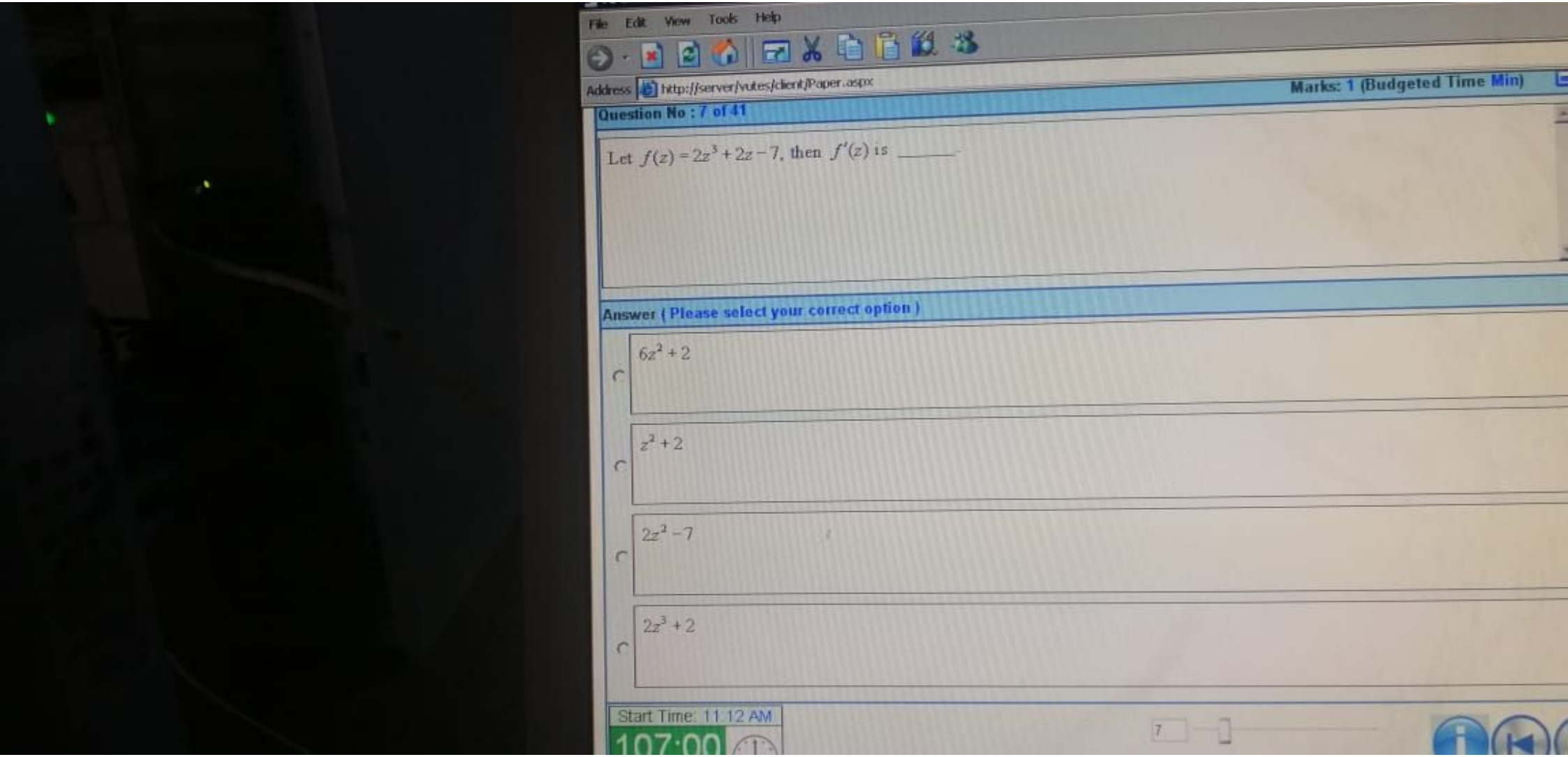
Question No : 6 of 41

Marks: 1 (Budgeted Time Min)

In the complex valued function $f(z) = \frac{1}{re^{i\theta}}$, the value of U_θ = _____.

Answer (Please select your correct option)

- $e^{i\theta}$
- $\sin\theta$
- $\frac{\sin\theta}{r}$
- $-\frac{\sin\theta}{r}$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Marks: 1 (Budgeted Time Min)
Question No : 7 of 41
Let $f(z) = 2z^3 + 2z - 7$, then $f'(z)$ is _____
Answer (Please select your correct option)
$6z^2 + 2$
$z^2 + 2$
$2z^2 - 7$
$2z^3 + 2$
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MC1804

MTH632 Complex Analysis and Differential Geometry

Marks: 1 (Budgeted Time Min)

Question No : 4 of 41

The absolute value of the number, $z = 1-i$ is _____.

Answer (Please select your correct option)

- 2
- $\sqrt{2}$
- 0
- 1

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MTH632 Complex Analysis and Differential Geometry **MC180403106** (SAD

Question No : 5 of 41 **Marks: 1 (Budgeted Time Min)**

The magnitude of $\exp(-2+3i)$ is _____.

Answer (Please select your correct option)

- exp(-2*3)
- exp(2*3)
- exp(3)
- exp(-2)

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MC180403106

MTH632 Complex Analysis and Differential Geometry

Question No : 3 of 41 **Marks: 1 (Budgeted Time Min)**

The conjugate of complex number $z=-3-3i$ is _____.

Answer (Please select your correct option)

- $3+3i$
- $-3+6i$
- $3-3i$
- $-3+3i$

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MTH632 Complex Analysis and Differential Geometry MC18

Question No : 2 of 41 **Marks: 1 (Budgeted Time Min)**

The modulus of $z = 3+2i$ is _____.

Answer (Please select your correct option)

- $\sqrt{5}$
- $\sqrt{6}$
- $\sqrt{12}$
- $\sqrt{13}$

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MTH632 Complex Analysis and Differential Geometry

Question No : 1 of 41 Marks: 1 (Budgeted Time M

The sum of two complex numbers, $z = 3 + i$, $w = -1 + 2i$ is, $z + w =$ _____.

Answer (Please select your correct option)

- $2 + 3i$
- $3 + 2i$
- $-1 + i$
- $-3 + 2i$

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MTH632 Complex Analysis and Differential Geometry **MC180**

Question No : 41 of 41 **Marks: 5 (Budgeted Time Min)**

Let C denote the positively oriented boundary of the square whose sides lie along the lines $x = \pm 2$ and $y = \pm 2$. Evaluate the integral $\int_C \frac{\tan(z/2)}{(z - x_0)^2} dz$, $(-2 < x_0 < 2)$ by using the extended Cauchy integral formula.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 40 of 41

Marks: 5 (Budgeted Time

Verify that $\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$ whenever $|z|<1$.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 39 of 41 **Marks: 5 (Bu**

Find a function w such that $w = u + iv$ is analytic, if $u = e^x \sin y$.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry **MC180**

Question No : 38 of 41 **Marks: 3 (Budgeted Time Min)**

Verify the Frenet formula for the relation, $\vec{n}' = \tau\vec{b} - k\vec{t}$, where $k > 0$ and τ are curvature and torsion of a curve respectively.

Answer (Please click here to Add Answer)

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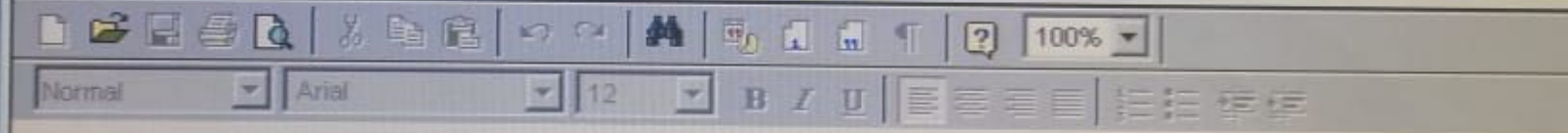
MTH632 Complex Analysis and Differential Geometry

Question No : 37 of 41 **Marks: 3 (Bu**

Evaluate the integral: $\int_0^{\pi/6} e^{i2t} dt$.

Answer (Please click here to Add Answer)



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MTH632 Complex Analysis and Differential Geometry

Question No : 36 of 41 **Marks: 3 (Budgeted Tim**

Prove that $u = e^{-y} \sin x$ is harmonic.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 35 of 41 **Marks: 3 (Budgeted Time**

Evaluate, $\lim_{z \to i} \frac{z^4 - 1}{z^2 + 1}$.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 32 of 41 **Marks: 2 (Budgeted**

Test whether the function $w = 2xy + i(x^2 - y^2)$ is analytic or not.

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry **MC1804**

Question No : 34 of 41 **Marks: 2 (Budgeted Time Min)**

If tangent vector along the helix $\vec{x}(t) = a(\cos t)\hat{e}_1 + a(\sin t)\hat{e}_2 + b(t)\hat{e}_3$, $a, b \neq 0$ is $\vec{x}'(t) = -a(\sin t)\hat{e}_1 + a(\cos t)\hat{e}_2 + b\hat{e}_3$.

Then evaluate the unit tangent vector $\vec{t}(s) = \dot{x} = \dfrac{\vec{x}'(t)}{|\vec{x}'(t)|}$.

Answer (Please click here to Add Answer)

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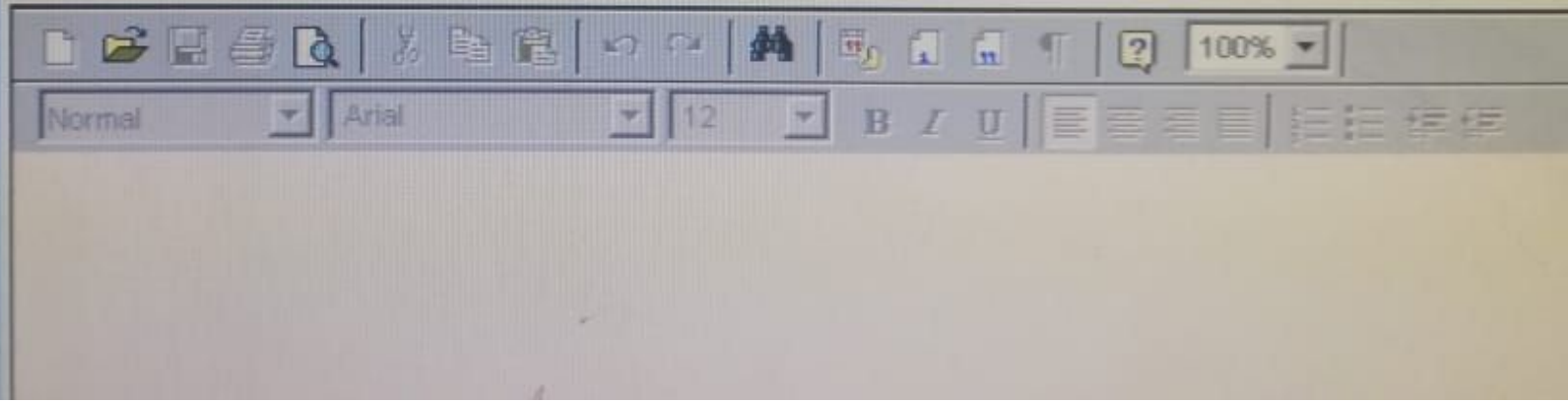
MTH632 Complex Analysis and Differential Geometry

Question No : 33 of 41

Show that $\frac{d}{dz}(\sin(z)) = \frac{1}{\sqrt{1-z^2}}$.

Answer (Please click here to Add Answer)



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MTH632 Complex Analysis and Differential Geometry

Question No : 31 of 41 MC18

Marks: 2 (Budgeted Time Min)

Evaluate $\lim_{z \to i} f(z)$, where $f(z) = 3z^2 - iz + 4 - 5i$ is a polynomial, it is continuous at all points in the complex plane.

Answer (Please click here to Add Answer)

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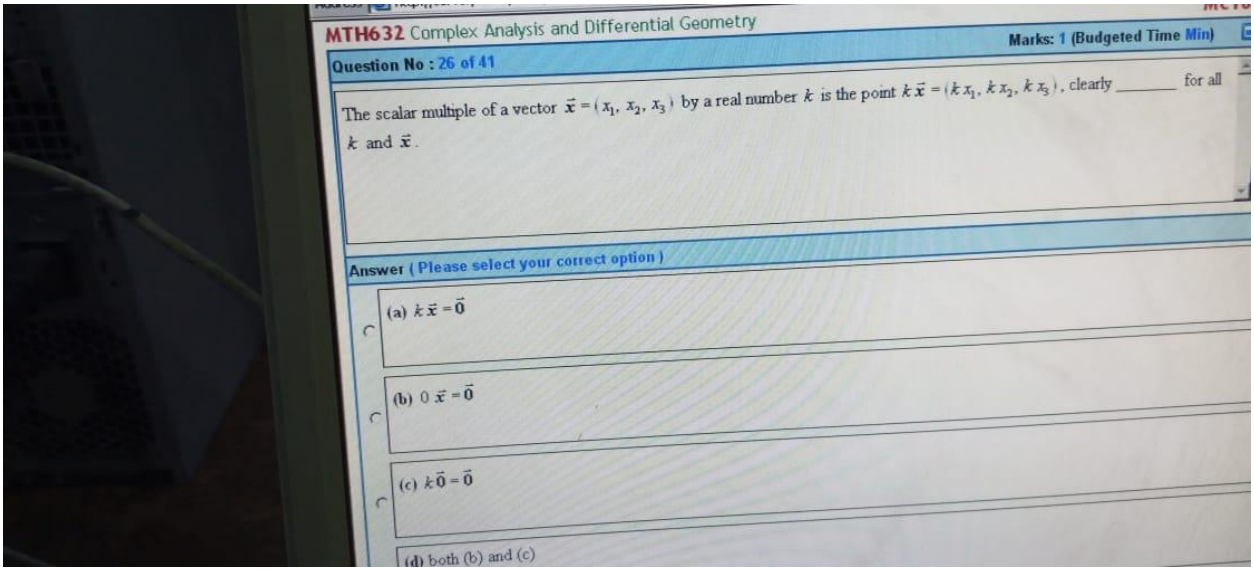


MTH632 Complex Analysis and Differential Geometry

Question No : 26 of 41

Marks: 1 (Budgeted Time Min)

The scalar multiple of a vector $\vec{x} = (x_1, x_2, x_3)$ by a real number k is the point $k\vec{x} = (kx_1, kx_2, kx_3)$, clearly _____ for all k and $\vec{x}$.

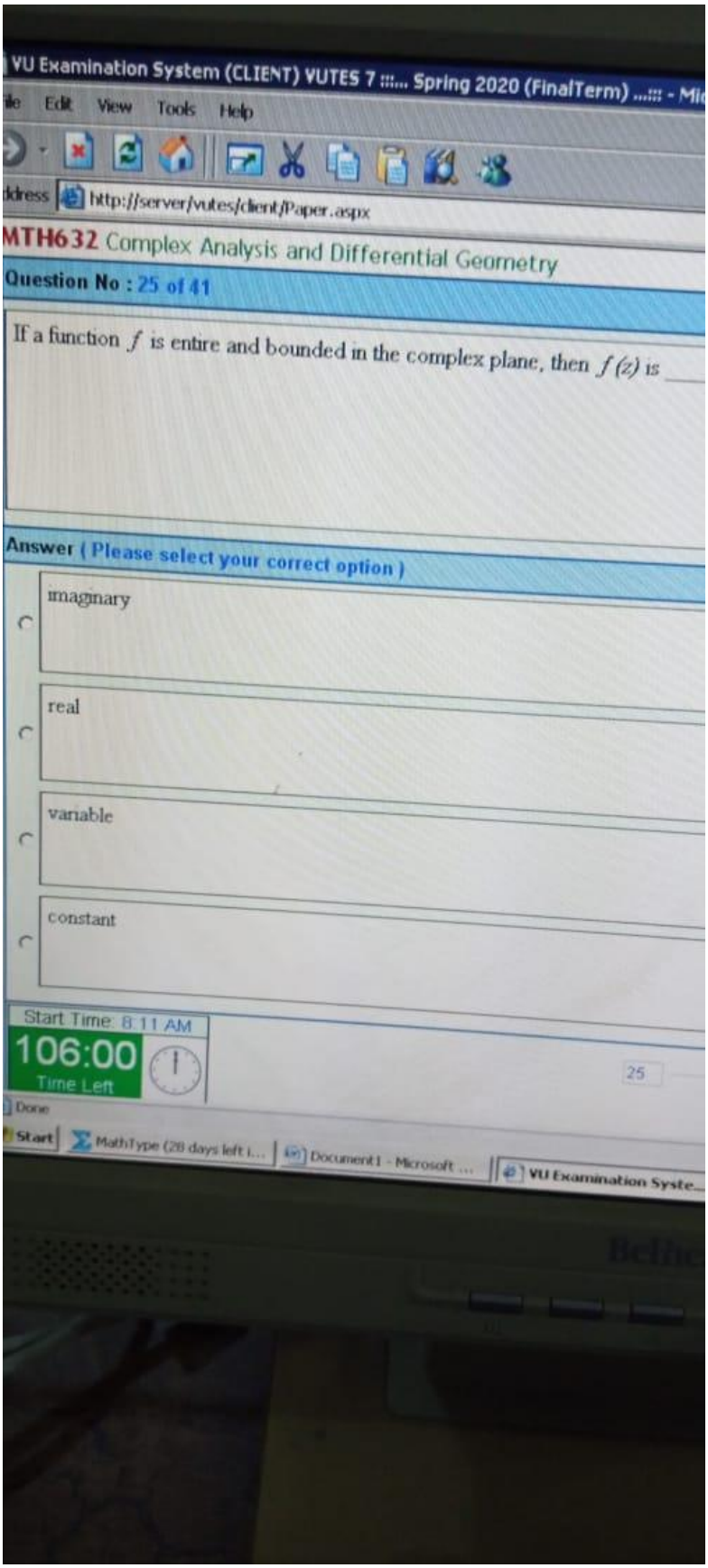
Answer (Please select your correct option)

(a) $k\vec{x} = \vec{0}$

(b) $0\vec{x} = \vec{0}$

(c) $k\vec{0} = \vec{0}$

(d) both (b) and (c)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MTH632 Complex Analysis and Differential Geometry
Question No : 25 of 41
If a function f is entire and bounded in the complex plane, then f(z) is ___
Answer (Please select your correct option)
imaginary
real
variable
constant
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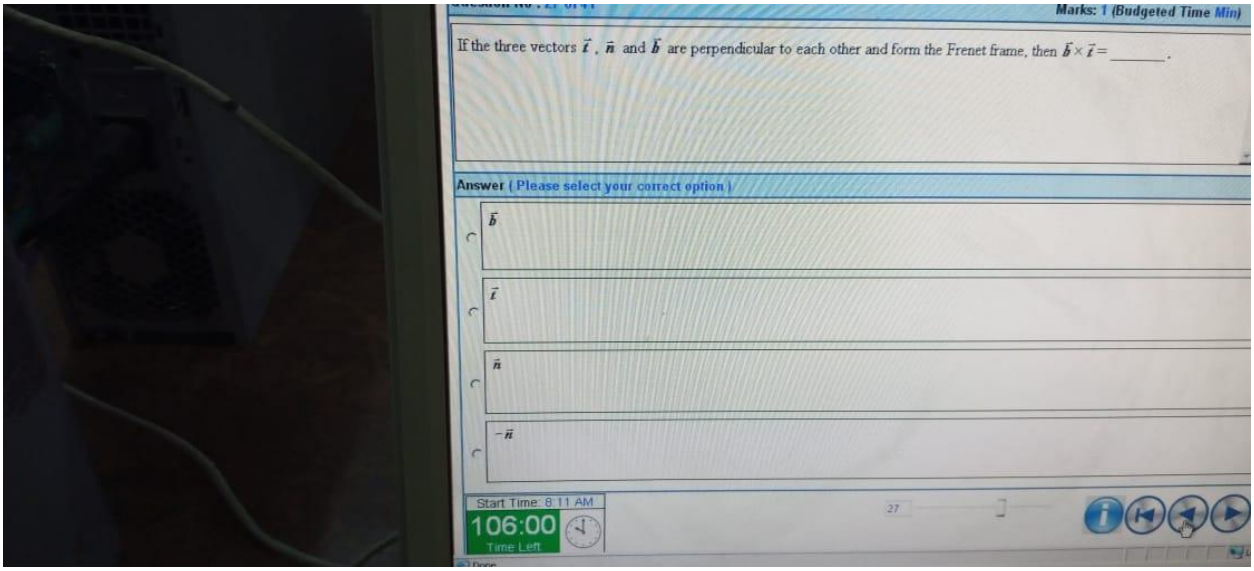
Marks: 1 (Budgeted Time Min)
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ _____.
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
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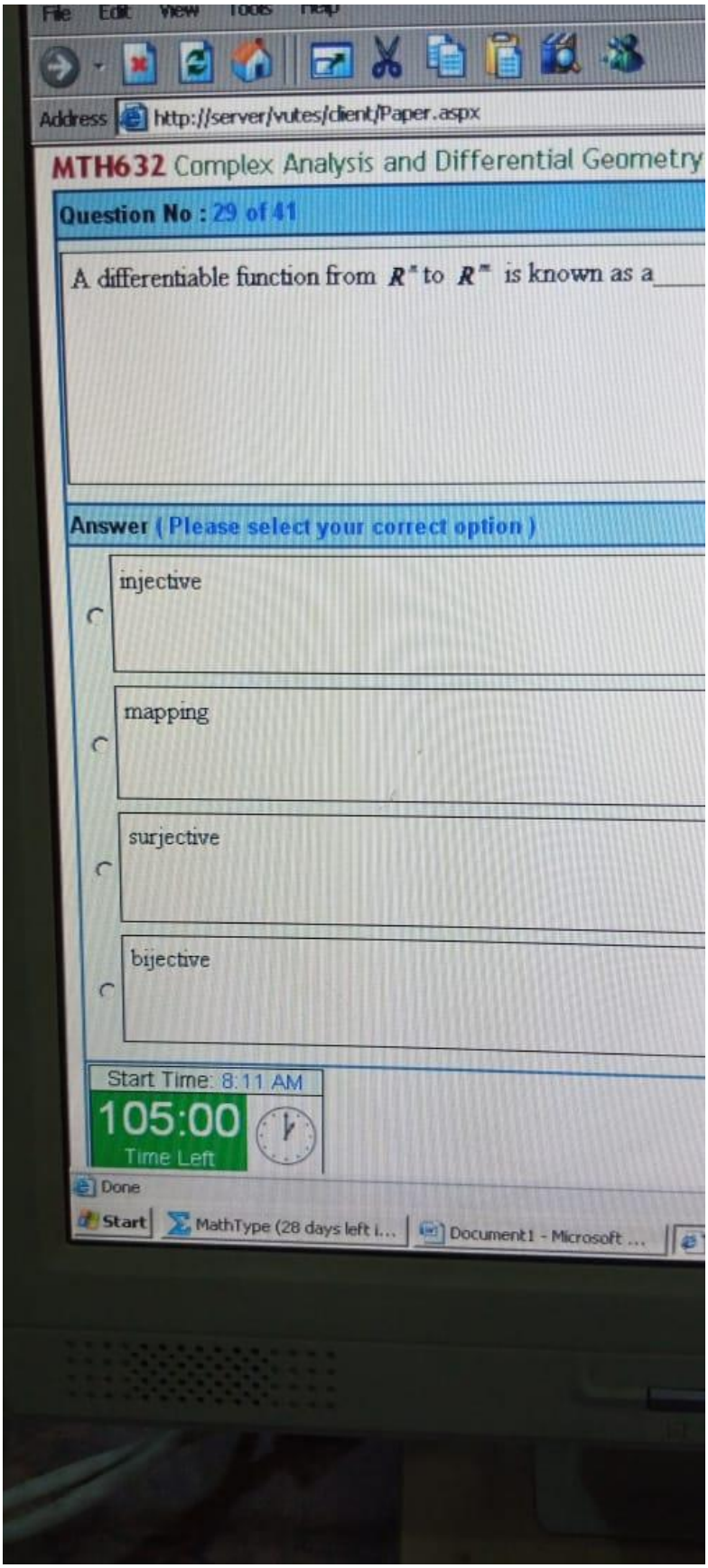
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MTH632 Complex Analysis and Differential Geometry
Question No : 29 of 41
A differentiable function from R^n to R^m is known as a____
Answer (Please select your correct option)
injective
mapping
surjective
bijective
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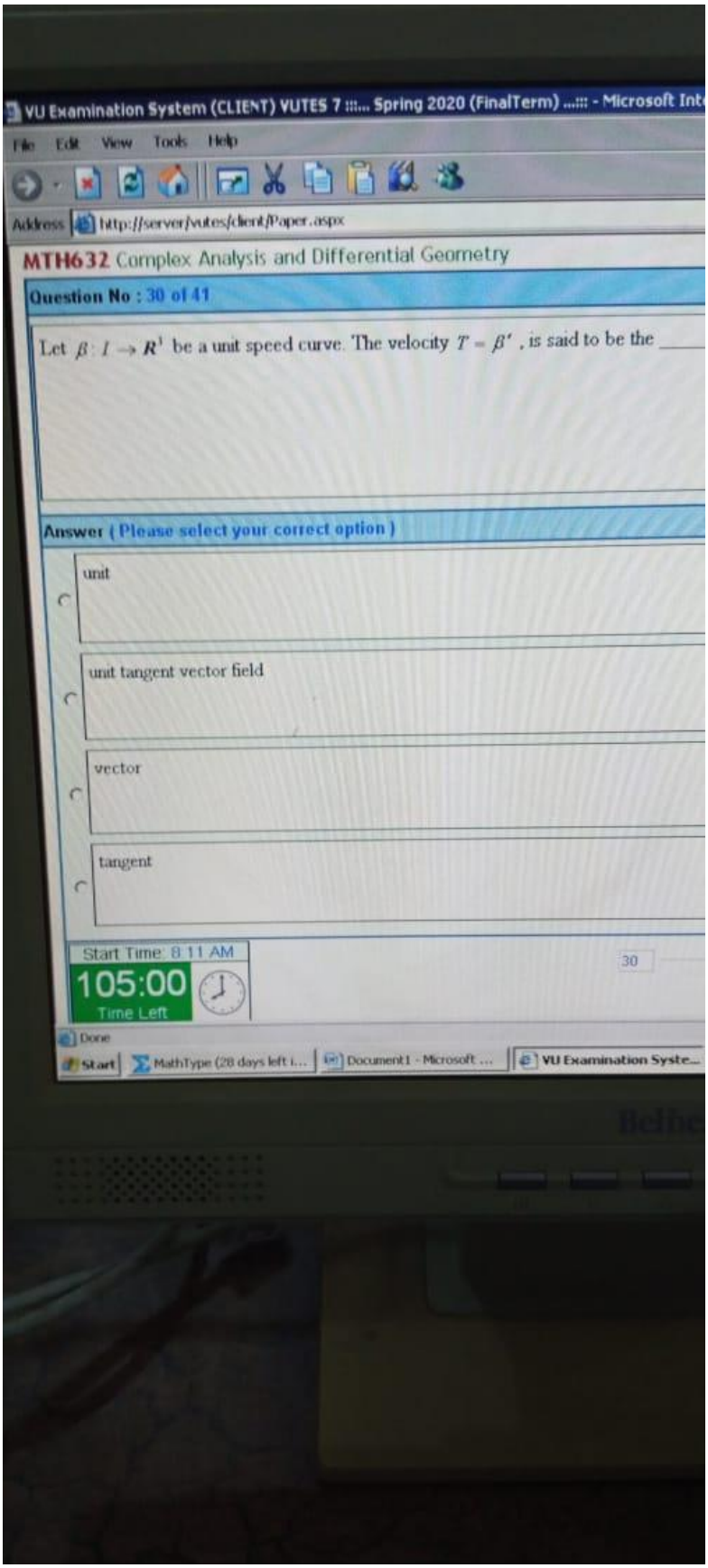
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MTH632 Complex Analysis and Differential Geometry
Question No : 30 of 41
Let $\beta : I \to R^1$ be a unit speed curve. The velocity $T = \beta'$, is said to be the ____
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
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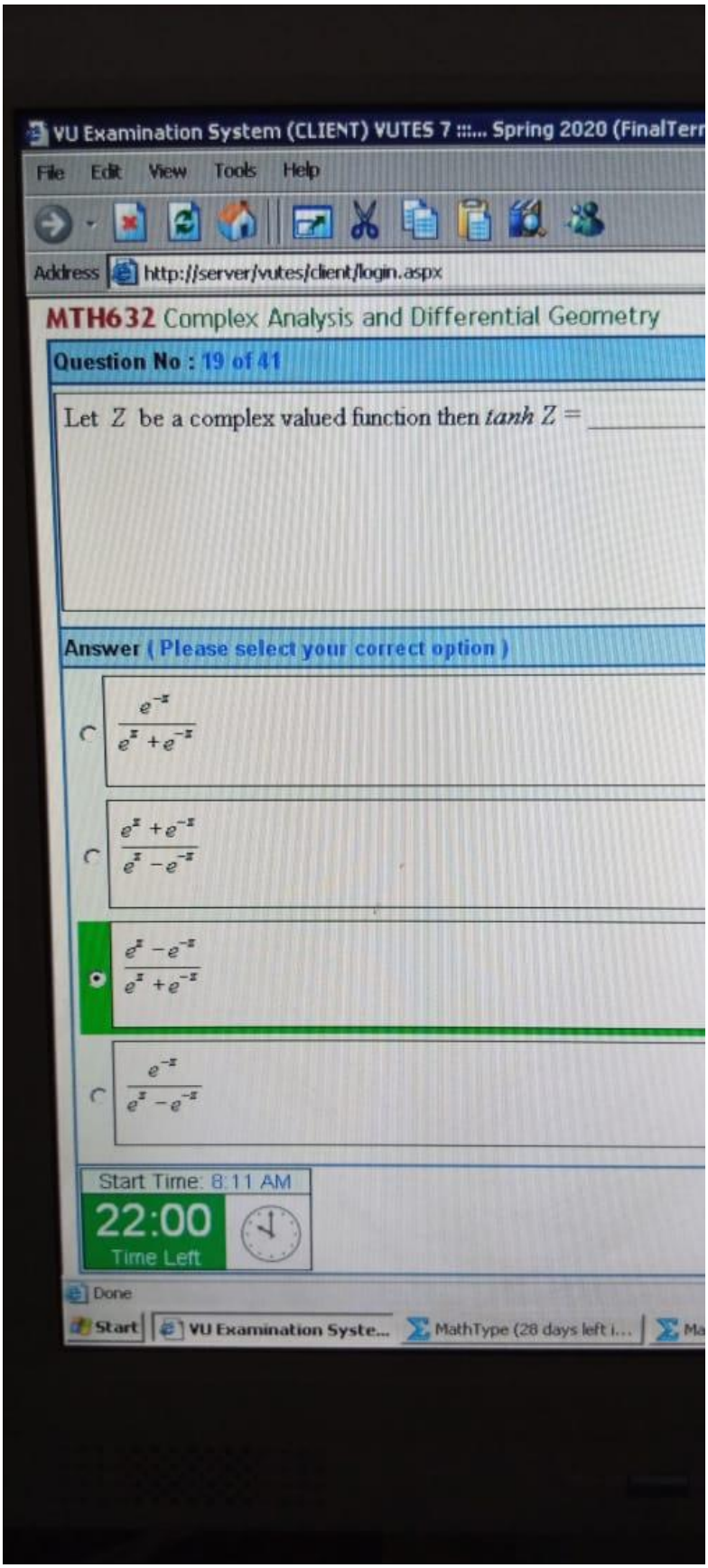
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Address http://server/vutes/client/login.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 19 of 41
Let Z be a complex valued function then tanh Z = _____
Answer (Please select your correct option)
$\frac{e^{-z}}{e^{z}+e^{-z}}$
$\frac{e^{z}+e^{-z}}{e^{z}-e^{-z}}$
$\frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$
$\frac{e^{-z}}{e^{z}-e^{-z}}$
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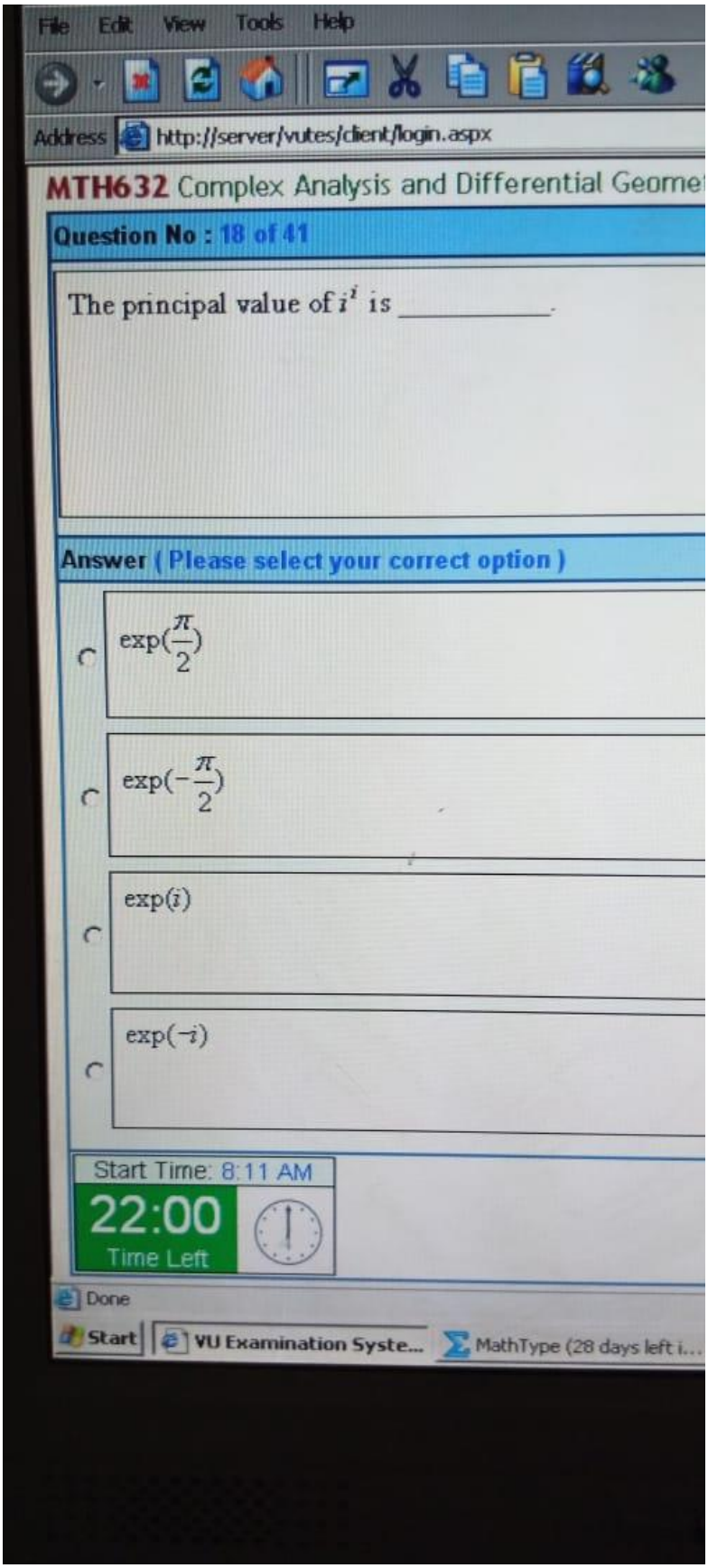
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Address http://server/vutes/client/login.aspx
MTH632 Complex Analysis and Differential Geome
Question No : 18 of 41
The principal value of i^i is _____.
Answer (Please select your correct option)
$\exp(\frac{\pi}{2})$
$\exp(-\frac{\pi}{2})$
$\exp(i)$
$\exp(-i)$
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Question No : 22 of 41 Marks: 1 (Budgeted Time

If $w(t)$ is a piecewise continuous complex-valued function defined on an interval $a \le t \le b$, then _____.

Answer (Please select your correct option)

- $\left|\int_a^b w(t)dt\right| \le \int_a^b |w(t)|dt$
- $\left|\int_a^b w(t)dt\right| \ge \int_a^b |w(t)|dt$
- $\left|\int_a^b w(t)dt\right| = \int_a^b |w(t)|dt$
- $\int_a^b w(t)dt \ge \left|\int_a^b |w(t)|dt\right|$

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MTH632 Complex Analysis and Differential Geometry

Question No : 41 of 41 MC1

Marks: 5 (Budgeted Time Min)

Let C_1 denote the positively oriented boundary of the square whose sides lie along the lines $x = \pm 1, y = \pm 1$ and let C_2 be the positively oriented circle $|z| = 4$ (given in Fig. below). Explain why, $\int_{C_1} f(z)dz = \int_{C_2} f(z)dz$ where $f(z) = \dfrac{z+2}{\sin(z/2)}$?

Answer (Please click here to Add Answer)

Address http://server/vutes/client/Paper.aspx

MTH632 Complex Analysis and Differential Geometry

Question No : 40 of 41 M

If $u = \log(x^2 + y^2)$ find v and $f(z)$ such that $f(z) = u + iv$ is analytic.

Answer (Please click here to Add Answer)

Address http://server/vutes/client/Paper.aspx

MTH632 Complex Analysis and Differential Geometry

Question No : 39 of 41

Show that

(a) $Log(1+i)^2 = 2Log(1+i)$ and (b) $Log(-1+i)^2 \neq 2Log(-1+i)$

Answer (Please click here to Add Answer)

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MTH632 Complex Analysis and Differential Geometry

Question No : 38 of 41

Let f and g be functions, ϕ and ψ be 1-forms. Then show that $d(fg) = df\,g + f\,dg$.

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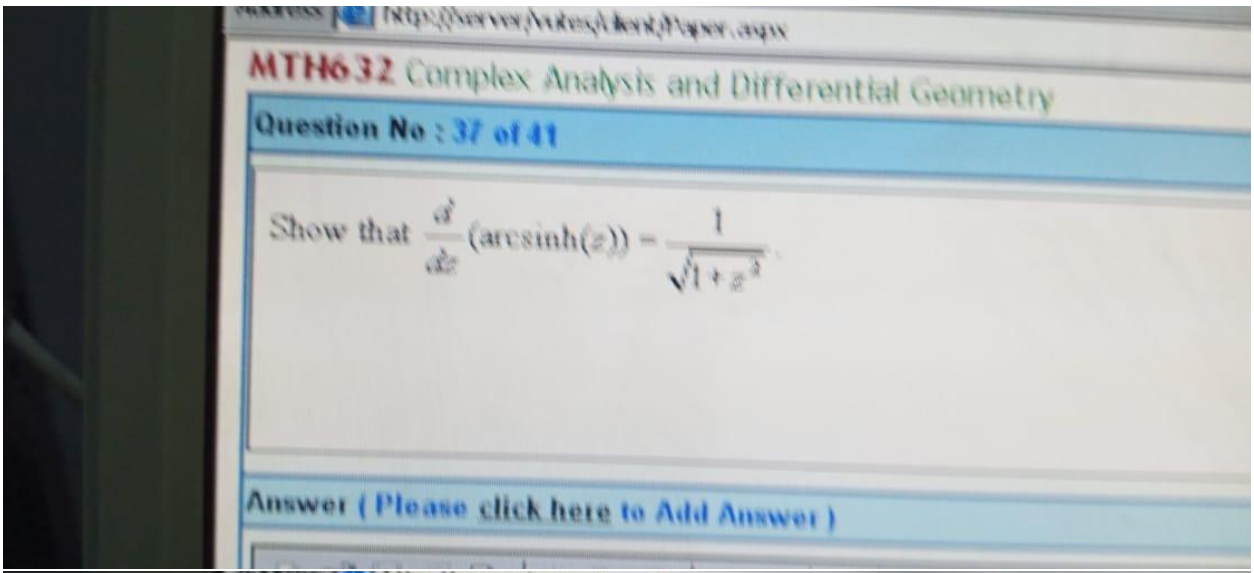
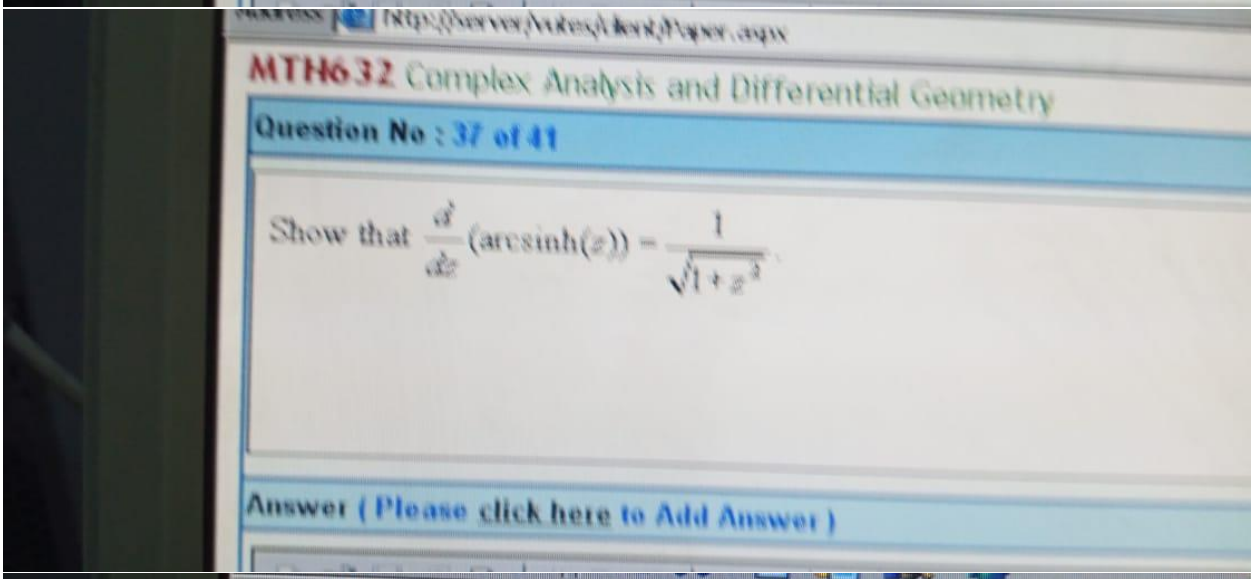
MTH632 Complex Analysis and Differential Geometry

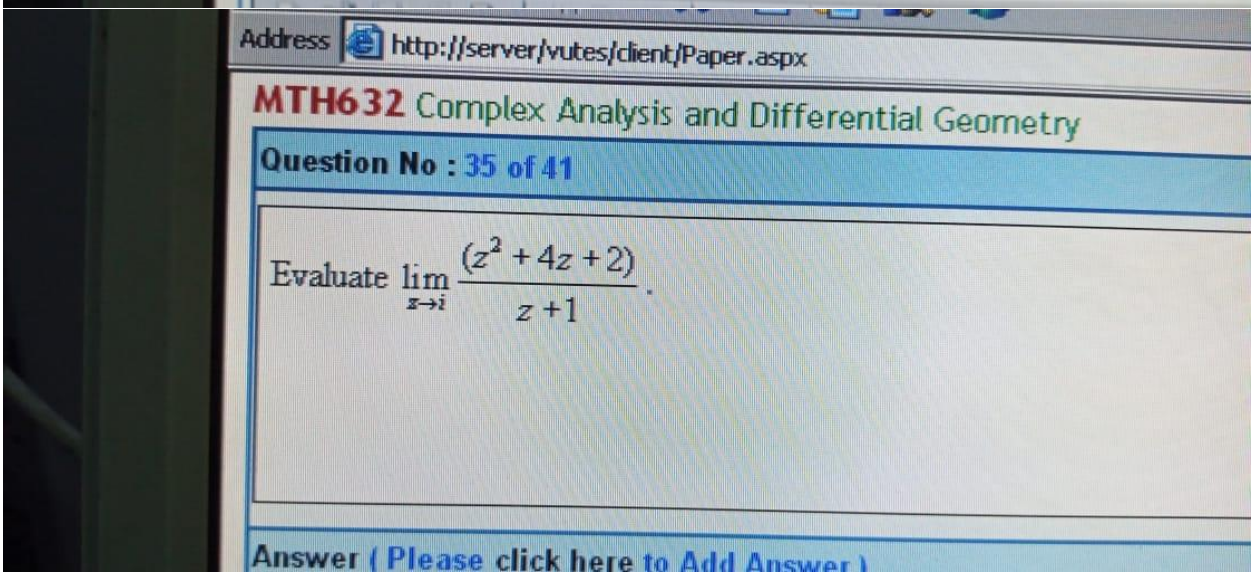
Question No : 36 of 41

If $\lim_{n\to\infty} z_n = z$ then by using the inequality, $||z_n| - |z|| \le |z_n - z|$, show that $\lim_{n\to\infty} |z_n| = |z|$.

Answer (Please click here to Add Answer)


MTH632 Complex Analysis and Differential Geometry
Question No : 37 of 41
Show that $\frac{d}{dz}(\operatorname{arcsinh}(z)) = \frac{1}{\sqrt{1+z^2}}$.
Answer (Please click here to Add Answer)


MTH632 Complex Analysis and Differential Geometry
Question No : 37 of 41
Show that $\frac{d}{dz}(\operatorname{arcsinh}(z)) = \frac{1}{\sqrt{1+z^2}}$.
Answer (Please click here to Add Answer)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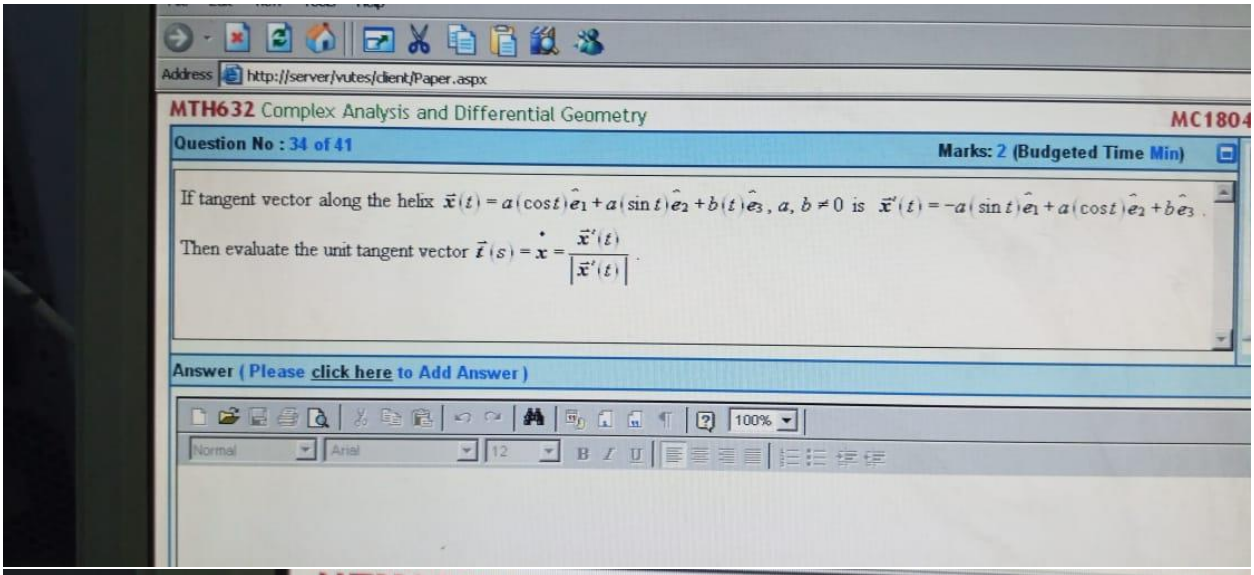
MTH632 Complex Analysis and Differential Geometry
Question No : 35 of 41
Evaluate $\lim_{z \to i} \frac{(z^2+4z+2)}{z+1}$.
Answer (Please click here to Add Answer)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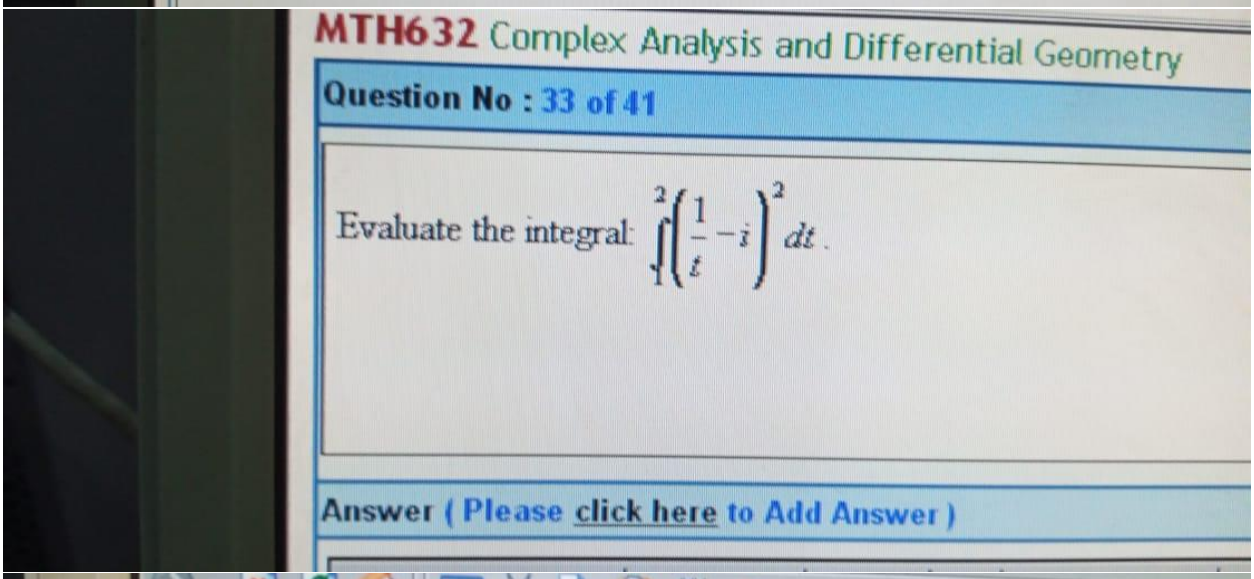
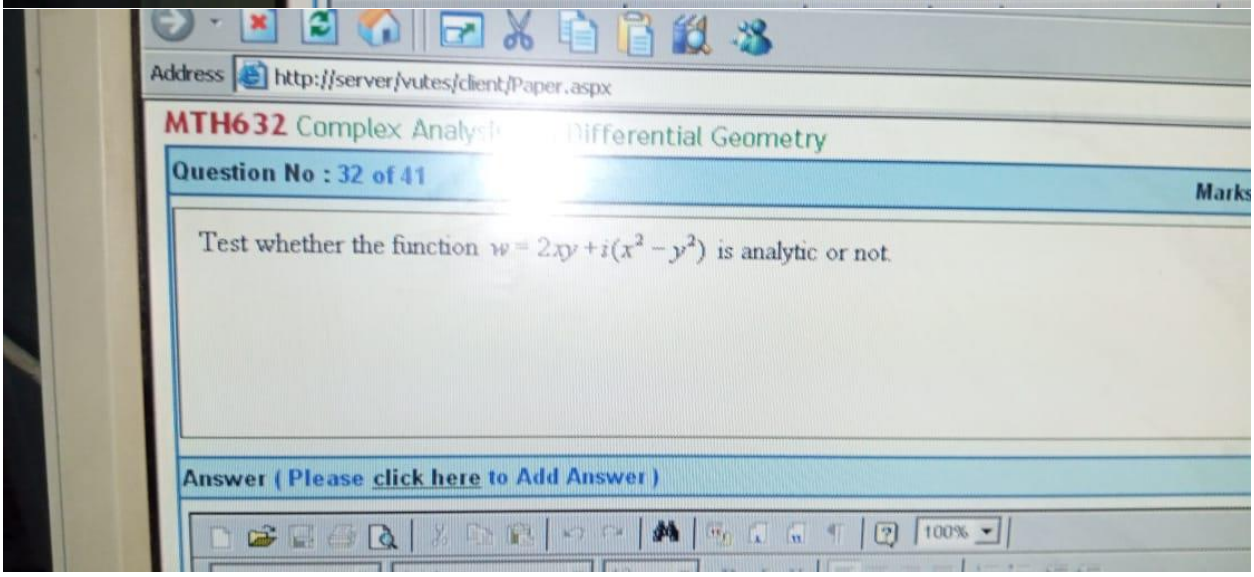
MTH632 Complex Analysis and Differential Geometry
MC1804
Question No : 34 of 41
Marks: 2 (Budgeted Time Min)
If tangent vector along the helix $\vec{x}(t) = a(\cos t)\hat{e}_1 + a(\sin t)\hat{e}_2 + b(t)\hat{e}_3$, $a, b \neq 0$ is $\vec{x}'(t) = -a(\sin t)\hat{e}_1 + a(\cos t)\hat{e}_2 + b\hat{e}_3$.
Then evaluate the unit tangent vector $\vec{t}(s) = \dot{\vec{x}} = \frac{\vec{x}'(t)}{|\vec{x}'(t)|}$.
Answer (Please click here to Add Answer)
100%
Normal
Arial
12


MTH632 Complex Analysis and Differential Geometry
Question No : 33 of 41
Evaluate the integral: $\int_1^2 \left(\frac{1}{t} - i\right)^2 dt$.
Answer (Please click here to Add Answer)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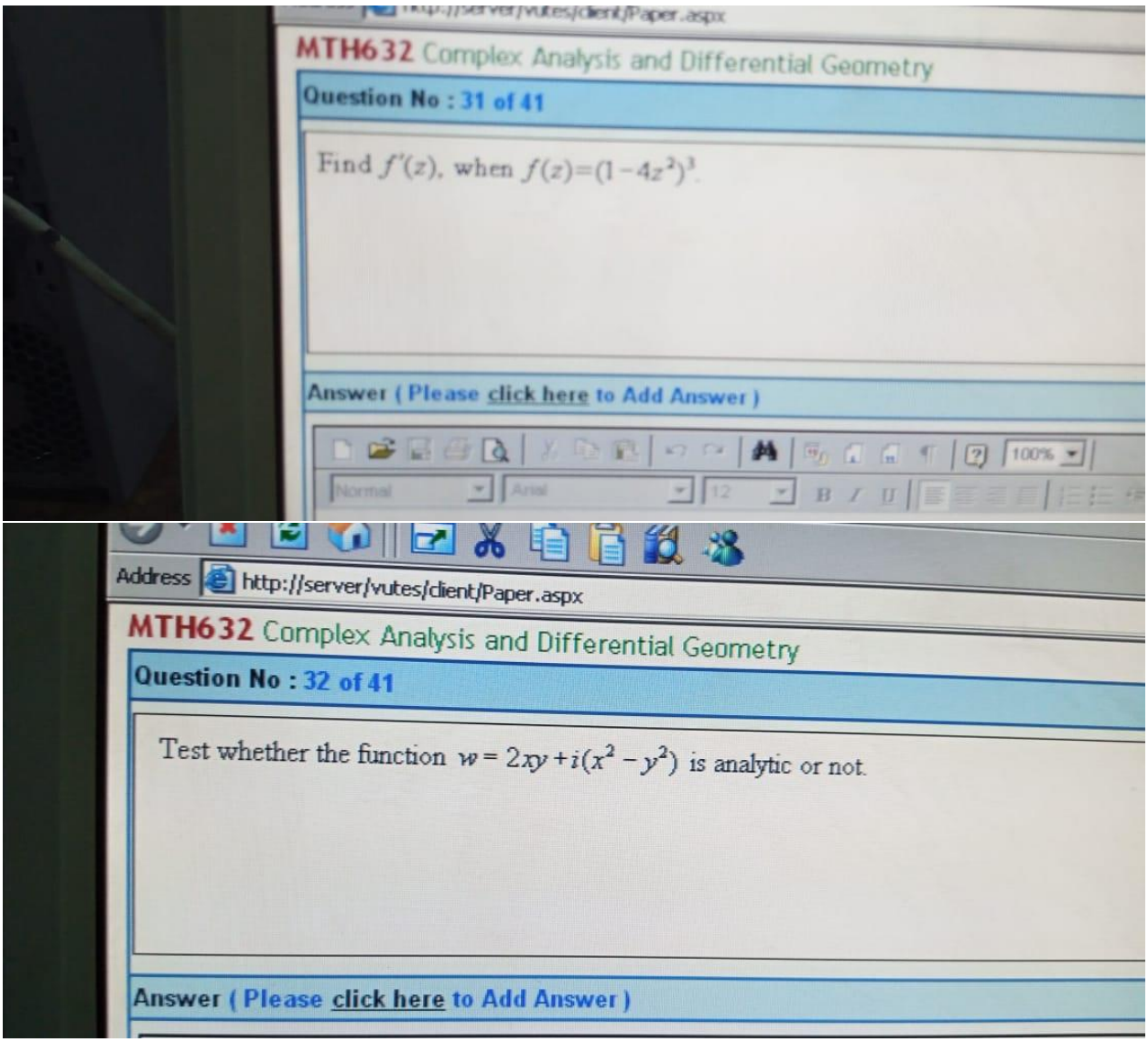
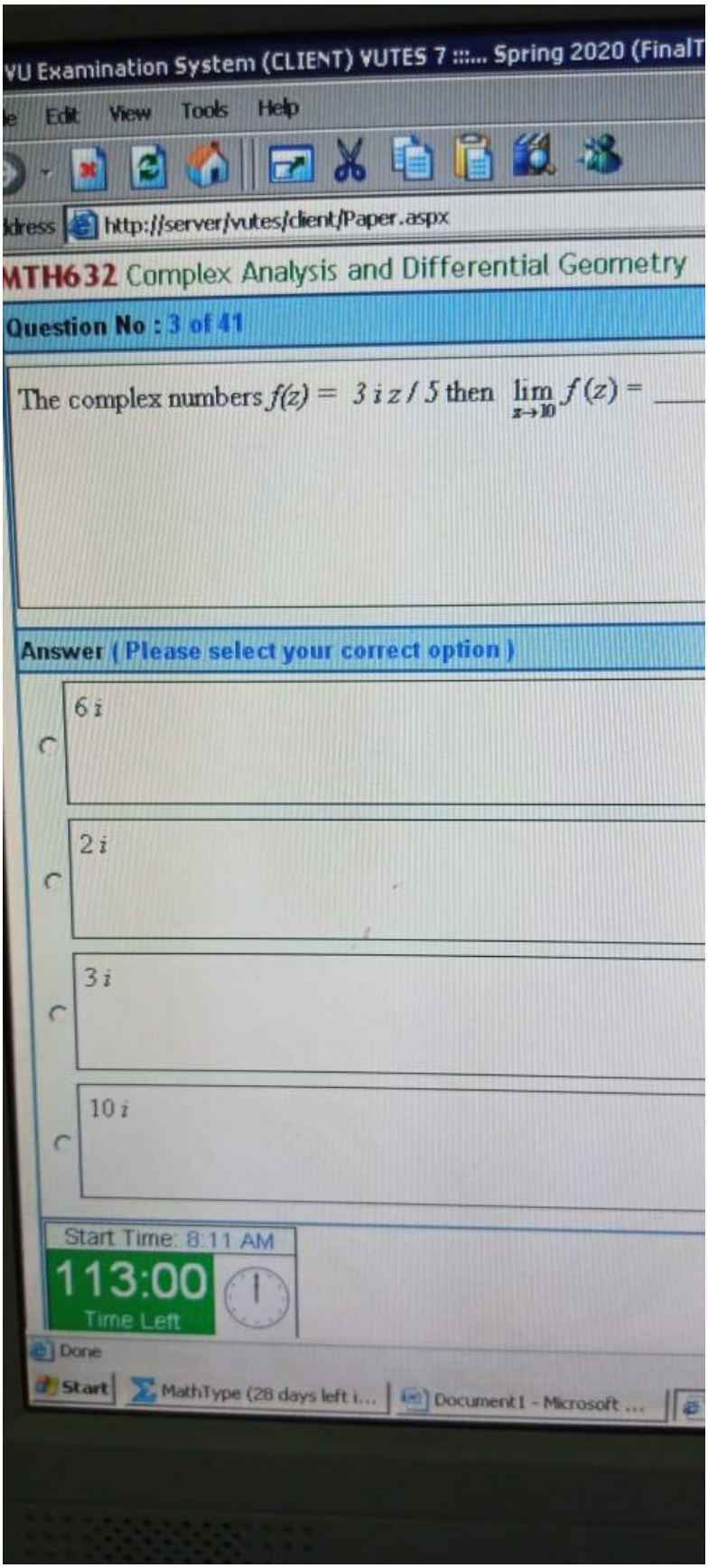
MTH632 Complex Analysis and Differential Geometry
Question No : 32 of 41
Marks
Test whether the function $w = 2xy + i(x^2 - y^2)$ is analytic or not.
Answer (Please click here to Add Answer)
100%


MTH632 Complex Analysis and Differential Geometry
Question No : 31 of 41
Find $f'(z)$, when $f(z)=(1-4z^2)^3$.
Answer (Please click here to Add Answer)
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 32 of 41
Test whether the function $w=2xy+i(x^2-y^2)$ is analytic or not.
Answer (Please click here to Add Answer)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MTH632 Complex Analysis and Differential Geometry
Question No : 3 of 41
The complex numbers $f(z) = 3iz/5$ then $\lim_{z \to 10} f(z) =$ ____
Answer (Please select your correct option)
$6i$
$2i$
$3i$
$10i$
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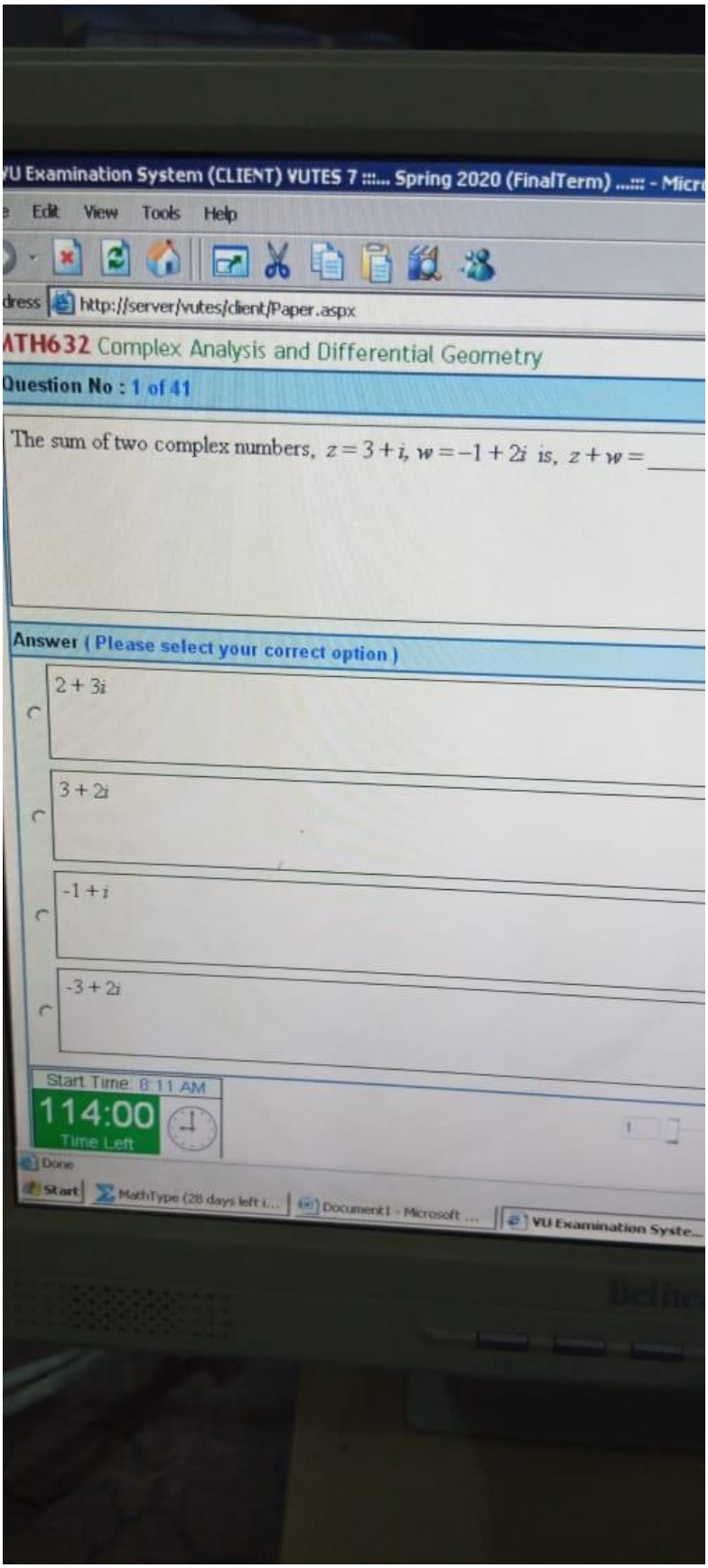
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ATH632 Complex Analysis and Differential Geometry
Question No : 1 of 41
The sum of two complex numbers, $z = 3 + i$, $w = -1 + 2i$ is, $z + w =$ ____
Answer (Please select your correct option)
2 + 3i
3 + 2i
-1 + i
-3 + 2i
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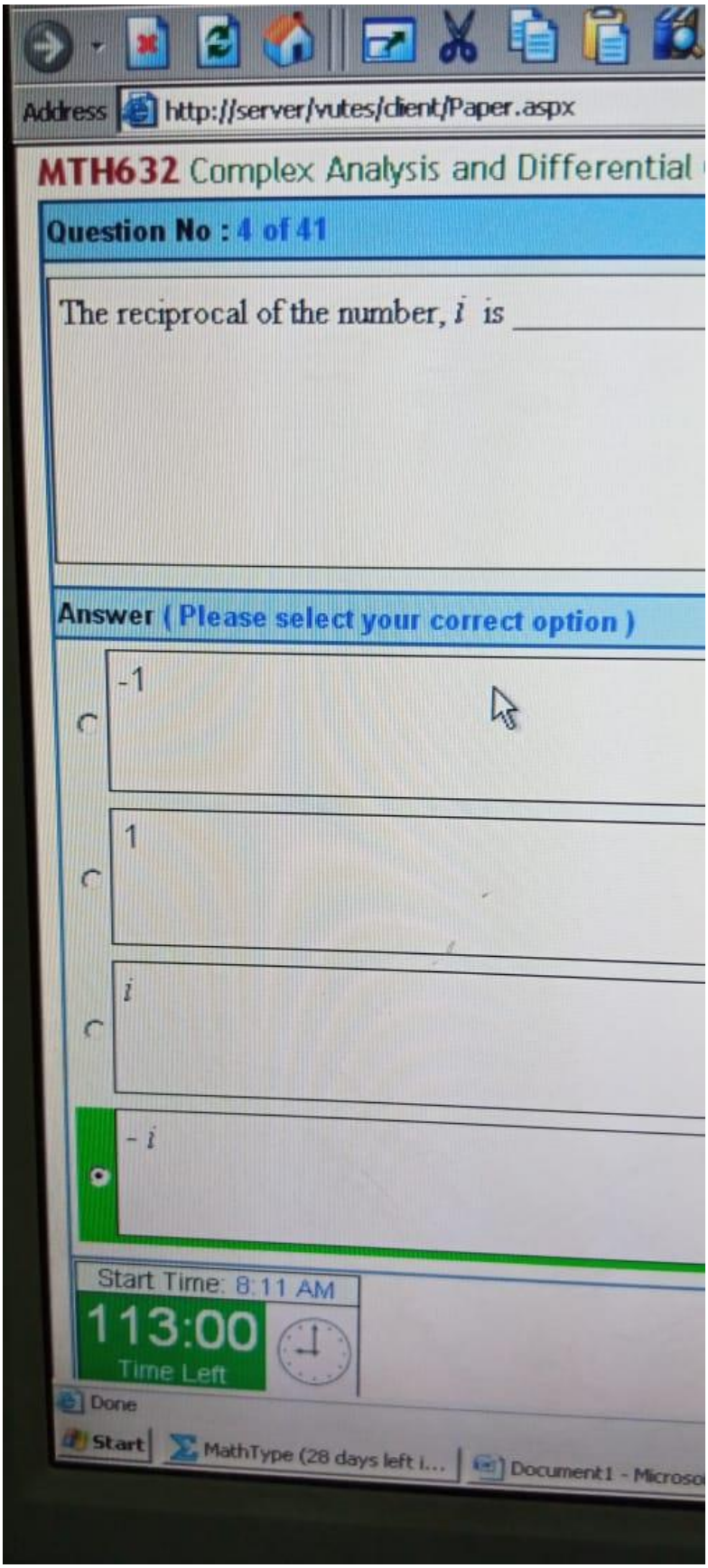
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MTH632 Complex Analysis and Differential
Question No : 4 of 41
The reciprocal of the number, i is _____
Answer (Please select your correct option)
-1
1
i
$-i$
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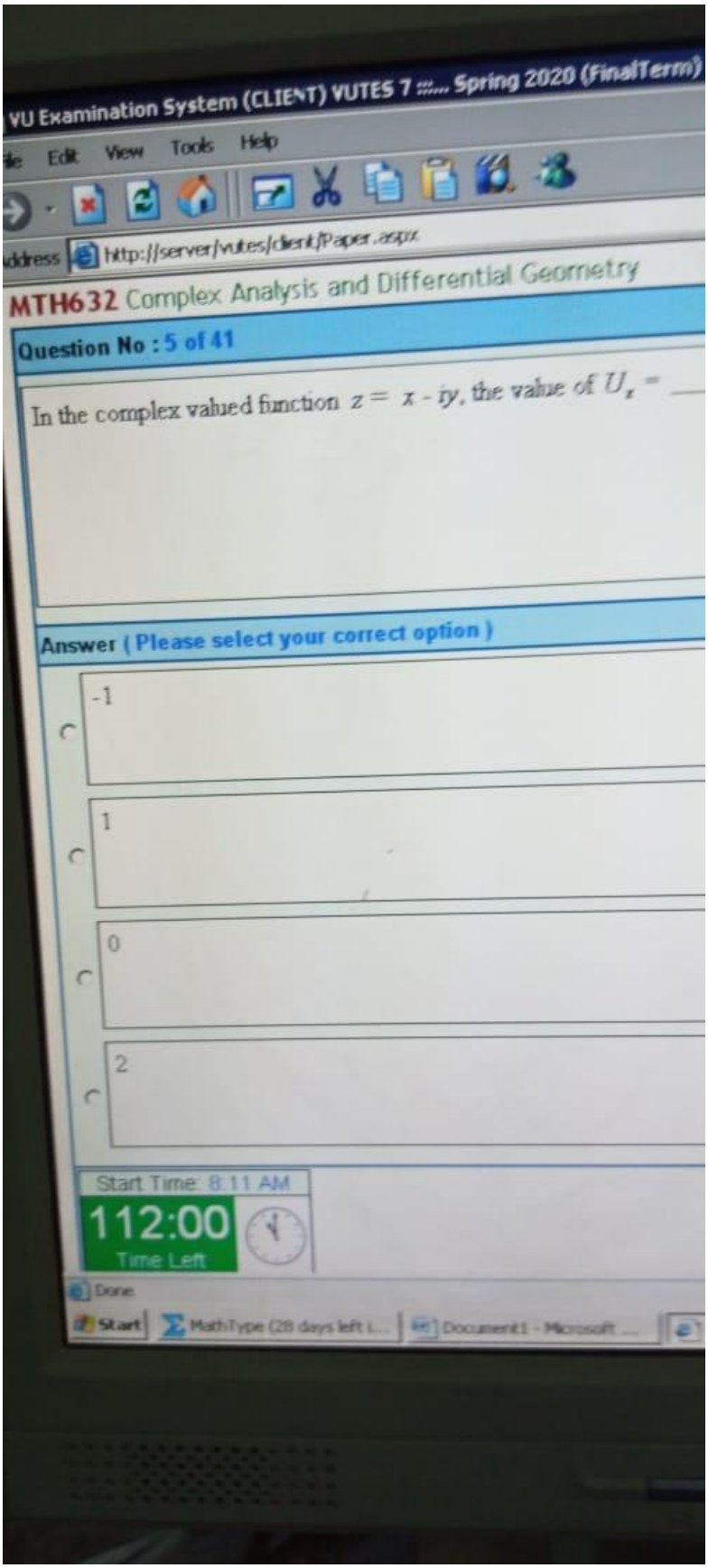
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MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 41
In the complex valued function $z = x - iy$, the value of $U_x =$ ___
Answer (Please select your correct option)
-1
1
0
2
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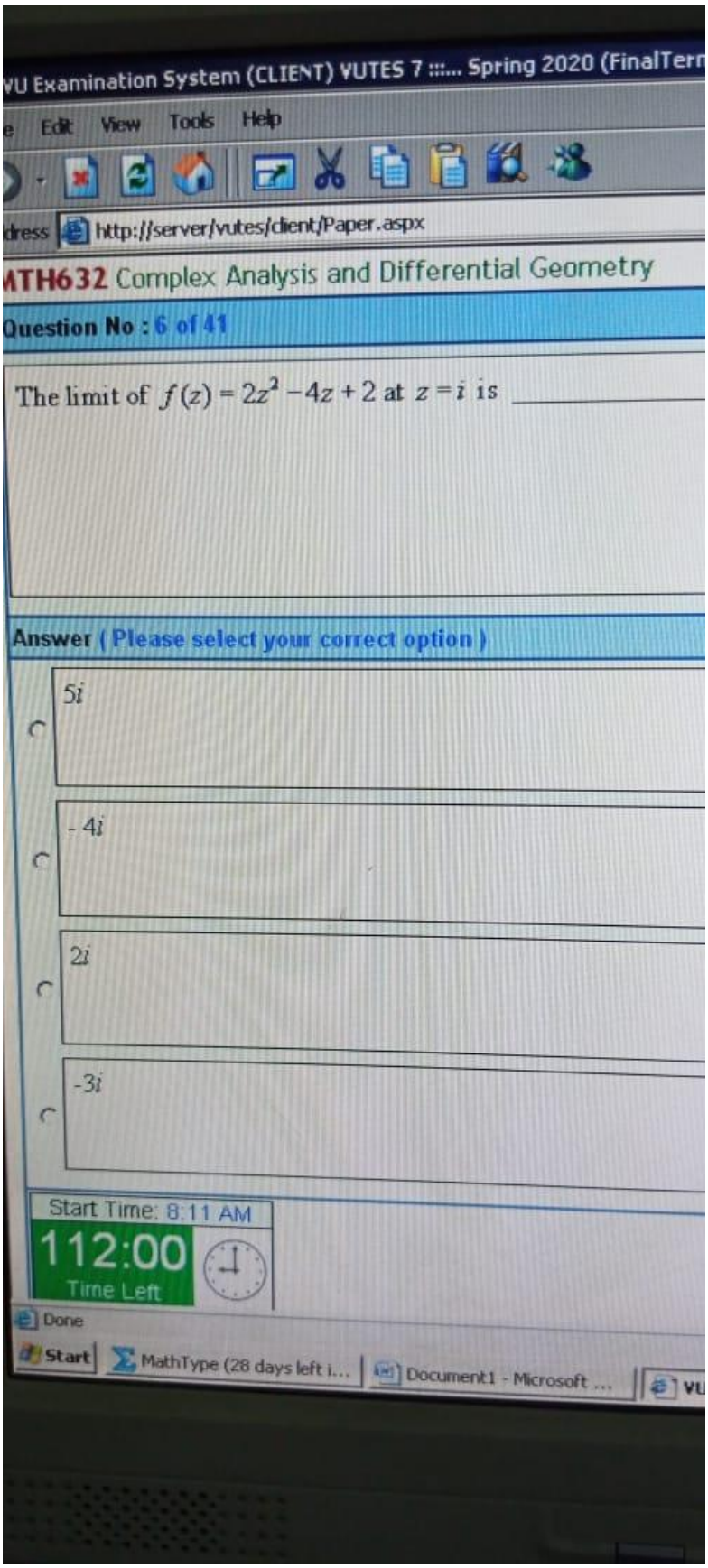
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MTH632 Complex Analysis and Differential Geometry
Question No : 6 of 41
The limit of $f(z) = 2z^2 - 4z + 2$ at $z = i$ is _____
Answer (Please select your correct option)
5i
- 4i
2i
-3i
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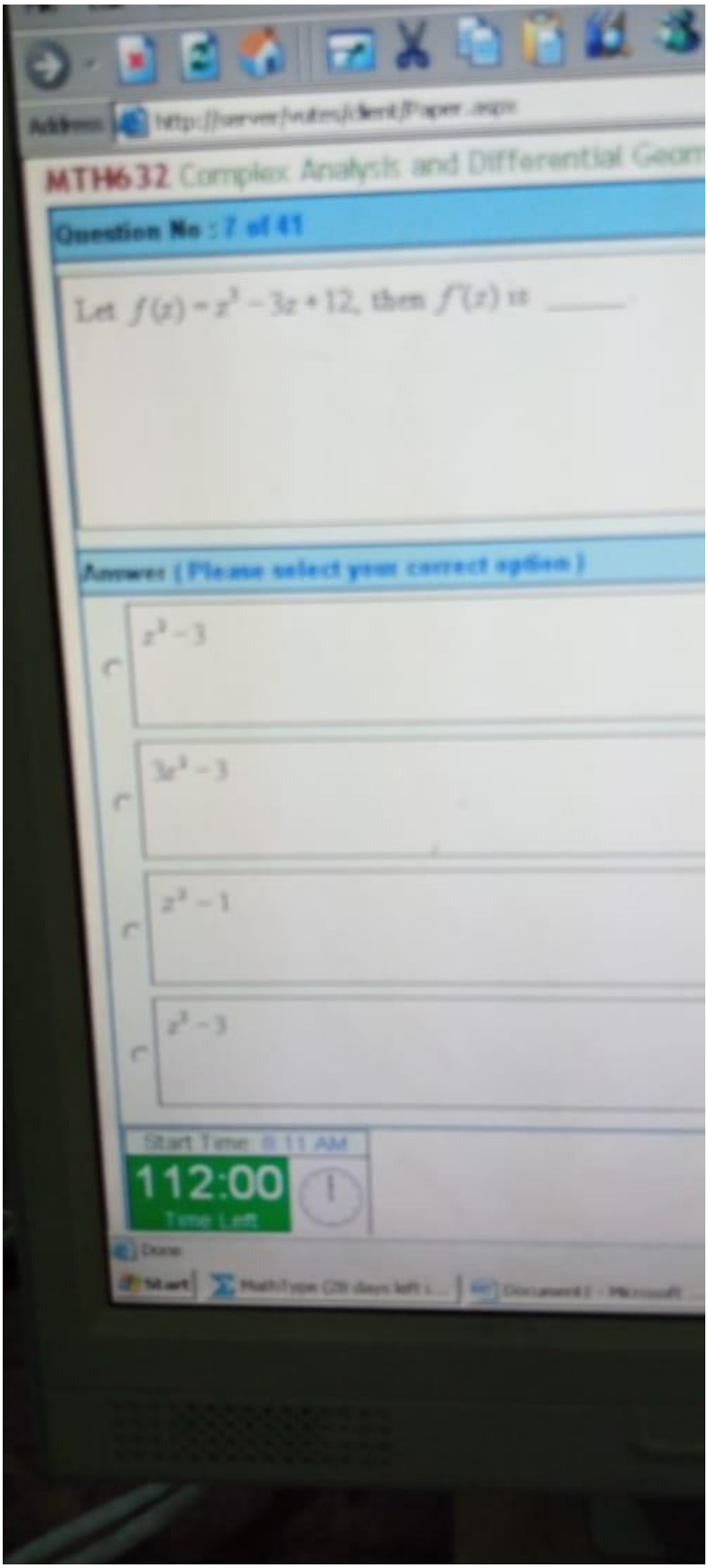
MTH632 Complex Analysis and Differential Geom
Question No : 7 of 41
Let $f(z) = z^3 - 3z + 12$, then $f'(z)$ is _____
Answer (Please select your correct option)
$z^3 - 3$
$3z^3 - 3$
$z^3 - 1$
$z^3 - 3$
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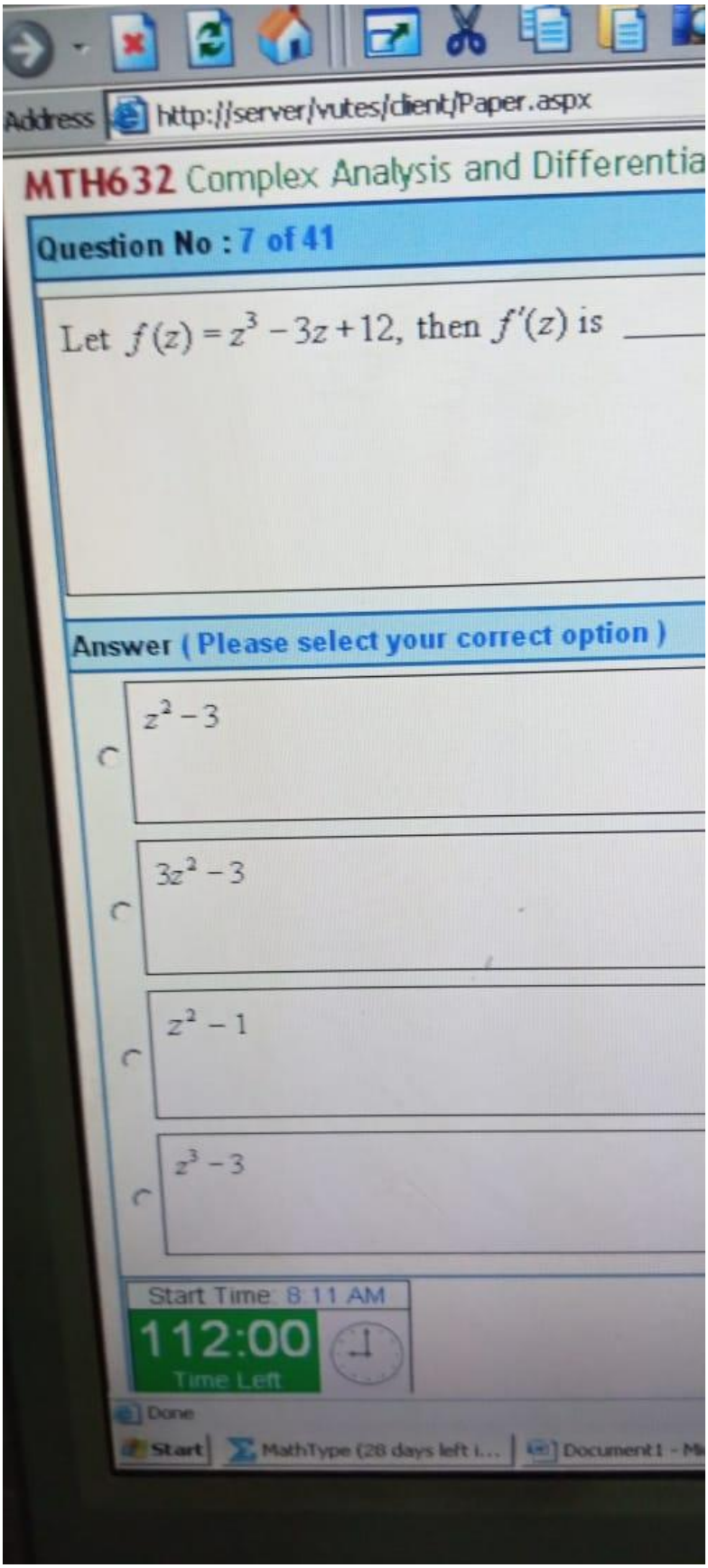
Time Left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MTH632 Complex Analysis and Differentia
Question No : 7 of 41
Let $f(z)=z^3-3z+12$, then $f'(z)$ is ____
Answer (Please select your correct option)
z^2-3
$3z^2-3$
z^2-1
z^3-3
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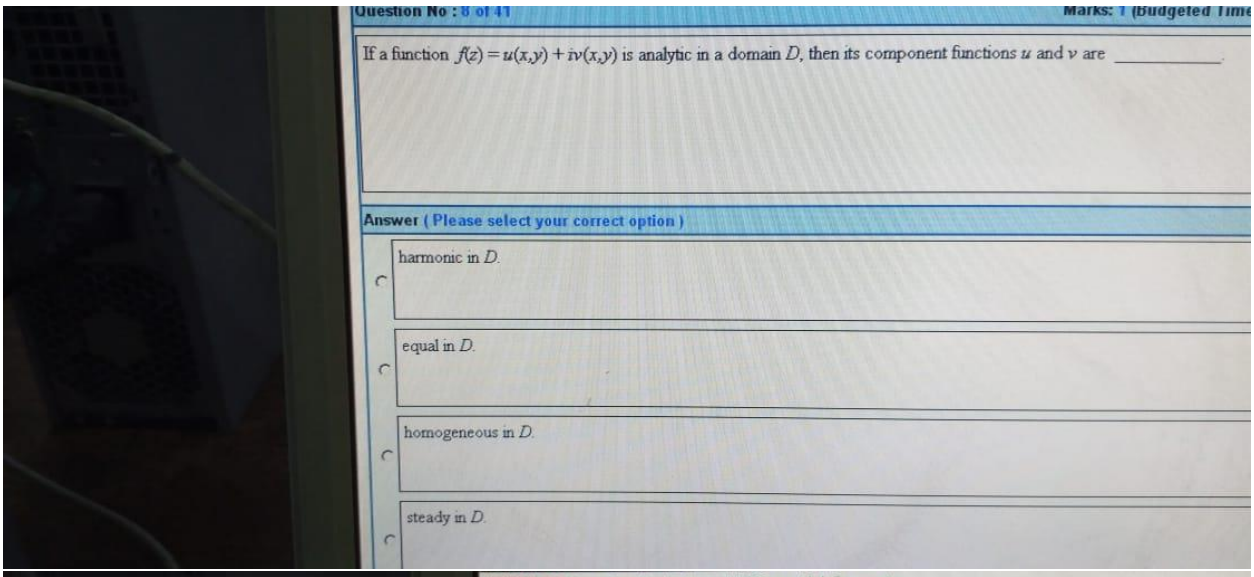
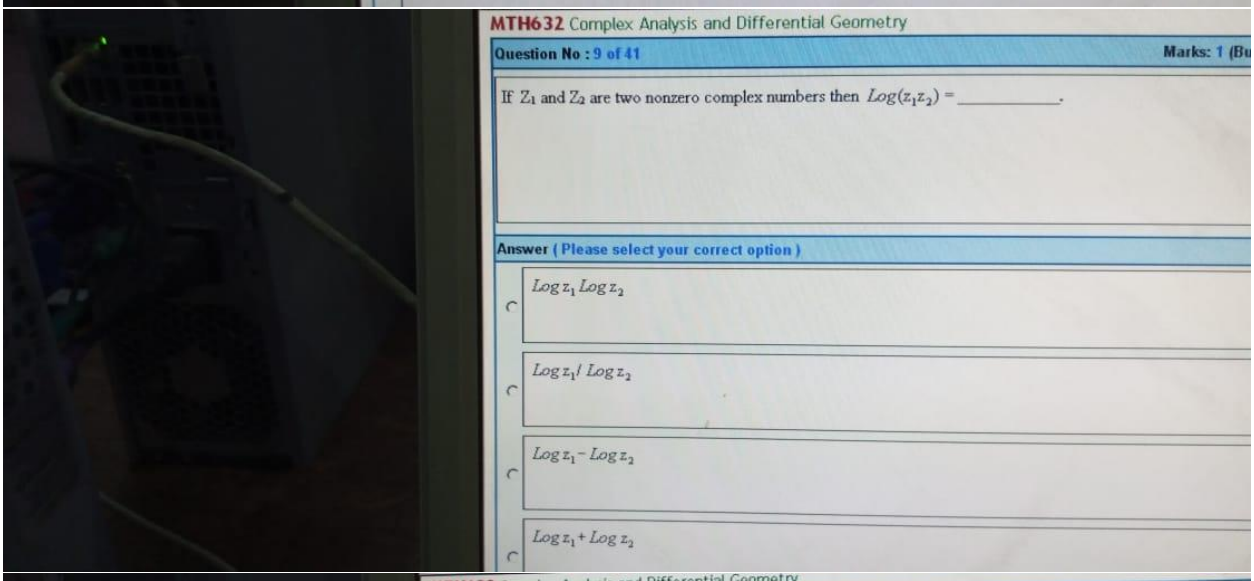
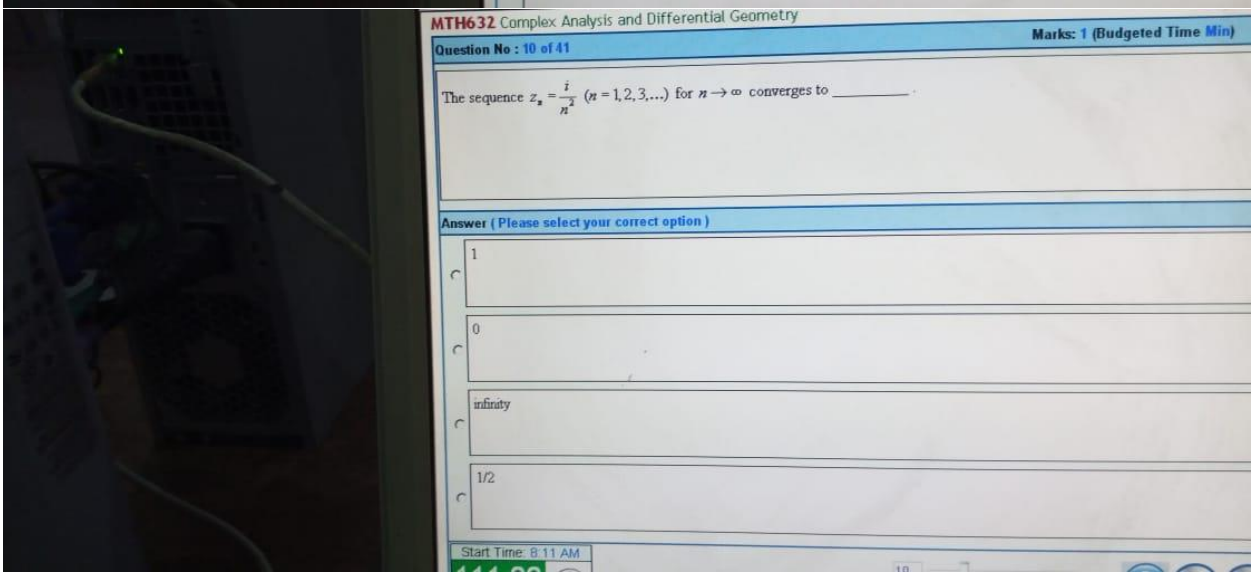

Question No : 8 of 41
Marks: 1 (Budgeted Time
If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____.
Answer (Please select your correct option)
harmonic in D.
equal in D.
homogeneous in D.
steady in D.


MTH632 Complex Analysis and Differential Geometry
Question No : 9 of 41
Marks: 1 (Bu
If Z_1 and Z_2 are two nonzero complex numbers then $Log(z_1z_2)$ = _____.
Answer (Please select your correct option)
$Log\, z_1\, Log\, z_2$
$Log\, z_1 / Log\, z_2$
$Log\, z_1 - Log\, z_2$
$Log\, z_1 + Log\, z_2$


MTH632 Complex Analysis and Differential Geometry
Question No : 10 of 41
Marks: 1 (Budgeted Time Min)
The sequence $z_n = \frac{i}{n^2}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
1
0
infinity
1/2
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MTH032 Complex Analysis and Differential Geometry

Question No : 12 of 41 **Marks: 1 (Budgeted Tim**

If a series of complex numbers converges, the n^{th} term converges to zero as n tends to _____.

Answer (Please select your correct option)

- infinity
- finite numbers

MTH032 Complex Analysis and Differential Geometry

Question No : 11 of 41 **Marks: 1 (Budgeted Time Min)**

If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____.

Answer (Please select your correct option)

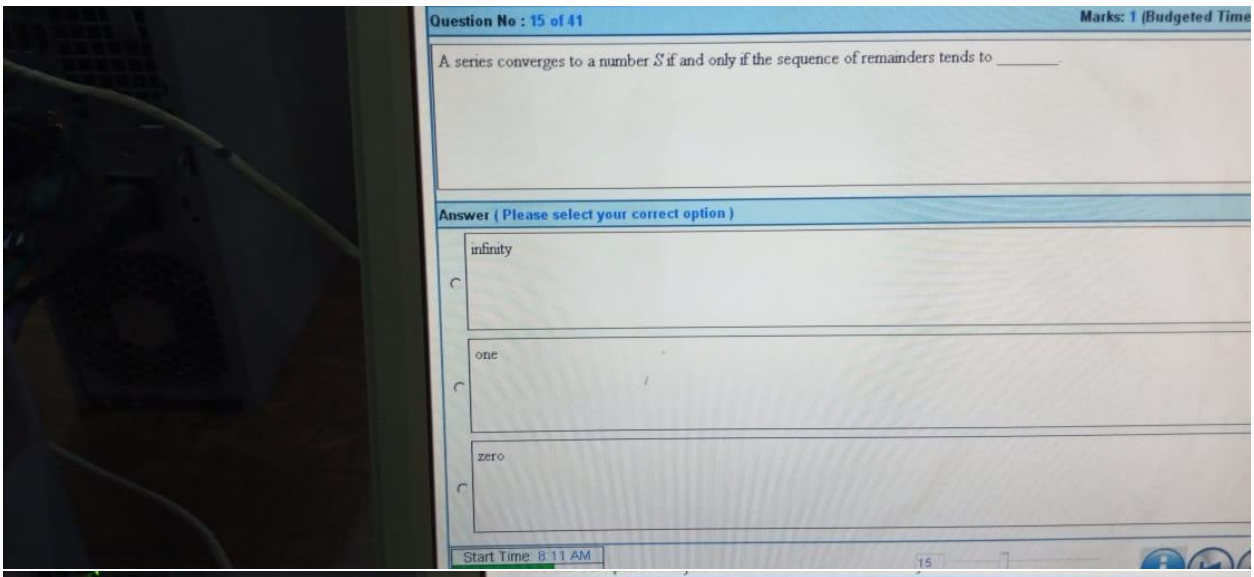
- $c=1$ and $a=-b$
- $a=1$ and $c=-b$
- $b=1$ and $a=-c$
- $a=b=c=1$

Question No : 13 of 41 **Marks: 1 (Budg**

If U and V are harmonic functions then $f(Z) = U + iV$ is _____.

Answer (Please select your correct option)

- analytic function.
- need not be analytic function.
- analytic function only at Z=0.
- none of the others.


Question No : 15 of 41
Marks: 1 (Budgeted Time
A series converges to a number S if and only if the sequence of remainders tends to _____.
Answer (Please select your correct option)
infinity
one
zero
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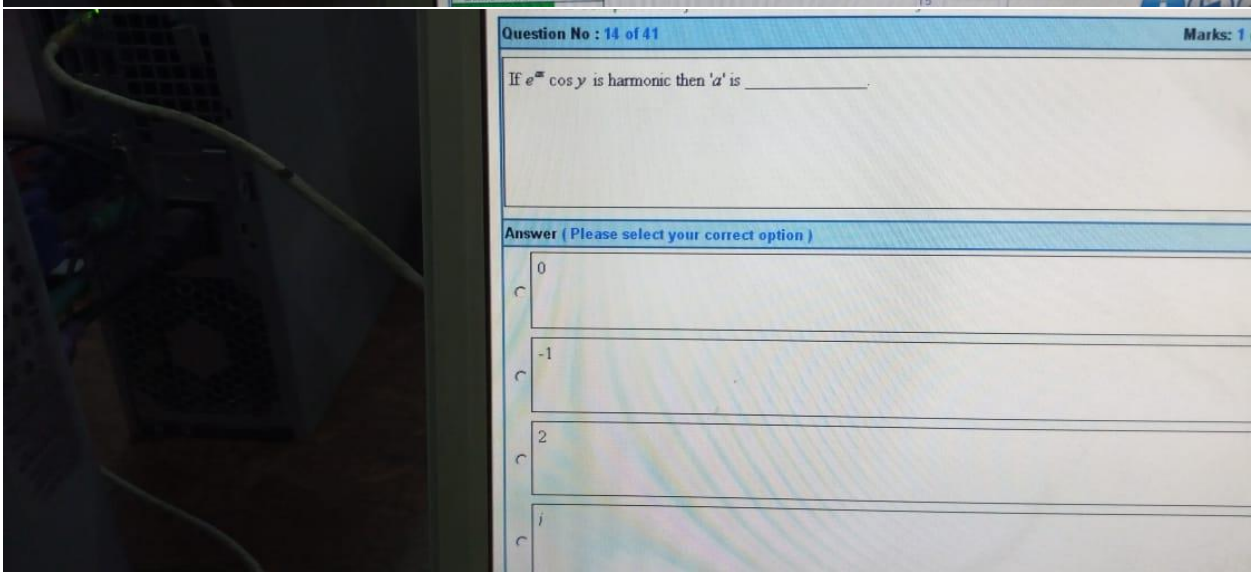
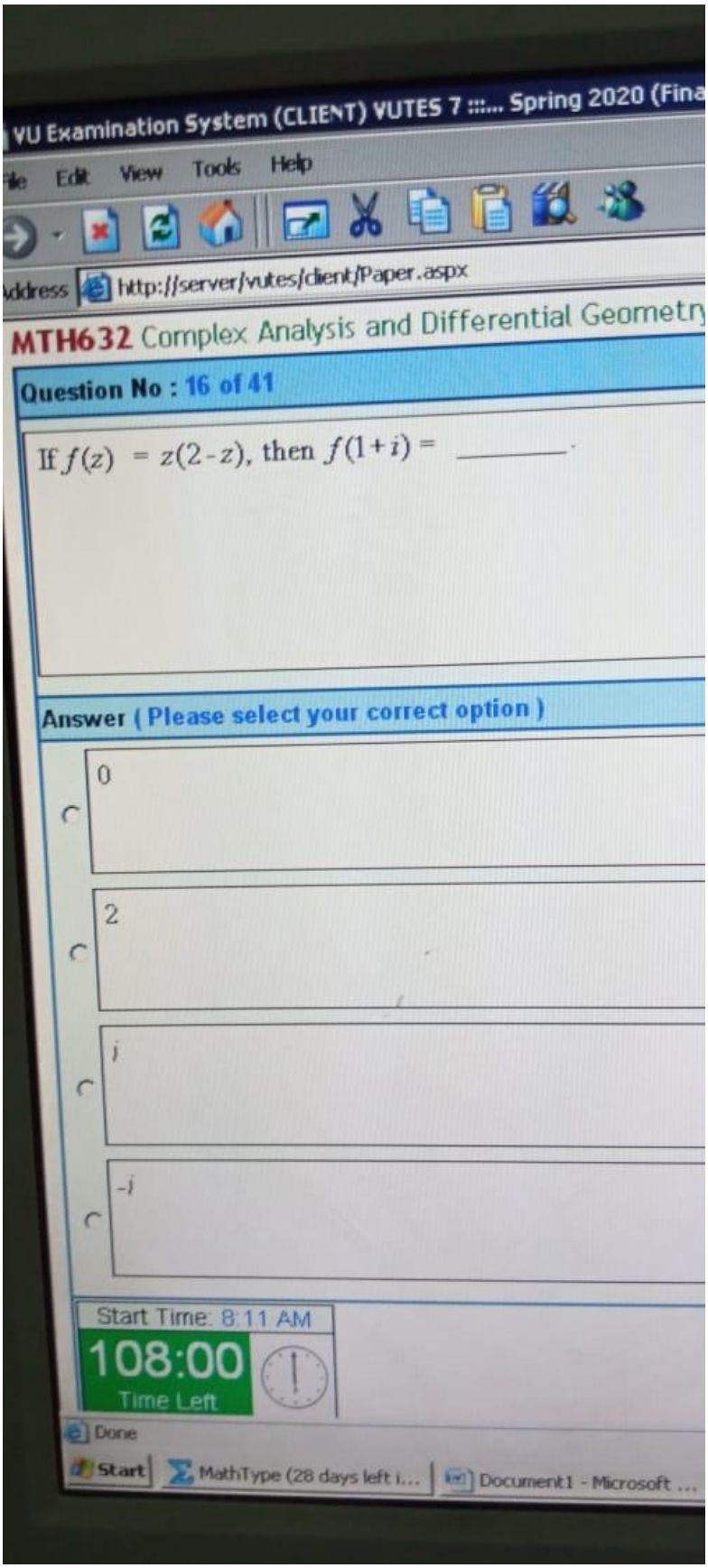

Question No : 14 of 41
Marks: 1
If $e^{ax}\cos y$ is harmonic then 'a' is _____.
Answer (Please select your correct option)
0
-1
2
i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MTH632 Complex Analysis and Differential Geometry
Question No : 16 of 41
If $f(z) = z(2-z)$, then $f(1+i) =$ _____.
Answer (Please select your correct option)
0
2
i
-i
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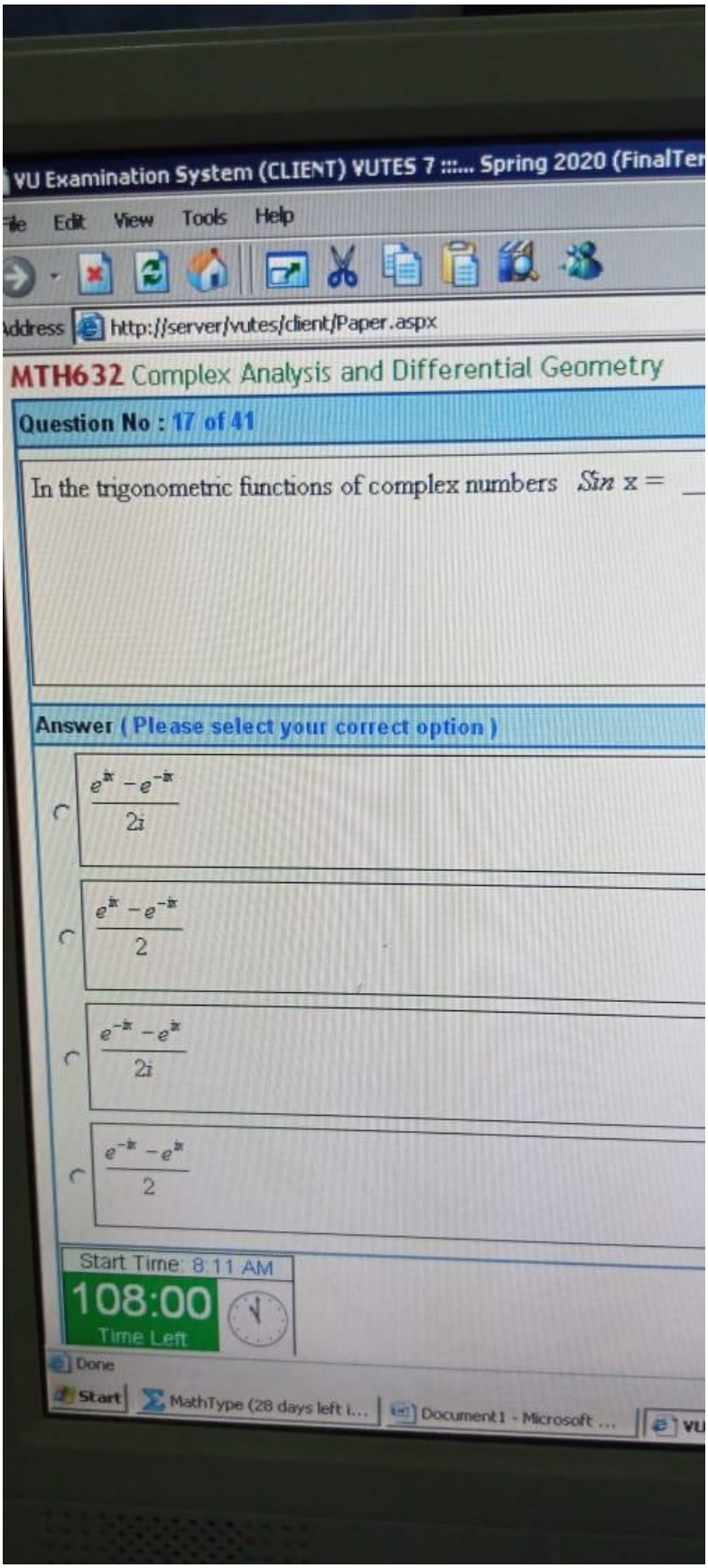
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MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 41
In the trigonometric functions of complex numbers Sin x =
Answer (Please select your correct option)
$\frac{e^{ix} - e^{-ix}}{2i}$
$\frac{e^{ix} - e^{-ix}}{2}$
$\frac{e^{-ix} - e^{ix}}{2i}$
$\frac{e^{-ix} - e^{ix}}{2}$
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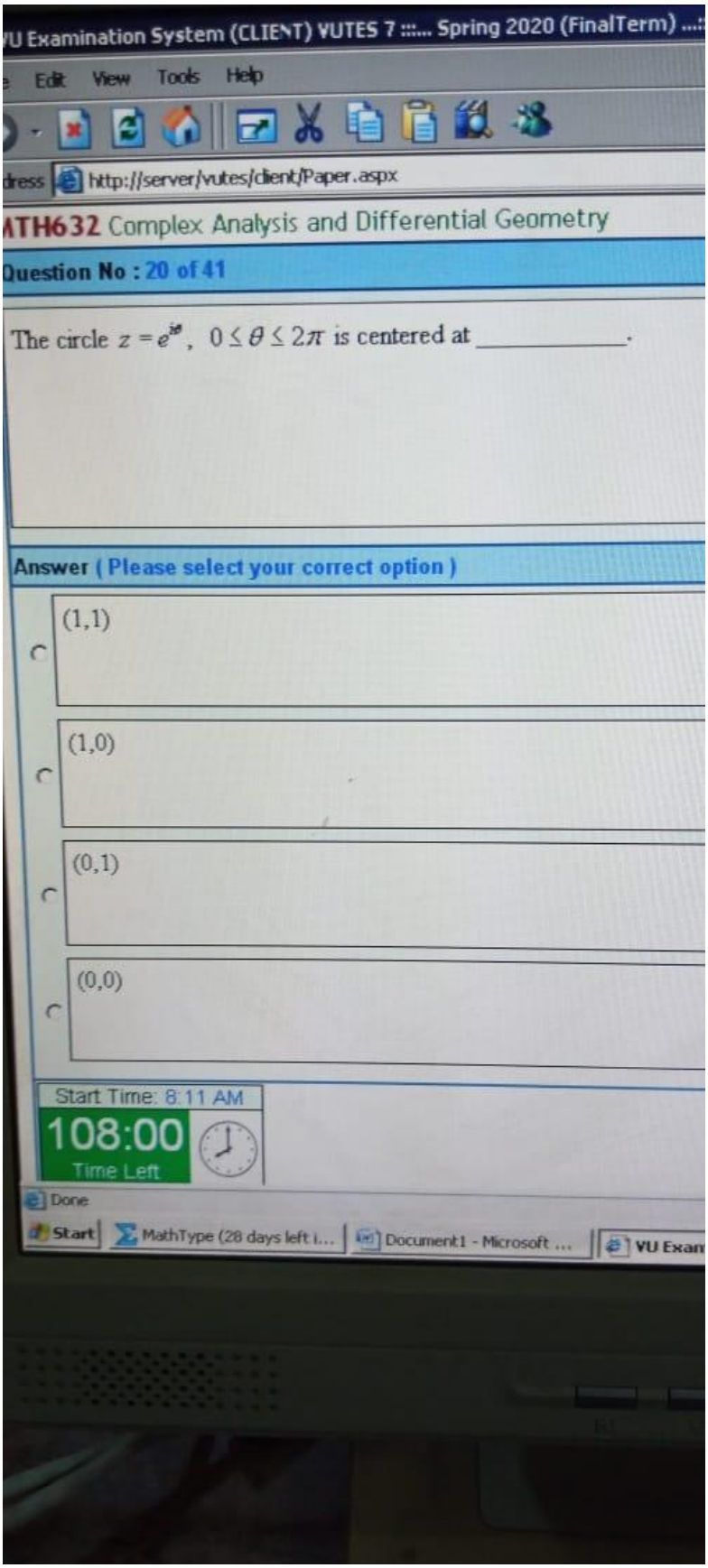
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TH632 Complex Analysis and Differential Geometry
Question No : 20 of 41
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
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Question No : 23 of 41 Marks: 1 (Budgeted Tir

Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C \frac{z^2}{z-3}dz =$ _____ .

Answer (Please select your correct option)

- 3
- 0
- 1
- -3

MTH632 Complex Analysis and Differential Geometry

Question No : 24 of 41 Marks: 1 (Budgeted Time Min)

Let C be the positively oriented circle $|z| = 2$. Since the function $f(z) = \frac{z}{9-z^2}$ is analytic within and on C and since the point $z_0 = -i$ is interior to C, then using the Cauchy Integral formula $\int_C \frac{z}{(9-z^2)(z+i)}dz =$ _____ .

Answer (Please select your correct option)

- $\frac{\pi}{9}$
- $\frac{\pi}{5}$
- 2π
- $2\pi i$